

Morphological Evolution of Galaxies with Deep Learning

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Abstract

Galaxies exhibit a wide range of characteristic shapes and structures, from spheroidal, featureless ellipticals to discs with prominent spiral arms. Understanding how these morphologies vary over cosmic timescales can reveal important insights into the physical processes that govern the formation and evolution of galaxies. Galaxy morphologies are typically classified manually through the visual inspection of galaxy images. However, this approach is laborious and is unable to effectively scale to larger datasets. In this thesis, we develop a suite of deep learning models that are capable of rapidly and efficiently classifying galaxies. In particular, we train convolutional neural networks (CNNs) to classify galaxies into up to four morphological types; elliptical, lenticular, spiral and irregular/miscellaneous. In addition to these morphologies, we further train separate CNNs to classify barred galaxies. Our CNNs utilise a new model architecture that outperforms existing architectures at morphological classification.

Our deep learning approach is not only efficient but highly versatile, enabling us to classify images of both real and simulated galaxies from multiple datasets. These include low-redshift images from the Sloan Digital Sky Survey (SDSS) and SAMI Survey, high-redshift images from the Cosmological Evolution Survey (COSMOS), and images of simulated galaxies from the EAGLE hydrodynamical cosmological simulation. These classifications directly allow us to understand the formation and evolution of barred galaxies in EAGLE. Our models also allow us to examine the redshift evolution of the fraction of lenticular galaxies in COSMOS, finding that lenticular galaxies are more common at lower redshifts. We also identify a sub-population of lenticular galaxies with properties that cannot be fully explained under the “faded spiral” formation scenario. Additionally, we apply our models to galaxies from the SAMI survey in order to establish the kinematic differences between different morphologies, including barred and unbarred galaxies. Lastly, we also extend our suite of deep learning models to include a model capable of morphological segmentation, and use it to estimate the physical lengths of stellar bars.

This thesis demonstrates the effectiveness of a deep learning approach to morphological classification and shows that deep learning models can be successfully utilised to study the evolution of galaxy morphology across multiple settings from simulations to observations, including at both low and high redshifts. Through consistently classifying the morphologies of thousands of galaxies across multiple epochs in time, we gain insights into the timescales and drivers of morphological change, and ultimately into the processes governing the evolution of galaxies. Our models are well positioned to tackle the huge volumes of image data anticipated from future large-scale surveys.

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Declaration

I, Mitchell Kee Cavanagh, certify that this thesis has been substantially accomplished during enrolment in the degree.

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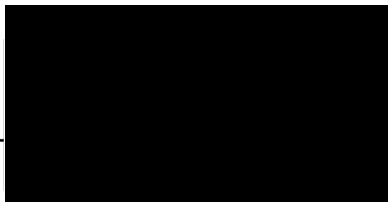
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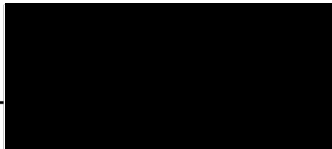
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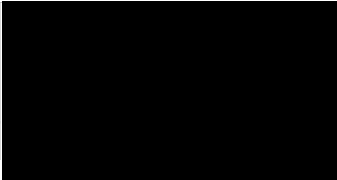
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Introduction

Our Universe contains an incomprehensibly vast number of galaxies, with modern estimates proposing a total number in the order of hundreds of billions at least. Observations have shown that galaxies are incredibly diverse, displaying a range of different characteristic shapes and structures from round, featureless spheroids, to discs with vibrant, well-defined spiral arms, to galaxies with prominent, central stellar bars (Hubble, 1936; de Vaucouleurs, 1963). The overall visual shape and structure of a galaxy is referred to as its morphology. Accounting for the processes that govern the formation and evolution of such diverse morphologies is an ongoing challenge in astronomy with crucial theoretical ramifications.

It is well established that a galaxy’s morphology is a fundamental property, intimately linked to its physical and dynamical properties (Conselice, 2014). For instance, spiral galaxies are typically rotationally-supported discs exhibiting high rates of ongoing star formation, as manifested in their sweeping spiral arms. This stands in stark contrast to elliptical galaxies, which are mostly quiescent spheroids with high degrees of velocity dispersion and large bulge components. Lenticular galaxies, also known as S0 galaxies, exhibit both a bulge and disk component, but lack any clear spiral arms and otherwise look visually similar to ellipticals, especially when viewed face on (Buta, 2013). As such, they are considered an intermediate type between ellipticals and spirals. The majority of spiral galaxies contain centrally-positioned, rectangular structures known as stellar bars (Aguerri et al., 2009), however there are many open theories as to how such bars form. Understanding how prevalent different galaxy morphologies are in the present-day Universe, as well as how the fractions of galaxies in different morphological types vary over cosmic timescales, is paramount to gleaning insights into the processes that govern the formation and evolution of galaxies (Dressler, 1980; Dressler et al., 1997; Couch et al., 1998; Conselice, 2014). Furthermore, observations of galaxy morphologies serve as important constraints on the fidelity of cosmological simulations (Springel et al., 2005; Somerville & Davé, 2015; Vogelsberger et al., 2020).

Galaxies are typically classified into certain morphological types via visual inspection, typically by groups of expert astronomers (Nair & Abraham, 2010a; Buta, 2013; Moffett et al., 2016) or through the combined efforts of crowdsourcing and citizen science (Lintott et al., 2011; Willett et al., 2013; Masters et al., 2021). However, this process is laborious and quickly becomes intractable for large volumes of data. Modern astronomy sits at the cusp of generational leap in the capabilities of large-scale surveys, which are expected to yield data on hundreds of thousands, if not billions of galaxies. Examples include the Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST, Ivezić et al. 2019) and the Square Kilometre Array. For volumes of data at this scale and beyond, morphological classification by visual inspection is

infeasible (Silva et al., 2018). In this thesis, we tackle this issue by developing a suite of models, based on deep learning, which are capable of automatically classifying galaxy morphologies in a rapid, efficient and consistent manner, simply by analysing their images. We demonstrate that these models can readily scale to high volumes of data through carrying out large-scale studies into the physical properties and evolution of different galaxy morphologies in both observations and simulations. Our deep learning techniques enable us to examine the evolution of barred galaxies in the EAGLE cosmological simulation (Crain et al., 2015; Schaye et al., 2015), establish the growth of lenticular galaxies over time through classifying over 85,000 galaxies in the Cosmic Evolution Survey (Scoville et al., 2007), distinguish different morphologies based on their kinematics in the SAMI survey (Croom et al., 2021a), and even estimate the physical lengths of the bars in barred galaxies with a specialised model capable of morphological segmentation.

1.1 Galaxy Morphology

1.1.1 Morphological Classification

The first step to studying different morphologies is to utilise a logical and consistent system with which to categorise them. Traditionally, such classification systems define and distinguish different morphologies based on their qualitative, visual appearance (Sandage, 2005; Nair & Abraham, 2010a; Buta, 2013; Vázquez-Mata et al., 2022). One of the earliest visual classification systems used to study morphology is the Hubble sequence (also known as the Hubble Tuning Fork) (Hubble, 1936), which organises galaxies into three main morphological types. As illustrated in Figure 1.1, the Hubble sequence represents morphology along

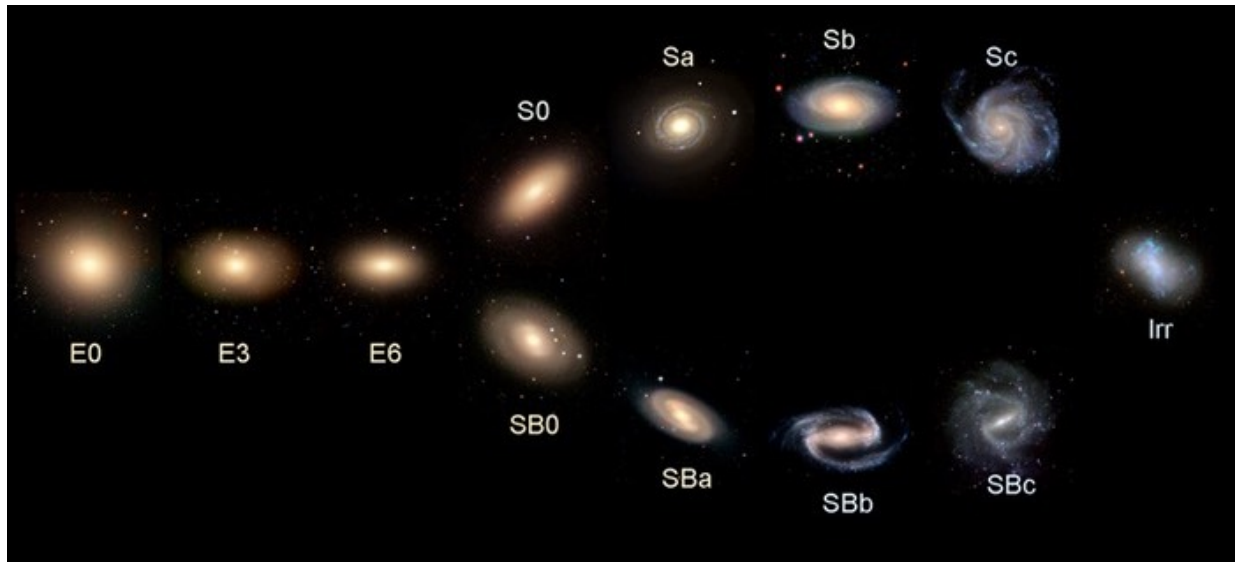


Figure 1.1: An illustration of how galaxy morphologies are categorised and represented with the Hubble Tuning fork. Image credit: University of Iowa.

a main axis, with elliptical galaxies at one extreme (subdivided by varying ellipticities), spirals on the opposite extreme, and lenticular galaxies occupying the middle. Spirals are further split into two branches depending on whether they contain a stellar bar (some variations of the Hubble sequence also include a similar split for lenticulars, as in Figure 1.1). Irregular galaxies, that is galaxies with non-standard

morphologies that do not fit into any of the other classes, are not explicitly included in the original Hubble sequence, but are typically added for completeness as an additional class appended past the spiral branches. The Hubble sequence has since been massively refined and expanded to include additional subtypes (see [de Vaucouleurs 1963](#); [de Vaucouleurs et al. 1991](#); [Buta 2013](#); [Buta et al. 2015](#)). One important extension is the quantitative “T-Type” system, in which numerical values are assigned to different morphological subtypes. In general, positive values denote spiral morphologies (with higher numbers denoting later-stage spirals), whilst negative values and values close to 0 indicate elliptical and lenticular morphologies respectively ([de Vaucouleurs et al., 1991](#)). While the Hubble sequence remains the dominant classification scheme, it is not without its limitations. Common criticisms argue that the Hubble sequence does not encode physical properties ([Irwin, 1995](#)), that it is ill-suited to describing very high redshift galaxies ([Abraham & van den Bergh, 2001](#)) (although preliminary results from the JWST may counter this; see [Ferreira et al. 2022a](#)), and that it is ultimately too subjective ([Roberts & Haynes, 1994](#)). More recent classification systems utilise alternative definitions based on more objective criteria such as physical, kinematic and/or spectroscopic properties ([Cappellari et al., 2011](#); [Kalinova et al., 2017](#); [Graham, 2019](#); [van de Sande et al., 2021](#)), or instead define morphology in terms of hierarchical decision trees ([Willett et al., 2013](#); [Walmsley et al., 2021](#)).

Although it is possible to distinguish between many morphological types purely from visual inspection ([Willett et al., 2013](#); [Buta, 2013](#)), some morphologies are easier to classify than others. For instance, lenticular galaxies are notoriously difficult to classify. In general, it is important to note that morphological classification by visual inspection is associated with a high degree of uncertainty. It is common even for expert astronomers to disagree on a substantial fraction of samples ([Kelvin et al., 2014](#); [Cortese et al., 2016](#); [Moffett et al., 2016](#); [Pfeffer et al., 2023](#)). This inherent uncertainty is a consequence of the largely subjective definition of morphology and should therefore not be ignored when devising a model for automated morphological classification. In this thesis, we focus our attention on four primary morphological types: ellipticals, lenticulars, spirals and irregular/miscellaneous. These are defined based on galaxy T-Types from the [Nair & Abraham \(2010a\)](#) morphological catalogue; exact details are provided in Chapter 3. It is worth noting that our irregular/miscellaneous class (which we will refer to as irregular for short) also contains examples of interacting galaxies. In addition to developing deep-learning based models to classify these primary morphologies, we also develop separate models to classify barred galaxies. Each of these models are trained to classify such morphologies using only the images of galaxies as their inputs. Figure 1.2 shows a selection of quintessential galaxies in each of the four morphological types, plus an example of a barred galaxy.

1.1.2 The Significance of Morphology and its Evolution

Morphology is far from merely a label with which to organise galaxies. Studies have long established the importance of a galaxy’s morphology to its physical, dynamical and chemical properties. This link between a galaxy’s morphology and its properties makes morphology a crucial observable for studying galaxy evolution ([Peebles & Nusser, 2010](#)). Galaxy morphology is known to align with stellar mass ([Moffett et al., 2016](#)), luminosity ([Kormendy et al., 2009](#)), colour ([Skibba et al., 2009](#)), star formation activity ([Kennicutt, 1998](#)) or lack thereof ([Martig et al., 2009](#)), gas dynamics and gas accretion ([Maiolino et al., 1997](#); [Fanali et al., 2015](#)), metallicity ([Calura et al., 2009](#)) and stellar kinematics ([Cappellari et al., 2007](#)). Morphology also depends strongly on environment ([Martel et al., 1998](#); [Tanaka et al., 2004](#); [Peng et al., 2010](#)). One prominent

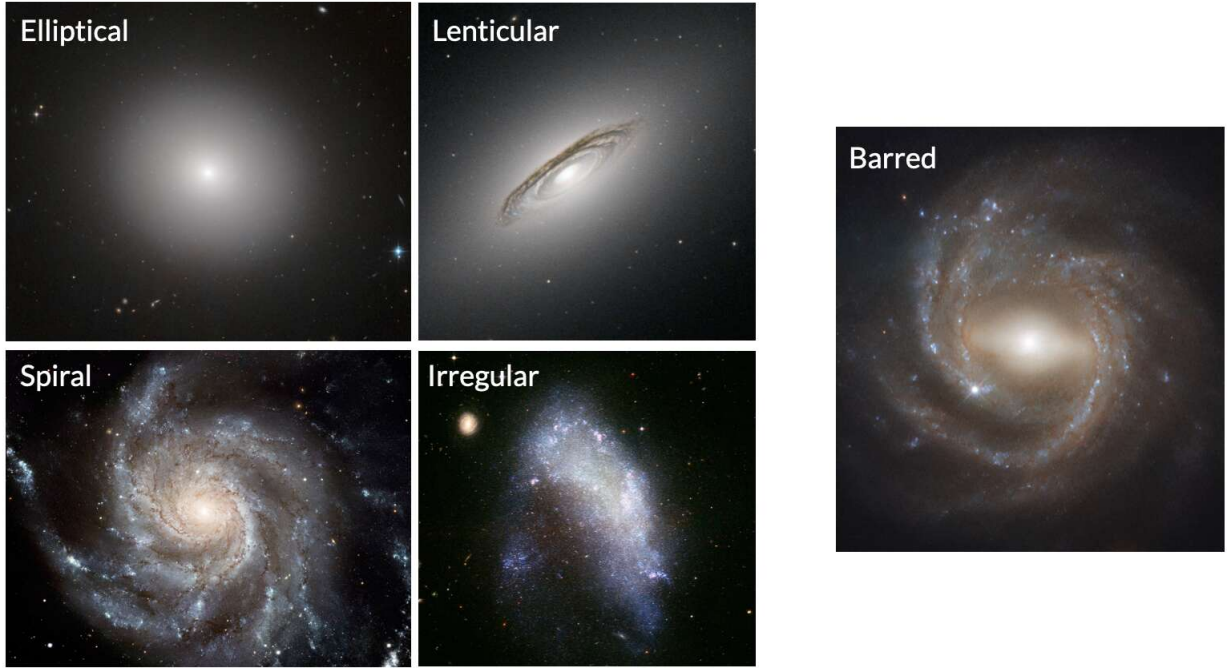


Figure 1.2: Quintessential examples of elliptical, lenticular, spiral, irregular and barred galaxies. Objects featured (clockwise from top left): IC 2006, NGC 6861, NGC 7773, NGC 1427A, Messier 101 (Pinwheel Galaxy). Image credit: NASA, ESA, ESO, Hubble Space Telescope.

example is the morphological-density relation (Dressler, 1980; Dressler et al., 1997), in which dense regions such as clusters host significantly more early-type (elliptical and lenticular) galaxies compared to spirals (Postman et al., 2005). This suggests that environment plays a role in driving the growth of early types in dense environments. Studies based on observations and simulations have analysed many environmental processes that shown that they strongly affect galaxy morphology, such as tidal interactions (Toomre & Toomre, 1972), ram pressure stripping (Abadi et al., 1999) and galaxy mergers (Barnes & Hernquist, 1992). Examining galaxy morphologies in clusters can also reveal information on the nature of how their parent clusters have formed and evolved (Kravtsov & Borgani, 2012).

Galaxies are constantly evolving. An important aim in astronomy is to understand the processes that drive this evolution. The overall distribution of galaxy morphologies is known to have varied significantly over cosmic timescales (Peng et al., 2010; Mortlock et al., 2013; Conselice, 2014). Robust morphological classification systems are essential for characterising these changes and enabling targeted studies into the physical processes responsible for driving morphological transitions from past to present (Roberts & Haynes, 1994; Abraham & van den Bergh, 2001; Buta, 2013). This is since the distributions of present-day morphologies serve as important signatures of their past evolutionary histories. For instance, the most massive galaxies in the present-day Universe are predominantly ellipticals (Robotham et al., 2014; Moffett et al., 2016). Elliptical galaxies, especially those with large bulge components, are believed to have been formed through cumulative mergers over time (Aguerri et al., 2001; Bournaud et al., 2007; Kaviraj, 2014; Martin et al., 2018). This mass-assembly process, believed to be the dominant formation pathway for ellipticals, is consistent with the hierarchical nature of galaxy formation and growth under the standard Λ CDM cosmological paradigm (Cole et al., 2000; Springel et al., 2005; De Lucia et al., 2006). Determining

the redshifts at which these massive ellipticals form is a key challenge for theories of galaxy formation, with studies suggesting that they predominantly formed at high redshifts beyond $z > 1$ (Wang et al., 2012), possibly through the merger of progenitor spiral galaxies (Naab & Ostriker, 2009).

Part of the enduring appeal of the Hubble sequence (in spite of its original simplicity) lies in its representation of galaxy evolution whereby, in an overarching sense, galaxies are believed to naturally transition from predominantly star-forming late types (spirals) to quiescent early types over time (Peng et al., 2010; Ilbert et al., 2010; Fang et al., 2013; Conselice, 2014; Bremer et al., 2018). Observational evidence shows that early type morphologies continue to grow. The present-day fraction of spirals is known to be lower than it was in the past. A wide range of studies have so far established that a significant fraction of galaxies have transformed from predominantly star-forming spirals at higher redshifts to more quiescent early types in the present epoch (Dressler et al., 1997; van Dokkum & Franx, 2001; Ilbert et al., 2010; Buitrago et al., 2014). One of the key processes accounting for this reduction is that many spirals have faded and transitioned into lenticulars (Moran et al., 2007b; Bekki & Couch, 2011; Johnston et al., 2014; D’Onofrio et al., 2015). This trend has been observed for galaxies in both dense, cluster environments (Desai et al., 2007; Poggianti et al., 2009; Coccato et al., 2022), and isolated, field galaxies (Oesch et al., 2010) where star formation naturally ceases over time as galaxies deplete their cold gas reservoirs (Rizzo et al., 2018). Another important mechanism for the morphological transformation of galaxies is galaxy merging (Martin et al., 2018). Tracking the changes in the prevalence of different morphologies over time provides key observable constraints on the nature and extent of these physical processes.

Galaxy morphology continues to be a central focus in present-day astronomical research. Recently, studies have analysed a unique morphology known as ultra-diffuse galaxies, believed to be a subtype of dwarf ellipticals (Conselice, 2018). Curiously, some galaxies of this morphological type lack sufficient dark matter (van Dokkum et al., 2018; Sellwood & Sanders, 2022), which is problematic for the accepted Λ CDM cosmological model. The properties of some galaxy types, such as star-forming lenticulars (Rathore et al., 2022), cannot be fully explained by the prevailing theory surrounding their formation and evolution. These are just a few examples of many observations that continue to pose challenges for our understanding of morphology. With the advent of the JWST, very high-redshift galaxies are now able to be studied at vastly improved resolutions. While the extremely high-redshift Universe is commonly thought to be dominated by irregular morphologies, preliminary JWST imaging has revealed a rich diversity in galaxy morphology including discs previously classified as compact based on HST imaging (Ferreira et al., 2022a; Kartaltepe et al., 2022; Robertson et al., 2022). Indeed, the SKA is expected to bring forth a similar leap in capability, allowing radio sources to be characterised and studied in unprecedented detail (Koribalski et al., 2020). Overall, these findings underscore the importance of analysing the changes in galaxy morphology over cosmic timescales. Morphology is critical to the study of both the intrinsic, secular processes (Kormendy & Kennicutt, 2004; Cheung et al., 2013; Sellwood, 2014) and the extrinsic, environmental processes (Blanton & Moustakas, 2009; Calvi et al., 2012) that govern the evolution of galaxies.

1.2 Barred Galaxies

So far, we have discussed the broad morphological features that characterise entire galaxies. However, galaxies are known to exhibit a large variety of substructures and finer morphological sub-features

(Buta, 2013), some of which are often treated as different morphological subtypes in their own right (de Vaucouleurs et al., 1991). One such substructure is a stellar bar, also known as a galactic bar. Bars are centrally-located rectangular stellar structures embedded within a galaxy’s stellar disc. They are known to have a significant impact on the evolution of their host galaxies (Sellwood & Wilkinson, 1993; Cheung et al., 2013; Vera et al., 2016). For instance, bars are powerful drivers of angular momentum redistribution throughout the entire stellar disc (Weinberg, 1985; Athanassoula, 2005), and also influence dynamical interactions between the disc and its surrounding dark matter halo (Valenzuela & Klypin, 2003). This redistribution of angular momentum is even thought to influence the formation of other morphological features such as ansae, rings and even spiral arms (Rautiainen et al., 2002; Martinez-Valpuesta et al., 2007; Athanassoula, 2013). Bars are also known to buckle due to vertical instabilities (Lokas, 2019), causing them to appear peanut or X-shaped when viewed edge on (Raha et al., 1991). Bars are also key drivers of gas dynamics, especially radially-directed gas inflows (Shlosman & Noguchi, 1993; Berentzen et al., 2007; Fanali et al., 2015). These inflows can significantly alter a galaxy’s properties, as they fuel central star formation and starbursts (Jogee et al., 2005; Ellison et al., 2011; Coelho & Gadotti, 2011; Lin et al., 2020), as well as affect AGN activity (Shlosman et al., 1989; Alonso et al., 2014; Emsellem et al., 2014) and the properties of stellar bulges (Kormendy & Kennicutt, 2004).

1.2.1 The Bar Fraction

Bars are present in the majority of spiral galaxies, with various studies estimating the present-day bar fraction in spirals to be at least 50% to as high as 70% (Eskridge et al., 2000; Aguerri et al., 2009; Saha & Elmegreen, 2018). Similarly, the present-day bar fraction out of all galaxies is estimated to be around 25% to 30% (Nair & Abraham, 2010b; Melvin et al., 2014). The bar fraction is dependent on many factors, from intrinsic properties such as mass and colour (Skibba et al., 2012; Erwin, 2018), to extrinsic properties such as local density and environment (Elmegreen et al., 1990; Marinova et al., 2009; Lee et al., 2012). It is important to note that the bar fraction varies over cosmic time. Studies have established that there are more bars now than there were in the past: that is, the bar fraction declines with increasing redshift (Sheth et al., 2008; Cameron et al., 2010; Melvin et al., 2014). Analysing and quantifying this change in the bar fraction with redshift, ideally in a large-scale population of galaxies, is an important foundational step to studying the physical processes that drive bar formation and evolution. This is also critically important in cosmological simulations (Algorry et al., 2017; Rosas-Guevara et al., 2019; Zhou et al., 2020; Rosas-Guevara et al., 2022). This is where models capable of consistently classifying barred galaxies over multiple epochs (based on only their images) are inherently useful. Indeed, this is the goal of Chapter 4 of this thesis, in which we examine the bar fraction of galaxies in the EAGLE cosmological simulation using our deep learning models.

1.2.2 Classifying Bars

As with galaxy morphology in general, bars have long been classified by visual inspection (Nair & Abraham, 2010b; Masters et al., 2011; Buta et al., 2015). However, owing to their unique shape and properties, many automated techniques have been developed and successfully utilised to classify them. One such automated approach is through the Fourier analysis of a galaxy’s azimuthal light profile, in which a bar

can be detected by comparing the relative amplitudes of the $m = 2$ and $m = 0$ Fourier modes (Elmegreen & Elmegreen, 1985; Ohta et al., 1990; Odewahn et al., 2002; Odewahn, 2004). Fourier-based methods can also be used to detect bars in deprojected galaxy images (García-Gómez et al., 2017). Other approaches include applying ellipse fitting codes to obtain galaxy isophotes (lines of equal brightness) (Abraham et al., 1999; Laine et al., 2002; Menendez-Delmestre et al., 2007; Barazza et al., 2008), or by analysing photometric decompositions (Reese et al., 2007; Durbala et al., 2009). These automated techniques all have key weaknesses. In particular, they rely on auxiliary information from analytical post-processing routines, and also rely heavily on thresholds (such as the ratio of Fourier modes) or free parameters to define the presence of a bar, which may or may not be applied consistently from paper to paper. This can lead to higher or lower bar fractions compared to visual inspection (Barazza et al., 2008; Aguerri et al., 2009; Masters et al., 2011; Cheung et al., 2013). Different methods can lead to wildly different bar fractions even on the same sample (Lee et al., 2019). An automated classification technique based on deep learning can overcome some of these shortcomings. As these methods require only the image of a galaxy in order to obtain classifications (and as they are trained to recognise bars in such imagery), they can be applied more consistently, bypassing issues such as different Fourier amplitude criteria or different free parameters for ellipse fitting. A deep learning approach can therefore offer a more direct means of comparing barred galaxies in different surveys, and indeed for comparing bar in simulations with those in observations, as explored in Chapter 4.

1.2.3 Bar Strength

When classifying bars, studies commonly group bars into two categories: strong and weak. These terms are largely phenomenological, originally describing the relative size and prominence of the bar (de Vaucouleurs, 1963) in relation to the host galaxy, or simply as an indicator of how easy it is to visually identify the bar (Melvin et al., 2014). Bar strength can also be characterised more concretely in terms of its physical parameters, such as its diameter, thickness, ellipticity, intensity of star formation and luminosity (Elmegreen & Elmegreen, 1985; Rozas et al., 1998; Laurikainen & Salo, 2002), although the exact definitions and thresholds for such criteria vary considerably between studies (Díaz-García et al., 2016). Studies have also used the relative amplitude of the Fourier modes of the azimuthal light profile to define bar strength (Das et al., 2008; Aguerri et al., 2009), as well as ellipse fitting (Marinova & Jogee, 2007). Previous studies have also used the vote fractions for bars in the Galaxy Zoo 2 survey as a means of defining bar strength (Masters et al., 2012; Willett et al., 2013; Melvin et al., 2014); strong bars for galaxies with higher vote fractions, weak bars for galaxies with lower vote fractions. In this sense, strong and weak correspond to the collective confidence of the various respondents that the given galaxy is, in fact, barred. In Chapter 4, we explore using the probabilities outputted by our deep learning models as a similar, confidence-based proxy for bar strength.

The fraction of strong bars in the local Universe is understood to be much less than the total bar fraction. Sheth et al. (2008) found a strong bar fraction of 30% compared to an overall bar fraction of 65% in spirals. The recent Galaxy Zoo DeCALS survey found a strong bar fraction of 15% and weak bar fraction of 28% (Walmsley et al., 2021). Studies analysing simulations have found that there are a roughly equal number of strong bars as there are weak bars (Algorry et al., 2017; Rosas-Guevara et al., 2019). There is also evidence to suggest that bar strength correlates with morphology, with previous studies concluding

that bars in lenticulars tend to be weaker than bars in spirals (Laurikainen et al., 2007; Aguerri et al., 2009). This discrepancy is likely due to the growth of bulges in lenticulars (Kormendy & Kennicutt, 2004), which reduces the gravitational torques of the bar (Buta et al., 2010).

Although the terms “strong” and “weak” have been used widely throughout the literature, it is important to note that the exact nature of bar strength is still a topic for debate. In particular, the recent study by G eron et al. (2021) argues that bar strength is better defined on a continuous spectrum based on physical properties, rather than with discrete and inherently subjective terminology. It is worth noting that, in general, automated classification techniques tend to obtain higher bar fractions compared to visual classification (Masters et al., 2011; Melvin et al., 2014) (though Lee et al. 2019 is a notable exception). This could be due to visual classification favouring strong bars, while weaker bars are harder to identify (Masters et al., 2011) and are therefore more likely to instead be detected by automated methods.

1.2.4 Bar Formation and Evolution

The overwhelming majority of the current, theoretical understanding of the physical processes governing the formation and evolution of stellar bars has come from simulations. Studies have established two dominant formation scenarios for bars; they can either form internally as a result of intrinsic, secular processes, or can they be induced by external interactions. The pioneering N-body simulation of Hohl (1971) was among the first to demonstrate how bars can arise, seemingly out of nowhere, in an isolated galaxy. Bars are believed to be formed in this manner due to the slow, dynamical evolution of large-scale, non-axisymmetric instabilities within the galaxy’s disc itself (Goldreich & Lynden-Bell, 1965; Lynden-Bell, 1979; Combes & Sanders, 1981; Pfenniger & Friedli, 1991; Sellwood & Wilkinson, 1993; Athanassoula, 2003). Such instabilities form resonances and induce positive feedback, driving the formation of the bar. 3D simulations have shown that these instabilities are especially noticeable in the vertical axis, perpendicular to the disk (Pfenniger, 1984). There are various processes that can inhibit the formation and subsequent evolution of bars formed by means of dynamical instability. For instance, bulges are known to suppress bar formation by increasing the stability of the disc (Kormendy & Kennicutt, 2004; Sellwood, 2014) (however, this does not entirely preclude strong bars from forming, see Kataria & Das 2018). Studies have shown that disc-halo interactions, namely angular momentum exchanges, can in some cases support the growth of bars (Athanassoula, 2002), but can in other cases serve to hinder their overall, long-term growth (Long et al., 2014). Vertical buckling instabilities (Raha et al., 1991; Martinez-Valpuesta et al., 2006; Xiang et al., 2021) also play a significant role in the long-term evolution of bars, with Lokas (2019) demonstrating their capacity to weaken bars.

Despite being well studied in simulations, this spontaneous, self-interacting mechanism of bar formation cannot fully account for the properties of all barred galaxies, especially those in clusters (Elmegreen et al., 1990). Therefore, an additional mechanism must be at play, namely an external mechanism that induces bar formation. Indeed, many studies have established that bars can form as a consequence of tidal interactions (Noguchi, 1987; Miwa & Noguchi, 1998; Lokas et al., 2014; Moetazedian et al., 2017; Peschken & Lokas, 2019). Simulations have also shown that galaxy flybys can also trigger the formation of bars (Lang et al., 2014; Zana et al., 2018). Another formation scenario is galaxy merging, where bars can form from the remnants of galaxy merging (Peirani et al., 2009; Cavanagh & Bekki, 2020). One way to distinguish bars formed from dynamical self-interactions with those formed from tidal interactions is to

examine their rotational frequency, also known as the pattern speed (Tremaine & Weinberg, 1984; Aguerri et al., 2009, 2015; Garma-Oehmichen et al., 2019). Bars formed via tidal interactions have been shown to rotate more slowly compared to those formed through dynamical self-interaction (Miwa & Noguchi, 1998; Martinez-Valpuesta et al., 2017).

It is important to note that the increase in the bar fraction over time does not necessarily imply that all barred galaxies remain barred. Studies have proposed that bars are transient features with the potential to dissipate and regenerate over time. Indeed, very strong gas inflows are capable of destroying bars (Friedli & Benz, 1993), while other processes such as accretion, cold gas inflow and tidal interactions can regenerate bars (Sellwood & Moore, 1999; Berentzen et al., 2004). Studies based on simulations have shown that galaxies can undergo multiple episodes of bar formation within the timescale of the age of the Universe (Bournaud & Combes, 2002; Romano-Díaz et al., 2008), with a mean lifetime of around one to two billion years (Bournaud et al., 2005). Simulations by Kraljic et al. (2012) found two phases of bar growth: bars that formed at high redshifts are comparatively short-lived due to destructive mergers (see also Romano-Díaz et al. 2008), while those that formed at lower redshifts $z < 1$ continue to persist to the present epochs. However, studies based on simulations have reached different conclusions regarding exactly when bars begin to form. For instance, some simulations suggest that bars form relatively early in their host galaxy’s lifetime at redshifts $z \gg 1$ and remain fairly stable thereafter (Curir et al., 2008), while others suggest that bars predominantly form later at lower redshifts below $z < 1$ (Kraljic et al., 2012; Spinoso et al., 2017). Studies have also demonstrated that the timescales for bar formation may differ substantially depending on the gas content of the galaxy, with longer timescales for gas-rich and late-type galaxies (Villa-Vargas et al., 2010; Athanassoula et al., 2013). This is attributed to the effects that gas dynamics have on reducing instabilities, especially vertical instability (Berentzen et al., 1998).

Although bars have been linked to many physical processes, there are still many open questions regarding their exact nature and origins. As such, the study of barred galaxies remains an ongoing endeavour in both observations and simulations. One such notable mystery is the role they play in quenching star formation. Previous studies have proposed that bars play an important role in quenching their host galaxies (Kormendy & Kennicutt, 2004; Kruk et al., 2018; Fraser-McKelvie et al., 2020b; Geron et al., 2021). Indeed, many of the largest, most prominent bars ever observed are found in early type galaxies (Erwin, 2005), and many studies show bars to be more common in redder, more massive and gas-poor galaxies (Masters et al., 2011, 2012; Skibba et al., 2012; Vera et al., 2016; Cervantes Sodi, 2017) (although other studies have shown bars may be just as common in gas-rich late-type galaxies c.f. Erwin 2018; Tawfeek et al. 2022). However, it is not completely clear if bars quench galaxies, or if bars are simply more likely to be found in quenched galaxies (Fraser-McKelvie et al., 2020b). This “chicken or the egg” problem is an ongoing subject of research, especially in cosmological simulations (Rosas-Guevara et al., 2019; Zhou et al., 2020). The recent study by Roshan et al. (2021), based on high-resolution cosmological simulations, identified bars exhibiting unusually fast rotational frequencies (also known as pattern speeds, see Tremaine & Weinberg 1984) that differ substantially from observations. It is worth noting that cosmological simulations tend to produce fewer bars compared to observations (Algorry et al., 2017; Peschken & Lokas, 2019; Zhao et al., 2020; Reddish et al., 2022), a key reason being the technical constraints of limited spatial resolutions. As future large-scale observational surveys probe ever higher redshifts, they will undoubtedly probe deeper into the evolutionary history of bars. Consequently, the study of

the formation and evolution of stellar bars, and understanding the impacts that they have on galaxy morphology as a whole, remains a key objective for observations and simulations alike.

1.3 Lenticular Galaxies

At the centre of the classical Hubble sequence lie lenticular galaxies, also referred as S0 galaxies (Hubble, 1936). As with spirals, lenticular galaxies contain clear bulge and disc components. However, they also lack any characteristic spiral arms, and their stellar populations more closely resemble those found in ellipticals (Ziegler et al., 2001). This dual nature has led many studies and morphological classification systems to treat S0s as an intermediate, transitional morphological type (de Vaucouleurs et al., 1991; Nair & Abraham, 2010a; Buta, 2013). The exact nature and origin of lenticular galaxies remains unclear (Roberts & Haynes, 1994; Burstein et al., 2005; Moran et al., 2007a; Cortesi et al., 2013). It is still a matter of current debate as to whether lenticulars are more closely related to ellipticals or spirals, or if they are merely a transitional morphology (Poggianti et al., 2001; Bremer et al., 2018; Phillipps et al., 2019).

Many lenticulars tend to be uniform in appearance with mostly featureless discs (Buta, 2013), making them very difficult to distinguish from elliptical galaxies, especially when viewed face-on. Although lenticulars may look similar to ellipticals, their physical and kinematic properties are vastly different (Moran et al., 2007a). Furthermore, studies have found that lenticulars can play host to a rich variety of morphological substructures. These include dust lanes, stellar bars, ansae, lenses and even stellar rings (Laurikainen et al., 2009; Finkelman et al., 2012; Laurikainen et al., 2013; Buta et al., 2015; Sil'chenko et al., 2018; Tous et al., 2022). Previous studies have demonstrated that S0s exhibit a diverse range of physical and kinematic properties, ranging from varying bulge ratios and luminosities (Poggianti et al., 2001; Drory & Fisher, 2007; Barway et al., 2009; Laurikainen et al., 2010; Barway et al., 2013), to large differences in metallicities, colour and star formation rates (Sansom & Northeast, 2008; Kannappan et al., 2009; Amblard et al., 2014; Tous et al., 2020; Rathore et al., 2022), as well as kinematics and stellar angular momentum (Graham et al., 2018).

Recent studies examining S0s have found that they exhibit a bimodality in both their velocity dispersion and stellar populations with respect to stellar mass and star formation rates (Fraser-McKelvie et al., 2018; Domínguez Sánchez et al., 2020; Tous et al., 2020). In particular, low mass S0s contain mostly younger, metal-poor stars and have a relatively flat velocity dispersion profile, while high-mass S0s predominantly contain older, metal-rich stars and are more strongly dispersion-supported. This wide range of observed properties is indicative of a multi-faceted evolutionary history, with studies proposes multiple potential formation pathways for S0s (Deeley et al., 2020; Johnston et al., 2020; Coccato et al., 2022). As such, understanding how the fraction of lenticulars varies over redshift in different environments and across different mass ranges can provide crucial insights into the nature of their formation and subsequent evolution.

1.3.1 The S0 Fraction

It is generally accepted that the fraction of lenticular galaxies decreases with increasing redshift; S0s are more prevalent now than they were in the past. This increasing S0 fraction has been observed in both dense, cluster environments (Dressler et al., 1997; Desai et al., 2007; Poggianti et al., 2009; Vulcani et al.,

2011), as well as for lenticulars in the field (Oesch et al., 2010; Huertas-Company et al., 2015). However, some studies have instead found very little to no change in the fraction of lenticulars with redshift (van der Wel et al., 2007; Holden et al., 2009).

The S0 fraction is also known to have a strong environmental dependence. It is well established that lenticular galaxies are much more prevalent in dense environments, usually to the direct detriment of spiral galaxies (Dressler, 1980; Giovanelli et al., 1986; Fasano et al., 2000; Postman et al., 2005; Poggianti et al., 2009; Cappellari et al., 2011). This increase in prevalence with increasing density is true for early type morphology in general, as per the morphology-density relation (Dressler, 1980; Dressler et al., 1997; Treu et al., 2003; van der Wel et al., 2007). Quantitative surveys have produced a range of values for the S0 fraction in both different environments and at different redshifts. van den Bergh (2009) found the fraction of S0 galaxies to be as high as 28% in the dense Virgo cluster, to below 10% in the field, while Poggianti et al. (2009) found the S0 fraction in nearby clusters to be around 44%, while dropping to less than half this value for high redshift clusters. Given that S0s are more prevalent in high density regions, they are typically regarded to have transformed from spiral galaxies, having largely lost their spiral arms due to truncated star formation (Poggianti et al., 2001; Barr et al., 2007; Moran et al., 2007b; van den Bergh, 2009; Laurikainen et al., 2010; D’Onofrio et al., 2015; Johnston et al., 2020). However, this “faded spiral” formation scenario is merely one possible formation pathway for lenticular galaxies.

1.3.2 Formation Scenarios

The origin of lenticular galaxies remains a key focus of current research. Studies have proposed many mechanisms and physical processes to account for the formation of lenticular galaxies (Fraser-McKelvie et al., 2018; Coccato et al., 2020; Deeley et al., 2020; Coccato et al., 2022). These largely fall into two distinct categories, based on the main pathway to formation. The first of these is the “faded spiral” pathway, describing the aforementioned morphological transformation from spirals to lenticulars as a result of the suppression of star formation. Studies have indicated that this is the dominant formation scenario in dense, cluster environments (Johnston et al., 2014; Mishra et al., 2019). There are many physical processes that are capable of suppressing star formation in spirals, such as the results of tidal interactions (Bekki & Couch, 2011), ram pressure stripping (Gunn & Gott, 1972), gas starvation (Larson et al., 1980), galaxy harassment and strangulation (Moore et al., 1996; Peng et al., 2015), or, more generally, any process capable of removing gas (Eliche-Moral et al., 2013; Rizzo et al., 2018). However, studies have argued that the faded spiral pathway by itself is insufficient to describe the broad range of physical and kinematic properties in observed S0 populations, especially for S0s whose properties differ greatly from those of their supposed progenitor spirals (Burstein et al., 2005; Williams et al., 2010; Falcón-Barroso et al., 2014; Méndez-Abreu et al., 2018).

The second major formation pathway involves galaxy merging (both major merging and repeated minor merging), as well as accretion (Bekki, 1998; Arnold et al., 2011; Prieto et al., 2013; Borlaff et al., 2014; Querejeta et al., 2015; Tapia et al., 2017; Diaz et al., 2018; Eliche-Moral et al., 2018). Studies have also found that different formation mechanisms may be more favourable at different mass ranges, with high mass S0s having predominantly formed through the merger pathway, while low-mass S0s more closely resemble faded spirals (Kannappan et al., 2009; Fraser-McKelvie et al., 2018; Domínguez Sánchez et al., 2020) (the latter having also been studied in simulations, see Bellstedt et al. 2017.)

Although these two main pathways are largely dependent on environmental processes, S0s are known to also inhabit low-density, field environments. Indeed, studies have shown that such isolated S0s could nevertheless have evolved from spirals as a result of intrinsic secular evolution, bulge-growth and disc fading (Kormendy & Kennicutt, 2004; Oesch et al., 2010; Kormendy, 2013; Maltby et al., 2015), or simply through the passive consumption and exhaustion of gas needed to support star formation (Rizzo et al., 2018). Other potential mechanisms include AGN activity (van den Bergh, 2009) and fragmentation resulting from disc instabilities (Saha & Elmegreen, 2018). Studies have also suggested that certain S0 formation mechanisms may be more efficient than others in different environments (Johnston et al., 2020), with Coccato et al. (2022) showing that the faded spiral formation mechanism is more efficient than mergers in dense environments. It has been suggested that S0s simply cannot be completely accounted for with one single formation pathway, whether that be a faded spiral or merger (Deeley et al., 2021). Instead, a mix of multiple formation pathways is necessary to account for the diverse range of properties of observed lenticulars (Fraser-McKelvie et al., 2018).

1.3.3 Star Forming Lenticulars

Recent studies have found increasing evidence for populations of star forming S0s; more specifically, lenticulars with higher star formation rates than otherwise expected under the faded spiral formation scenario (Bait et al., 2017; Tous et al., 2020; Rathore et al., 2022). While the notion of a star-forming S0 seems to be an oxymoron, they are known to exist, albeit they are rare and mostly confined to low masses below $M_{\star} < 10^9 M_{\odot}$ (Im et al., 2001; Kannappan et al., 2009; Schawinski et al., 2009). So far, there have been few studies regarding the origins and nature of star forming S0s. Rathore et al. (2022) suggest that such S0s could currently be undergoing rejuvenation; star formation having been restarted. In particular, more than half of the star forming S0s identified by Rathore et al. (2022) show signs of recent interactions, and were likely quenched in the past. It is known that both minor merging and accretion from gas-rich companions is capable of triggering starbursts and/or accelerating star formation (Di Matteo et al., 2007; Wei et al., 2010; Marino et al., 2011; George, 2017). It is interesting to note that these star forming S0s are predominantly low mass lenticulars (Tous et al., 2020; Rathore et al., 2022), since previous studies have established that quenching is generally more efficient for high-mass galaxies (Peng et al., 2010). Further studies are needed in order to definitively ascertain the origin of star forming S0s. However, it is worth noting that previous studies have identified similar classes of low-mass galaxies called “blue ellipticals” or “blue compact dwarfs” (Cairós et al., 2010; Lazar et al., 2023) that exhibit unusually high star formation rates in spite of their early-type morphology.

1.4 Towards Automated Morphological Classification

1.4.1 The Importance of Scalability

Large-scale studies into the evolution of galaxy morphology are reliant on the availability of morphological classifications. As discussed in the beginning of this Chapter, such morphological catalogues are typically obtained through visual classification, often by expert astronomers (Nair & Abraham, 2010a; Buta et al., 2015; Moffett et al., 2016). Despite its accuracy, manual visual inspection is, by its very nature, highly

laborious. This therefore places strict upper limits on the sizes of datasets that can be feasibly classified for a given amount of time. The first efforts to scale to larger datasets capitalised on the potential of crowdsourcing and citizen science. These collective efforts have improved the efficiency of visual classification, and have subsequently allowed studies to tackle much larger datasets (Willett et al., 2013; Simmons et al., 2017; Masters et al., 2021; Walmsley et al., 2021). However, it is important to note that this gain in efficiency is achieved simply through increasing the number of classifiers, rather than addressing the speed of the classification process itself. Future large-scale observational surveys are expected to yield enormous amounts of data that will far exceed the current capacity for visual classification (Silva et al., 2018), causing ever-greater demand for crowdsourcing. There is thus a need for highly efficient automated morphological classification. In this thesis, we develop models capable of automatically classifying galaxy morphologies in an extremely rapid and consistent manner. To achieve this, we make use of deep learning.

1.4.2 A Brief Introduction to Deep Learning and its Historical Context

Deep learning is a relatively nascent subfield of machine learning (itself a branch of artificial intelligence) that has seen tremendous growth over the last decade. This is thanks to the rise of Big data, the development of advanced training algorithms and general-purpose machine learning frameworks such as TensorFlow (Abadi et al., 2016), and the increased computing capabilities of *graphical processing units* (GPUs). Although deep learning is an umbrella term that encompasses a wide range of models and techniques, the model that is most commonly adopted, to the point where it is near-synonymous with the field, is the convolutional neural network (LeCun et al., 2015). CNNs are specialised neural networks capable of extracting highly abstract features from data. They are especially adept at processing images. Deep learning has been applied to all manner of tasks including classification, clustering and regression, and has increasingly been deployed in myriad fields and disciplines (Pouyanfar et al., 2019; Dargan et al., 2020; Alzubaidi et al., 2021). Examples of widespread applications include facial recognition, object detection, medical imaging, financial analytics, content recommendation (used by Facebook, Netflix and Spotify, among many others), natural language processing, geophysics, and astronomy. A key driver of deep learning is the increasing volumes and availability of raw data; an especially relevant consideration for modern and future astronomical surveys. We will discuss the background theory and structure of CNNs in more detail in Chapter 2, however it is worth briefly discussing where deep learning sits in the wider context of machine learning and artificial intelligence, as this reveals why deep learning is well suited to tackle the challenge of classifying galaxy morphologies.

Figure 1.3 illustrates how deep learning has historically evolved, building upon the cumulative advancements in both artificial intelligence and machine learning. Artificial intelligence (AI) is a broad, interdisciplinary field that, at its most fundamental level, is concerned with the study of intelligent agents; the capacity to be able to perform tasks and solve problems autonomously (Russell & Norvig, 2021). Among the first commercially viable and successful agents were rules-based expert systems, which operated based on an internal set of instructions (if-then statements). A notable example of an expert system that garnered much publicity at the time was IBM’s *Deep Blue*, the first chess computer to beat a reigning world chess champion in Garry Kasparov. However, the key disadvantage with expert systems is their reliance on knowledge bases: hard-coded rules. The field of machine learning addresses this issue by adopting a completely different approach. Rather than relying on explicitly pre-programmed rules, machine learning

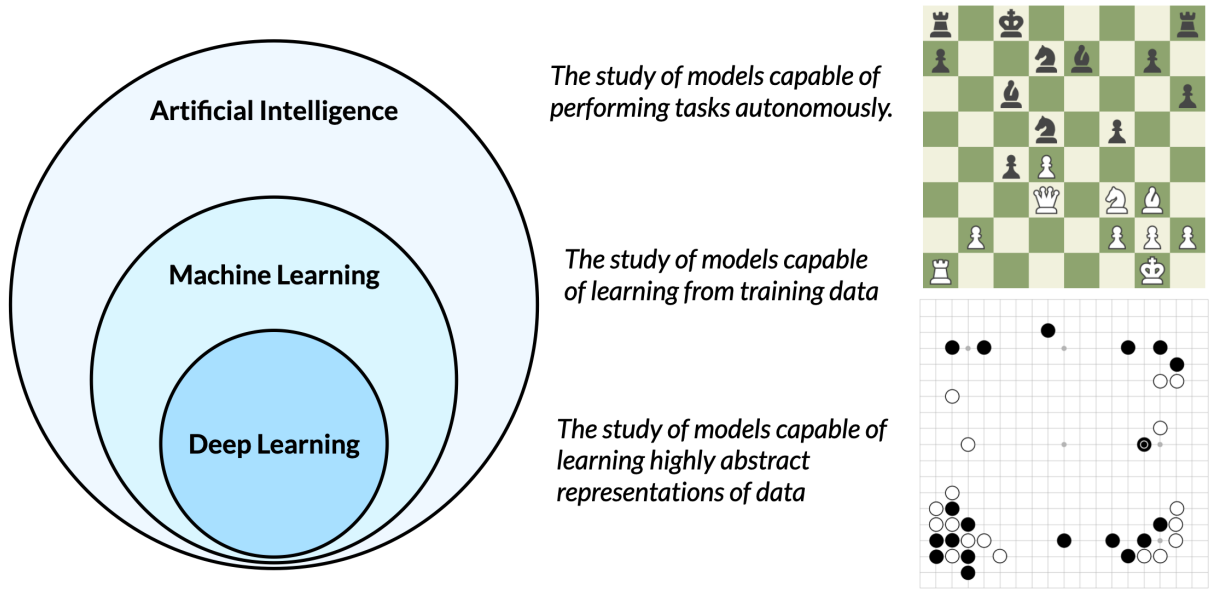


Figure 1.3: An illustration of where deep learning sits in the wider context as subfields of both machine learning and artificial intelligence. There is a brief summary of each field, from which it can be seen how each subfield is an extension of its predecessor. The top-right image shows the final board position in *Deep Blue* vs. Garry Kasparov, Game 6, 1997 (Image Credit: chess.com). The bottom-right shows the famous "Move 37" as played by Deepmind's *AlphaGo* in its second match against Lee Sedol in 2016 (Image Credit: DeepMind).

models automatically learn to perform a given task simply by analysing large quantities of examples (i.e. training data). One such machine learning model is a neural network, partly inspired by the synaptic organisation of the brain and the manner in which humans learn to perceive patterns (Fukushima, 1980; Haykin, 2009).

Deep learning takes this data-driven approach one step further, leveraging the power of advanced neural networks (*especially* CNNs) to extract high-level features and thus learn highly abstract representations of the given training data (see Schmidhuber 2015 for a historical review). This technique has seen tremendous success in general-purpose image recognition, where this hierarchical feature extraction process is well suited to analysing the innate structure and spatial hierarchies of images (Krizhevsky et al., 2012; LeCun et al., 2015; Chollet, 2021). It is this capacity for highly abstract representation learning that saw deep learning finally tackle the game of Go (see Figure 1.3), a game known for its extreme branching factor that all but renders traditional brute-force algorithms intractable. Abstract feature extraction makes deep learning an ideal tool to study galaxy morphologies, especially given how galaxies exhibit such a diverse range of features, shapes and finer substructures (Buta et al., 2015).

1.4.3 A Deep Learning Approach

In this thesis, we develop a suite of deep learning models to classify galaxy morphologies using convolutional neural networks. One major difference between the models developed in this thesis compared to existing models in the literature is that existing models are predominantly binary classifiers, for example

dividing galaxies into early and late types (Domínguez Sánchez et al., 2018; Vega-Ferrero et al., 2021; Cheng et al., 2021b). Our models go further, directly classifying galaxies into up to four morphological types at a time. Overall, this deep learning approach to morphological classification has several important benefits, as well as several drawbacks. One of the most appreciable benefits is that of speed and efficiency. Once trained, CNNs can classify images extremely quickly. Tens of thousands of galaxies are able to be classified in mere minutes; hundreds of galaxies on the order of seconds. This classification speed greatly surpasses human capabilities, and is a key reason as to why deep learning can easily scale to larger datasets that would otherwise be intractable, even by the largest pools of citizen scientists (Willett et al., 2013; Masters et al., 2021). Indeed, Vega-Ferrero et al. (2021) used a deep learning approach to classify tens of millions of galaxies. Another major benefit is that CNNs are model-independent; all that’s needed to train them is the training data itself.

Analytical approaches, such as those involving isophote fitting, Fourier analysis or parametric modelling, necessarily involve hard-coded thresholds and prior assumptions regarding the model itself (Odehahn et al., 2002; Aguerri et al., 2009; Díaz-García et al., 2016; Tarsitano et al., 2018). On the contrary, CNNs are not reliant on prior knowledge; they learn automatically through training simply by analysing the training data. Once trained, CNNs can be rapidly applied to consistently classify images of galaxies. This consistency is a key strength of CNNs, especially in the context of morphological classification. This is since CNNs will output identical classifications for a given set of images regardless of how many times it is run; something that cannot be so easily said when it comes to visual inspection by human classifiers. Lastly, CNNs are extremely versatile and can be quickly fine-tuned to perform other tasks, such as classifying images from a different survey to the one it was trained on, via transfer learning. Furthermore, as a result of this internal consistency, classifications obtained by CNNs are directly comparable. This is what enables us to classify and compare galaxies in simulations with observations (as with EAGLE in Chapter 4), in addition to observations at different redshift ranges (as with SDSS, COSMOS and SAMI in Chapters 3, 5 and 6 respectively).

The main drawbacks with a deep learning approach largely revolve around the nature of the training itself, along with its heavy reliance on suitable training data. For our purposes, our CNNs are predominantly trained on datasets of galaxy images such that each image is associated with a label denoting its known morphology (although in Chapter 6 we will see that labels can also be images in their own right). During training, the CNN is optimised so that its output (predicted morphologies, for instance) matches the desired, true outputs as closely as possible. This is achieved through repeated refinement by looping over the full training data over several iterations. When it comes to deep learning, training data is paramount. In general, CNNs can only ever be as good as the data that they are trained on (LeCun et al., 2015; Goodfellow et al., 2016; Chollet, 2021). Importantly, this means that any uncertainties, errors and/or biases in the training data will necessarily carry through into the CNN. This includes selection biases inherent to the training data, as well as uncertainties with the labels. There are also potential issues and biases that arise from the training of the CNN itself. We will discuss some of these latter biases, along with techniques used to rectify and/or mitigate them, in Chapter 2. Another important issue is the risk of overfitting, a situation where a model is too closely fit to the training data, to the point where it fails to properly generalise to unseen examples. As such, it is crucial that the training data is suitably large for the task at hand, which in our case means that it ought to contain many, diverse examples of different galaxy morphologies.

Fortunately, the NA10 catalogue indeed contains many unique morphologies divided by T-Type, and one of its explicit use-cases is to enable the development of machine learning models. One final consideration is that of computational resources. Although training CNNs in the first place is relatively expensive (in our case, several hours on a modern GPU per model), it pales in comparison to the time saved through their use compared to visual classification.

1.5 The Growth of Deep Learning in Astronomy

In recent years, machine learning techniques – especially deep learning – have become increasingly more common in astronomy, and have been applied to perform a rich variety of tasks (see [Baron 2019](#); [Fluke & Jacobs 2020](#); [Huertas-Company & Lanusse 2023](#) for recent reviews). Machine learning itself can be further subdivided into separate yet broad families based on their methodology: statistical learning, based on probabilistic models and techniques, and deep learning, based primarily on neural networks. Studies in astronomy have utilised a broad range of statistical learning methods, including random forests, Bayesian inference, K-nearest neighbours, complex networks (graphs) and support vector machines (among many other methods, see [Cheng et al. 2020](#); [Feigelson et al. 2021](#)). Applications include classifying galaxy morphology ([Sreejith et al., 2018](#); [Beck et al., 2018](#); [Martin et al., 2020](#); [Baqui et al., 2021](#)), estimating photometric and spectroscopic redshifts ([Kügler et al., 2015](#); [Leistedt et al., 2016](#); [D’Isanto & Polsterer, 2018](#)), galaxy SED fitting and flux estimation ([Dobbels et al., 2020](#); [Gilda et al., 2021](#)), detecting gamma rays ([Saz Parkinson et al., 2016](#)), studying the structure of spiral arms ([Silva et al., 2018](#)), analysing cosmological simulations ([Villaescusa-Navarro et al., 2021](#)), and even modelling the cosmic web ([Tsizh et al., 2020](#)).

Deep learning techniques can be applied to solve many types of problems, including classification, regression, clustering and segmentation. One of the key aspects that distinguishes most deep learning applications from statistical learning is that deep learning methods typically utilise raw pixel data (images) as inputs, whereas statistical methods tend to use parameters as inputs. It therefore comes as no surprise that CNNs are the most commonly used deep learning model for image classification ([LeCun et al., 2015](#)). However, it is important to note that classic, artificial neural networks have also been applied to analyse galaxy images ([Storrie-Lombardi et al., 1992](#); [Naim et al., 1995](#); [De La Calleja & Fuentes, 2004](#); [Banerji et al., 2010](#)). In observational astronomy, CNNs have been used to classify and study galaxy morphologies across a range of different surveys ([Dieleman et al., 2015](#); [Domínguez Sánchez et al., 2018](#); [Zhu et al., 2019](#); [Barchi et al., 2020](#); [Walmsley et al., 2020](#); [Vega-Ferrero et al., 2021](#); [Cheng et al., 2022](#)). CNNs have also been used to classify barred galaxies ([Abraham et al., 2018](#)), detect merging and anomalous galaxies ([Ackermann et al., 2018](#); [Ćiprijanović et al., 2020](#)), conduct galaxy surface brightness fitting ([Tuccillo et al., 2018](#)), detect sources of gravitational lensing ([Schaefer et al., 2018](#)) and even investigate similar morphological types through representation learning ([Walmsley et al., 2022](#)). Morphological classification is not restricted to just optical sources; studies have also used CNNs to classify radio sources ([Lukic et al., 2019](#); [Wu et al., 2019](#)), an application that is expected to grow significantly with the advent of the Square Kilometre Array ([Koribalski et al., 2020](#)).

Deep learning techniques have also been used to identify and characterise exoplanets ([Pearson et al., 2018](#); [Shallue & Vanderburg, 2018](#); [Osborn et al., 2020](#)), and have also been applied to planetary science, including detecting and studying craters on Mars ([Lee, 2019](#)) and analysing the lunar surface ([Bickel et al.,](#)

2021). Deep learning has also been extensively applied in cosmological simulations, including for analysing galaxy morphology (Huertas-Company et al., 2019). Other applications include analysing the masses of clusters (Yan et al., 2020), upscaling low resolution density maps (Li et al., 2021), analysing galaxy mergers (Ferreira et al., 2022b), classifying different S0 formation mechanisms (Diaz et al., 2019), classifying barred galaxies (Cavanagh & Bekki, 2020), and studying the filaments of large-scale cosmic structure (Inoue et al., 2022). In the emerging field of multi-messenger astronomy, deep learning techniques have been highly successful at detecting and analysing gravitational waves (George & Huerta, 2018; Chatterjee et al., 2019; Cuoco et al., 2020). These myriad applications demonstrate the versatile capabilities of deep learning.

Recent studies in astronomy have also used CNNs for the task of image segmentation. Here, CNNs are used to deconstruct an input image, often to highlight specific regions in the image that, for example, correspond to different classes. One such example is Morpheus (Hausen & Robertson, 2020), a model which colours each pixel in an image based on its immediate morphology: spheroid, disc, irregular, background and point-source. Another example is the image segmentation of radio sources in order to highlight regions of ionised and neutral hydrogen (Bianco et al., 2021). The most common model architecture for image segmentation is the U-Net. We will describe U-Nets in more detail in Chapter 6 when we discuss our image segmentation application, namely using CNNs to extract the spiral arms and stellar bars from an image of a galaxy.

Lastly, another noteworthy, emerging application of deep learning methods in astronomy with a remarkably versatile set of applications is the use of generative deep learning, in which models are trained not simply to analyse data, but to create entirely new data from scratch. One popular model is the variational autoencoder (VAE, Kingma & Welling 2013), which uses a probabilistic approach to learning a latent probability distribution that best approximates the underlying distribution of the given training data. New data can be synthesised simply by sampling from this latent distribution. The learning (encoding) and generation (decoding) components are separate CNNs. VAEs have a broad range of applications, from image denoising and deblending (Arcelin et al., 2020; Gheller & Vazza, 2021), detecting gravitational waves (Gabbard et al., 2022), detecting pulsars (Lousto et al., 2021) and enabling exploratory studies into the clustering of galaxy morphology (Cheng et al., 2021a). Another model is the generative adversarial network (GAN, Goodfellow et al. 2014), which uses a game-theoretic approach to synthesising highly realistic samples: one CNN generates artificial (fake) data, while another CNN determines if this new data is real or fake. GANs are notoriously difficult to train, however, and their use in astronomy is less widespread compared to other generative techniques. That said, GANs have been successfully applied to simplify very compute-heavy tasks, such as creating weak lensing maps (Mustafa et al., 2019), 3D density matter fields (Feder et al., 2020) and constructing dark matter halo merger trees (Robles et al., 2022) in simulations, and for advanced image processing, such as colouring, upsampling and denoising astronomical images (Reiman & Göhre, 2019; Kalvankar et al., 2021).

1.6 Thesis Overview

In this thesis, we propose an automated approach to morphological classification utilising techniques from the field of deep learning. We develop, train and apply a suite of convolutional neural networks (CNNs) to rapidly and efficiently classify galaxies into up to four main morphological types (elliptical, lenticular,

spiral and irregular/miscellaneous) in order to study the evolution of galaxy morphology. We also train a CNN to classify galaxies as either barred or unbarred in order to study the properties of barred galaxies, as well as the evolution of the bar fraction with redshift. These CNNs utilise new model architectures specifically designed and repeatedly refined for this very task. In particular, these architectures outperform existing models at directly classifying galaxy morphologies into multiple morphological types.

Our deep learning approach allows us to classify galaxies in an efficient and consistent manner. This consistency is crucial, for it means that we can directly compare our CNN classifications across different datasets. We subsequently apply these models to classify galaxies from a variety of datasets in both simulations and observations, including for nearby, low-redshift galaxies, as well as more distant, high-redshift galaxies. These diverse applications are possible due to the inherent versatility of CNNs, which work purely by analysing images. They can therefore be applied without the need for any post-processed data such as light curves or spectra or manually-defined thresholds that would otherwise be required with traditional, parametric classification methods such as ellipse fitting. Furthermore, CNNs can also be easily adapted and refined to classify images from different datasets using transfer learning in cases where a direct application is not possible, such as where there is a dramatic difference in image quality.

These models enable us to study the large-scale morphological evolution of galaxies in datasets from both cosmological simulations and observational surveys. In particular, we study the evolution of barred galaxies in EAGLE (classifying over 35,000 unique images), showing that the bar fraction remains roughly constant from the present epoch $z = 0$ out to redshift $z = 0.5$, before exhibiting a general decline out to redshift 1. We also classify the general morphologies of galaxies in EAGLE and, in doing so, find that the bar fraction is highest in spiral galaxies. Furthermore, by tracking galaxies over multiple snapshots in redshift, our model allows us to detect when bars form, and when they are destroyed. We find instances of galaxies that undergo multiple episodes of bar formation and destruction; with bars having an overall mean lifetime of around . Through tracking these events, and seeing whether the morphologies of the host galaxies change, we can comment on the potential mechanisms behind both the formation and destruction of bars over time.

Through utilising transfer learning, we are able to enhance the capabilities of our existing deep learning models to allow them to tackle the significantly noisier COSMOS imaging. This allows us to examine the redshift evolution of different morphologies in the COSMOS dataset out to $z = 1$ (over 85,000 images), with a particular focus on the fraction of lenticular galaxies in COSMOS. In doing so, we show that the fraction of lenticulars declines as redshift decreases, and that this decline varies depending on the mass range of lenticulars. By examining the physical properties of our classified lenticulars, we find evidence for two distinct subpopulations: a more massive, redder quiescent population, as well as a less massive, bluer star-forming population, the latter of which cannot be accounted for solely based on the archetypical “faded spiral” mechanism of S0 formation.

We also apply our models to study the kinematic properties of different galaxy morphologies, including barred and unbarred galaxies, in the SAMI dataset. We directly apply our deep learning models to high-resolution HSC imaging, demonstrating their remarkable versatility. We show that it is possible to differentiate between different morphologies based on the galaxy’s spin parameter proxy λ_{R_e} when plotted against both ellipticity ϵ and velocity dispersion σ . As an additional application showcasing the potential of deep learning in astronomy, and which draws from the same methods and techniques used in our

main work, we further develop a model capable of morphological segmentation and use it to highlight the extent of bars and spiral arms within galaxies. We apply this model to selection of barred galaxies in both the NA10 and SAMI datasets in order to estimate the lengths of their stellar bars, and subsequently explore how these bar lengths vary with respect to several properties including the mass and morphology of the host galaxy.

1.6.1 Thesis Structure

This thesis is structured as follows:

- In Chapter 2, we discuss the methods, techniques and background theory behind our deep learning approach. This includes a discussion on traditional, fully-connected neural networks and how they are trained, followed by details on convolutional neural networks and their various structural layers and components. We discuss our model architectures and how they were devised and subsequently refined through hyperparameter tuning. This is followed by a discussion of the overall training and evaluation workflow for our models, including details on preprocessing, data partitioning, data augmentation, and the core training procedure itself. Chapter 2 also includes some visualisation techniques that illustrate how a CNN processes an image of a galaxy, as well as how it represents morphology internally. This Chapter then concludes with details on model adaptation with transfer learning.
- In Chapter 3, we discuss the technical development, training, evaluation and preliminary applications of our core suite of convolutional neural networks. This includes further details on our newly-devised C2 architecture, and how its layout, training and performance compares to several previous architectures. Although the main focus is on the development of the direct 3-way and 4-way models, we also develop a suite of binary classifiers (e.g. elliptical vs. spiral), and apply them in an indirect, hierarchical manner. In particular, we show that our direct 3-way and 4-way classifiers are more accurate than an indirect, hierarchical classification approach. The models were trained on images of nearby, low-redshift galaxies from the NA10 dataset with redshifts $z < 0.1$. We evaluate the performance of our models through recovering the physical properties of the galaxies in our final NA10 test set, and also explore some applications to an independent sample of galaxies from the Galaxy Zoo survey, showcasing the model’s versatility.
- In Chapter 4, we apply our suite of deep learning models to classify galaxy morphologies in the EAGLE simulation. In particular, we classify mock SDSS g -band images of the simulated galaxies, hence enabling the direct application of our existing models. We further develop a new model, using the same C2 architecture, to classify barred galaxies. We use these models to study the evolution of the bar fraction with redshift in the EAGLE cosmological simulation. Based on these classifications, we study the changes in the bar fraction in different morphological types, examine the rates of bar creation and destruction, investigate scenarios of bar regeneration, and also investigate possible mechanisms that drive the creation and destruction of bars. Thanks to the versatility and consistency of CNNs, this study allows us to directly compare bars in observations with those in simulations.

- In Chapter 5, we apply our suite of deep learning models to classify the morphologies of high-redshift galaxies in COSMOS up to $z < 1$ in order to study the redshift evolution of the lenticular fraction. In particular, we discuss our use of transfer learning and data augmentation in order to enhance and adapt our existing models to better classify the noisier COSMOS images. This includes a discussion of the new C3 architecture, which was designed with this robustness and efficiency in mind. We apply an ensemble of these noise-augmented models to classify over 85,000 galaxies across a range of redshifts to track the redshift evolution of the lenticular fraction, including across different mass ranges. Based on these classifications, we identify two distinct populations of S0s with markedly different physical properties: a high-mass, red, quiescent population; and a low-mass, blue, star-forming population. We discuss this result along with its implications within the wider context of S0 formation and evolution.
- In Chapter 6, we apply our suite of deep learning models to classify morphologies of galaxies from the SAMI survey. Here we demonstrate the versatility of our models, as we apply it to high-resolution HSC images, which differ from the quality of the SDSS imaging that the models were originally trained with. We discuss the kinematic properties of our classified galaxies with respect to their morphologies, focusing on both the four primary morphological types in addition to barred and unbarred galaxies. We further discuss the development of a U-Net, a specialised deep learning model suited to performing morphological segmentation. We train this U-Net using crowdsourced bar and spiral masks from the Galaxy Zoo 3D survey, and subsequently apply this U-Net to extract masks highlighting the bars and spiral arms of galaxies using only their images as inputs. Lastly, we design and implement an algorithm to automatically estimate the length of stellar bars based on the bar masks predicted by our U-Net. We apply this algorithm to bar masks of known bars from NA10 and predicted bars from SAMI, and explore the changes in bar length with respect to various physical properties.
- In Chapter 7, we conclude this thesis with a summary of the key results and a discussion on the outlook and future potential of this work, in addition to promising future applications of deep learning to the field of astronomy as a whole.

Methodology

In this Chapter, we explore the fundamental theory behind the operation of convolutional neural networks, including how CNNs interpret and represent data, and discuss in detail the techniques used to design, optimise, train, evaluate and adapt our CNNs. The material and methods explored in this Chapter are integral to all subsequent Chapters. Before delving into CNNs, however, it is worth briefly discussing the relevant software packages and frameworks used to facilitate the creation, training and application of the models in this thesis. To develop our models, we used the general-purpose machine learning framework TensorFlow (Abadi et al., 2016) along with the high-level deep learning API Keras (Chollet et al., 2015), the latter of which we use to construct the model itself. These models were trained on Nvidia GPUs with both the CUDA Toolkit and cuDNN accelerator frameworks installed. Throughout this chapter, we will refer to various layers and components of CNNs. We use regular text (e.g. "convolutional layer") to refer to the concept, while we use monotype (e.g. Conv2D) to explicitly refer to the actual Keras layer object, such as when referencing code or the model architectures.

2.1 Neural Networks

A neural network is a type of machine learning model loosely inspired by the synaptic organisation of neurons in the human brain. In its most fundamental form, a neural network consists of several, sequential layers of nodes, such that each node in a given layer is connected to all the nodes in both the preceding and succeeding layers (Haykin, 2009). A network connected in this manner is termed "fully-connected", and a layer of fully-connected nodes is also referred to as a **Dense** layer. Each node in a neural network outputs a value known as its **activation**, which is determined as follows:

$$x_i^l = \phi \left(\sum_{j=1}^{N_{l-1}} w_{i,j} \cdot x_j^{l-1} + b_i \right) \quad (2.1)$$

where x_i^l refers to the i th node in the current layer l , x_j^{l-1} to the j th node in the preceding layer $l - 1$, $w_{i,j}$ denotes the weight assigned to the edge connecting x_i^l and x_j^{l-1} , b_i is a bias term, and ϕ denotes an activation function. The output of a node is hence equal to the weighted sum of its input nodes (plus a bias term), as passed to an activation function. Note that for the input layer the activations are simply the inputs to the network. This recursive equation is used to progressively calculate the outputs of each layer (this is known as a forward pass), and eventually the outputs of the neural network itself.

To train a neural network, we must supply it with a suitably large set of training data. Such data typically consists of tuples of data and labels; in our case, for instance, images of galaxies paired with a label (such as an integer) denoting its morphology. This is an example of supervised learning, since each example in the training data consists of both an input and a desired output (as denoted by the label). A neural network is trained so that its predicted output y' is as close as possible to the desired, true output y . This is akin to approximating a function that maps from an input x to the true output y . More specifically, the network is trained to minimise the statistical loss, as defined by some loss function, between y' and y . In practice, this is achieved through tweaking the weights and biases in proportion to their gradients with respect to this loss. The exact amount of tweaking is determined by both the learning rate and the optimisation algorithm, which is typically some flavour of gradient descent. This tweaking is usually conducted with random batches of data rather than on a per-sample basis for better computational efficiency. One complete iteration over the entire dataset (i.e. over all batches) is referred to as an *epoch*. This training loop continues over many epochs such that the weights and biases eventually converge to their optimum values, and the loss is minimised as much as possible. It is very important to note that the weights and biases are initially assigned random values without any prior assumptions regarding the training data itself. As such, neural networks learn by themselves via training. The above description is a high-level summary of the key concepts and terminology. We will return to these concepts more explicitly when discussing our model training workflow in Section 2.4.3, but for the scope and purpose of this thesis it is not necessary to go into any further depth. That said, for a more in-depth, general treatment of training neural networks, see [Haykin \(2009\)](#); [Goodfellow et al. \(2016\)](#); [Chollet \(2021\)](#).

2.2 Convolutional Neural Networks

Convolutional neural networks are an important class of neural networks that utilise convolutional filters to process feature maps. These convolutional filters all have their own kernel weights. Just as the weights and biases in a fully-connected network are tweaked during training, so too are these kernel weights tweaked whilst trained. This is how CNNs learn to automatically extract features (see [LeCun et al. 2015](#) for a review). CNNs are typically structured into two separate parts, as illustrated in Figure 2.1. The first part consists of several alternating convolutional and pooling layers, and the outputs consist exclusively of feature maps. It is here where the feature extraction process occurs. A sequence of convolutional layers terminated by a pooling layer is also referred to as a convolutional block and often treated as a single unit. The second part consists of a traditional, fully-connected neural network where the outputs of the first part have been flattened to a one-dimensional array. This part typically contains one or more fully-connected, dense layers, immediately followed by the final layer: the output layer.

As shown in Figure 2.1, the input to our CNN consists of an image, and the outputs consist of a single value for each category, which are to be interpreted as classification probabilities. In this context, these probabilities are also referred to as confidences, and we will use these terms interchangeably. The final classification is simply defined as the category with the highest output probability; in the case of Figure 2.1, this is a spiral. Note that while it is possible to configure the output layer as a single node so as to directly output the index of the predicted category, such a setup is not ideal for classification¹ as it

¹Such single-value outputs are better suited to regression.

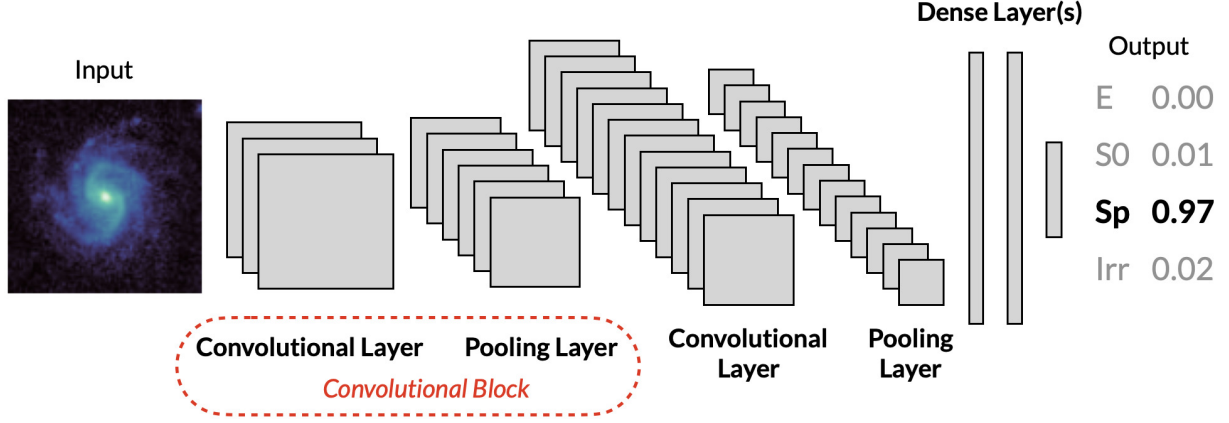


Figure 2.1: A simplified diagram showing the basic structure of a typical convolutional neural network. In this example, the outputs correspond to classification probabilities.

removes otherwise useful information about uncertainties. Indeed, as our models return probabilities for each class, we can use them not just to classify morphologies, but to also investigate which morphologies have higher degrees of uncertainties, and hence understand where misclassifications occur.

2.2.1 The Convolutional Layers

Convolutional layers are used to process feature maps through applying a discrete, linear convolution using a learnable filter (also known as a kernel). The kernel itself is typically much smaller than the size of the input, and the feature map is obtained through repeatedly applying the filter to patches of the input (akin to a sliding window) (see Dumoulin & Visin 2016 for a thorough selection of arithmetic examples). Each convolutional filter produces a single feature map, and the same filters weights are shared across all patches of the input (LeCun et al., 2015). Importantly, the weights for these filters are initially random, and are instead *learned* as the network is trained. It is also important to note that the same filter weights are applied to each patch of the image; this ensures a translation to invariance, in that the same features will be extracted regardless of their location in the image. We can express this convolution operation more explicitly as follows:

$$X_l^{(m)} = f \left(\sum_{n=1}^N W_l^{(n,m)} * X_{l-1}^{(n)} + b_l^{(n)} \right) \quad (2.2)$$

Here f denotes the activation function, $W_l^{(n,m)}$ denotes the filter weights associated with the m th output feature map in the current layer l , $X_{l-1}^{(n)}$ denotes the values of the n th feature map from the previous layer $l - 1$, and $b_l^{(n)}$ is a bias matrix. In practice both X and W are four-dimensional tensors; one dimension for the batch size, two dimensions for width and height, and one for the number of feature maps.

2.2.2 The Pooling Layers

Pooling layers are used to down-sample feature maps. Not only does this improve computational efficiency, but this necessarily enforces a degree of dimensionality reduction that improves the overall robustness of the network. There are two commonly used types of pooling: max pooling, which outputs the maximum

value for a given pool of values (this can be thought of as a means to highlight dominant features); and average pooling, which outputs the average value for a given pool of values (this can be thought of as akin to smoothing features). Of these, max pooling enjoys the most widespread use and tends to perform better empirically; maximal presence tends to be more informative than average presence (Chollet, 2021). Pooling layers also introduces a degree of invariance to small translations. For our models, we exclusively utilise max pooling layers.

2.2.3 The Dense Layers

Dense layers refer to fully-connected layers of nodes and edges, as with the traditional neural networks discussed in Section 2.1. In a typical CNN, after the last convolutional or pooling layer the feature maps are flattened into a one-dimensional array (in Keras this is achieved through the use of a dedicated Flatten layer). This one-dimensional array then passes through one or more dense layers, which serve as intermediate, hidden layers before the final output layer. To summarise in a somewhat metaphorical sense, it is with these layers that the network makes sense of the feature maps.

2.2.4 Activation Functions

Activation functions play a crucial role in shaping and constraining layer outputs. They are fundamentally important for neural networks since they introduce non-linearities. These non-linearities lie at the heart of the universal approximation theorems; theorems demonstrating that neural networks of arbitrary width and/or depth can approximate any continuous function (Cybenko, 1989; Zhou, 2020). Without them, neural networks would only ever be able to learn linear features (as implied by the linearity of both Equations 2.1 and 2.2). There are numerous activation functions in common usage (see Dubey et al. 2022 for a review). For the purposes of our models, we use the rectified linear unit (also known as ReLU) activation function, defined as

$$\text{ReLU}(x) = \max(0, x)$$

Activation functions are also very important for the output layer. In general, the choice of activation function for the output depends on both the nature of the task as well as the desired format of the output, which is often dictated by the format of the training data. Our task involves classification, and our outputs thus correspond to classification probabilities for each category. For this reason, we use the softmax activation function for our output layer, as this ensures all output values sum to 1.

2.2.5 Dropout Regularisation

Neural networks are inherently prone to overfitting. Many regularisation techniques have been developed to ameliorate this. One of the simplest such techniques is dropout regularisation (or simply, dropout; Srivastava et al. 2014), a generic technique that mitigates overfitting by introducing artificial sparsity. In practice, this is implemented with a Dropout layer, which randomly ignores a given fraction of its inputs during training. The next layer thus has access to less information, forcing it to become more robust. This fraction, known as the dropout rate, is typically around 50%. Dropout is most effective with larger neural networks.

2.2.6 Batch Normalisation

Batch normalisation (Ioffe & Szegedy, 2015) is a regularisation technique designed to address the issues of internal covariate shift; that is, the changes to the activations as the network’s parameters are updated during training. In practice, this is achieved by normalising the means and variances of the inputs for each batch. A consequence of this is that it permits higher learning rates without negatively impacting convergence and stability. As such, networks with batch normalisation are less reliant on dropout for mitigating overfitting.

2.3 Model Architectures

The general layout of a neural network – the number and arrangement of its constituent layers – is referred to as its architecture (other terms include topology, or simply layout). In this thesis, we developed a new convolutional neural network architecture (code-named C2) that outperforms existing architectures at classifying galaxy morphologies into multiple morphological types. This original architecture has undergone continued, iterative improvements, with a major redesign resulting in a much more efficient model (C3) that is faster to train and adapt. There is, of course, an important question regarding any architecture: how was it obtained? Machine learning is, by its very nature, an experimental discipline; many design patterns and established best practices are based on empirical results, yet heuristics can only go so far. Indeed, it is possible to conjure up all manner of architectures simply by testing different combinations of layers and parameters until one model so happens to work (a process pejoratively referred to as "gradient student descent"). However, this is in and of itself a tedious, inefficient procedure with no sound basis as to whether a model is truly optimal². There is a vastly better way to design and refine CNN model architectures: hyperparameter tuning.

2.3.1 Hyperparameter Tuning

A hyperparameter refers to any parameter that affects the architecture of a model, or the manner in which it is trained. Examples include the number of convolutional filters in a given layer, kernel sizes, the number of successive convolutional layers in a given block, the number of nodes in a fully-connected Dense layer, and the learning rate. Hyperparameters do not necessarily need to be numerical; even the types of pooling layers, optimisers and activation functions can be considered to be hyperparameters.

The general process can be summarised as follows. It is first necessary to define some base architecture, specifying the hyperparameters and defining their full range of allowed values. For instance, one may require that the number of filters in the first convolutional layer be between 32 and 64 inclusive in increments of 8. These various ranges altogether define a search space. The next step is to conduct the actual search itself using a search algorithm, ideally one that is not simply a brute-force grid search or naïve random search. In the course of this thesis, we used both the KerasTuner and Optuna frameworks to conduct hyperparameter tuning (O’Malley et al., 2019; Akiba et al., 2019). For each framework, we used the default search algorithm. For KerasTuner this is the bandit-based search algorithm Hyperband, which

²It would be a special case of irony if, in the process of designing a model to remove the need to have to manually inspect galaxies, one had to manually inspect CNN architectures.

uses adaptive resource allocation to swiftly evaluate and discard poor hyperparameter configurations (Li et al., 2016). For Optuna this is the Tree-Structured Parzen Estimator, a Bayesian approach in which future parameters are chosen based on their likelihood of improving the objective (Bergstra et al., 2011). In each case, the goal of the search algorithm (i.e. the objective) was to maximise the CNN’s validation accuracy: the accuracy with respect to some unseen validation data. These search algorithms run multiple trials, each with different combination of hyperparameters. In practice, however, the search is largely constrained by computational time. Over the course of this thesis, the total real-world processing time dedicated to hyperparameter tuning amounted to three and a half weeks.

When conducting hyperparameter tuning, it is especially important to ensure that the full training data has been appropriately partitioned. Crucially, the data used to evaluate the individual trial models should be independent of the test data used to evaluate the final model. This is since hyperparameter tuning is, by its very nature, implicitly biased to the data used to evaluate the individual trials. It is also easily susceptible to confirmation bias when the hyperparameter search space is later refined. An additional set of unseen data – completely removed from the hyperparameter tuning process – is therefore required in order to get a true, unbiased evaluation of the final model. For this reason, the complete training data is commonly partitioned into three independent sets:

- A **training set**, used for training the model.
- A **validation set**, used to evaluate the model during hyperparameter tuning.
- A **test set**, used to evaluate the final model (and *only* the final model).

Note that it is commonplace to use the terms validation set and test set interchangeably; what is important is that separate sets are used for hyperparameter tuning and for evaluating the final model. Furthermore, when training the final model it is acceptable to recombine the training and validation sets into a single, larger training set; the key being that the test set remains independent of both.

2.3.2 The C2 and C3 Architectures

The first new architecture devised as part of this project was initially inspired by the model architecture used in Dieleman et al. (2015). It adopts a similar three-block layout composed of alternating convolutional and pooling layers, with an additional convolutional layer in the middle conventional block (see Figure 3.1). The initial, candidate architecture consisted of many hyperparameters. These include the number of filters in each convolutional layer; kernel sizes for each convolutional layer; the number of nodes in each dense layer; the dropout rate; whether to use max pooling or average pooling; and finally the learning rate. We used KerasTuner (O’Malley et al., 2019) to conduct hyperparameter tuning in order to determine the optimal values. Full details of the C2 architecture are provided in Section 3.2.3.

Over the course of the thesis, this base, C2 architecture went through further iterative refinements, culminating in a major redesign benefiting from extensive hyperparameter tuning using the general-purpose framework Optuna (Akiba et al., 2019). This new architecture – code-named C3 – utilises five convolutional blocks of alternating convolutional and max pooling layers, and a single dense layer instead of two (see Figure 5.2). Another major change is the absence of any dropout layers; this is since the hyperparameter tuning found that a dropout rate of 0 performed best. The C3 architecture performs

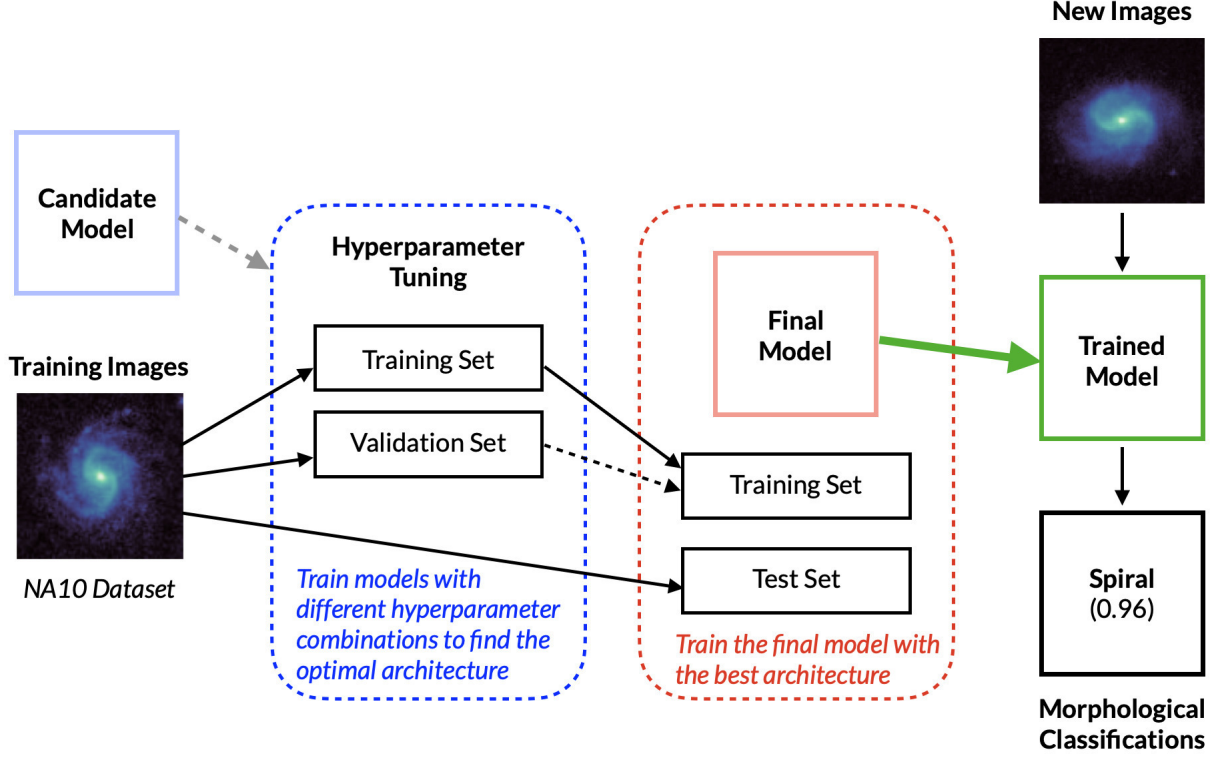


Figure 2.2: A diagram summarising the overall workflow for developing our suite of CNN models. A candidate architecture is first optimised with hyperparameter tuning, before being used to train the final model. This finalised model is then applied to new images in order to obtain morphological classifications. Note the three-way partition of the original training data.

marginally better than its predecessor architecture, yet the most important improvement is that C3 is significantly faster to train. This makes C3 more suitable for use with transfer learning, as demonstrated in Chapter 5. Full details of the C3 architecture are provided in Section 5.2.2.

2.4 Model Training

Figure 2.2 gives a schematic overview of the overall approach taken to develop our suite of deep learning models. The same approach is used for both the C2 and C3 architectures. We initially start with a candidate, base model architecture and conduct extensive hyperparameter tuning (as described in the previous section), eventually adopting the optimal architecture to train the finalised models. We then apply these models to new data (SDSS and Galaxy Zoo in Chapter 3; EAGLE in Chapter 4; COSMOS in Chapter 5; and SAMI and Galaxy Zoo 3D in Chapter 6) to classify galaxy morphologies.

2.4.1 The Training Data

For this thesis, we use the [Nair & Abraham \(2010a\)](#) (NA10) morphological catalogue of expert visual classifications for 14,034 galaxies. In particular, we used g-band SDSS images at a fixed physical size of 50kpc to train our CNNs. The galaxies themselves are classified according to numerical T-Types, however we combine galaxies from multiple T-Types together into up to four morphological categories (see Section 3.2.1

for full details). The first three morphological categories – elliptical, lenticular and spiral – are used to train the 3-way model (also referred to as the 3-class model), with the fourth irregular/miscellaneous category also included to train the 4-way (a.k.a. 4-class) model. Our labels are encoded using one-hot encoding (a.k.a. dummy coding). For example, in the 4-way case, an elliptical is encoded as $[1, 0, 0, 0]$, a lenticular as $[0, 1, 0, 0]$, and so on.

Importantly, we did not directly input the raw images themselves; instead, the images underwent additional preprocessing to make them suitable for use with CNNs. In particular, the images were all resampled via bi-cubic interpolation to a fixed pixel resolution of 100×100 pixels (corresponding to the input resolution of our CNN) using the Python package `Pillow`. Note that this image resizing only affects the pixel resolution; the physical resolution remains unchanged, at 50 kpc. The images were all linearly normalised such that all pixels have values strictly within the range 0 and 1 inclusive. No further alterations such as clipping or cut-off thresholds were made. Adopting a fixed range of inputs by normalising the input images is important for ensuring that the CNN can be applied consistently to new images. It also helps to improve the efficiency and stability of training.

As discussed in Section 2.3.1, it is important to partition the data into separate independent sets for training, validation with hyperparameter tuning, and for evaluating the final model. We partitioned the full NA10 training data into a training set and test set according to an 85:15 split for the initial C2 model, and 80:20 split for all subsequent models. For hyperparameter tuning, the training set was further split into separate training and validation sets with the same ratios. When partitioning the full dataset, it is important to ensure that each class shares similar fractional representations across each of the subsets (this is referred to as a stratified split). For example, if 50% of the training set consists of spiral galaxies, then the test set should also ideally contain 50% spirals to ensure a fair evaluation. We found the 80:20 split to be better balanced, especially since our sample contains very few irregular galaxies.

In an ideal setting, the training data for a given classification task should include an equal number of samples for each intended category. If instead there are an unequal number of samples in different categories, this may cause the model to overfit towards the overrepresented categories, and hence underperform on the underrepresented categories. This is referred to as categorical bias, and is an extremely ubiquitous, often unavoidable issue with training data across many disciplines. When developing our CNNs to classify galaxies as either barred or unbarred (see Chapter 4), we deliberately used an equal number of barred galaxies and unbarred galaxies from the NA10 catalogue, the latter being randomly sub-sampled from the more numerous, overrepresented pool of unbarred galaxies.

2.4.2 Data Augmentation

When it comes to deep learning, the more training data, the better. However, existing datasets (especially those with labelled data) are often carefully curated, and cannot be easily expanded to incorporate new samples. Fortunately, there exists a straightforward, practical set of methods that can be applied to any dataset to artificially increase the number of training samples simply by transforming existing samples. This is referred to as data augmentation. Among the simplest and most commonly used data augmentation methods are those involving affine transformations. These include rotations, mirroring, translation, rescaling (which essentially equates to zooming in and out given CNNs have a fixed input size), as well as shear, although shear is seldom used in practice. Other techniques include cropping, adding

Gaussian noise (something we will use heavily in Chapter 5), as well as a suite of colour augmentation techniques including changing brightness, contrast, saturation and hue (see Shorten & Khoshgofaar 2019 for a comprehensive review).

In this thesis, we primarily utilised rotations and mirroring. For the C2 model, we used 90° rotations and mirroring to increase the number of samples by at least a factor of 8, in addition to cropping (four corners plus centre, see Section 3.2.4 for details) to increase the number of samples by a total factor of 40. The rotations and mirroring were found to improve on the accuracy of the CNN, but cropping had comparatively little benefit and was subsequently abandoned when training later models given its high memory usage and computational cost. For the C3 model, we used the in-built Keras rotation and flipping layers (RandomRotation and RandomFlip respectively). Working in tandem, these preprocessing layers can transform the training data into a practically infinite number of unique images. The rationale behind these particular data augmentation methods is that image orientation should not matter; after all, an image of a spiral galaxy will remain an image of a spiral galaxy regardless of whether it is rotated or flipped. What matters is the content of the image, not its presentation. We will return to this point in both Section 2.6 and Chapter 5 where we examine data augmentation as a means of emulating data.

2.4.3 Training and Evaluation

All model architectures were coded and trained with the high-level, machine learning API Keras (Chollet et al., 2015), which also utilises general-purpose machine learning framework TensorFlow (Abadi et al., 2016). Figure 2.3 illustrates the core steps involved in training loop, which is entirely handled by Keras³. Apart from the training data and the model itself, there are two other important components. The first of these is the loss function, which defines how to interpret and quantify the statistical distance (i.e. the loss) between the predictions y' and true targets y (the ultimate objective being to minimise this distance). For our models we use categorical cross-entropy, which is a measure of how close an observed probability distribution is to the expected, true distribution. Minimising the cross-entropy is similar in formulation to maximising the log likelihood (Murphy, 2012). The second component is the optimiser, which is used to update the trainable parameters in the CNN itself (see Figure 2.3). For our models we utilise the Adaptive Moment Estimation optimiser (a.k.a. Adam; Kingma & Ba 2014), an especially efficient gradient-based optimiser.

We train our models to minimise the validation loss with respect to the final test set. To prevent overfitting and ensure optimal results, we utilise two Keras callbacks (Chollet, 2021). The first is ReduceLROnPlateau, a callback which reduces the learning rate by some given factor (0.1 by default) if the validation loss⁴ has not improved over a certain amount of epochs (this period over which to measure is referred to as the *patience*). The second is EarlyStopping, a callback which ceases training completely if, over the past n epochs where n is the *patience*, there has been no improvement to the validation loss. We also use the ModelCheckpoint callback to save the best-performing model weights.

There are several metrics with which to evaluate the performance of a CNN, including the overall accuracy, as well as the precision, recall (this is also less formally referred to as the per-class classification accuracy) and/or F1 scores for each individual class (Murphy, 2012; Chollet, 2021). The C2 architecture

³See https://keras.io/api/models/model_training_apis/.

⁴In general, these callbacks can work for any arbitrary metric/objective.

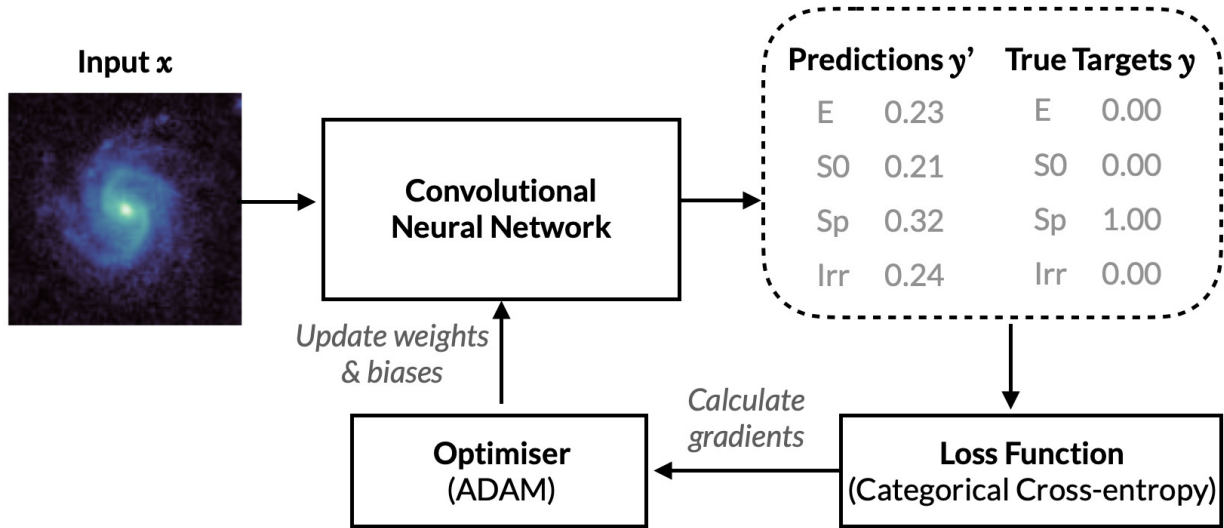


Figure 2.3: Diagram showing the core training loop for a CNN. The difference between the current output y' and true targets y is characterised in terms of a loss function. The optimiser then tweaks the weights and biases of the CNN in order to minimise this loss. This loop continues until terminated.

has overall accuracies of 83% (3-way) and 81% (4-way), with up to 98% as a binary classifier (elliptical vs. spiral). The C3 architecture has marginally better accuracies of 84% and 82% in the 3-way and 4-way case respectively. A common means of visualising the uncertainties of CNN classifications is with a confusion matrix. In this thesis we adopt the standard convention used in machine learning (specifically that used with `scikit-learn`; Pedregosa et al. 2012), where rows correspond to true categories, and columns to the predicted categories. Confusion matrices for the primary models trained with the C2 architecture are shown in Figures 3.6 and 3.7 (indicating the 3-way and 4-way models respectively), with Figure 4.2 showing the confusion matrix for the bar model. Confusion matrices for the models based on the C3 architecture are shown in Figure 5.4.

2.5 Model Visualisation and Interpretability

Convolutional neural networks are often treated as “black boxes”; in goes some input data, and out comes some result. For the most part, the inner workings are left unseen. This is partly by virtue of the inherent nature of CNNs, especially as the convolutional filter weights are tweaked automatically during training. Yet there is much that can be gleaned and understood from visualising how CNNs process their inputs, and how they represent the training data as a whole.

2.5.1 Feature Visualisation

Convolutional neural networks work by extracting features. Successive convolutional layers extract progressively more abstract features; it is this approach which enables CNNs to learn highly abstract representations of arbitrary data (LeCun et al., 2015; Goodfellow et al., 2016; Chollet, 2021). One of the immediate ways to visualise how a CNN works is to simply examine the feature maps, a.k.a. intermediate

outputs. This process yields a direct insight into how the input image is transformed as it progresses through the CNN. Another useful visualisation method is to focus on the convolutional filters themselves. In this filter visualisation technique, we iteratively transform a neutral, grey input image (pure Gaussian noise) into an image that maximises the response of a particular filter in some given layer (essentially running a gradient ascent loop on the filter). The resulting patterns are typically very abstract, but nevertheless provide an intuitive viewpoint of what the filters have been trained to respond to.

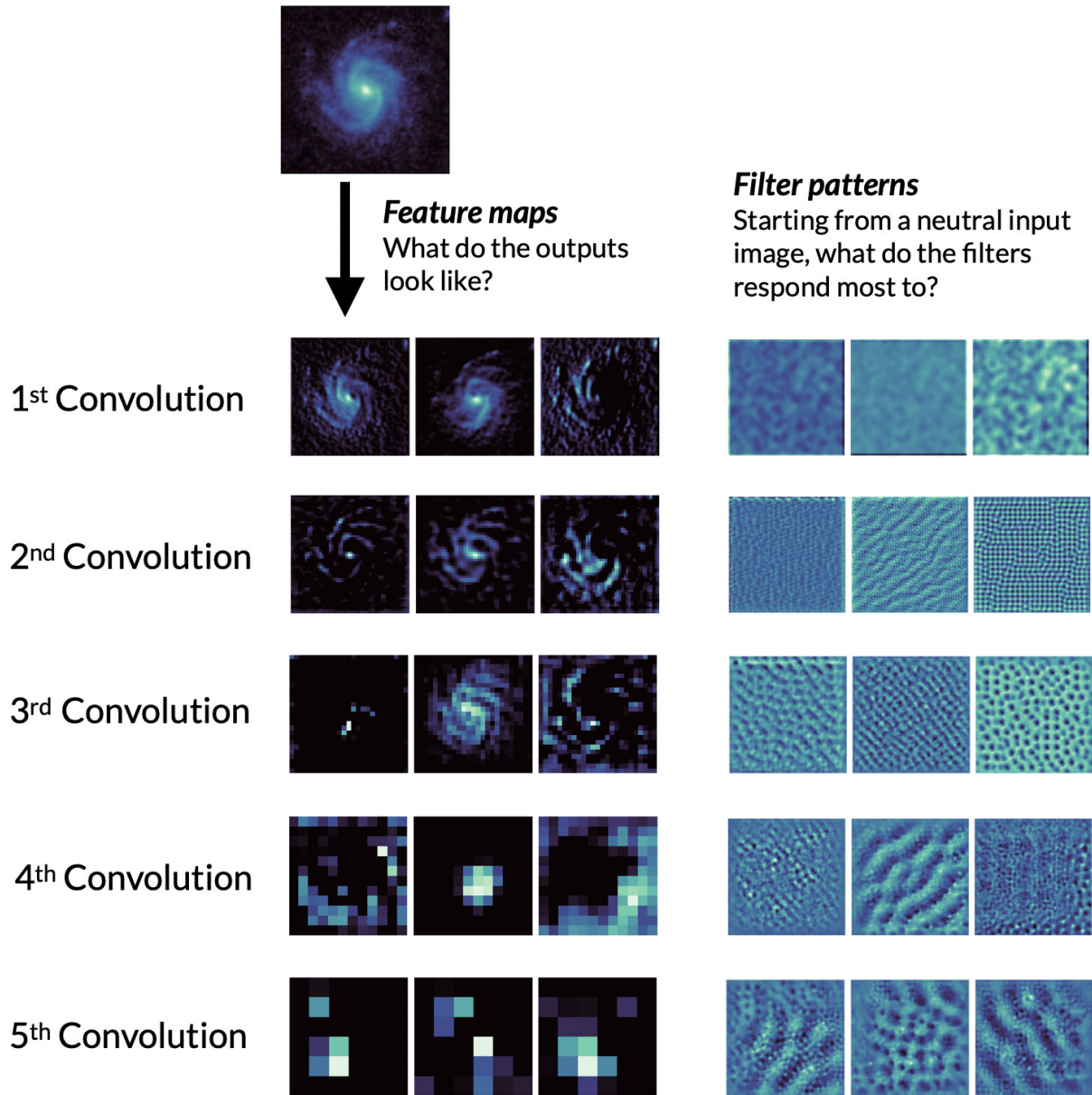


Figure 2.4: Selection of example feature maps and filter patterns for the five convolutional layers in the C3 model. The feature maps are obtained through inputting an example spiral galaxy. The filter patterns are obtained from using gradient-ascent on an initially neutral image to maximise the response of that filter. Note only a fraction of the total feature maps / filter patterns are shown.

Figure 2.4 shows an example of both feature map visualisation and filter visualisation for the C3 model architecture, where the former feature maps are obtained from an initial image of a spiral galaxy. As we progress through the five layers of the network, we can see how the feature maps become progressively more abstract. In the first couple of layers, the feature maps still resemble a spiral galaxy, yet by the fifth layer they all but resemble noise (at least to a human classifier). At the same time, we can see from the filter visualisation that as we progress from layer to layer, the filters progressively respond to finer, increasingly intricate details (with respect to some neural, grey input image). Figure 2.4 demonstrates how CNNs are extremely well suited to classifying galaxy morphologies, with the filters capable of extracting highly detailed features and sub-features, such as highlighting the edges of spirals.

2.5.2 How CNNs Represent Morphology

We have referred to training a neural network as akin to approximating some function f that maps from an input x to an output y . Indeed, the universal approximation theorems show that this is ultimately possible for any continuous function f . As we have seen, deep learning models achieve this simply through exposure to many examples. The success of deep learning lies in its ability to learn some abstract representation of the example data in order to generalise to unseen inputs. Fundamentally, the secret to this success lies in the nature of this underlying data representation.

The manifold hypothesis is an important and fundamental assumption that underpins much of machine learning. It proposes that everyday, high-dimensional data (such as images of galaxies) actually lie on some latent manifold within some embedded, lower-dimensional latent space. The key to deep learning – and, more generally, machine learning – is being able to approximate this latent manifold as closely as possible. This is essentially an exercise in high-dimensional curve fitting where the points are determined from the examples in the training data (Chollet, 2021). By fitting a manifold, a deep learning model is therefore able to generalise to completely new, unseen inputs simply via interpolating along the manifold’s surface. This also underscores the importance of training data; after all, there must be a sufficiently dense sample of points for the model to fit.

CNNs work by learning highly abstract representations of the data they are trained on. We can therefore get an intuitive sense for how the CNN interprets galaxy morphology by mapping the training data to points on the CNN’s latent manifold. For this example, we will use the penultimate layer of the C3 model as our latent space. Recall, at this specific layer, an input image is mapped to an array of 320 values, i.e. as 320-dimensional latent space. Therefore, all the images in the training set can be represented as embeddings within this 320-dimensional space. We can use a general-purpose dimensionality reduction routine such as principal component analysis, t-SNE or UMAP to transform these embeddings into two dimensions so that they can be more readily visualised. The result is shown in Figure 2.5, having been obtained using UMAP (McInnes et al., 2018).

It can be seen from Figure 2.5 that the morphological classes are fairly well separated. Of course, there are several caveats with the above analysis: the axes are arbitrary, and the shape of the reduced manifold is highly dependent on the dimensionality reduction algorithm. Many studies in machine learning, especially pertaining to image generation, have focused on exploring the latent spaces of deep learning models in order to interpret their fundamental structure. One example involves finding dimensions or principal components that control for specific features in some given dataset of images, such as hair colour and

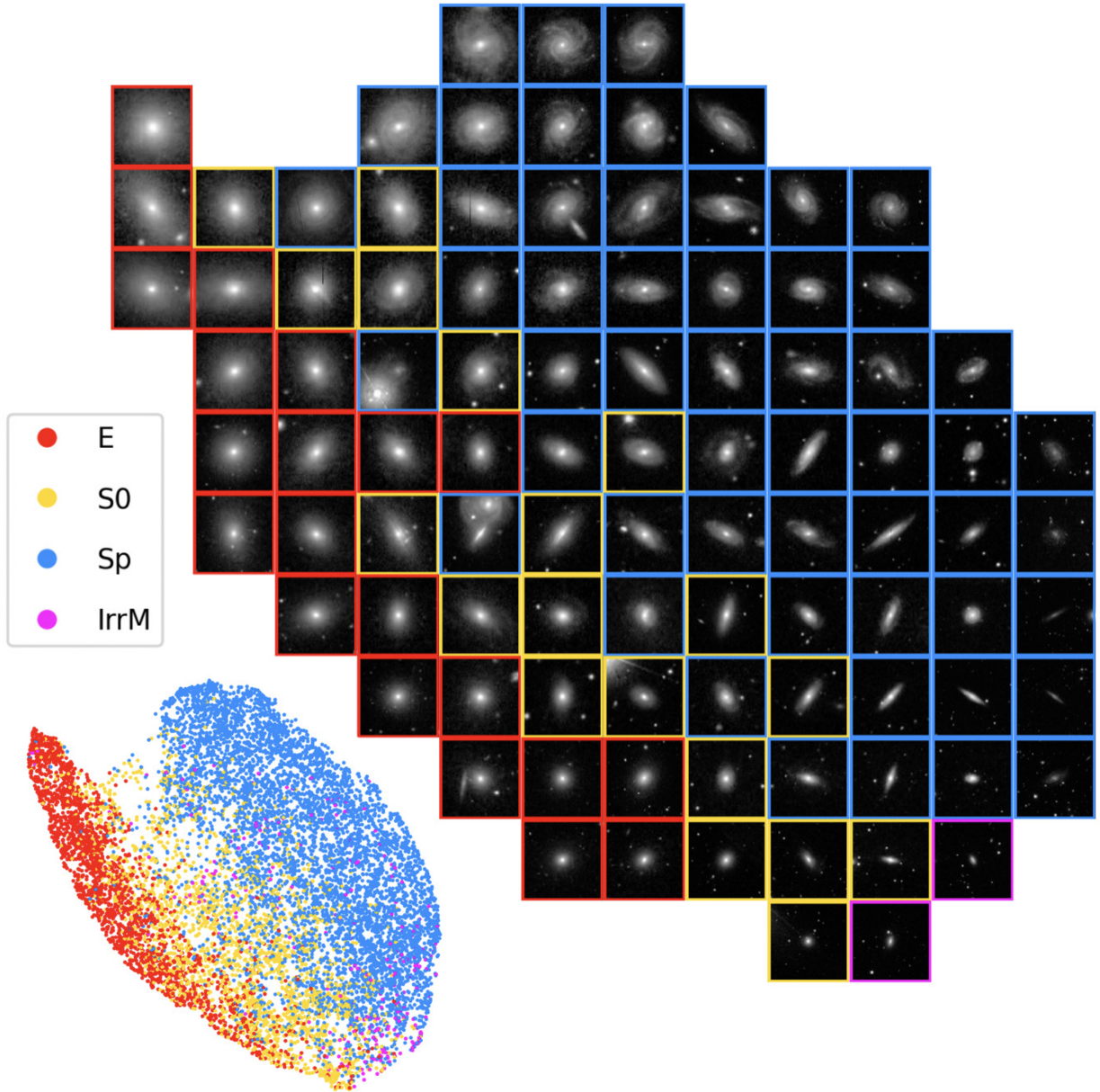


Figure 2.5: An image atlas showing example galaxies from the training data embedded in a reduced 2D manifold obtained through transforming the outputs of the penultimate layer of the C3 model (a 320-dimensional vector) with UMAP. The coloured outlines correspond to the morphological class. The full manifold consisting of all training samples is also shown in the bottom left. It can be seen that the manifold divides roughly into two halves, with ellipticals on one side and spirals on the other, while lenticulars are mostly located in between ellipticals and spirals. Irregulars are mostly spread out throughout the manifold but cluster slightly at the bottom right of the manifold. Images are shown exactly as they appear to the CNN.

facial expressions (Shen et al., 2020) or even different lighting conditions (Härkönen et al., 2020). Past studies in astronomy have explored the manifolds of galaxy properties using principal component analysis (Whitmore, 1984; Djorgovski, 1992). With the power of modern deep learning, such exploratory analysis techniques can be applied at scale to images of galaxies to interpret the fundamental structure of galaxy morphology (Cheng et al., 2021a), identify and group together similar morphologies (Walmsley et al., 2022), or even study the clustering of galaxy properties (Sarmiento et al., 2021). Indeed, Figure 2.5 hints at an underlying, global structure, with the y -axis roughly corresponding to physical size, and the x -axis corresponding to morphology. It can thus be seen that CNNs are useful tools not just for classifying galaxy morphologies, but also for understanding the nature of morphology itself.

2.6 Transfer Learning

In general, deep learning models are computationally expensive to train from scratch. One way to reduce this computational burden is to instead modify and fine-tune an existing, pretrained model. This can be achieved through the use of transfer learning, one of many subfields of machine learning (see Weiss et al. 2016 for a review). The core aim of transfer learning is that of domain adaptation. The goal is to adapt some base model, initially trained to perform a particular task in some source domain S , to instead perform a new task in some different yet related target domain T . In an intuitive sense, transfer learning aims to leverage the pre-existing capabilities of a pretrained model such that it can be applied to related domains. As a result, transfer learning can be used to train new models with considerably fewer data than would otherwise have been required to train from scratch. In practice, there are several ways to explicitly (re)train a model with transfer learning. One common method involves freezing part of the base model, and simply fine-tuning the remaining layers. Here, freezing a layer simply means preventing its weights and biases from updating; the intuition being that it preserves all the pretrained values. In this scenario, the convolutional layers are typically frozen to preserve the learned convolution filters for feature extraction, while the remaining dense layers are left alone to be tweaked. Another method is to leave the entire base model as-is and simply fine-tune the entire model, albeit with a considerably lower learning rate compared to the original learning rate used for the base model (typically by at least a factor of 10). This ensures that the pretrained weights are gradually refined rather than being immediately overwritten. In this thesis, we adopt the complete-model fine-tuning approach.

As will be discussed in detail in Chapter 5, we primarily use transfer learning for the purpose of adapting our existing models, initially trained to classify SDSS images from the NA10 morphological catalogue (source domain), to instead classify images of galaxies from the COSMOS survey (our target domain). However, there is still the requirement for a set of training data with which to train the new model. In this situation, there are two options. The first is to use some labelled COSMOS data (i.e. images of COSMOS galaxies with known morphologies), however there are no such suitable datasets that utilise the same labelling schema as that of NA10⁵. Another method is to modify the images from our existing NA10 training data such that they resemble images from the COSMOS dataset. The key advantage of emulating the full dataset in this manner is that it allows us to simply re-use the existing labels (i.e. the

⁵It is by no means a general condition for transfer learning that both models use consistent labels (this would preclude many applications). It is, however, a consideration in our case as we wish for our models to classify the same morphological types.

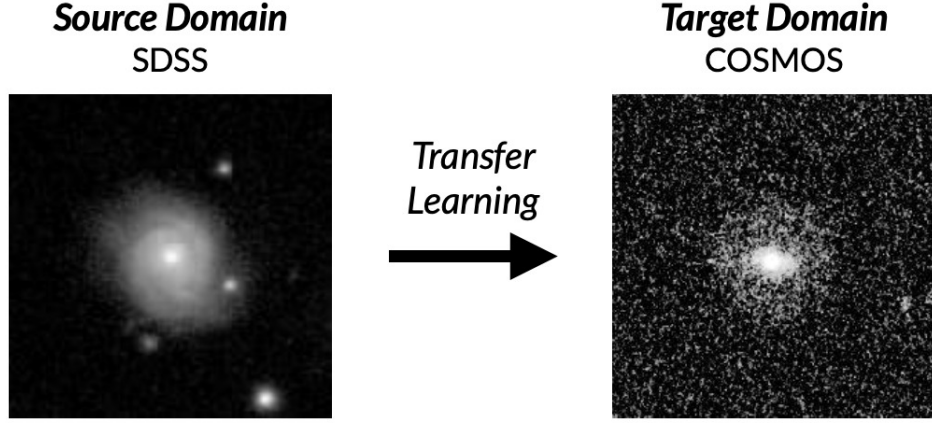


Figure 2.6: Example galaxy images from the SDSS (left) and COSMOS (right) datasets. In the case of transfer learning, we wish to adapt models trained to classify SDSS images to instead classify COSMOS images.

morphologies). Both of these transfer learning techniques have been shown to be appropriate for adapting models to different surveys, and even for emulating galaxies at higher redshifts (Domínguez Sánchez et al., 2019; Vega-Ferrero et al., 2021).

As part of this thesis, we explore using transfer learning to adapt models initially trained to classify low-redshift images to instead classify high-redshift images. This is achieved through adapting our existing models trained on galaxies from the NA10 dataset ($z < 0.1$) to classify galaxies from the COSMOS dataset ($z < 1$). The biggest hurdle for such a domain shift is the dramatic difference in image quality. Figure 2.6 shows the dramatic difference in noise levels. Noise has a material impact on the accuracy of CNNs, particularly for models not explicitly trained on noisy images. Modifying inputs with artificial noise is a known strategy for data augmentation and for improving the robustness of CNNs (Jin et al., 2016; Shorten & Khoshgoftaar, 2019). As such, to carry out the transfer learning, we aim to reproduce the characteristic noise levels of the galaxy images in our COSMOS sample. We then utilise data augmentation to apply this noise into the SDSS images, the goal being to emulate the visual appearance of the COSMOS images. This is achieved through the use of a custom Keras layer that applies these noise levels during training itself, ensuring that the CNN is trained over a wide range of noise levels rather than a few predefined levels that could be easily overfitted to. An important caveat with this approach is that it assumes that low-redshift galaxies are representative of high-redshift galaxies; in particular, that there is no evolution over time. This is a significant issue at very high-redshifts where galaxy morphologies are markedly different, however to limit this we only consider COSMOS galaxies up to $z = 1$.

In astronomy, previous studies have successfully explored and adopted various techniques of transfer learning to adapt pretrained models to carry out new tasks. These include adapting models to classify images from different surveys (Domínguez Sánchez et al., 2019), and even to adapt models trained on images of simulated galaxies to instead classify images of real galaxies (Ghosh et al., 2020; Eriksen et al., 2020). Interestingly, Ackermann et al. (2018) successfully adapted pretrained models trained on images from the general-purpose ImageNet image classification dataset (Deng et al., 2009) to instead classify galaxy mergers. This is a notable example, since ImageNet does not contain any images of galaxies, let

alone any astronomical objects. Indeed, adapting large-scale pretrained models is a common practice in modern deep learning, most notably in natural language processing with large language models such as GPT-3, as well as in image classification and generation (see [Han et al. 2021](#) for a review). Transfer learning will likely be an integral part of future applications of deep learning in astronomy, particularly in the applications of existing CNNs (including our deep learning suite developed in this thesis) to new images from future surveys.

The morphological classification of galaxies with deep learning

The first step on the road to analysing galaxy morphologies is the development of AI models that can automatically classify galaxy morphologies in an efficient, accurate and consistent manner. While existing machine learning classifiers in astronomy are predominantly binary classifiers, for example distinguishing between elliptical vs. spiral, we wish to develop a model that can directly differentiate between more than two morphologies at a time. In this Chapter, we design a new convolutional neural network architecture, and use this to train two models capable of classifying galaxies into one of multiple morphological types. In particular, we train a 3-way model to classify galaxies as either elliptical, lenticular or spiral, as well as a 4-way model that adds a fourth morphological class for irregular and/or miscellaneous galaxies. These models are among the first to include a dedicated morphological class for lenticular galaxies alongside spirals and ellipticals. We find that our new architecture outperforms previous architectures at morphological classification. To see how our 3-way and 4-way models compare with binary classifiers, we developed several binary classifiers and applied them in a hierarchical manner akin to a decision tree. We find that our direct 3-way / 4-way classifier approach performs better than the hierarchical approach. To verify that our models are making physically sensible predictions, we compare the physical properties of the galaxies from our test dataset based on our predicted morphologies with those based on the true morphologies, and find that there is a good level of overall agreement. We further demonstrate the robustness of these models by classifying an independent sample of low-redshift galaxy images from the Galaxy Zoo dataset.

This chapter is based on the manuscript “Morphological classification of galaxies with deep learning: comparing 3-way and 4-way CNNs”, which was published on June 2, 2021 in the journal Monthly Notices of the Royal Astronomical Society, Volume 506, Issue 1, pp. 659–676. There are some minor textual and formatting changes compared to the final, published version. The background theory on convolutional neural networks and details of the development of our models were substantially expanded upon in Chapter 2 of this thesis.

Abstract

Classifying the morphologies of galaxies is an important step in understanding their physical properties and evolutionary histories. The advent of large-scale surveys has hastened the need to develop techniques for automated morphological classification. We train and test several convolutional neural network architectures to classify the morphologies of galaxies in both a 3-class (elliptical, lenticular, spiral) and 4-class (+irregular/miscellaneous) schema with a dataset of 14034 visually-classified SDSS images. We develop a new CNN architecture that outperforms existing models in both 3 and 4-way classification, with overall classification accuracies of 83% and 81% respectively. We also compare the accuracies of 2-way / binary classifications between all four classes, showing that ellipticals and spirals are most easily distinguished (>98% accuracy), while spirals and irregulars are hardest to differentiate (78% accuracy). Through an analysis of all classified samples, we find tentative evidence that misclassifications are physically meaningful, with lenticulars misclassified as ellipticals tending to be more massive, among other trends. We further combine our binary CNN classifiers to perform a hierarchical classification of samples, obtaining comparable accuracies (81%) to the direct 3-class CNN, but considerably worse accuracies in the 4-way case (65%). As an additional verification, we apply our networks to a small sample of Galaxy Zoo images, obtaining accuracies of 92%, 82% and 77% for the binary, 3-way and 4-way classifications respectively.

3.1 Introduction

Galaxies exhibit a variety of morphologies throughout cosmic time (Conselice, 2014), ranging from broad families of spirals to large ellipticals. A galaxy’s morphology is related to its internal physical processes (Sellwood, 2014), such as its dynamical and chemical evolution, star formation activity (Kennicutt, 1998), and can reveal insights into its past evolutionary history, including mergers (Mihos & Hernquist, 1996) and interactions with its environment (Sol Alonso et al., 2006). A galaxy’s morphology is often determined by visual inspection, and while this is sufficient to delineate a rich array of morphological types in the nearby Universe (Buta, 2013), there is the inherent issue of scalability as future surveys yield far greater magnitudes of data.

Visually classifying the apparent morphologies of galaxies through the manual inspection of images is a laborious, time-consuming task. Recently, the use of citizen science has seen great success in classifying larger datasets (Lintott et al., 2011; Willett et al., 2013; Simmons et al., 2017), however it is important to keep in mind that this was achieved by increasing the order of magnitude of human classifiers, while the speed of the actual classification itself is more or less unchanged. Future large-scale surveys are expected to yield enormous amounts of data, with *Euclid* alone expected to survey at least to the order of 1 billion galaxies. The scale of surveys such as these will overwhelm the capacity of human volunteers (Silva et al., 2018), necessitating the development and deployment of techniques for automated classification.

Automated classification techniques, in particular those utilising neural networks (Dieleman et al., 2015; Domínguez Sánchez et al., 2018; Zhu et al., 2019), have the ability to revolutionise the speed at which samples can be individually classified. Such classification techniques range from analytical approaches, such as the Fourier analysis of luminosity profiles (Odewahn et al., 2002) and isophote decompositions (Méndez-Abreu et al., 2017), to statistical learning methods (Sreejith et al., 2018), random forest classifiers (Beck et al., 2018) and ensemble classifiers (Baqui et al., 2021), which are all part of a broader suite of

machine learning methods (Cheng et al., 2020). Neural networks have been utilised in this field for some time, but only recently has the rapidly growing field of deep learning seen widespread applications in astronomy (Baron, 2019). One of the key architectures behind the success of deep learning, particularly in applications involving image recognition and computer vision, is the convolutional neural network (hereafter CNN), introduced by LeCun et al. (1998). Unlike the typical multiplayer perceptrons present in a normal neural network, CNNs are specifically designed to extract features in data through multiple convolutions (LeCun et al., 2015). Each convolution can be thought of as a layer of abstraction, with the extracted high-level features used to learn a generalised representation of data. Importantly, CNNs provide a model-independent means of feature extraction, hence their versatile range of applications. Although having originally been developed to recognise handwritten characters, modern CNNs are capable of general image recognition across myriad image types (He et al., 2015). Indeed, image classification is one of the main astronomical applications of CNNs (Cheng et al., 2020).

CNNs have been successfully utilised to detect quasars and gravitational lenses (Pasquet-Itam & Pasquet, 2018; Schaefer et al., 2018), study bulge/disk dominance (Ghosh et al., 2020), detect stellar bars (Abraham et al., 2018) and classify radio morphologies (Wu et al., 2019). CNNs have also been widely used with simulations, including cosmological simulations (Mustafa et al., 2019) and mock surveys (Ntampaka et al., 2020), as well as to develop tools for galaxy photometric profile analysis (Tuccillo et al., 2018). Recent studies have utilised CNNs for the purpose of binary classification (Ghosh et al., 2020), or classifying between general morphological shapes (Zhu et al., 2019). Fewer studies have looked at 3-way classification between distinct morphological types, though some works have explored classifying between ellipticals and barred/unbarred spirals (Barchi et al., 2020), or between ellipticals, spirals and irregulars (De La Calleja & Fuentes, 2004)

In this work, we train and test several different CNN architectures for the purpose of initially distinguishing between elliptical (E), lenticular (S0) and spiral (Sp) galaxies, before extending the schema to include a fourth irregular and miscellaneous category (Irr+Misc). We base our training data on the visually classified catalogue of Nair & Abraham (2010a), consisting of 14034 samples from the SDSS Data Release 4 (Adelman-McCarthy et al., 2006), and apply a series of data augmentation techniques to increase the number of training samples for use with the CNN. Our best accuracies are consistent with previous studies, surpassing others when it comes to per-class accuracies. We dedicate a section of our discussion to focusing on the physical implications of the CNN classifications (both correct and incorrect), as well as how these are reflective of the inherent uncertainty in the training data, how such uncertainties affect CNN accuracies, and how CNNs can be used to address these uncertainties.

The structure of the paper is as follows. In Section 3.2 we briefly outline the theory of CNNs and its key concepts and terminology, discuss the CNN architectures including our new C2 network, discuss the augmentation and training methodology, and discuss the overall performance of the 3-way and 4-way classification tasks for each of the four CNN architectures. In Section 3.3 we present our key results, starting with binary classifications between morphological classes, before presenting the best results for the 3-way and 4-way classifications and analysing these by considering the physical properties of the samples for each classification. In Section 3.4 we discuss the accuracy of the CNN. We examine the physical properties of classified samples and show that there are common trends in the misclassified samples across several physical properties. We showcase binary hierarchical classification as an alternative

Table 3.1: Definition of our four morphological categories in terms of Hubble morphologies and numerical T-Types schemas. Ellipticals are defined as c0 to E+ (Nair T-Type -5 only), S0s defined as S0- to S0/a (Nair T-Types -3 to 0), Spirals as Sa to Sm (Nair T-Types 1 to 9), and Irr+Misc as Im onwards (Nair T-Types 10 and 99).

Category	Elliptical			S0				Spiral									Irr+Misc	
Class	c0	E0	E+	S0-	S0	S0+	S0/a	Sa	Sab	Sb	Sbc	Sc	Scd	Sd	Sdm	Sm	Im	Misc
RC3	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10	-
Nair	-5	-5	-5	-3	-2	-2	0	1	2	3	4	5	6	7	8	9	10	99

to the direct multi-class CNNs, exploring five different classifiers and their key differences to our main CNN. We present several examples of images that were correctly and incorrectly classified, commenting on inherent uncertainties and the challenge of unambiguously classifying samples, whether by eye or by CNN. As a final, independent verification of our network, we apply our models to a small sample consisting of 1,000 randomly chosen ellipticals and 1,000 randomly chosen spiral galaxies from the Galaxy Zoo dataset. Lastly, we summarise our findings in Section 3.5.

3.2 Methods

3.2.1 Dataset

The [Nair & Abraham \(2010a\)](#) catalogue, hereafter NA10, contains 14,304 visually classified galaxies from the SDSS DR4 ([Adelman-McCarthy et al., 2006](#)) with redshifts $z < 0.1$. These are all single-band, g-band images with spatial resolutions of 50 kpc. Of key interest to this work are the Hubble T-Type classifications, which range from -5 to 12 with a miscellaneous 99 category for samples with no clear category. The miscellaneous category also includes interacting galaxies and mergers. One of the goals of the catalogue is to serve as a dataset with which to calibrate automated classification techniques.

For this work, we consider up to four morphological categories; ellipticals (E), lenticulars (S0), spirals (Sp) and irregulars plus miscellaneous types (Irr+Misc). The samples in the Nair catalogue have T-Types that range from -5 to 12, with a miscellaneous T-Type 99. Table 3.1 outlines our definition of the four morphological classes, along with the NA10 T-Types and the classic RC3 ([de Vaucouleurs et al., 1991](#)) schema for reference. Using these definitions, 2723 samples are categorised as ellipticals, 3215 as S0s, 7708 as Spirals and 388 as Irr+Misc.

3.2.2 Convolutional Neural Networks

In this section we briefly outline the theory of neural networks, introduce some key terminology and explore how CNNs differ from standard neural networks. A standard, fully-connected, feed-forward artificial neural network consists of multiple layers of nodes, such that each node is connected to all the nodes in its preceding and succeeding layers ([Haykin, 2009](#); [Goodfellow et al., 2016](#)). The input data is mapped to the nodes in the input layer, with the nodes in the final layer corresponding to the outputs. Each node has an activation value x that is obtained as follows:

$$x_i^l = \phi \left(\sum_j^{N_{l-1}} x_j^{l-1} w_{i,j}^l + b_i^l \right)$$

where l denotes the current layer, x_i^l denotes the current node, N_{l-1} is the number of nodes in the previous layer, x_j^{l-1} is the activation of the j th node in the previous layer, $w_{i,j}^l$ is the weight assigned to the edge connecting the current node x_i^l to the previous node x_j^{l-1} , b_i^l is a bias, and ϕ is an activation function. To train a neural network, we supply it with at least two similarly representative sets of data; a training set, and a test set. A third validation set, independent of the test set, is used to tune the hyperparameters of the network. The neural network is trained by adjusting the weights w^l and biases b^l for all nodes in every layer l to minimise the error (with respect to an objective or loss function) with the desired outputs in the training set. This is usually achieved via a learning algorithm such as gradient descent over multiple iterations of training, where each iteration is referred to as an epoch. Once trained over a sufficiently large number of epochs, the final success rate is determined via an evaluation on an unseen test set. The training is typically conducted in batches, with the weights and biases updated for each batch.

CNNs are a specialised type of neural network designed to extract high-level features from the provided input data through the application of convolutions. Each 2D convolutional layer takes an $s \times s$ input array and applies m filters/weights using a $k \times k$ kernel (a.k.a window), where $k < s$, in order to obtain m unique feature maps. In particular, if the input to a convolutional layer is a set of N matrices $\mathbf{X}_{l-1}^n$ for $n = 1, \dots, N$, then the feature maps are given as

$$\mathbf{X}_l^{(m)} = f \left(\sum_{n=1}^N \mathbf{W}_l^{(n,m)} * \mathbf{X}_{l-1}^{(n)} + b_l^{(n)} \right)$$

where m denotes the m th feature map, $\mathbf{W}_l^{(n,m)}$ are the filters/weights for layer l , input map n and feature map m , $b_l^{(n)}$ is a vector of biases, and f is a (multi-dimensional) activation function. Here $*$ denotes a linear convolution. Note that the inputs and feature maps need not be square. Importantly, each feature map is convolved from the input image with a different set of filters/weights. These weights $\mathbf{W}_l^{(n,m)}$ are learnable and are tweaked during training. Key to the success of CNNs is that they learn multiple filters in parallel, which enables the network to not only obtain different abstract representations of the data, but also use these in order to distinguish between data of different types.

Unlike in a normal neural network, where each node's edges are free parameters, feature maps use shared weights by applying a single convolutional filter across the entire input, rather than separate filters for individual parts of the input. Thus, in order to minimise the loss with the training data, the CNN directly adjusts these filter weights, hence obtaining different feature maps. As such, CNNs provide a model independent means to classify images, relying solely on self-taught convolution weights to extract the most meaningful features in data. Successive convolutional layers are used to extract higher-level features. Generally, later convolutional layers extract more feature maps with smaller kernels.

CNNs also contain another important layer known as a pooling layer. These layers are used to down sample feature maps (Zhou et al., 2015). For this work, we use max pooling. This method of pooling preserves the maximal elements, helping to highlight dominant features. This also helps to introduce a degree of translational invariance and also serves to reduce noise. After the features have been extracted via convolutional and pooling layers, all the feature maps are flattened into a single 1D array. These are then fully connected to subsequent layers of nodes (referred to as dense layers), ultimately ending with the output layer which contains a total number of nodes equal to the total number of output categories. These layers are functionally identical to a normal neural network. After flattening the feature maps, it is

common to remove a fraction of randomly selected nodes. This is an efficient regularisation technique known as Dropout, and is used to reduce the impacts of overfitting and improve generalisation (Srivastava et al., 2014).

3.2.3 Model Architectures

In this work we consider four principal CNN architectures, each with different complexity and originally designed for different input image sizes. All architectures are designed to accept single-band images as their input. Figure 3.1 displays a schematic illustration of each architecture. Each architecture is built using Keras (Chollet et al., 2015), a high-level machine learning interface. The first CNN, referred to as C1, is based on our previous work (Cavanagh & Bekki, 2020), and was originally designed for the binary classification of 50×50 imagery. This architecture consists of a single block of convolution and pooling layers (two Conv2D plus a MaxPooling), a penultimate Dense layer with 256 nodes, and the output Dense layer which contains either three or four nodes for 3-way or 4-way classification.

The second CNN architecture is based on the work of Dieleman et al. (2015). This architecture contains multiple convolutional blocks, starting with two blocks of alternating, single Conv2D and MaxPool layers, then a block of two successive Conv2D layers followed by a final MaxPool. There are also two Dense layers before the output layer. Key differences to the original implementation in Dieleman et al. (2015) is that we make use of the Adadelata adaptive learning rate algorithm for training (Zeiler, 2012), and we have also added Batch Normalisation layers at the end of every convolutional block. When training a network, the training data is usually processed in batches, with the weights and biases tweaked after each batch. Batch normalisation is used to enforce uniformity across all samples in a batch by keeping a consistent mean and standard deviation. This has the effect of smoothing the objective function (hence smoother gradients), leading to better performance, improved regularisation and generalisation (Ioffe & Szegedy, 2015; Santurkar et al., 2019).

The third architecture is based on AlexNet (Krizhevsky et al., 2012), a network designed for general image classification. This architecture has commonly been used in many classification problems, including most recently in astronomy (Ghosh et al., 2020). This is the most complex architecture out of the four, and is notable for its heavy use of dropout regularisation with two Dropout layers, each dropping 50% of the previous layer’s nodes. The convolutional layers also utilise a large number of filters. The original optimiser used for the training of this network is stochastic gradient descent with momentum, however the learning rate is computationally expensive to tune. We instead use the Adadelata adaptive learning rate algorithm. We also add batch normalisation layers after each convolutional block but before the max pooling, adding even more regularisation to the network.

The fourth architecture, named C2, is completely new from this work, designed as a three-block CNN with two dense layers, a single dropout layer, and batch normalisation throughout. Unlike previous models, we decided to use the Adam (Adaptive Moment Estimation) algorithm (Kingma & Ba, 2014), an efficient stochastic gradient descent algorithm that can be considered an extension of Adadelata. Unlike Adadelata, Adam’s learning rate must be pre-specified. To determine the optimal design for our C2 architecture, including the best learning rate, we performed hyperparameter tuning with the tool KerasTuner (O’Malley et al., 2019). A hyperparameter is any value that affects the architecture of a neural network (e.g. number of nodes) or the manner in which it is trained (e.g. learning rate). Hyperparameter tuning refers to the process

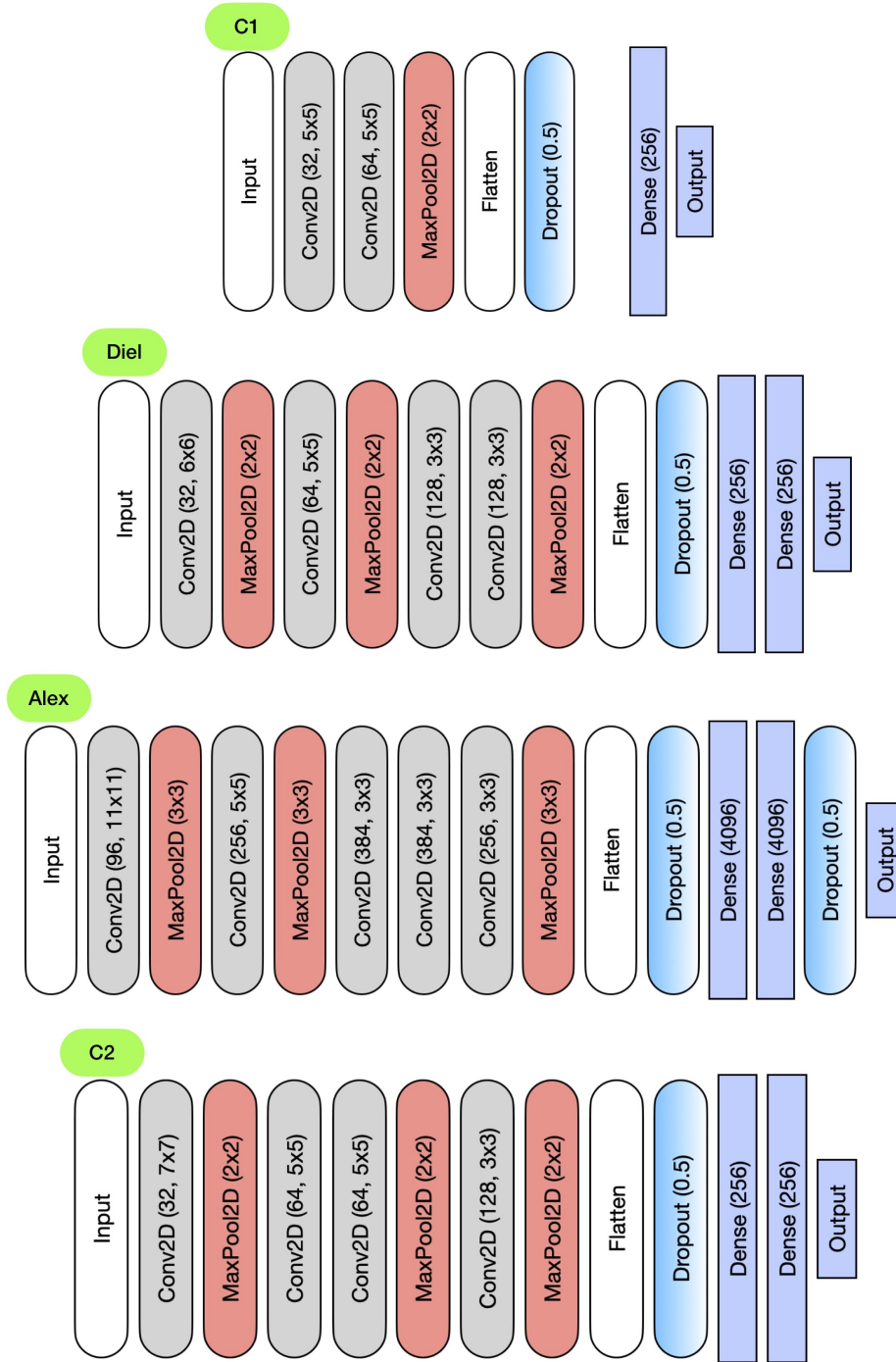


Figure 3.1: Schematic representation of all four principal CNN architectures used in this work: the CNN from Cavanagh & Bekki (2020) (C1), the CNN based on Dieleman et al. (2015) (Diel), the CNN based on Krizhevsky et al. (2012) (Alex), and our new CNN introduced in this work (C2). Layers are colour coded: grey for convolutional layers, red for pooling, light blue for dropout and blue for dense layers. Convolutional layers are annotated with the number of feature maps, followed by the kernel size. Pooling layers are annotated with the size of the pooling. Dropout layers are annotated with the dropout rate (the fraction of nodes ignored in the previous layer). Dense layers are annotated with the number of nodes.

of optimising the hyperparameters within the search space as specified by the range of desired values. In the case of C2, we started with the same sequence of layers in Figure 3.1, and used hyperparameter tuning to determine the optimal number of feature maps for each convolutional layer (out of a choice of 32 or 64 for the first convolutional layer, and 64 or 128 for all layers thereafter), width of the convolutional filter or kernel (7x7 or 5x5 for the first layer, 5x5 or 3x3 for all others), number of nodes in the dense layers (256, 512 or 1024), degree of dropout (0.3, 0.4 or 0.5), whether to use max pooling or average pooling, and finally the learning rate for the Adam optimiser, to be determined by sampling between 10^{-3} and 10^{-4} . The optimal parameters for the C2 architecture, as determined by hyperparameter tuning, are annotated in Figure 3.1. The final, tuned learning rate for the Adam optimiser was 2×10^{-4} .

Across all architectures, we used consistent activation functions, with ReLU for all convolutional and dense layers, and with a final softmax activation for the output layer. Softmax, a type of normalised exponential, is important as it converts the activations into probabilities that sum to 1 across all output categories. The predicted category, or simply the prediction, is just the category with the maximum value (in other words, the output node with the highest activation). This maximum value itself is often referred to as the probability or confidence; we will use both terms interchangeably.

3.2.4 Data Augmentation and Pre-Processing

The size of the NA10 dataset is relatively small given the complexity of the CNN architectures, and is thus prone to overfitting. In order to increase the size of the dataset, and subsequently improve the performance of the CNN, we apply several data augmentation techniques. These are listed as follows:

- Cropping. For an image of size $n \times n$ we can crop multiple $m \times m$ images for $n > m$. We have chosen to crop each image from each corner, as well as in the center, for a total of five cropped images. Cropping was done with an 110×110 image to yield five 100×100 images.
- Rotation. We rotate each image by 90° increments, yielding four images.
- Flipping / Mirroring. Yields two new images. Our implementation is equivalent to a flip about the axis $y = x$.

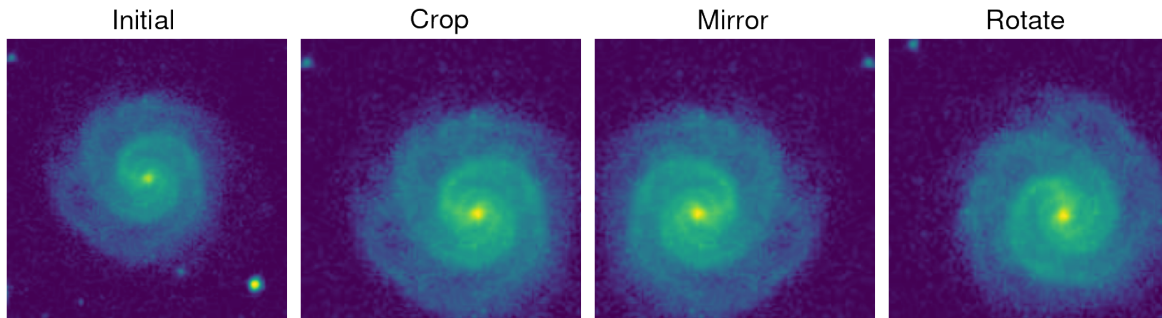


Figure 3.2: An example of our principal data augmentation techniques (cropping, mirroring and rotating) as applied to a spiral galaxy. The original image is on the left.

As implemented, these techniques combine to increase the number of images by a factor of 40, from 14,034 to a maximum of 561,360 images. An example of these techniques as applied to a sample spiral galaxy from our training set is shown in Figure 3.2. These data augmentation techniques also help to generalise the input data and enforce a degree of spatial and translational invariance. This is important, as the CNN must learn to identify the features themselves, and not simply that they are present in specific regions within the feature maps. Based on preliminary experiments with a small subset of the total data, we found that both rotating and flipping had a noticeable, positive impact on the overall training accuracy, albeit similar effects due to cropping were not as pronounced. We chose only to process five cropped images to maintain a reasonable file size while keeping within our computational limitations; any further cropping is a case of diminishing returns given the lack of any meaningful improvement to the overall accuracy.

In this study, our primary resolution for the input image data is 100×100 . We also perform some further runs at 200×200 to assess the impact of changing resolution, however it is important to note that the prohibitively expensive memory and computational requirements for the 200×200 runs restricted the degree of augmentation (in particular, no cropping). This is since the limited amount of GPU memory available necessarily places an upper limit on the total number of samples that can be trained with at a given resolution. Instead, we direct most of our focus to the fully augmented 100×100 data. The original imagery in the NA10 dataset comes in variable resolutions, the most common of which were between 100 and 130 pixels. These were all resized to the desired, target resolution using cubic interpolation with Pillow, a fork of the Python Imaging Data. The raw pixel data was extracted and reshaped into the correct format for loading into the CNN. All input images were linearly normalised such that pixel values are between 0 and 1. No further alterations were made to the images.

3.2.5 Training and Validation

The CNNs were constructed with the high-level API Keras (Chollet et al., 2015) and trained using the TensorFlow machine learning library running with nVidia’s GPU programming toolkit CUDA. Training was primarily conducted on ICRAR’s *Hyades* cluster, making use of a Nvidia GTX 1080 Ti graphics card, with smaller-scale runs and testing performed on an Nvidia GTX 1650 Ti laptop graphics card. The dataset was first partitioned into separate training and test sets according to an 85:15 split, i.e. 15% of the dataset is reserved for testing, with 85% used for training. The training set was further partitioned into separate training and validation sets (again with an 85:15 split) for the purpose of hyperparameter tuning. A third validation set, fully independent of the actual test set, is necessary for hyperparameter tuning in order to avoid biasing the choice of hyperparameters. Thus, when the optimal network is finally trained, it is evaluated on an independent, unseen dataset. This is an important test of the network’s ability to effectively generalise.

Training was conducted over a maximum of 100 epochs with a batch size of 400 for 100×100 data, and a reduced batch size of 200 for the 200×200 data due to memory limitations. To conduct these runs efficiently, we used early stopping; a method of ceasing training once there is no appreciable improvement in the validation loss over a threshold number of epochs. This threshold - the number of epochs over which to monitor whether the training is improving or not - is known as the patience. All runs were conducted with a patience of 7, and had generally converged and been stopped by the 20th to 25th epoch. Early stopping helps to avoid wasting unnecessary computations once the validation accuracy has plateaued,

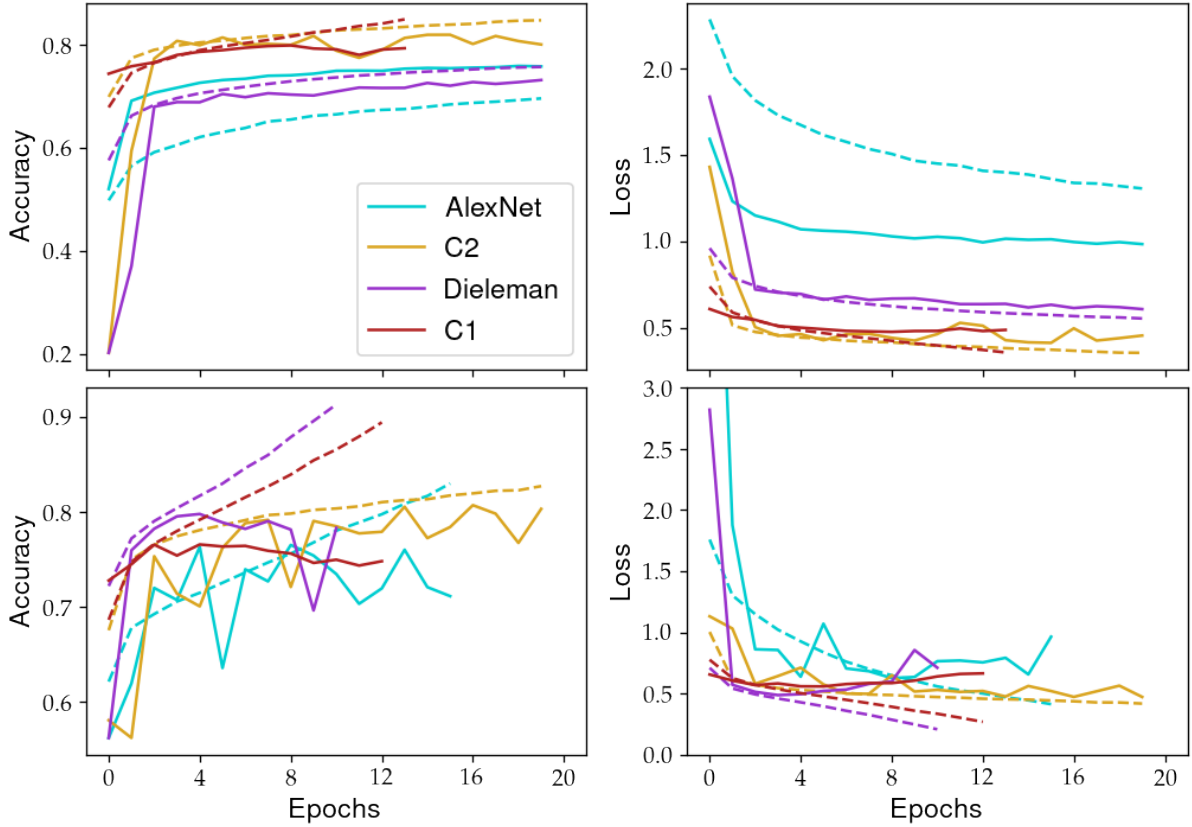


Figure 3.3: Accuracies and loss for all four CNN architectures for the 3-way (top panels) and 4-way (bottom panels) classification schemas as functions of the number of epochs, and as trained on the 100×100 data. The training accuracy/loss is represented with a dotted line, while the validation accuracy/loss is represented with a solid line. Architectures are represented with C1 in red, C2 in yellow, AlexNet in cyan and Dieleman in magenta.

and subsequently prevents overfitting. The best weights were stored using Keras’ ModelCheckpoint callback function. We make use of the categorical cross-entropy loss function for all CNNs.

Figure 3.3 shows all training and final validation accuracies/loss for all four CNN architectures. Note here that the validation accuracy/loss is merely a metric based on evaluations on the test set. All the weights and biases are initially randomised at the start of training. It can be seen that the validation accuracies initially rise rapidly before plateauing. Similarly, the validation loss gradually falls before becoming constant; indeed, early stopping is used to halt the training when the loss no longer decreases. Were early stopping not employed, the validation loss would continue to increase (see the C1 and AlexNet 4-way plots in the bottom-right panel of Figure 3.3), further widening the gap with the training loss and resulting in overfitting. The accuracy and loss values exhibit large random fluctuations, especially in the 4-way case, which is another sign of overfitting. As can be seen in Figure 3.3, our C2 network is the best overall performer. It is less affected by overfitting and achieves lower validation losses compared to the other models, which is likely reflective of its extensive hyperparameter tuning. Note the validation accuracy and loss are purely metrics corresponding to repeated evaluations of the network with the independent test set at the conclusion of every epoch. In Keras, these steps do not affect any of the network parameters.

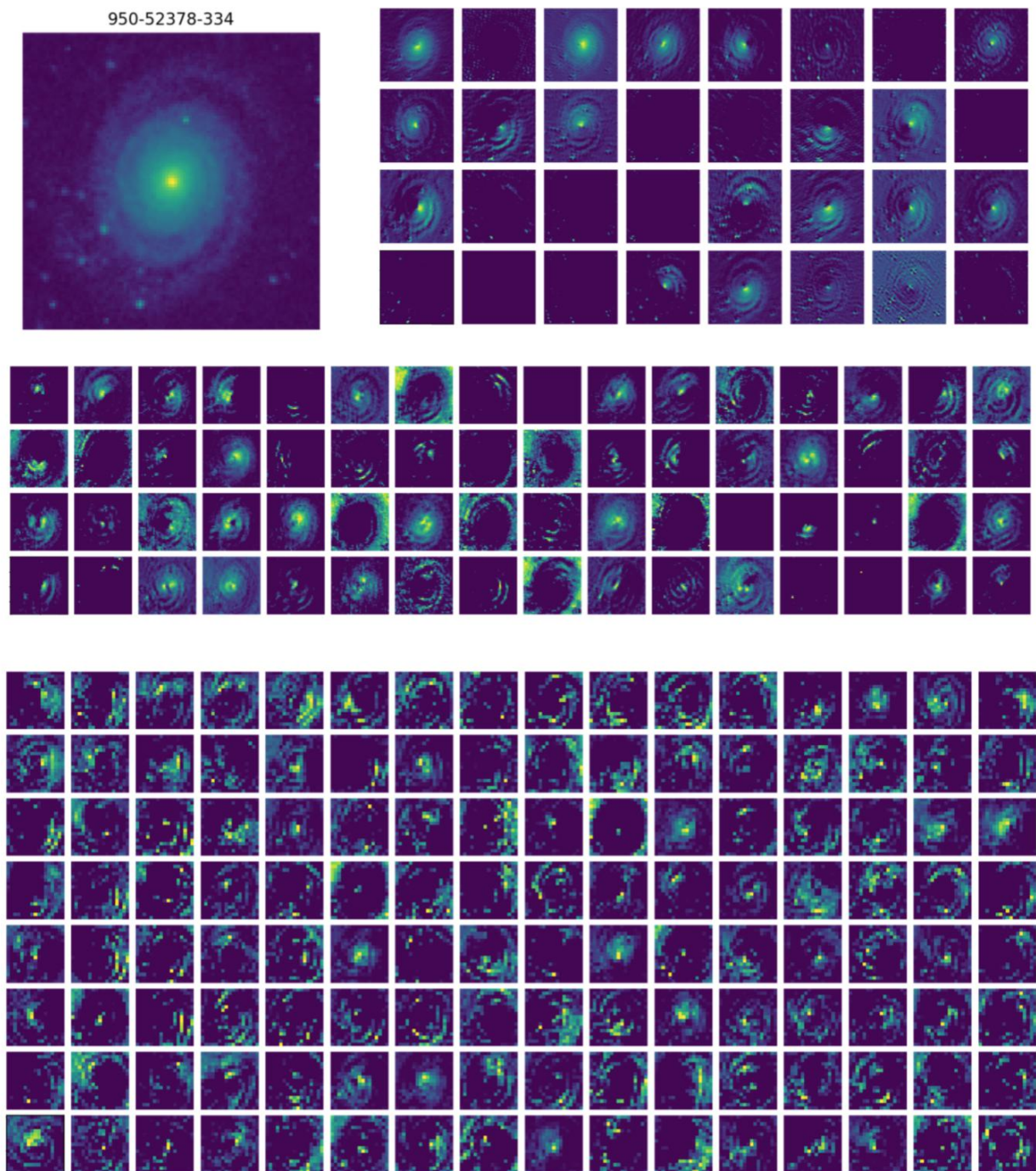


Figure 3.4: Plot of each individual feature map in the first, third and final convolutional layers of the C2 network, as generated for the example spiral galaxy shown in the top-left.

It is useful to visualise the feature maps within the convolutional and pooling layers in order to understand how the network is interpreting the input data; in particular, the features that are being extracted via the learnable filter weights. Figure 3.4 shows every individual feature map in the first, third and fourth convolutional layers of the C2 CNN architecture (see Figure 3.1) for the shown spiral galaxy in the 3-way case. It can be seen that, after more and more convolutions and poolings, the features maps

become more abstract. Through the use of these convolutions, the CNN is able to extract relevant features from the input image, such as spiral arms, the central bulge, and even a trace of the outline of the galaxy itself. It is important to stress that the filters/weights for each convolution are initially random and have instead been learned by the network. Thus visualising the feature maps enables us to see the results of these convolutions, and in the case of Figure 3.4, the images that the CNN has learned to interpret as a spiral galaxy.

3.3 Results

3.3.1 Binary classifications

Before classifying galaxies as either ellipticals, lenticulars, spirals or irregulars/misc, it is useful to first consider binary classifications between all four of these classes. This allows us to anticipate which distinctions are most problematic and likely to be sources of confusion. This will further serve to assist in interpreting the confusion matrices of the 3-way and 4-way classification.

We conducted two-way classifications between all four morphological classes (E vs S0, E vs Sp, ...) using our newly developed C2 network architecture. We further conducted two-way classifications between combinations of classes: Sp vs E and S0 (this can be considered an early vs. late type classifier), E vs. S0 and Sp, and irregulars vs. regulars, where regulars are subsampled from all E, S0 and Sp samples. Since there are an extremely low number of irregulars in the training set compared to the other classes, the 2-way classifications involving irregulars used a training set with an equal number of samples from each class, hence the over-represented class was randomly subsampled. In these cases, the performance is dependent on the quality of the subsample (insofar as its resemblance, or lack thereof, to the full sample), so each run was repeated multiple times with different subsamples, with the mean accuracy taken as the final result. The “regulars” training set used for the irregular vs. regular run was created with an equal number of samples from the E, S0 and Sp categories. All other runs not involving irregulars (e.g. E vs Sp) were conducted with all the available data, hence needing only a single run. All datasets were partitioned according to an 85:15 split into training and testing sets.

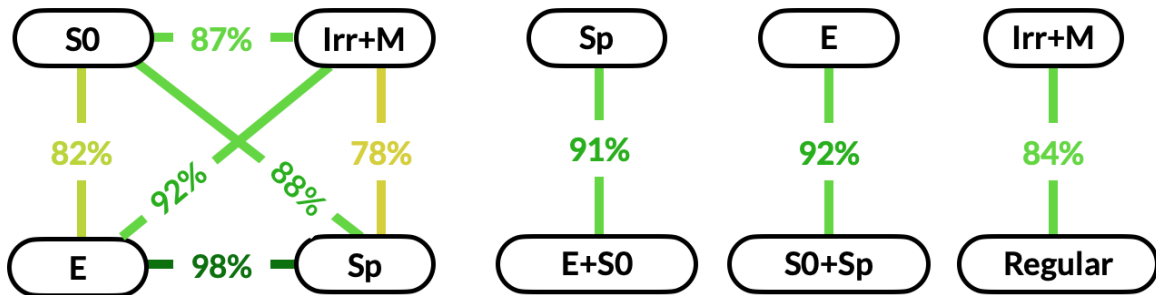


Figure 3.5: Binary classification accuracies for various combinations of morphological types as trained and tested with the C2 network. Accuracies are coloured according to relative accuracy, with dark green corresponding to the strongest distinction (E vs. Sp at 98%) and olive corresponding to the weakest distinction (Sp vs. Irr+Misc at 78%).

Table 3.2: Best classification accuracies of the principal CNN architectures for different input sizes and number of output categories, with 3-way distinguishing between E, S0 and Sp, and 4-way distinguishing between E, S0, Sp and Irr+Misc.

Network	Input Size	Best 3-Way	Best 4-Way
C1	100×100	79%	75%
Dieleman	100×100	82%	76%
	200×200	82%	79%
AlexNet	100×100	75%	74%
	200×200	84%	83%
C2	100×100	83%	81%
	200×200	81%	80%

Figure 3.5 shows the classification accuracies for all the aforementioned binary classifications. Ellipticals and spirals are able to be distinguished with a very high accuracy (>98%), spirals and lenticulars at 88%, ellipticals and irregulars at 86%, spirals and irregulars at 79%, ellipticals and lenticulars at 78%, and lastly lenticulars and irregulars at 70%. We also find that spirals can be disassociated from Es and S0s combined with a high accuracy (91%); similarly for Es vs. S0s and spirals (92%). Lastly, irregulars vs. regulars had an accuracy of 84%, although it is important to keep in mind that this used a considerably smaller training set as there were so few irregulars. These results are consistent with expectations from previous, established studies, e.g. [De La Calleja & Fuentes \(2004\)](#) obtained a 95% accuracy for E vs Sp.

3.3.2 3-way and 4-way classifications

Many previous studies utilising CNNs in astronomy have used binary classifications. In this study, we directly train a CNN to classify firstly between three classes, and again between four. Table 3.2 presents a summary of the overall classification accuracies for all CNN architectures with both 100×100 and 200×200 data. Note that memory limitations prevented the C1 architecture being used with the 200×200 data. Our C2 architecture outperforms all other models with the 100×100 data, with only the 200×200 4-way AlexNet run beating it. We obtain accuracies of 83% and 81% for 3-way and 4-way classifications respectively with 100×100 data. Increasing the resolution to 200×200 has a marginal effect on the overall classification accuracies, with only the AlexNet architecture producing a consistent gain. Given the resultant four-fold increase in computationally complexity and memory requirements for little comparative gain, we instead base our analysis on the 100×100 data. Our achieved accuracies are comparable to recent CNN applications, e.g. [Barchi et al. \(2020\)](#) with 83% for 3-class classification (albeit for ellipticals and barred/unbarred spirals).

Accuracy alone is not sufficient to judge the overall performance of a CNN. When it comes to multi-class classification, it is necessary to consider how the classifications are distributed. The confusion matrix is one way of comparing the true or correct labels to the predicted labels. As is conventional for machine learning, rows correspond to the actual, true classes, while columns correspond to the predicted classes (i.e. the label returned by the neural network). With this convention, the cell (E, S0) refers to true ellipticals classified by the network as lenticulars. The confusion matrix of a perfectly accurate CNN is an identity matrix.

Figures 3.6 and 3.7 show confusion matrices for both the 3-way and 4-way classifications respectively, using the best-performing C2 neural network architecture, and trained on 100×100 resolution images. In

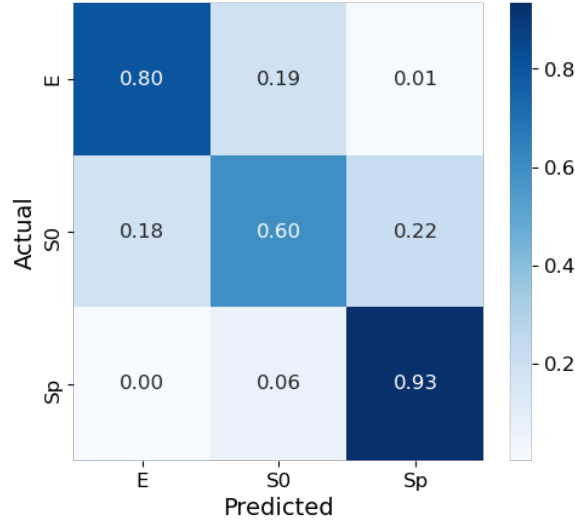


Figure 3.6: The confusion matrix of our C2 network for the 3-way classification between ellipticals (E), lenticulars (S0) and spirals (Sp), coloured according to the accuracy of the classification.

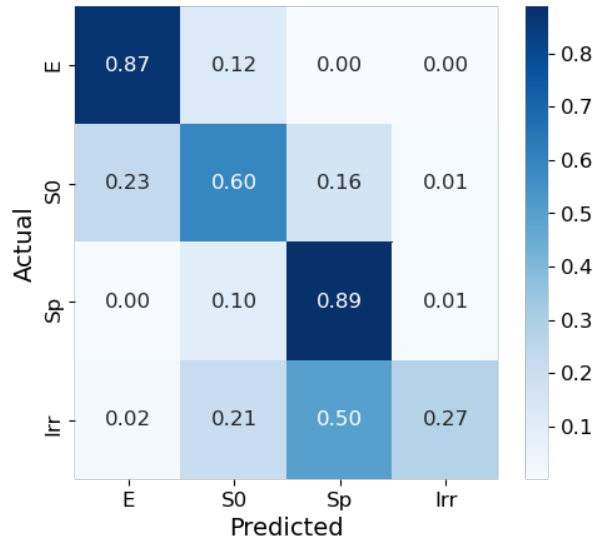


Figure 3.7: The confusion matrix of our C2 network for the 4-way classification between ellipticals (E), lenticulars (S0), spirals (Sp) and irregular/misc samples (Irr), coloured according to the accuracy of the classification.

the case of the 3-way classification, spirals are most accurately classified (93%), following by ellipticals (80%) and S0 (60%). S0s are almost equally misclassified as E or Sp (18% and 22% respectively). Almost all the misclassified E were misclassified as S0, with just 1% misclassified as spirals. Similarly, just 6% of true spirals were misclassified as S0. Thus it is clear that ellipticals and spirals are the easiest morphological classes to distinguish between, while S0s are considerably more difficult.

In the 4-way classification case (Figure 3.7), we find that both ellipticals and spirals have strong prediction accuracies (87% and 89% respectively). The accuracy of classifying S0s is unchanged, while the fourth irregular/miscellaneous class has a very poor accuracy only slightly higher than random. It is likely

that the low accuracy of the irregular class is a contributing factor to the accuracy and loss fluctuations in Figure 3.3. The majority of irregular samples are misclassified as spirals (50%) or S0s (21%). This is likely due to a combination of low-population categorical bias (there are only 388 irregular/misc samples out of 14034 total samples), and the fact that many miscellaneous samples show evidence of mergers, interaction and/or fine structure that may be interpreted as spiral or S0-like. This will be further analysed and discussed in Section 3.4, where we discuss hierarchical classification.

Another means of assessing the performance of CNNs is to consider the overall F1-scores for each category. The F1-score is the harmonic mean of the precision and recall. In the context of machine learning, the precision of a neural network is defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

where TP and FP denote true positives and false positives respectively. Precision, in this context, is equivalent to the positive predicative value or PPV. Recall is defined as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

where FN denotes false negatives. It is equivalent to the sensitivity or true positive rate, as it measures the fraction of correctly identified positives. The F1-score is thus simply

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

and has a value between 0 and 1. The higher the F1-score, the better. Tables 3.3 shows the precision, recall and F1-scores for our 3-way and 4-way classifications respectively. In general, spirals and ellipticals perform best. Although the precision is quite high for the irregular/miscellaneous samples, the recall is only slightly higher than random. This is indicative of the challenge in accurately classifying low populations of irregulars.

Table 3.3: Precision, recall and F1 scores for the best 3-way and 4-way classifications with our C2 network

Class	Precision	Recall	F1-Score
E	0.79	0.80	0.79
S0	0.65	0.60	0.62
Sp	0.91	0.93	0.92
Weighted Avg	0.82	0.83	0.83
E	0.75	0.87	0.81
S0	0.61	0.60	0.60
Sp	0.91	0.89	0.90
Irr+M	0.48	0.27	0.35
Weighted Avg	0.80	0.80	0.80

It is also useful to analyse the distribution of the confidence of each sample; that is, the probability that it belongs to the predicted class. As we have used a softmax activation function for the output layer, the activation values all sum to 1 across all categories. The predicted class is simply the category with the largest probability, and the value of that largest probability is known as the confidence. Figure 3.8 shows the distribution of these confidences for all samples in both the 3-way and 4-way classifications. Most

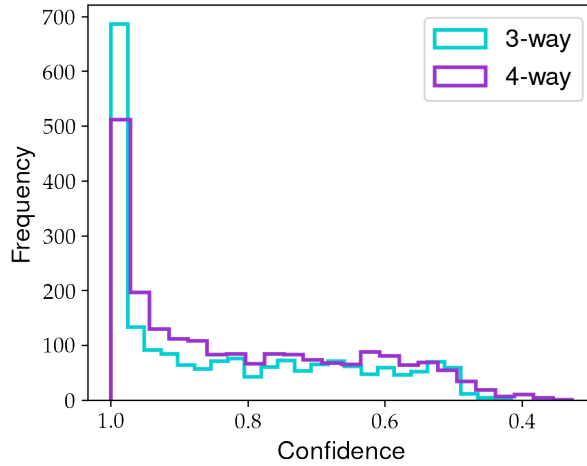


Figure 3.8: Distribution of the confidences for each classified sample for both the 3-way (blue) and 4-way (purple) CNNs

samples are classified with a high degree of confidence, however the fact that the distribution is smooth at high to mid probabilities indicates that the CNN has not been unduly overfitted. In a highly overfitted CNN, the confidences would all cluster around 1 with little to no tail.

3.4 Discussion

3.4.1 Classification Accuracies

In general, our binary classification accuracies (Figure 3.5) are comparable to those of other studies. Our accuracy of 98% for ellipticals vs. spirals is in line with other studies (Domínguez Sánchez et al., 2018; Barchi et al., 2020), and also compares well with other early vs. late-type classifiers (de Diego et al., 2020). The key issue when it comes to training a binary classifier CNN is that there are an unequal number of samples for each class. This is most prominent with the irregular/miscellaneous samples, a mere 2.8% of the total sample. With such a disparity, it is necessary to choose a subset of the over-represented category in order to equalise the number of samples in each category, and hence reduce categorical bias. However, given the smaller overall dataset, there is a higher risk of overfitting, which the data augmentation is able to partially ameliorate.

Our best, direct 3-way classification accuracy of 84% is also consistent with recent studies (Barchi et al., 2020), though we note few studies have attempted to classify specifically between ellipticals, lenticulars and spirals, so a direct comparison is lacking. What is considerably promising is that our accuracy was achieved with a network trained only on single-band imagery, while the results of other studies are achieved with full colour images. This is especially advantageous when it comes to high-redshift surveys, or for galaxies imaged in only a single band. Figure 3.6 shows that spirals and ellipticals are easiest to identify, while true lenticulars are spread between all three categories. It is important to note that there is a greater proportion of true ellipticals misclassified as S0 compared to true spirals. This is reflective of the lower E vs. S0 binary classification accuracy in Figure 3.5. Similarly, given the comparatively higher S0 vs. Sp accuracy, we expect fewer spirals to be misclassified as S0.

Our direct 4-way classification accuracy of around 81% is hampered by the difficulty in accurately classifying the irregular and miscellaneous samples. As seen in Figure 3.7, most of the irregulars are classified as spirals, with one fifth also classified as S0s. The S0 accuracy is unchanged, while both ellipticals and spirals have slightly lower accuracies. The irregular class, which includes miscellaneous samples, and is particularly hampered by its low relative population. These factors all combine to affect the classification accuracies. We will discuss more about the impact of morphology in §4.4.

3.4.2 Physical Confusion Matrices

One way to glean insights into the nature of the CNN classifications is to consider the physical properties of each sample based on the cell that they occupy in the confusion matrix. These “physical confusion matrices” are shown in Figure 3.9 where we are calculating the median age, redshift and stellar mass for each sample based on their classification, e.g. median age of all true ellipticals classified as E, S0, Sp, all true spirals classified as E, S0, Sp, etc. All values are from the NA10 tables, with the values for physical quantities based on the MPA/JHU catalogues. As expected, true ellipticals have higher redshifts, masses and are older than their spiral counterparts. While this is expected given the general pathway of morphological evolution (Conselice, 2014), this is also reflective of the sample biases at higher redshifts

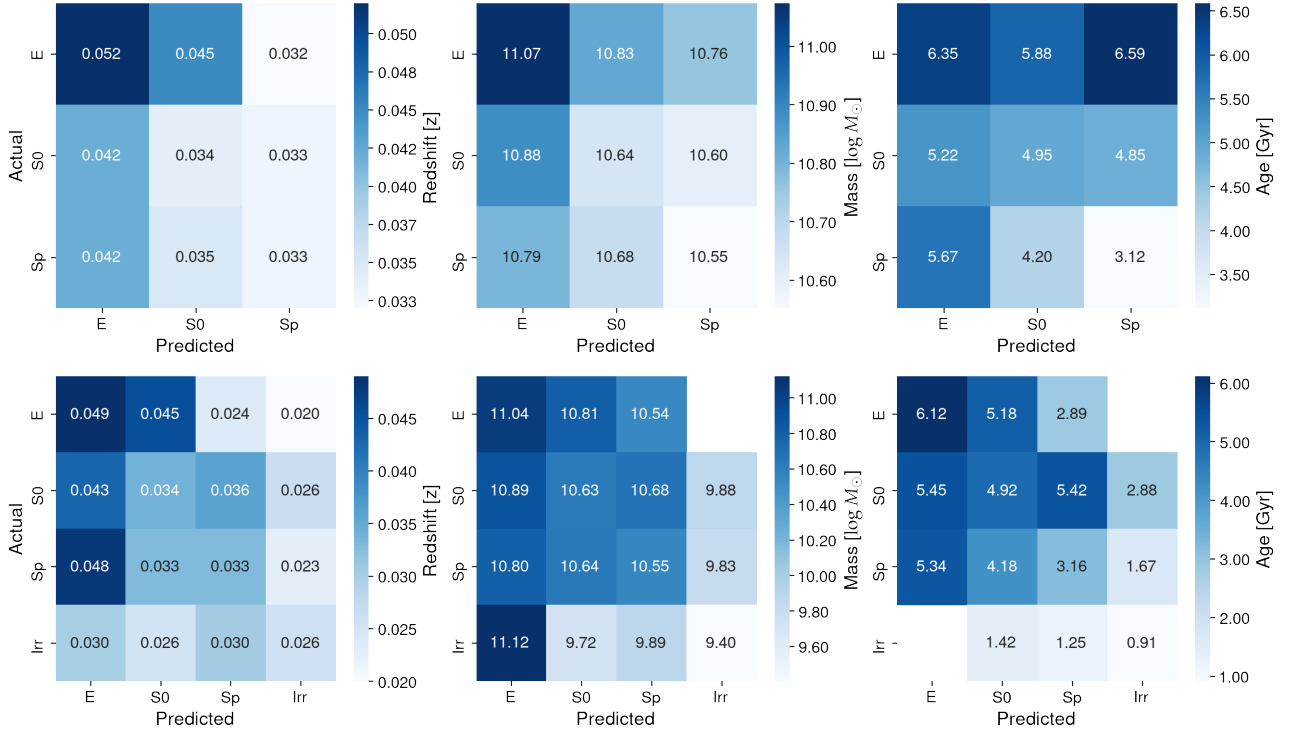


Figure 3.9: Matrix plots, similar to the confusion matrices in Figures 3.6 and 3.7, where instead each cell shows the median redshift (left), stellar mass (center) and age (right) for each of the samples in the 3-way (top) and 4-way (bottom) CNN classifications of Figures 3.6 and 3.7 respectively. As an example, the cell (E, S0) in the top-left most panel shows the median redshift of all true ellipticals that were misclassified as lenticulars.

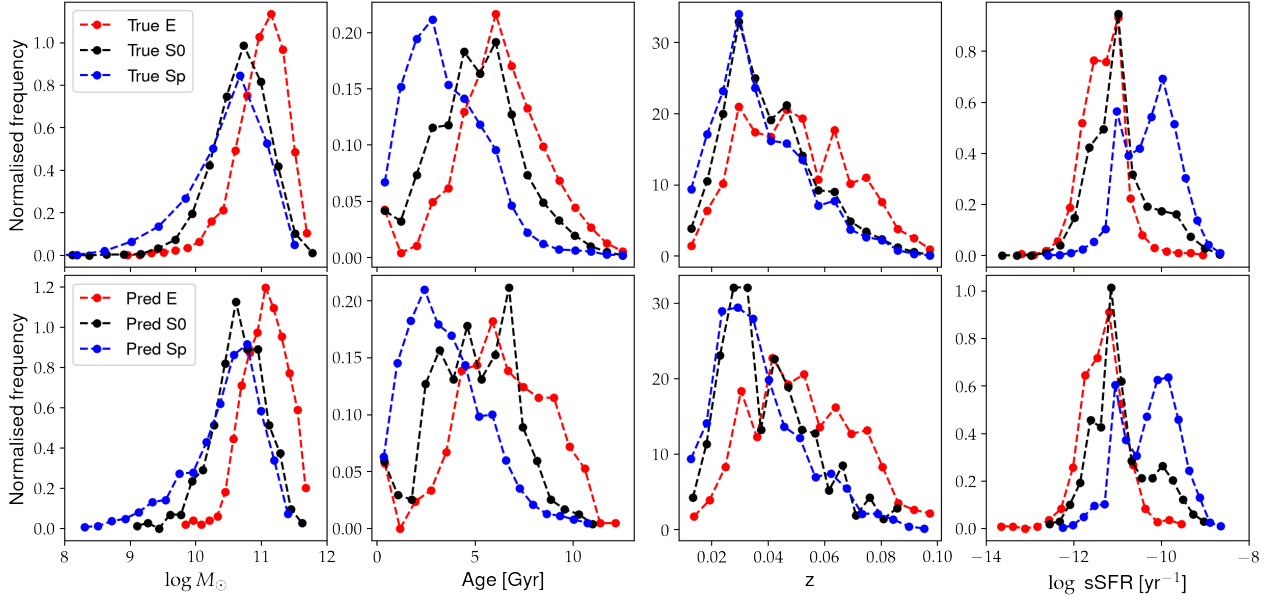


Figure 3.10: Plots of the normalised frequency for the E, S0 and Sp morphological classes as functions of stellar mass $M_{\odot}$, age (in Gyr), redshift z and specific star formation rate $\log r_{\text{SFR}}$. The top row is for all true morphological samples in the complete dataset, while the bottom panels are for the predicted morphologies of the samples in the testing set as classified by the 3-class CNN as either E (red), S0 (black) or Sp (blue).

where bright, compact objects like elliptical galaxies are more easily detected.

Given the values for the confused samples, there is evidence that even the misclassifications are physically meaningful. In the case of redshift, true ellipticals misclassified as S0s have (1) a lower median redshift than correctly classified ellipticals, and (2) a higher median redshift than correctly classified true S0s. Similarly, the median redshift is higher for both true S0s classified as ellipticals and true spirals classified as ellipticals. These trends are also observed for the other two properties: true ellipticals misclassified as S0 are less massive and younger than correctly classified true ellipticals, and are also more massive and older than correctly classified true S0s. These trends are also seen in the 4-way physical confusion matrices for all but the irregular class, where the low number of samples makes it difficult to establish any trends. Note that empty cells in some of the four-way confusion matrices are where no physical data is available for the sample(s) present. Importantly, the S0 and spiral samples misclassified as ellipticals are more distant, more massive and older than true S0s and spirals. Each of these characteristics are indeed to be expected of an elliptical galaxy. This thus demonstrates that the CNN has learned to identify features that discriminate between the different morphological types. If the misclassifications were random then we would not see such a trend across multiple physical properties.

Another way to visualise these trends is by comparing the distribution of the physical properties for the predicted morphological types with that of the true morphological types of the entire dataset. This is to see whether the CNN classifications are physically reasonable, as Figure 3.9 contends. Figures 3.10 and 3.11 show plots of the (normalised) density distribution of the stellar mass, age, redshift and specific star formation rate for both the true morphological types of all samples (top panels), and predicted morphological types of all the test set samples (bottom panels). Both the distributions are well matched,

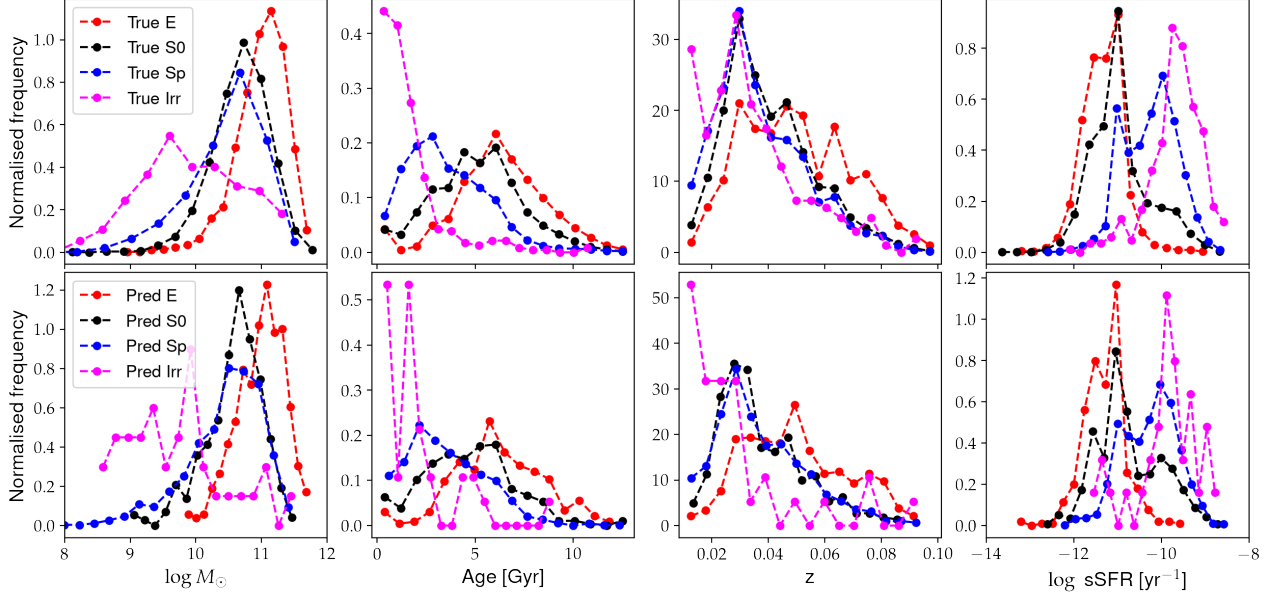


Figure 3.11: Similar plot as in Figure 3.10, albeit with the addition of the fourth morphological class. The top row plots the true morphologies of the complete dataset, while the bottom plots predicted morphologies for samples in the testing set as classified by the 4-class CNN as either E (red), S0 (black), Sp (blue) or Irr+Misc (magenta).

suggesting that the predicted morphologies share the same general distributions of physical properties as the actual morphologies. As expected, ellipticals are more massive, older, more distant and are largely quenched. Likewise, the distribution of the physical properties of predicted spiral galaxies is consistent with a quintessentially low-mass, young, low-redshift, actively star-forming population. The particular case of the irregular / miscellaneous samples in Figure 3.11 is more problematic given the low population and its broad categorisation. This categorisation, which includes interacting galaxies / mergers, is reflected in the wider distribution in stellar masses. However, the distribution of irregulars does peak at the youngest overall age, in both true and predicted morphology.

3.4.3 Hierarchical Classification

In this work, we have primarily trained CNNs to directly classify galaxy samples directly as either one of three or four classes. It is thus useful to compare this direct comparison with an indirect classification method involving hierarchical classification. This indirect method was also explored in an attempt to obtain better classification accuracies for the irregulars. Here, we use multiple CNNs to construct a binary decision tree, with which to classify samples. This approach has been used in many previous works, fundamentally as part of the Galaxy Zoo (Lintott et al., 2008), as well as in other studies in astronomy involving machine learning (Huertas-Company et al., 2011; Domínguez Sánchez et al., 2018; Huertas-Company et al., 2019). Here we simply process each sample in the test set by passing it through a series of binary CNNs (see Figure 3.5 for the classification accuracies of each). We can perform the 3-class classification using two different hierarchical classifiers; first, classify between E and S0+Sp, then, given the sample is not an elliptical, between S0 and Sp. The second route is to first classify between E+S0 and

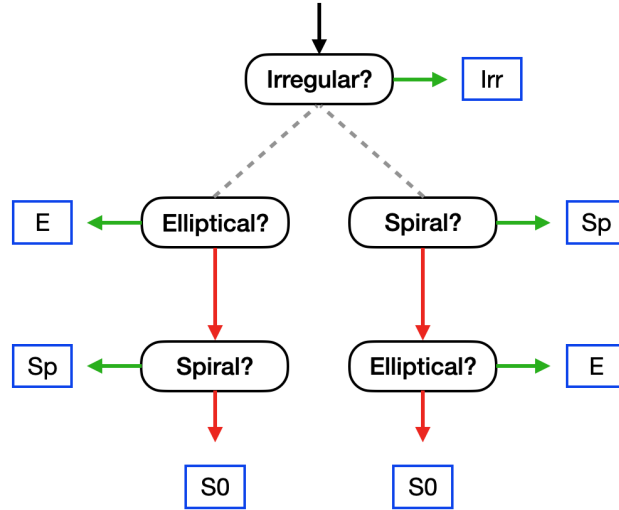


Figure 3.12: Diagram showing the two different paths in our hierarchical classification tree. In the case of 3-way classification, the first step is ignored. To classify between E, S0 and Sp, there are two routes depending on whether to first look for ellipticals or look for spirals. The subsequent step is then to discriminate between the other two categories.

Table 3.4: Table of classification accuracies for all five hierarchical classifiers.

Path	Best 3-way Accuracy	Best 4-way Accuracy
E-first	82%	63%
Sp-first	83%	63%
Reg+3-way	N/A	65%

Sp, and then, given the sample is not a spiral, between E and S0. In the case of the 4-class classification, the first binary CNN is simply E+S0+Sp (Regular) vs. Irr+Misc. Figure 3.12 illustrates these two different pathways. We will refer to the pathway that commences with E vs. S0+Sp as E-first, and E+S0 vs. Sp as Sp-first. We also classify between the four classes first by applying the Regular vs. Irr+Misc binary classifier, and then applying our direct 3-way CNN (thus two overall steps instead of three). We refer to this as our Reg+3-way classifier. Table 3.4 lists the best overall classification accuracies for each of the five hierarchical classifiers.

Figure 3.13 shows the confusion matrices for the hierarchical 3-way classification with both the E-first and Sp-first pathways, from which it is clear that order matters. While both have similar overall accuracies, the Sp-first path best matches the direct 3-way CNN (Figure 3.6) with a better 65% S0 accuracy and comparable E and Sp accuracies. The E-first path, while excellent at classifying ellipticals, has a much poorer S0 accuracy, with over a third of true S0s misclassified as ellipticals. This is despite the fact that both the E vs. S0+Sp and E+S0 vs. Sp binary classifiers have similar >90% accuracies. The disparity is reflective of a deeper issue: the fact that ellipticals and S0s are inherently much harder to disentangle, compared to distinguishing between spirals and S0s (compare the 82% and 88% binary classification accuracies in Figure 3.5). This may seem counter-intuitive; after all, if Sp are able to be accurately picked up, then the remaining E+S0 samples should have a worse accuracy. The key point is that this is dependent on the percentage of true spirals. Spirals are able to be differentiated between E and S0 with 98% and 88%

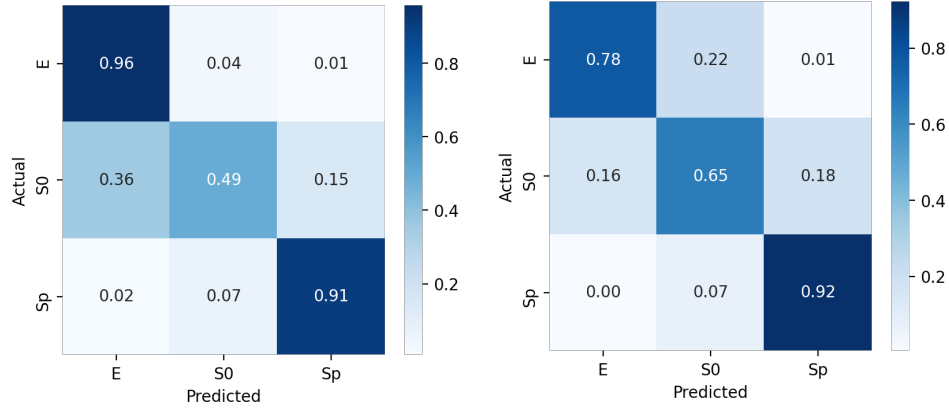


Figure 3.13: Confusion matrices for the 3-way hierarchical classification of ellipticals (E), lenticulars (S0) and spirals (Sp) for the E-first pathway (left) and Sp-first pathway (right) (see Figure 3.12 for an overview of these pathways).

Table 3.5: Table of per-class classification accuracies for the three 4-way hierarchical classifiers (E-first, Sp-first and Reg+3-way).

CNN	E	S0	Sp	Irr
E-first	91%	33%	63%	90%
Sp-first	76%	45%	64%	90%
Reg+3-way	80%	46%	65%	90%

accuracies respectively, so fewer false positives will progress to the next stage in the hierarchical tree. However, because E vs. S0 has a lower accuracy, more false positive ellipticals (in particular, true S0s falsely classified as E) will emerge from the first E vs. S0+Sp stage. Hence the E-first path results in a greater spread of samples, while the Sp-first path is more accurate at picking up true S0s. For both the Sp-first pathway in Figure 3.13 and the direct 3-way CNN in Figure 3.6, more true ellipticals are misclassified as S0s compared to true spirals.

Table 3.5 shows the per-class classification accuracies for each of our 4-way hierarchical CNNs. Note from Table 3.4 that the E-first and Sp-first had similar overall accuracies (63%), with a slightly higher 65% for the Reg+3-way CNN. Adding an explicit step to classify irregulars has dramatically improved the classification accuracy from 27% to 90%, but this has come at the severe cost of reduced accuracies for other classes. This suggests that the more steps there are in the pathway, the harder it is to classify the classes in the later steps. This is to be expected given that, by design, the hierarchical classifiers rely on passing galaxies through multiple binary classifiers, leading to greater uncertainties through error propagation. Although the E-first 4-way CNN can classify ellipticals with a 91% accuracy, the S0 accuracy is the lowest out of the three (33%). The overall Sp and Irr classification accuracies are similar across all three classifiers, with the Irr accuracy unchanged. As with the 3-way hierarchical classifiers, the Sp-first approach provides a better spread of accuracies across all the categories. The Reg+3-way CNN offers a further marginal improvement.

These results show that indirect, hierarchical classification can offer comparable accuracies to direct 3-way classification, but performs poorly when compared to direct 4-way classification. Importantly, the

direct classification uses just a single model, and is thus far more practical to apply and computationally efficient to train. Although the direct CNNs outperform the hierarchical classifiers, the 3-way Sp-first classifier offers a 5% improvement in S0 classification accuracy.

3.4.4 Galaxy Morphology

Galaxies exhibit an array of fine morphological structure, from spiral arms, dust lanes and stellar bars, to tidal tails and/or tidal interaction (Conselice, 2014). The power of CNNs lie in their ability to extract features in images. As can be seen in the example in Figure 3.4, the convolutional layers are able to outline, trace and detect the presence of spiral arms. The process of visual classification relies on being able to detect and categorise such morphological features (Buta, 2013). In general, S0 galaxies are difficult to visually distinguish (Blanton & Moustakas, 2009), in part since they exhibit morphological characteristics that, insofar as solely examining an image is concerned, are near identical to ellipticals. S0s can also display prominent dust lanes, rings and large disc components, which make them visually similar to spirals, especially in the edge-on case. It is thus expected that the overall CNN classification accuracies are lower for S0s. Although the networks were trained on single-band images (g-band), for the purpose of illustrations all examples in the figures of this section are the RGB colour image cut-outs taken directly from SDSS DR4. Figure 3.14 shows a random selection of galaxies, from each morphological class, that were correctly identified by the CNN. Most of these samples were classified with a near-certain confidence (>0.99). Although the ellipticals and S0s in Figure 3.14 look visually similar, the CNN is able to distinguish between the two types, possibly by being able to detect the more prominent bulge and/or evidence of disc-like structure as seen in the S0s. Figure 3.15 shows a selection of fifteen randomly chosen samples that were misclassified with a relatively high confidence, at least $>70\%$. The first three samples (starting from the top-left and moving across to the right) are all S0s that were misclassified as spirals. All three exhibit fine morphological structure, such as a prolate bulges and stellar bars, and there are hints of faint spiral arms in each. Whether such arms are truly well established enough to consider the galaxy a spiral is debatable, but these examples show that the CNN is able to both detect their presence and classify accordingly. J100944.44+525738.87, an elliptical classified as an S0, demonstrates the difficulty in distinguishing between these two types by image alone. There are also examples of clear confusion; the first two samples in the middle row, J023153.52+011234.22 and J082351.91+421319.51, are both barred galaxies. They have a dataset classification of S0 and Sp respectively, but have been classified by our CNN as opposites (Sp and S0, respectively). The CNN has also misclassified all the irregular samples in Figure 3.15 as spirals. Since the network is only trained on g-band images, the prominent blue colour of the irregulars that enables the naked eye to easily distinguish that morphological type is instead lost to the CNN. One other interesting example is that of J112459.22+641850.57, a spiral galaxy with extremely faint arms that arguably looks like a stellar ring, which the CNN has classified as an S0.

Figure 3.16 shows a random selection of samples that were classified with confidences less than 70%. These examples include samples that were both correctly and incorrectly classified, and can help reveal where most of the confusion is based. In general, compared to Figures 3.14 and Figures 3.15, there is far less variety in large-scale morphology, and instead distinctions must lie in finer-scale morphologies as well as foreground/background features. The sample images containing foreground objects in Figure 3.16 tend to result in lower confidences, most spectacularly with the first example, J075634.20+301032.46. The

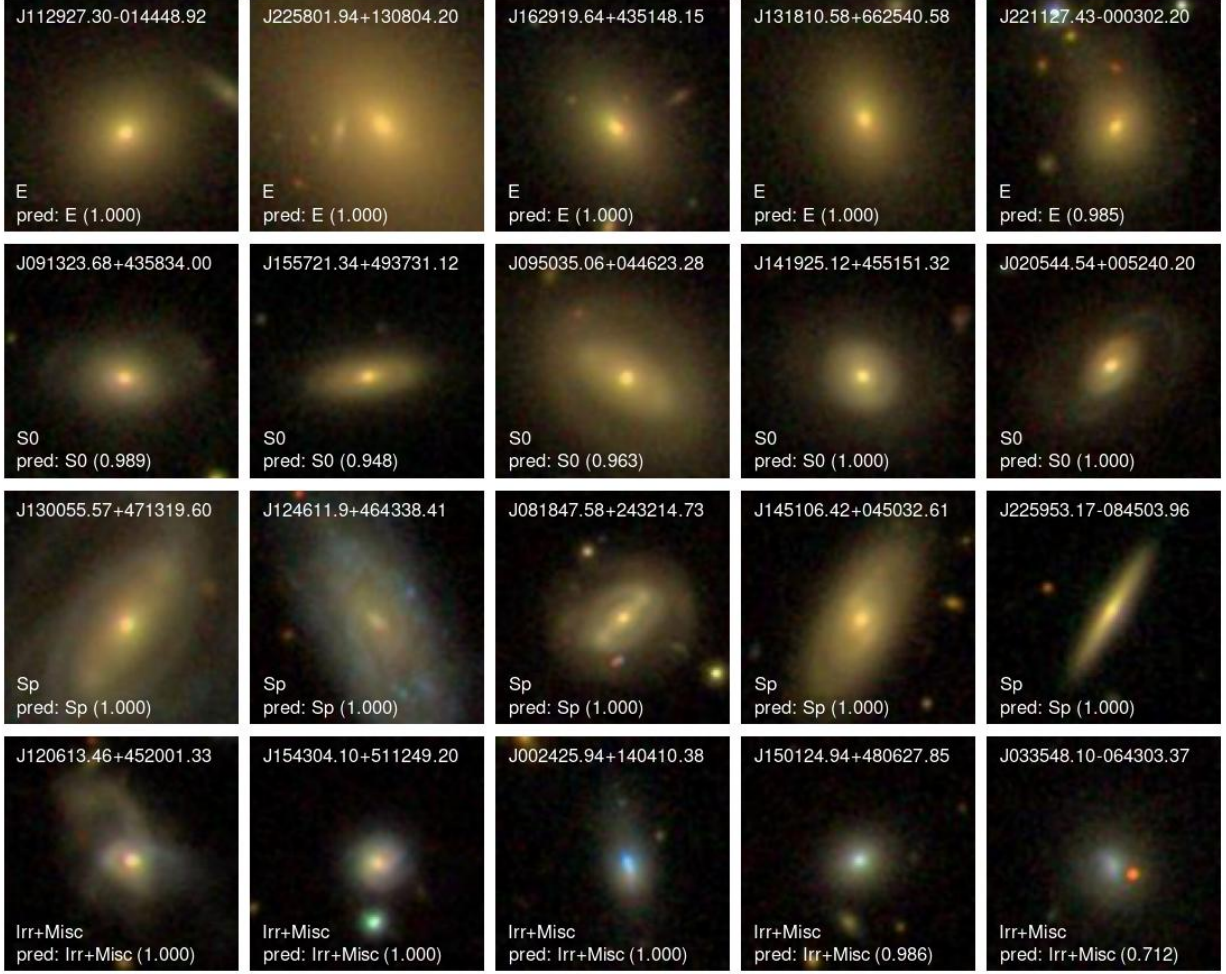


Figure 3.14: Randomly selected galaxy samples from each of the four morphological classes that were correctly classified by the CNN. Each image is annotated, from top to bottom, with its SDSS JID, actual morphological class, and predicted morphological class, with the CNN prediction confidence given in parentheses.

last two examples in the second row, J004227.71-084550.76 and J144725.16+032627.94, both near visually identical, have dataset classifications of S0 and E respectively, but both have been classified by the CNN as ellipticals with 49% and 51% confidence respectively. What is also important to note is the uncertainty that surrounds the classification of edge-on galaxies. In particular, the middle image in the top row of Figure 3.16 and the first image in the second row (J172421.13+583341.57 and J115613.87+034017.46) are both edge on galaxies, and they have both been classified as spiral galaxies. Indeed, since many spiral galaxies have prominent discs, from pure population statistics alone it is more likely that an edge-on galaxy is a spiral, rather than an elliptical. This is also problematic for S0 galaxies, many of which also have a prominent disk and are difficult to distinguish from spirals when viewed edge on. This is where other analytical techniques are needed to definitively tell the two apart (e.g Laurikainen et al. 2005).

A further way to check whether our CNN classifications are morphologically sensible is to compare our predicted morphologies with the probabilistic Galaxy Zoo morphologies for all the NA10 samples in our test set. This is important, as it allows us to compare the CNN classification confidences with the

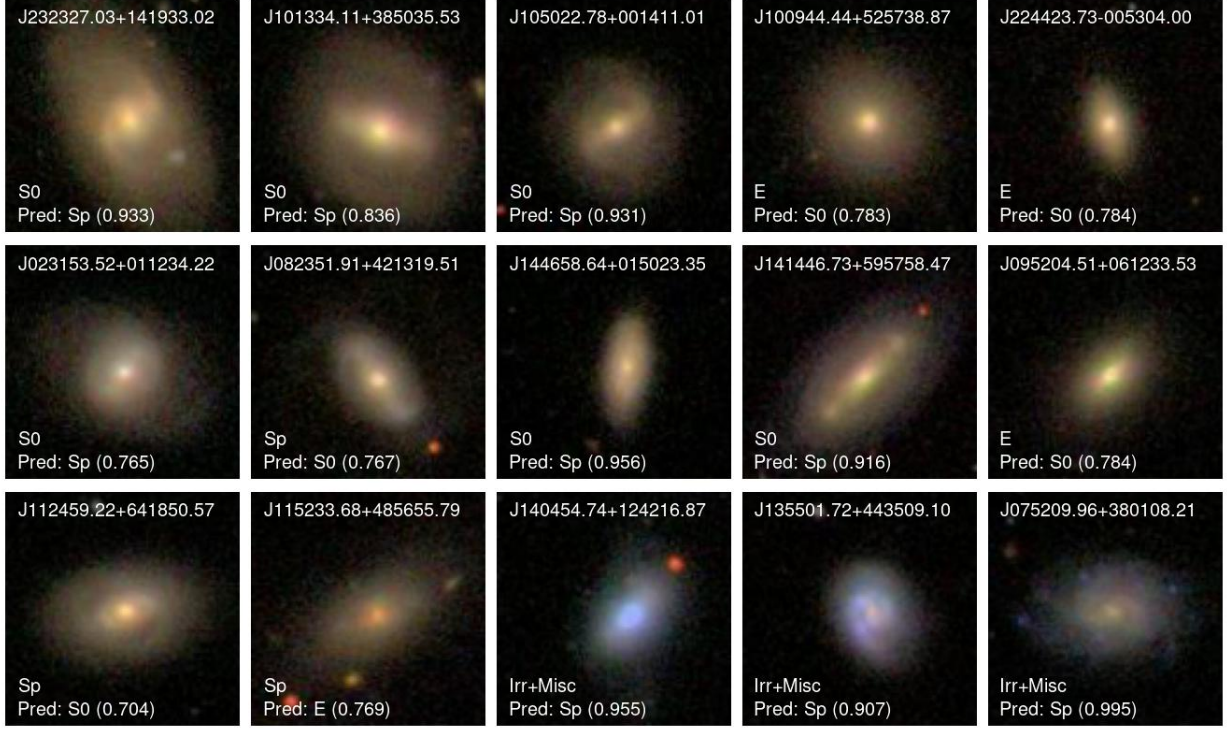


Figure 3.15: Randomly selected galaxy samples from each of the four morphological classes that were incorrectly classified by the CNN with a relatively high confidence ($>70\%$). Each image is annotated with similar text as in Figure 3.14

Galaxy Zoo probabilities that a sample is an elliptical or spiral. Here, we performed a cross-match and obtained the Galaxy Zoo probabilities P_{EL} , the probability of being an elliptical, and P_{CS} , the combined probability of being a spiral (Lintott et al., 2011). Figure 3.17 gives the normalised distributions of the samples in our NA10 test set that were classified as ellipticals (red), S0 (green) and spirals (blue), in terms of Galaxy Zoo probability of being an elliptical or spiral. As expected, all NA10 predicted spirals have a very low probability of being an elliptical, and very high probability of being a spiral. Similarly, all NA10 predicted ellipticals have high probabilities of being an elliptical, and low probabilities being a spiral. The S0s have significantly more spread, albeit are grouped more towards being an elliptical. This much broader distribution is consistent with the difficulty in definitively classifying S0s, as implied by the low confidence examples in Figure 3.16. This also further justifies why so many ellipticals were classified as S0s in the 3-way and 4-way classifications in Table 3.6.

3.4.5 Galaxy Zoo Sample Test

In addition to our tests with the NA10 dataset, we further apply five of our networks to a small yet unique sample of Galaxy Zoo images (Lintott et al., 2008). In particular, we chose a random selection of 1,000 ellipticals and 1,000 spiral galaxies from the Galaxy Zoo catalogue that were not included in the NA10 dataset. These images were classified via the collective contribution of citizen scientists as either elliptical, spiral or uncertain. Table 3.6 gives the number of samples classified in each morphological class, as well as the overall accuracy, for each of the five networks: binary elliptical vs. spiral, direct 3-way CNN, the

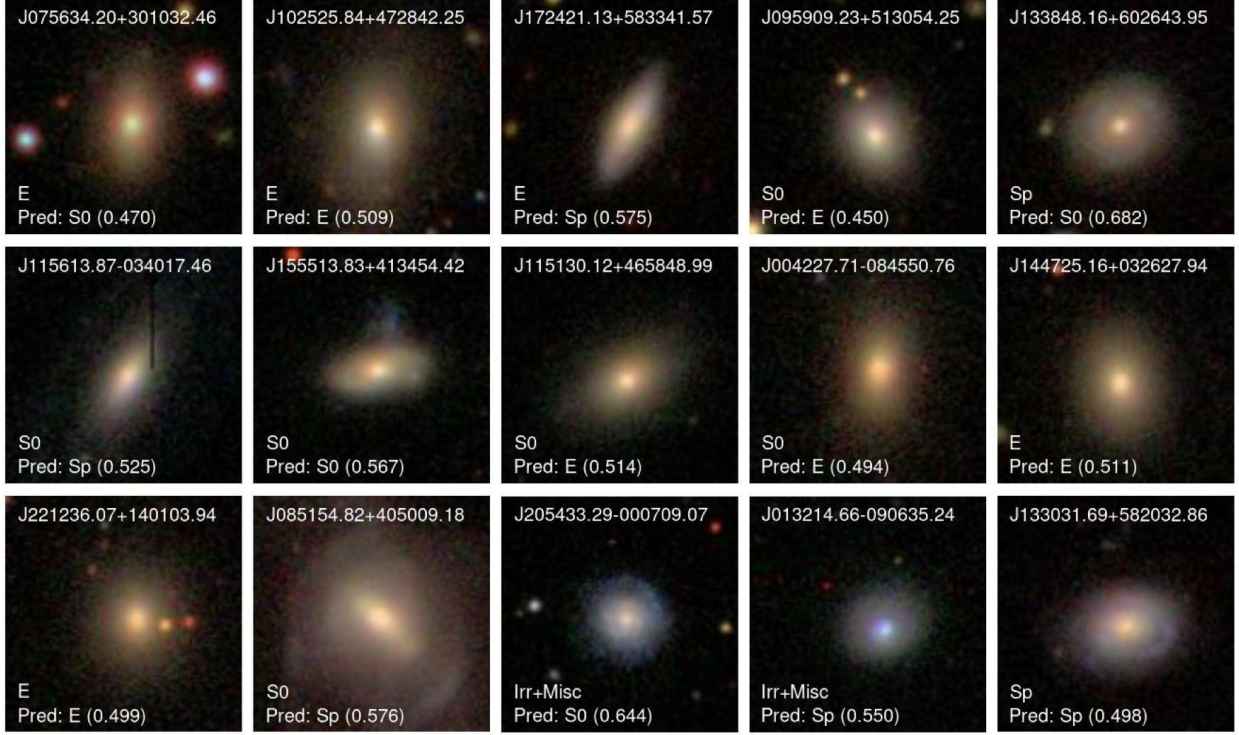


Figure 3.16: Randomly selected galaxy samples from each of the 4 morphological classes that were classified by the CNN with a relatively low confidence ($<70\%$, mostly $<60\%$). Each image is annotated with similar text as in Figure 3.14

two 3-way hierarchical classifiers (E-first and Sp-first), and the direct 4-way CNN. All CNNs excelled at classifying spirals with very high accuracies. However, the performance for the ellipticals is not as spectacular, with 85% for the binary CNN, dropping to 64% and 58% for the direct 3-way and 4-way CNNs

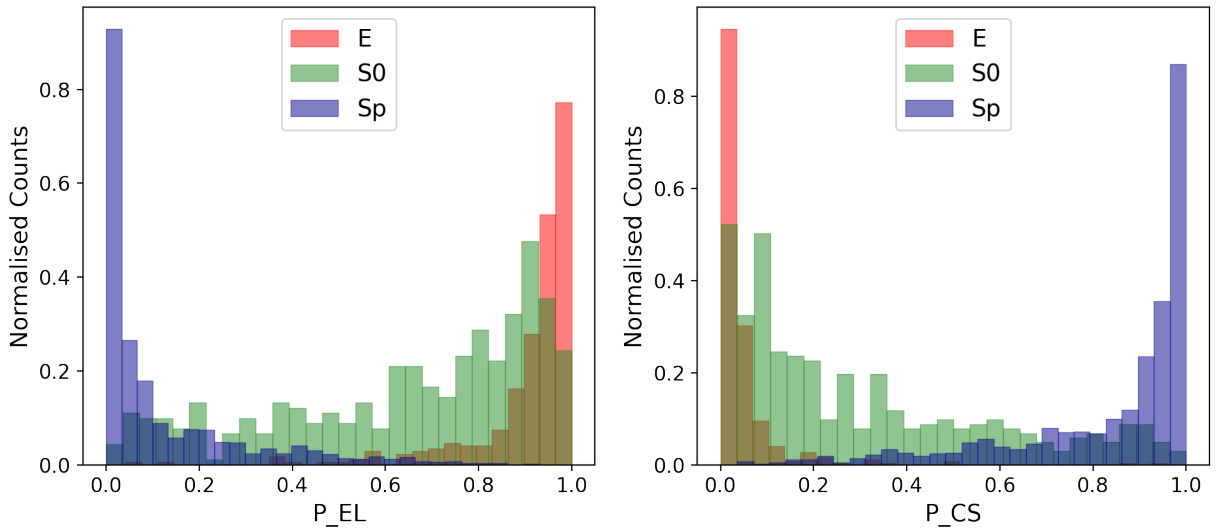


Figure 3.17: Distribution of all samples in our main NA10 test set, classified by the CNN as either elliptical (red), S0 (green) or spiral (blue), plotted with respect to the Galaxy Zoo probability of being an elliptical.

Table 3.6: Classification spreads and accuracies for the binary, direct 3-way, hierarchical 3-way and direct 3-way CNNs when applied to samples of 1,000 ellipticals and 1,000 spirals.

Model	1,000 Elliptical Images				1,000 Spiral Images				Accuracy (Total Images)
	E	S0	Sp	Irr	E	S0	Sp	Irr	
E vs. Sp	851		149		4		996		92%
Direct 3-way	642	251	107		2	7	991		82%
3-way E-First	529	368	103		6	29	965		75%
3-way Sp-First	612	178	210		2	7	991		80%
Direct 4-way	580	221	142	57	0	14	963	23	77%

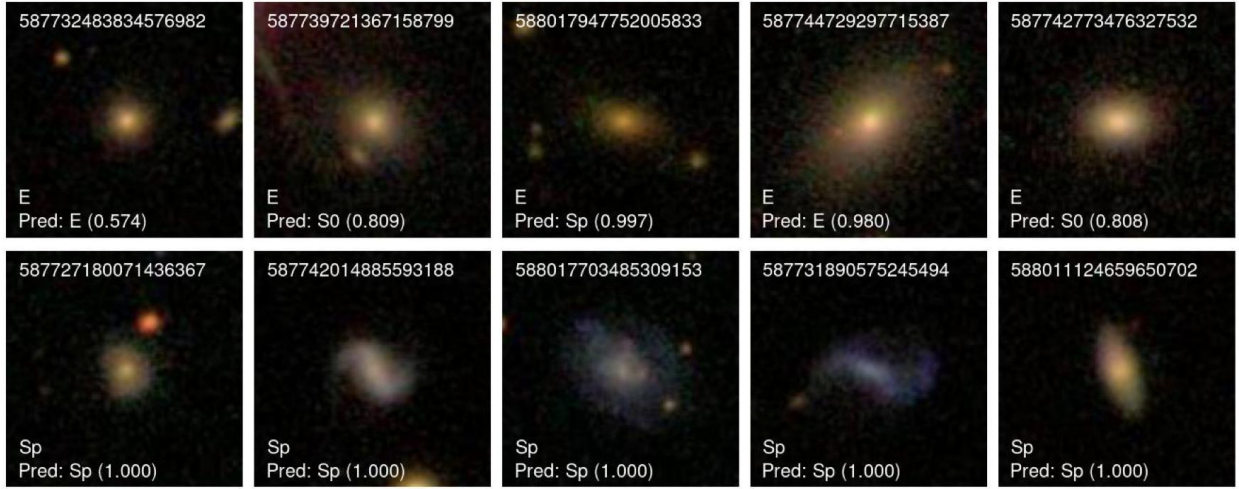


Figure 3.18: Randomly selected elliptical and spiral galaxies from our Galaxy Zoo subsample. Images are annotated with their Galaxy Zoo OBJID at the top, along with the Galaxy Zoo class, followed by the CNN predicted class with the confidence of the prediction given in parentheses.

respectively. There are two main reasons for this. Firstly, spirals have the highest per-class accuracy out of all the classes; this is consistent across all networks (see Figures 3.6, 3.7, 3.13, etc). Secondly, the Galaxy Zoo elliptical categorisation does not explicitly distinguish between elliptical and S0 galaxies. For the direct 3-way CNN, as seen in Table 3.6, out of the 1000 “ellipticals”, 251 are classified as S0s. This is even worse in the E-first hierarchical classifier, with 368 classified as S0s. These problems become apparent for 3-way classification and above. However, throughout all the CNNs, there are some ellipticals that have been misclassified as spirals. It is likely that these are edge-on ellipticals. Figure 3.18 shows a random selection of these elliptical and spiral samples, along with their predicted classes and confidence values as classified by our direct 3-way CNN. All spirals are correctly classified with 100% certainty, while the ellipticals generally have lower classification confidences.

3.4.6 Efficiency and Limitations

With the advent of large-scale structure surveys comes the need to process enormous amounts of data efficiently and accurately. This is crucial in the context of classifying galaxy morphologies, for which visual classification is intractable. The key strengths of automated classification techniques, such as our CNN approach, ultimately lie in their speed and ability to generalise. Although training a CNN can be a computationally expensive undertaking, the speed with which it can classify galaxies once trained is orders of magnitude greater than what could ever be possible with manual classification. Our C2 network processed the test set in under five seconds using a GTX 1080Ti GPU; for comparison, it took a group of classifiers at least five minutes to reach a consensus on the samples in Figure 3.15. The ability to generalise is another key strength of CNNs as, by design, they offer a model-independent means of extracting relevant features from arbitrary input data. Trained CNNs can also be easily adapted to classify other datasets through the use of transfer learning (see Domínguez Sánchez et al. 2019). It is also worth noting there is typically a high degree of uncertainty associated with visual classification, often resulting in disagreements even among groups of expert astronomers (Kelvin et al., 2014; Cortese et al., 2016; Moffett et al., 2016). This is an inevitable consequence of the subjective nature of morphology. As such, an overall accuracy of around 80% is typical for visual classification, and our CNNs are consistent with this level of uncertainty.

Despite these advantages, there are some drawbacks to our current implementation. The use of single-band images necessary limits the amount of information available to the CNN. Although this work has achieved comparable overall accuracies to other studies using colour images, combining multiple bands will make it easier to distinguish between different morphologies, especially irregulars that tend to be bluer. Our current CNN approach relies solely on images, yet kinematic data is useful at distinguishing between morphological types, particularly between E and S0 (Cappellari et al., 2007). Studies based on simulations have successfully combined density and velocity maps for improved classification (Shen & Bekki, 2020), and so the incorporation of kinematic data is a promising area of future study.

3.5 Conclusion

In this work, we trained and tested several convolutional neural networks to classify galaxy morphologies. This work is among the first to explicitly classify ellipticals, S0 galaxies and spirals with a direct 3-class CNN, in addition to a fourth irregular class with a direct 4-class CNN. The main outcomes of this work are summarised as follows:

- We have introduced a new CNN architecture that can classify between ellipticals, lenticulars and spirals with an accuracy of 84%, and between E, S0, Sp and irregulars/miscellaneous with an accuracy of 81%. These accuracies outperform the results of existing models at 3-way classification, and have equivalent accuracy for 4-way classification.
- We have demonstrated that a network using our new architecture has equivalent performance on single-band imagery compared to other studies that have utilised colour imagery. Although this is largely domain-specific, it gives confidence that CNNs can be applied to datasets that only contain information for a single-band. This is especially important for high-redshift surveys.

- We have also used our new architecture to train several binary classifiers to classify between the four chief morphological types. We have found that our CNN is best able to distinguish between ellipticals and spirals with an accuracy of 98%, while spirals and irregulars were the hardest to distinguish between, with an accuracy of only 78%.
- We have showed that the direct 3-class CNN has a marginally better accuracy compared to an indirect, hierarchical classifier, but that the direct 4-class CNN is more accurate.
- Through an analysis of the confusion matrices, and the physical properties of the misclassified samples, we have shown that even the misclassifications are physically sensible. In particular, samples predicted by our CNN to be ellipticals tended to have higher redshifts, higher stellar masses and were older in age, regardless of whether the classification was actually correct. This trend was observed in the case of S0s misclassified as E (higher z , more massive and older compared to true S0s), and Sp classified as E (compared to true spirals). The distributions of the physical properties of the predicted morphologies also match well with the true morphologies.
- Through inspecting several samples classified with low confidences, we found that ellipticals and S0s, often with a mostly featureless face, were harder to distinguish between. We found that many edge-on galaxies were often being associated with spiral galaxies, and this is likely due to the the effect of population statistics inherent in the overall dataset. We found that the low overall number of irregular samples also contributed to difficulties in the 4-way classification, with most irregular and miscellaneous samples classified as spirals, likely due to their myriad non-standard morphologies.

The evolution of barred galaxies in the EAGLE simulation

Having developed our core deep learning models and classified test samples of galaxy images in Chapter 3, we now turn our attention to large-scale applications. Cosmological simulations are extremely important tools with which to study the intricacies of how galaxies form and evolve. This is especially true for barred galaxies, where much of the theory regarding the formation of stellar bars has come from extensive studies of simulations. In this Chapter, we build on the success of our morphological classifiers and develop a new CNN specifically to classify barred galaxies. We apply our models to classify over 35,082 unique images of 3,964 simulated galaxies from the EAGLE cosmological simulation across ten different snapshots, with a focus on the evolution of the bar fraction from redshift $z = 1$ to $z = 0$. We apply our models to mock SDSS g -band images of the simulated galaxies, hence allowing our existing models (trained on SDSS imaging) to be directly applied without the need for large-scale transfer learning. Our classifications enable us to study not only how the bar fraction evolves for different morphological types, but also examine the rates of bar creation and destruction, the mean lifetimes of bars, and even examine potential mechanisms for the formation and destruction of bars. This work demonstrates the versatility of our deep learning approach, for our models, trained on images of real galaxies, are able to successfully classify images of simulated galaxies. This ability for self-consistent classification across simulations and observations is a key advantage of our deep learning approach.

This chapter is based on the manuscript “The evolution of barred galaxies in the EAGLE simulations”, which was published on January 4, 2022 in the journal Monthly Notices of the Royal Astronomical Society, Volume 510, Issue 4, pp. 5164–5178. There are some minor textual and formatting changes compared to the final, published version.

Abstract

We study the morphologies of 3,964 galaxies and their progenitors with $M_* > 10^{10} M_\odot$ in the reference EAGLE hydrodynamical simulation from redshifts $z = 1$ to $z = 0$, concentrating on the redshift evolution of the bar fraction. We apply two convolutional neural networks (CNNs) to classify 35,082 synthetic g -band images across 10 snapshots in redshift. We identify galaxies as either barred or unbarred, while also classifying each sample into one of four morphological types: elliptical (E), lenticular (S0), spiral (Sp), and irregular/miscellaneous (IrrM). We find that the bar fraction is roughly constant between $z = 0.0$ to $z = 0.5$ (32% to 33%), before exhibiting a general decline to 26% out to $z = 1$. The bar fraction is highest in spiral galaxies, from 49% at $z = 0$ to 39% at $z = 1$. The bar fraction in S0s is lower, ranging from 22% to 18%, with similar values for the miscellaneous category. Under 5% of ellipticals were classified as barred. We find that the bar fraction is highest in low mass galaxies ($M_* \leq 10^{10.5} M_\odot$). Through tracking the evolution of galaxies across each snapshot, we find that some barred galaxies undergo episodes of bar creation, destruction and regeneration, with a mean bar lifetime of 2.24 Gyr. We further find that incidences of bar destruction are more commonly linked to major merging, while minor merging and accretion is linked to both bar creation and destruction.

4.1 Introduction

Galactic bars are central, rectangular-shaped, internal morphological structures present in at least 25% to 30% of all galaxies (Nair & Abraham, 2010b; Masters et al., 2011; Melvin et al., 2014) and around half to more than two thirds of all spiral galaxies (Eskridge et al., 2000; Aguerri et al., 2009; Saha & Elmegreen, 2018; Lee et al., 2019). The prevalence of bars throughout the Universe – the bar fraction – is sensitive to many factors such as environment, colour, mass and gas content (Lee et al., 2012; Masters et al., 2012; Skibba et al., 2012; Vera et al., 2016; Erwin, 2018), and is known to evolve with redshift, with a lower overall bar fraction at higher redshifts compared to the present day (Sheth et al., 2008; Melvin et al., 2014). Bars are important drivers of galaxy evolution (Cheung et al., 2013; Conselice, 2014). They affect many key galaxy processes including the distribution of angular momentum (Weinberg, 1985; Athanassoula, 2005), secular evolution (Kormendy & Kennicutt, 2004; Sheth et al., 2005; Sellwood, 2014), interactions with the stellar disc and surrounding dark matter halo (Athanassoula, 2002; Valenzuela & Klypin, 2003), gas dynamics and metallicities (Bournaud & Combes, 2002; Berentzen et al., 2007; Fanali et al., 2015; Fraser-McKelvie et al., 2019), star formation, especially in the centres of galaxies (Ellison et al., 2011; Lin et al., 2020), AGN activity (Alonso et al., 2014; Galloway et al., 2015) and quenching (Masters et al., 2012; Kruk et al., 2018; Fraser-McKelvie et al., 2020b; Geron et al., 2021).

Given the evolutionary importance of bars, there is a motivation to examine bars in cosmological simulations. Studies have examined bars in several simulations, such as EAGLE (Schaye et al., 2015), Illustris/TNG (Vogelsberger et al., 2014; Pillepich et al., 2018) and NewHorizon (Dubois et al., 2021). These have determined a large variation in simulated bar fractions, ranging from nearly 40% for EAGLE (Algorry et al., 2017), 40% to 55% for TNG100 (Rosas-Guevara et al., 2019; Zhao et al., 2020), to as low as 9% in Illustris-1 (Peschken & Łokas, 2019; Zhou et al., 2020) and around 7 to 1% for NewHorizon (Reddish et al., 2022). Such a range in bar fractions is reflective of the inherent challenge of modelling the physical processes governing the formation and evolution of galaxies (Somerville & Davé, 2015; Vogelsberger et al., 2020). Some studies focusing on simulated barred galaxies (e.g Emsellem et al. 2014; Goz et al. 2015;

Okamoto et al. 2015; Spinoso et al. 2017) have utilised zoom-in simulations in order to model the physical processes important to bars, especially baryonic feedback processes (Zana et al., 2019) and gas dynamics, in high detail. However, such zoom-in simulations are strictly limited to analysing only a relatively small number of galaxies, and thus lack the ability to study the impact of bars on galaxy evolution at cosmological scales. Other studies have instead examined the large-scale statistics of a population of barred galaxies in cosmological simulations, such as EAGLE (Algorry et al., 2017).

In this work, we massively expand upon the scale of the latter approach through the use of convolutional neural networks (CNNs) to rapidly classify tens of thousands of galaxies. We study the morphological evolution of simulated galaxies in the reference EAGLE simulation from $z = 1$ (lookback time of 7.93 Gyr) to $z = 0$, with a focus on the bar fraction and the properties of barred galaxies. Cosmological simulations offer important insights into the impacts that bars have on galaxy evolution over cosmic time (Okamoto et al., 2015), when and how bars form, whether bars form preferentially in certain galaxies over others, as well as the typical lifetimes of bars. To facilitate this analysis, we utilise CNNs to identify barred galaxies in each of the ten EAGLE snapshots between $z = 1$ and $z = 0$.

Identifying barred galaxies, especially via visual inspection, is a difficult task, yet is well suited to automated techniques such as machine learning. Traditional methods for automatically detecting and/or classifying barred galaxies include the Fourier analysis of azimuthal profiles (Ohta et al., 1990; Odewahn, 2004) and/or deprojected images (Garcia-Gómez et al., 2017), photometric decompositions (Reese et al., 2007; Durbala et al., 2009), ellipse fitting of isophotes (Abraham et al., 1999; Laine et al., 2002; Erwin, 2005; Consolandi, 2016). Another approach is that of visual inspection, either by expert classifiers (Nair & Abraham, 2010a; Buta et al., 2015) or collective volunteer citizen-science (Willett et al., 2013). More recently, there have been many applications of machine learning, particularly CNNs, to classify galaxy morphologies (Dieleman et al., 2015; Domínguez Sánchez et al., 2018; Zhu et al., 2019; Barchi et al., 2020; Martin et al., 2020; Walmsley et al., 2020; Vega-Ferrero et al., 2021), including galaxy bars (Abraham et al., 2018; Cavanagh & Bekki, 2020). Such is the versatility of CNNs that they have been applied to a broad range of wider topics within astronomy, ranging from detecting galaxy mergers (Ackermann et al., 2018), gravitational lensing (Schaefer et al., 2018), exoplanets (Osborn et al., 2020), classifying galaxy formation processes (Diaz et al., 2019), identifying radio sources (Lukic et al., 2019), to cosmological models (Lucie-Smith et al., 2018; Mustafa et al., 2019; Li et al., 2021).

For this work, we use two CNNs, each trained on g-band SDSS DR7 (York et al., 2000; Abazajian et al., 2009) images from the Nair & Abraham (2010a) (hereafter NA10) catalogue, to classify synthetic g-band galaxy images as either barred or unbarred, and as one of four morphological types; elliptical (E), lenticular (S0), spiral (Sp) and irregular/miscellaneous (IrrM). We refer to these CNNs as the “bar CNN” and “4-way CNN” respectively. The bar CNN is new to this work, while the 4-way CNN is based on our previous work (Cavanagh et al. 2021, hereafter C21). We classified synthetic images of EAGLE galaxies with stellar masses above $M_{\star} \geq 10^{10} M_{\odot}$ for all ten snapshots between $z = 1$ and $z = 0$. The synthetic images are set to mimic the SDSS g-band imaging. It is worth noting that the observed bar fraction is dependent on photometric band, with bars more prevalent in the infrared (Eskridge et al., 2000; Menendez-Delmestre et al., 2007; Buta et al., 2010). However, the primary focus of this work is on g-band classifications, as it is easily comparable to surveys such as the SDSS and Galaxy Zoo (Lintott et al., 2011; Willett et al., 2013), and since the CNN models are trained on g-band data.

The structure of our paper is as follows. In Section 4.2, we briefly summarise the EAGLE simulations. We describe how the synthetic EAGLE images were generated, and how these were processed for use with the CNNs. We further outline the CNN models used to perform the classifications, briefly describe the data processing workflow, and lastly highlight how the bar CNN model was trained and evaluated. In Section 4.3, we present our key results on the redshift evolution of bar fractions in terms of overall samples, as well as across different mass ranges and morphological types, and briefly relate these results to observed bar fractions. We also present the evolution of morphological fractions for each of the four morphological types. In Section 4.4, we focus on the evolution of bars and their physical properties, especially regarding the differences in barred lenticulars and barred spirals. We examine the creation and destruction of bars and the extent to which such events are associated with galaxy merging. We also examine the caveats and limitations of our machine learning approach, and discuss the overall performance of our CNNs. Lastly, we summarise our key results in Section 4.5.

4.2 Methods

4.2.1 The EAGLE Simulation

The ‘Evolution and Assembly of GaLaxies and their Environments’ (EAGLE) project (Crain et al., 2015; Schaye et al., 2015) is a suite of cosmological hydrodynamical simulations of galaxy and evolution formation in the Λ cold dark matter cosmogony. The simulations adopt the cosmological parameters $\Omega_m = 0.307$, $\Omega_\Lambda = 0.693$, $\Omega_b = 0.04825$ and $\sigma_8 = 0.8288$, consistent with a (Planck Collaboration et al., 2014) cosmology. The simulations were performed with a modified version of the N -body Tree-PM smooth particle hydrodynamics code GADGET 3 (Springel, 2005) and include subgrid routines for radiative cooling, star formation, stellar evolution, stellar feedback, black holes (BHs) and AGN feedback (for details, see Schaye et al. 2015). The stellar and BH feedback parameters were calibrated such that at $z \approx 0$ the simulations match the observed galaxy stellar mass function, galaxy sizes and BH masses (Crain et al., 2015). The EAGLE simulations have been shown to broadly reproduce many properties of the evolving galaxy population, such as stellar masses and specific star formation rates (Furlong et al., 2015), sizes (Furlong et al., 2017), colours (Trayford et al., 2015, 2017) and cold gas properties (Lagos et al., 2015; Crain et al., 2017).

Bound structures (subhaloes/galaxies) are identified in the simulations using the friends-of-friends (FoF, Davis et al., 1985) and SUBFIND algorithms (Springel et al., 2001; Dolag et al., 2009). Galaxies are linked between consecutive snapshots through merger trees using the D-Trees algorithm (Jiang et al., 2014; Qu et al., 2017). The galaxy merger trees are available from the EAGLE public data base (McAlpine et al., 2016).

In this work we use the EAGLE ‘reference’ simulation (Ref-L100N1504) of a periodic volume with a side length of 100 comoving Mpc. The volume contains 1504^3 dark matter and gas particles initially with particle masses of $m_{\text{dm}} = 9.7 \times 10^6 M_\odot$ and $m_g = 1.8 \times 10^6 M_\odot$, respectively. The softening length is 2.66 co-moving kpc up to a maximum of 0.7 proper kpc, which is reached at $z = 2.8$. For our purposes, the softening length remains fixed at 0.7 kpc since we are restricted to snapshots with $z \leq 1$. Given the constraints on the simulation parameters, we note that artificially thicker discs, as well as disc heating

due to particle scattering, may hinder the formation of bars through gravitational instabilities (Bauer & Widrow, 2019), which may result in lower simulated bar fractions compared to observations.

4.2.2 Synthetic Images and Data Processing

We first select all galaxies with stellar masses $M_\star > 10^{10} M_\odot$ at snapshots between $z = 0$ and $z = 1$. Such galaxies are well resolved with $\gtrsim 10^4$ stellar particles. We first orient the galaxies face on by calculating the spin vector for all stars between 2.5 and 30 kpc from the centre of potential of each galaxy. The radius limits ensure that the spin is dominated by the rotating disc component of each galaxy (if present) rather than central bulge regions (Trayford et al., 2017).

We calculate rest-frame SDSS g -band luminosities for each stellar particle using the `FRSPS` stellar population model (Conroy et al., 2009; Conroy & Gunn, 2010) with a (Chabrier, 2003) initial stellar mass function, Miles spectral library (Sanchez-Blazquez et al., 2006), Padova isochrones (Girardi et al., 2000) and assuming simple stellar populations. We assume stars with ages < 10 Myr are fully embedded within an optically thick cloud as a simple model for dust absorption (Charlot & Fall, 2000).

The synthetic images were generated using `SPHVIEWER` (Benitez-Llambay, 2015), where particles are distributed according to their local smoothing length. We adopt a method somewhat similar to Trayford et al. (2017), by resampling the young stellar populations, from both the existing young star particles (< 100 Myr) and star-forming gas particles, at higher resolution ($10^3 M_\odot$) based on the star-formation rate of the particles (assuming a constant formation rate). The resampled particles adopt the same position, metallicity and smoothing lengths as the particle from which they are sampled, while new ages are randomly assigned between 0 and 100 Myr. Young star particles (< 100 Myr) then adopt the local SPH smoothing length, while for older stars we recalculate a local stellar smoothing length using the nearest 64 neighbours. This method ensures that star formation in (e.g.) the spiral arms of galaxies is adequately sampled.

The convolutional neural networks utilised in this work were designed to classify monochromatic images with 100×100 pixels of size 0.5 kpc, corresponding to a 50 kpc^2 image. As the CNN was trained on SDSS g -band images, we similarly generate $50 \times 50 \text{ kpc}^2$ rest frame g -band images with pixel sizes of 0.5 kpc for the EAGLE galaxies in each snapshot. The pixel data was linearly normalised such that every pixel has a minimum and maximum values of 0 and 1 respectively. This is consistent with the normalisation applied to the data that was initially used to train the models, as per C21. It is important to ensure that inputs to a CNN are normalised, as this helps ensure optimal model convergence and improves the model’s ability to generalise (LeCun et al., 2015; Gu et al., 2018). As such, future applications of the models require normalised inputs. We generated a total of 35,082 g -band images for 3,964 unique galaxies. Of these, 2,146 galaxies were present for all ten snapshots between $z = 0$ and $z = 1$. There are an average of 3,508 galaxies per snapshot.

4.2.3 CNN Models

CNNs are powerful tools that are able to identify and extract relevant features of data in a model-independent manner, relying solely on a sufficiently large training set of data with which to learn from (LeCun et al., 2015; Goodfellow et al., 2016; Gu et al., 2018; Dhillon & Verma, 2020). CNNs are readily

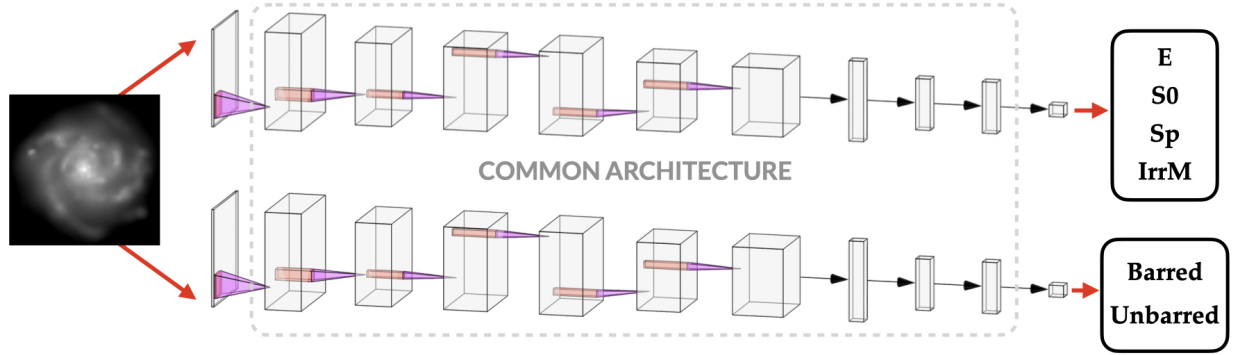


Figure 4.1: Schematic representation of our image classification workflow. Each image is classified by two CNNs. One has been trained to classify morphological types (either E, S0, Sp or IrrM), the other to determine whether the sample is barred. Each network shares the common C2 architecture from [Cavanagh et al. \(2021\)](#) (C21), differing only in the size of the output layer.

applicable to domains involving image classification, and have been increasingly utilised in astronomy (see [Baron 2019](#) for a review).

This work utilises two separate CNN models, each sharing a common architecture. One is trained to classify barred or unbarred galaxies (hereafter referred to as the bar CNN), the other is trained to classify samples into one of four morphological types; E, S0, Sp and IrrM (known hereafter as the 4-way CNN). The 4-way CNN is directly based on our previous work ([Cavanagh et al. 2021](#), C21) – this work also describes the common C2 architecture and how the four morphological types are defined in terms of NA10 catalogue. The bar CNN is a newly developed model trained specifically for this work, and is described in §2.4. Both models are trained on *g*-band SDSS images of galaxies from the NA10 catalogue. The 4-way model from C21 has an overall accuracy of 81%, however the per-class accuracy for irregular/miscellaneous samples was significantly lower than for the other three classes (27%) due to the extremely low number of irregular training samples.

Figure 4.1 represents the workflow for image classification. Each galaxy image is classified by the two CNNs to determine its morphological type and whether it is barred. We stress that each classification - bar and 4-way - is carried out independently, and the results of one model do not influence the other. Each model outputs an array of probabilities for each category. The final, predicted class is simply the category corresponding to the highest probability. In this context, the classification probabilities are also known as confidences, and we will use these terms interchangeably throughout the remainder of this work.

4.2.4 The Bar CNN

The bar CNN is trained to classify galaxies as either barred or unbarred. The data used to train the network consisted of *g*-band SDSS DR7 images of galaxies sourced from the NA10 catalogue of morphological classifications. Although the NA10 catalogue delineates bars into several other subcategories, our CNN does not distinguish between strong and weak bars; anything flagged as a bar is thus treated as a bar. The unbarred training samples were randomly selected from the pool remaining NA10 samples. To avoid categorical biases, we used an equal number of barred and unbarred training samples. Altogether, we used

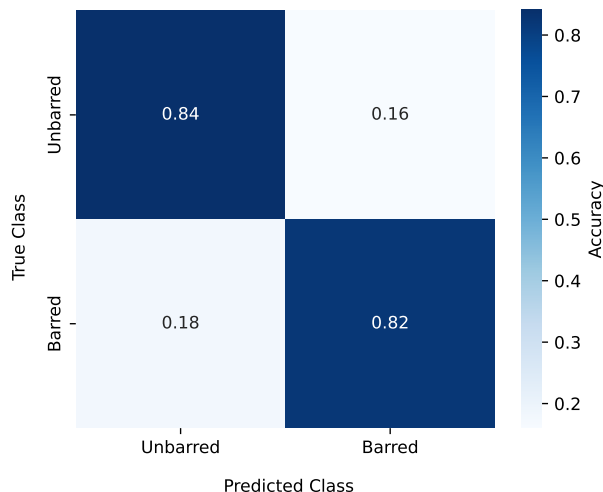


Figure 4.2: Confusion matrix for our bar CNN, shaded according to classification accuracy. The vertical axis corresponds to the true class while the horizontal axis corresponds to the predicted class.

2,612 barred galaxy samples combined with an equal number unbarred samples for a total of 5,224 unique samples. We increased the number of samples through data augmentation. Data augmentation is a general technique for artificially boosting the size of training data through transforming and manipulating the input images, ultimately improving a CNNs ability to generalise (see [Shorten & Khoshgoftaar 2019](#) for a review). In following with the methodology of C21, the initial resolution of the training images is 110×110 pixels. Each image is cropped five times (one from each corner, plus the centre) to obtain five 100×100 cropped images. These cropped images were then further augmented via flipping and 90° rotations, for a combined factor of 40 increase in the number of images. Hence, the final dataset consisted of 208,960 images, each of size 100×100 pixels, and each linearly normalised such that every pixel has a floating point value between 0 and 1.

We used a similar procedure as in C21 for developing the new bar CNN. The overall training data was partitioned into two independent sets - one for training the CNN, the other for testing it - according to an 80%:20% split. We used the Python machine learning library KERAS ([Chollet et al., 2015](#)), running on TENSORFLOW ([Abadi et al., 2016](#)), a general-purpose framework for machine learning. The networks were trained on a Nvidia RTX 1080Ti on ICRAR’s Pleiades cluster. Figure 4.2 shows the confusion matrix for the bar CNN. The overall accuracy of the network is 83%, which compares well with the accuracies of the models developed in C21. Unbarred galaxies are classified with a slightly higher accuracy of 84%, compared to the 82% accuracy of barred galaxies.

4.3 Results

All 35,082 images were independently classified by our CNNs as either barred or unbarred, and as either E, S0, Sp or IrrM. Figure 4.3 displays a random selection of barred galaxies that the bar CNN classified with very high confidence, i.e. $P(\text{bar}) > 0.95$. All the images in Figure 4.3 are the exact 100×100 images that comprised the input into the CNN. This is true for all future example illustrations. Among the provided examples are barred spiral galaxies, as well as strong bars within otherwise smooth discs.

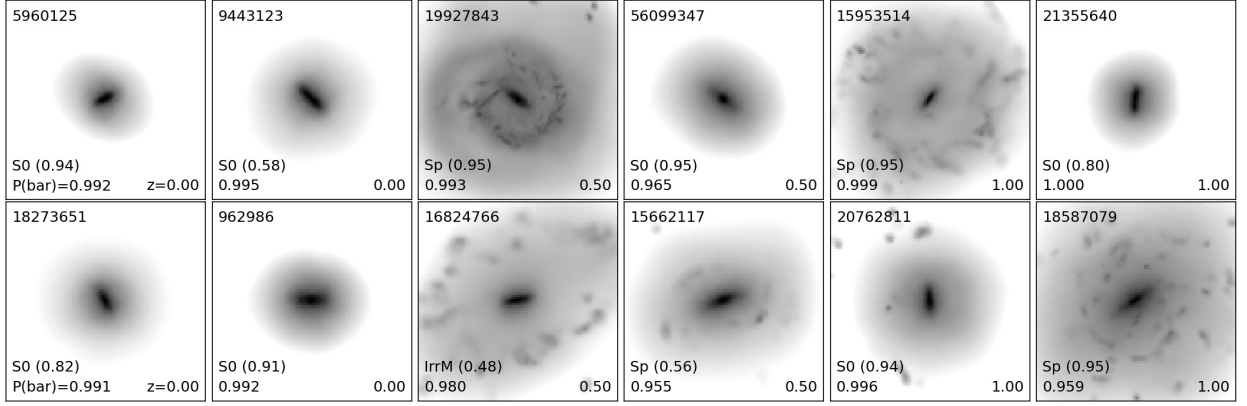


Figure 4.3: A random selection of barred galaxies for low, medium and high redshifts, with classification confidences $P(\text{bar}) > 0.95$. Each image has the same pixel size and normalisation as what was loaded directly into the CNN, as opposed to the raw synthetic image. Each image is annotated with their EAGLE CATID in the top left, the CNN morphological classification (with confidence in brackets) plus the probability of being barred in the bottom left and redshift in the bottom right.

4.3.1 Bar Fraction

We define the overall bar fraction as a function of redshift as simply the fraction of barred galaxies for all galaxies at a given redshift snapshot. Observations have concluded that the bar fraction generally decreases with increasing redshift (Sheth et al., 2008; Masters et al., 2011; Melvin et al., 2014). It is known that the bar fraction varies strongly with stellar mass (Cameron et al., 2010; Erwin, 2018). In particular, Nair & Abraham (2010b) found that a decrease in the observed bar fraction for higher mass samples, yet other studies (e.g. Melvin et al. 2014) found the bar fraction increases with stellar mass. To analyse the impact of stellar mass on the simulated bar fraction, we define three different mass ranges as follows: low-mass, from $10 \leq \log(M_*/M_\odot) < 10.5$, intermediate-mass for $10.5 \leq \log(M_*/M_\odot) < 11$, and lastly high-mass for $\log(M_*/M_\odot) > 11$.

Figure 4.4 shows the overall simulated bar fraction, i.e. the bar fraction across all morphological types, for all three mass regimes, in addition to the overall bar fraction for all samples. This overall bar fraction remains at a roughly constant 32-34% between $z = 0$ and $z = 0.5$, with a peak at about $z = 0.366$. Beyond $z = 0.5$, the bar fraction decreases to approximately 26% at $z = 1$. As for the separate mass ranges, we find that the bar fraction is highest in low-mass samples (10^{10} - $10^{10.5}M_\odot$), starting at around 37% for $z = 0$ and decreasing to approximately 27% for $z = 1$. The intermediate-mass bar fraction decreases from just under 30% at $z = 0$ to 23% at $z = 1$, though notably it reaches a minimum of around 21% at $z = 0.87$. The bar fraction in high-mass samples is considerably smaller, but actually increases with higher redshift to a final value of 18% at $z = 1$. In other words, bars in the most massive galaxies are less common now than they were in previous epochs. This suggests that the growth of new bars from $z = 1$ to the present epoch is largely confined to low and intermediate-mass samples. Importantly, the number of galaxies above $10^{11}M_\odot$ is substantially smaller than that in the other two mass ranges; at $z = 0$, just over 20 galaxies with $M_\odot \geq 10^{11}$ are barred, whilst over 800 in the low mass range are barred. As such, the high mass range is subject to low number statistics.

It is also worth noting the difference in morphologies across each mass range. Figure 4.5 shows that,

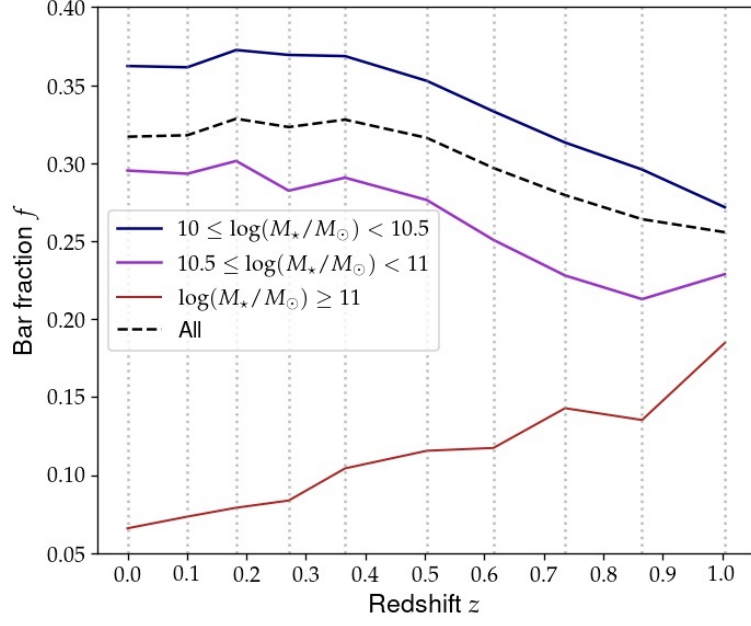


Figure 4.4: The overall bar fractions as functions of redshift for low, intermediate and high mass ranges. The dotted line denotes the bar fraction for all samples regardless of mass.

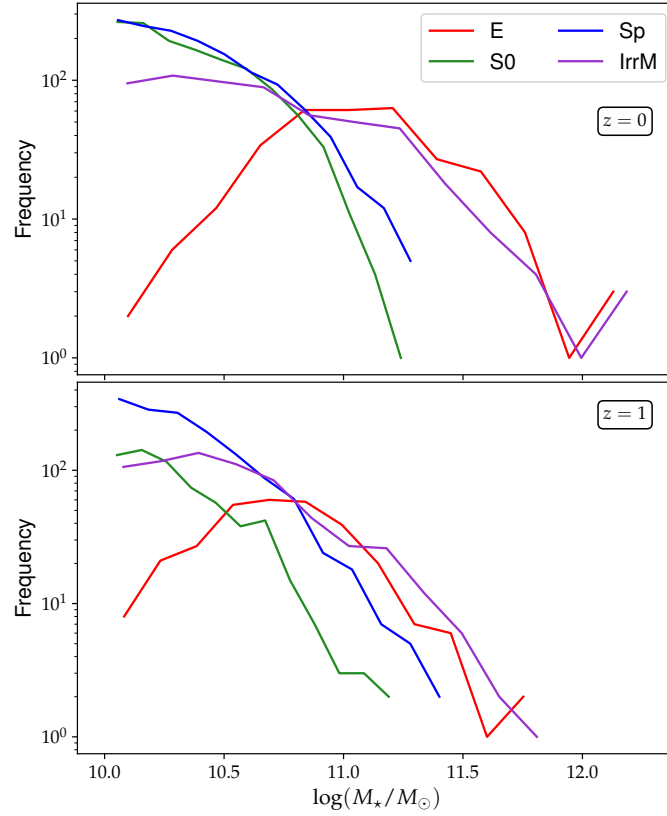


Figure 4.5: Frequency of each of the four morphological types in terms of stellar mass $\log(M_*/M_\odot)$ for $z = 0$ (top panel) and $z = 1$ (bottom panel).

in general, the high mass range predominantly consists of elliptical galaxies, while disc galaxies are more common at intermediate and low masses. We note that the bar fraction in all three mass ranges appears to converge at higher redshifts. This is since the mass distributions for each morphological type are more similar at $z = 1$. As the simulation progresses from $z = 1$ to $z = 0$, the high mass sample becomes more dominated by ellipticals, hence the reduced bar fraction at $z = 0$ for this mass range as seen in Figure 4.4. Since the low and intermediate mass range consists mostly of discs, this hints that the growth in the bar fraction in these mass ranges is confined to disc morphologies. We will discuss morphology in more detail in the next section.

Given the overall increase in bar fraction over time, it is worthwhile examining the change in overall mass distribution for barred and unbarred samples as a function of redshift, i.e. Figure 4.6. The median stellar mass for barred galaxies is roughly constant, with only a slight increase over time from $10^{10.28} M_{\odot}$ at $z = 1$ to $10^{10.30} M_{\odot}$ at $z = 0$. This could be driven by the growth in S0s as shown in Figure 4.5. Unbarred samples instead show a slightly more substantial increase in stellar mass over time, with the median stellar mass of an unbarred galaxy increasing from $10^{10.34} M_{\odot}$ to $10^{10.42} M_{\odot}$. The larger mass growth for unbarred galaxies is consistent with the hierarchical growth of early type galaxies seen in observations (Moffett et al., 2016). This is also consistent with the increase in the mass distribution of ellipticals from $z = 1$ to $z = 0$, as seen in Figure 4.5. Furthermore, Figure 4.4 shows that the bar fraction for high mass samples decreases slightly. This is consistent with the increase in the tail of the mass distribution for unbarred samples as z decreases, and that the tail of the mass distribution for barred samples remains constant for all z . We note that the individual particle masses within the reference EAGLE simulation are of order $10^6 M_{\odot}$ (Crain et al., 2015), which is an important limitation to consider when comparing to observations.

When it comes to comparing these results with observed bar fractions, it is important to be mindful of the different sample selection criteria used, as well as the method used to detect the bar (see Lee et al. 2019). As for mass ranges, Melvin et al. (2014) used a similar low/intermediate/high mass range approach but instead found that low mass galaxies had considerably lower observed bar fractions than high mass galaxies. Erwin (2018) obtained the opposite, showing a sharp decline in bar fraction for masses higher than $\log(M_*/M_{\odot}) \approx 10$. In terms of the values at the present redshift $z = 0$, the overall fraction of 32% compares well with established results based on observations, e.g. 30% (Nair & Abraham, 2010b), 29.4% (Masters et al., 2011), 30.4% (Lee et al., 2012).

4.3.2 Morphology

Bars are most commonly observed in spiral galaxies. Estimates for the observed bar fraction in spirals range from 55% (Aguerri et al., 2009) to as high as 70% (Eskridge et al., 2000; Sheth et al., 2008). To investigate the simulated bar fraction in different morphological types, we further classified all 35,082 with the 4-way CNN from C21. Figure 4.7 shows the bar fraction by morphological type for all galaxies regardless of stellar mass, in addition to the fraction of each morphology, as functions of redshift. We see that all morphologies, except for ellipticals, exhibit a decline in bar fraction over increasing redshift. As seen in observations (e.g. Aguerri et al. 2009; Barway et al. 2011), the bar fraction in spirals is higher than in all the other morphological classes, with a value of 49% at $z = 0$. Both lenticulars and irregular / miscellaneous samples have similar bar fractions, while the bar fraction in ellipticals remains very low (<5%) throughout all snapshots, with a value of 3% at $z = 0$.

Regarding how common each morphological type is, we see in the bottom panel of Figure 4.7 that the fraction of spiral galaxies rises with redshift from around 40% at $z = 0$, peaks at 49% at $z = 0.5$, then decreases slightly to 47% at $z = 1$. The rise in the fraction of S0s is considerably stronger at lower redshifts, with growth accelerating for $z < 0.5$. The fraction of ellipticals remains relatively flat throughout all snapshots, having dropped slightly from a peak of approximately 10% at $z = 1$ to 8% at $z = 0$. Although this slight reduction is not consistent with observations (Bremer et al., 2018), the ellipticals nevertheless become more massive over time. Irregulars similarly peak at $z = 1$, becoming less common over time to just over 15% at $z = 0$. The reduction in spirals from $z = 0.5$ to $z = 0$, and corresponding increase in S0s, is not reflected by any significant change in the evolution of the bar fraction; both the S0 and Sp bar fraction continue to increase slightly within redshift range. Since the overall fraction of irregulars and ellipticals remains constant for $z < 0.5$, galaxies are mostly transitioning from spirals to S0s. As the bar fraction in both S0s and spirals increases with decreasing z , this suggests that both barred spirals and unbarred spirals are transitioning to S0. Mechanisms known to facilitate such transitions include gas stripping, tidal interactions and mergers, and the fading of the disc and/or spiral arms (Barway et al., 2009; Bekki & Couch, 2011; Borlaff et al., 2014; Querejeta et al., 2015; Fraser-McKelvie et al., 2018; Rizzo et al., 2018). We will discuss possible avenues for both bar creation and destruction in more detail in Section 4.4.3.

We further investigated the bar fractions across the different mass ranges in each of the four mor-

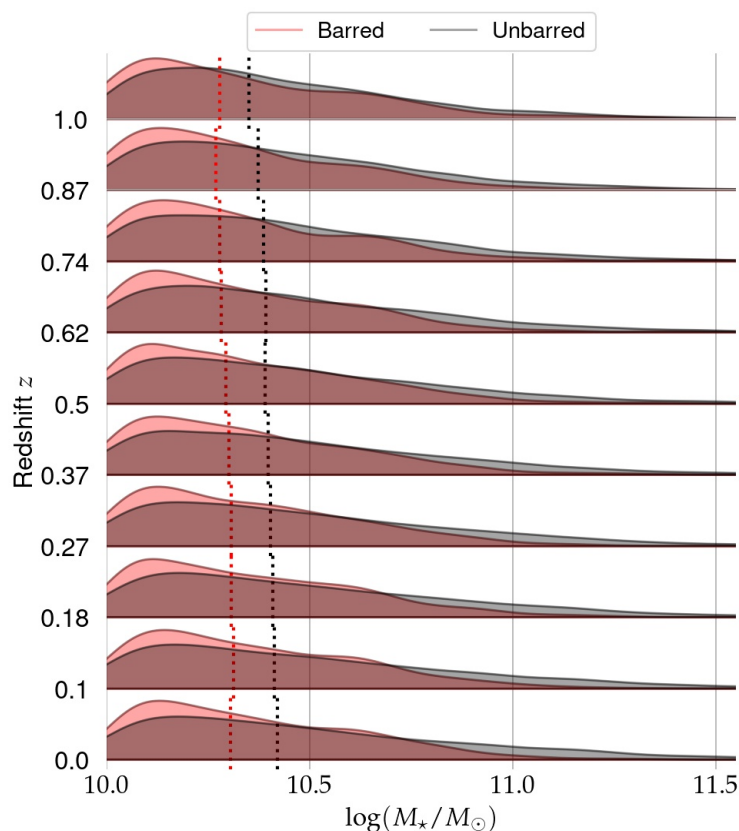


Figure 4.6: Ridgeplot of the distributions of stellar masses for barred (red shading) and unbarred (black shading) galaxies, in units of $\log(M_*/M_\odot)$, from $z = 1$ to $z = 0$. The dotted red and black lines denote the median stellar mass for barred and unbarred galaxies respectively.

phologies. We found, at $z = 0$, that the bar fraction in S0s is highest in the intermediate mass range (33%), while the low mass range S0 bar fraction (19%) is lower than the overall S0 bar fraction (22%). This is only observed for S0s: for all other morphologies, the bar fraction is highest in the low mass range. Our intermediate-mass range spirals have a bar fraction of around 41% at $z = 0$, comparing well with the value of 40% that [Algorry et al. \(2017\)](#) obtained for a selection of 296 EAGLE discs in the mass range $10.6 < \log(M_*/M_\odot) < 11$. We note that [Algorry et al. \(2017\)](#) used a Fourier analysis method to identify bars under a much stricter selection criteria, examining only disc galaxies. Our CNN method is applied to all galaxies within each snapshot, subject to the mass limits as outlined in Section 4.2.2, and so this approach allows us to examine both bar fractions by morphological type, as well as the overall bar fraction across all types. We further note that there is significant variation in the bar fractions obtained from simulations, e.g. 9% to 7% have been obtained from Illustris-1 and NewHorizon ([Zhou et al., 2020](#); [Reddish et al., 2022](#)), to 55% in TNG100 ([Zhao et al., 2020](#)).

Figure 4.8 shows a random selection of high-confidence galaxies classified in each of the four morphological classes for $z = 0$, $z = 0.5$ and $z = 1$. Visually inspecting a random sample of galaxies is a quick way to verify that the morphological classifications obtained with the CNN are appropriate, or if there are deeper issues with the models. As expected, ellipticals are more spheroidal in nature while lenticulars have a more prominent disk component, although in general it can be difficult to distinguish

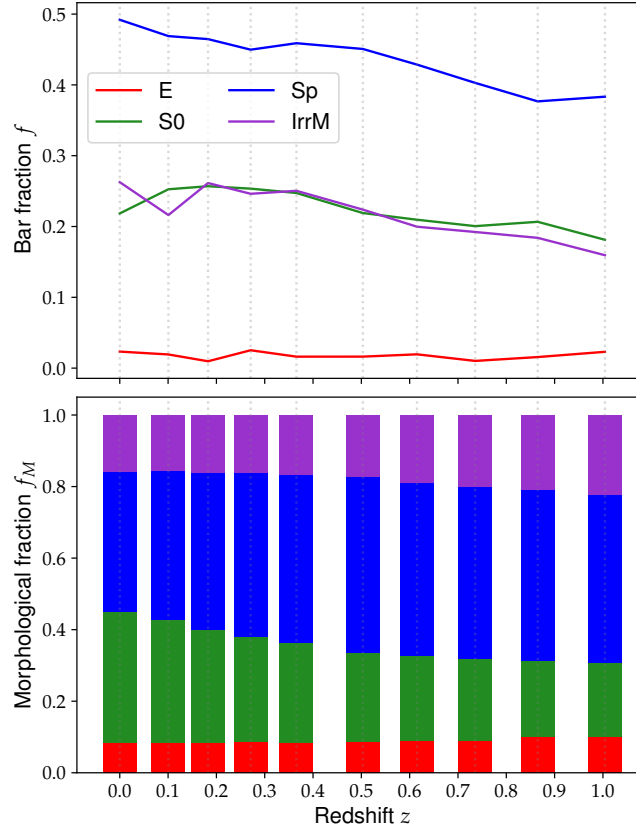


Figure 4.7: (Top) The overall bar fraction, for samples classified as either of the four morphological types, as a function of redshift. (Bottom) Stacked bar plot of the fraction of samples in each morphological type at each snapshot.

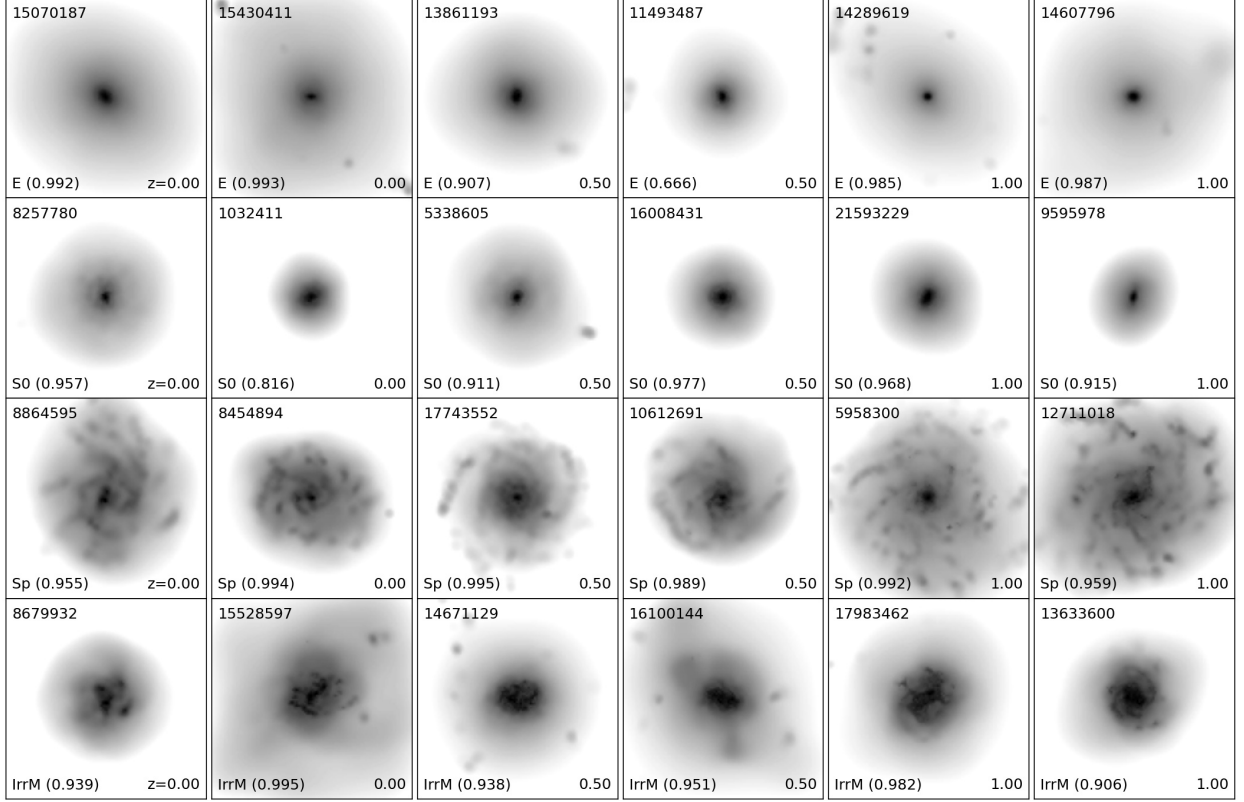


Figure 4.8: Example of randomly selected unbarred samples with high classification confidences for each of the four main morphological types.

between ellipticals and lenticulars based solely on an image (Blanton & Moustakas, 2009). Spirals are clearly distinguished, with the irregular / miscellaneous class examples clearly not fitting into any of the other three categories. We note that the miscellaneous samples from the NA10 dataset that the 4-way CNN was trained on contained several interacting galaxies, in addition to those with odd, ill-defined or clumpy shapes. This is likely why many galaxies with strong, turbulent gas flows are classified as irregular, even though they have a relatively symmetric and regular structure.

4.4 Discussion

4.4.1 Properties of Barred Galaxies

It is well known that galaxy bars have profound effects on the evolution of galaxies (Athanasoula, 2005; Vera et al., 2016; Fraser-McKelvie et al., 2020b; Geron et al., 2021). To determine whether there is any clear impact, we examine the physical and kinematic properties of barred and unbarred galaxies, focusing on the two morphological types (S0 and Sp) in which bars were found to be most prevalent. For this, we utilise the EAGLE physical and kinematic properties described in McAlpine et al. (2016) and Thob et al. (2019).

When it comes to disk galaxies, particularly barred galaxies, it is worth investigating the differences between spirals and lenticulars. Figures 4.9 and 4.10 display a series of histograms for various physical

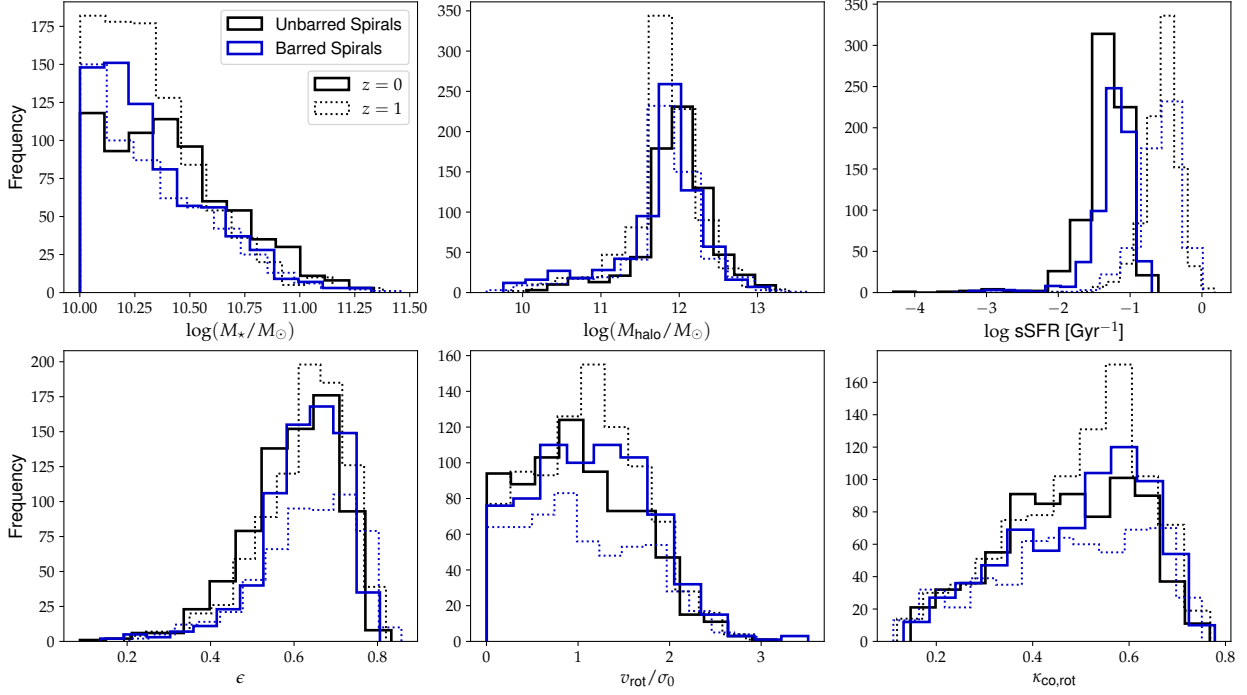


Figure 4.9: Frequency histograms for barred (blue) and unbarred (black) spiral galaxies for the stellar mass M_* , dark matter mass M_{halo} , specific star formation rate, ellipticity ϵ , rotation to dispersion ratio v_{rot}/σ_0 and the corotation parameter $\kappa_{\text{co,rot}}$. The solid and dotted lines indicate the distributions at $z = 0$ and $z = 1$ respectively. Kinematic properties are calculated from all stellar particles with spherical radius $r < 30$ proper kpc.

and kinematic properties for spirals and S0 galaxies respectively. These include stellar and dark matter masses, specific star formation rate, ellipticity (flattening) ϵ , the ratio of rotational velocity to dispersion v_{rot}/σ_0 , and the fraction of kinetic energy that is invested in corotation $\kappa_{\text{co,rot}}$ (Correa et al., 2017). The kinematic properties are calculated based on all stellar particles with a spherical radius $r < 30$ proper kpc (or pkpc). Note that ϵ is defined as $1 - b/a$ where a, b are the moduli of the major and minor axes of the ellipsoid corresponding to the spatial distribution of stars (see Thob et al. 2019 for details). Thus ϵ is inherently based on the 3D shape of the galaxy, and not just the ellipticity of the 2D face-on projection. We see in Figure 4.9 that barred spirals tend to be less massive than unbarred spirals, and slightly more star forming. There is no clear distinction between barred and unbarred spirals in terms of ellipticity, however barred spirals have slightly higher values of v_{rot}/σ_0 and $\kappa_{\text{co,rot}}$. Lenticulars, shown in Figure 4.10, clearly have a different distribution in these parameters. In particular, there is a near-flat stellar mass distribution from around 10^{10} to $10^{10.75} M_\odot$ for barred S0s. There are also more intermediate-mass barred S0s at $z = 0$ compared to $z = 1$. Barred S0s also have a flatter spread in values of specific star formation compared to unbarred S0s. Unlike spirals, there is a clear distinction between barred and unbarred S0s in terms of ellipticity. Furthermore, the distributions of $\kappa_{\text{co,rot}}$ for S0s is considerably more uniform than that for spirals.

With this in mind, we investigate whether there are any clear properties that can distinguish barred spirals between barred S0s. Table 4.1 shows the median values for various physical and kinematic properties

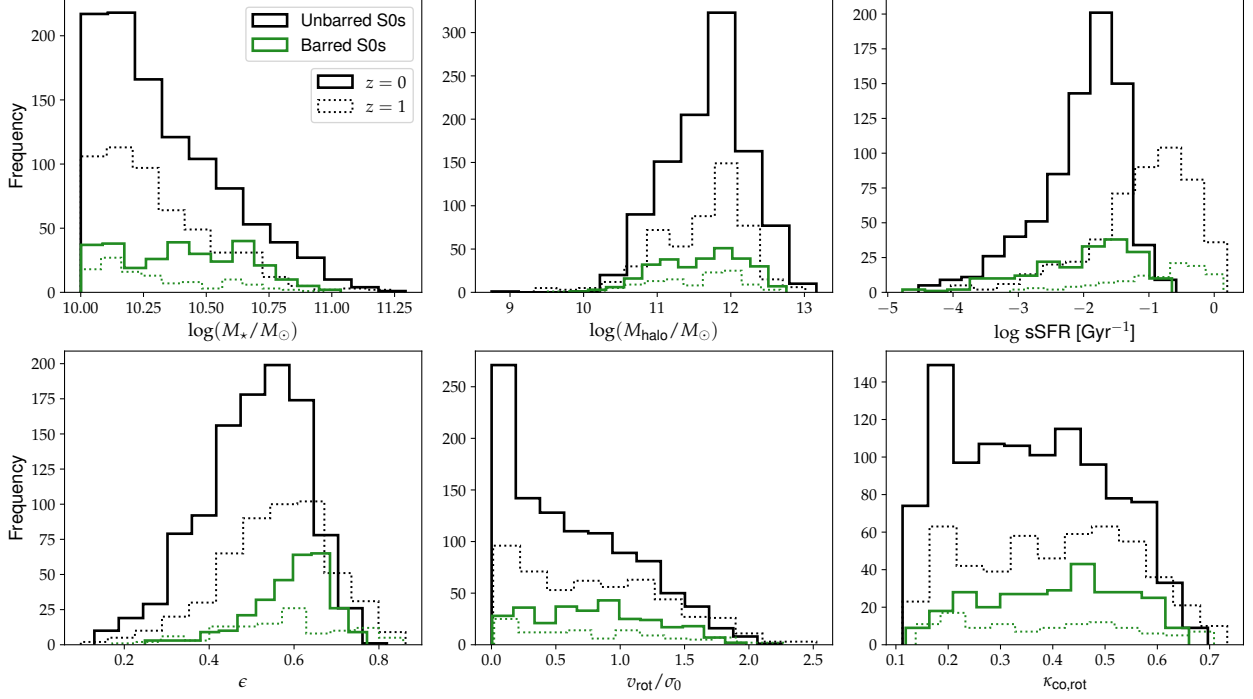


Figure 4.10: Frequency histograms for barred (green) and unbarred (black) lenticular galaxies, for the properties listed as in Figure 4.9. The solid and dotted lines indicate the distributions at $z = 0$ and $z = 1$ respectively.

for barred and unbarred spirals at the present epoch $z = 0$, and for a low and high mass range defined as less than and greater than $\log(M_*/M_\odot) = 10.5$ respectively. As expected, gas fraction $f_g \equiv M_{\text{gas}}/M_*$ is higher in spirals. However, while the gas fraction does not vary significantly among barred and unbarred spirals, for the case of S0s, barred S0s have lower gas fractions than their unbarred counterparts. This is considerably more apparent for the higher mass range. Star formation rates in S0s vary across mass range. High-mass barred S0s have lower star formation rates compared to unbarred S0s, while the reverse is true for low-mass S0s. Lower star formation rates in barred galaxies is consistent with bar-driven quenching and gas removal (Spinoso et al., 2017; Fraser-McKelvie et al., 2020b; Geron et al., 2021), but further analysis is required to investigate these mechanisms in detail. Across all types and mass ranges in Table 4.1, barred galaxies have higher corotation parameters $\kappa_{\text{co,rot}}$, with values higher in spirals than for S0s, consistent with more energetic corotation.

Lastly, Table 4.1 also lists the values of D/T , i.e. the disc-to-total mass fraction. Barred spirals have higher median D/T values compared to unbarred spirals, but the difference is significantly greater (up to 68%) for S0 galaxies. That is, barred S0s are more strongly disc dominated compared to unbarred S0s. Conversely, unbarred S0s have relatively larger bulge components. It is known that bulges can hinder and even prevent bar formation in disc galaxies (Kataria & Das, 2018), and that the observed bar fraction is comparatively lower in galaxies with prominent bulges compared to galaxies with prominent discs (Barazza et al., 2008; Aguerri et al., 2009). It is also known that S0s have higher bulge fractions compared to spirals (Mishra et al., 2018), that these are predominantly classical bulges compared to pseudobulges (Kormendy & Kennicutt, 2004), and that bulges are more common in higher mass S0s (Fisher & Drory,

Table 4.1: Table of median values for physical and kinematic properties of barred and unbarred spirals and S0s for different stellar mass ranges.

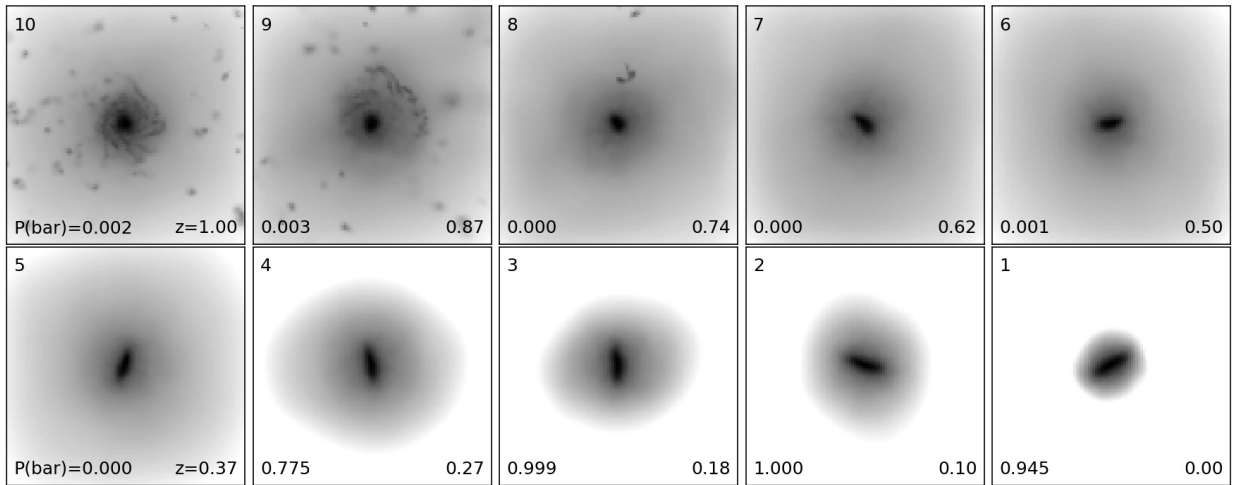
Type	Mass Range $\log(M_*/M_\odot)$	Bar	$\log f_g$	$\log \text{sSFR}$ Gyr^{-1}	$\kappa_{\text{co,rot}}$	D/T
S0	10-10.5	Y	-1.052	-1.578	0.439	0.554
S0	10-10.5	N	-0.924	-1.757	0.362	0.382
S0	>10.5	Y	-0.848	-2.682	0.386	0.535
S0	>10.5	N	-0.056	-2.455	0.307	0.318
Sp	10-10.5	Y	0.157	-1.157	0.524	0.593
Sp	10-10.5	N	0.148	-1.249	0.475	0.523
Sp	>10.5	Y	0.323	-1.313	0.550	0.680
Sp	>10.5	N	0.385	-1.380	0.516	0.606

2011). This likely explains why the high mass S0s in Table 4.1 has the lowest median D/T value of 0.318. We note in Table 4.1 that the D/T values of barred S0s are similar to those of unbarred, low mass spirals.

4.4.2 Bar Evolution

Our results have thus far demonstrated that the bar fraction in EAGLE is in a state of flux, with bars more common at $z = 0$ than they were at $z = 1$. Given that EAGLE uses consistent IDs, and our CNN can consistently classify galaxies across snapshots, it is possible to track the evolution of every unique galaxy throughout all snapshots. In total, 2,146 unique galaxies are present for all ten snapshots between $z = 1$ to $z = 0$. These are the galaxies with $M_* > 10^{10} M_\odot$ at $z = 1$ and which are not lost through merging with other galaxies. It is these galaxies that will form the basis of our analysis in the rest of this subsection.

Figure 4.11 shows the evolution of the galaxy with ID “10” at $z = 1$ through to $z = 0$. This galaxy initially starts out as an unbarred spiral, and ends up evolving into a compact barred irregular. Of note here is that this galaxy is undergoing tidal stripping from $z = 0.27$ onwards, resulting in a more compact

Figure 4.11: Snapshots of a unique galaxy between $z = 1$ to $z = 0$ showing the evolution and growth of a galaxy bar.

size at $z = 0$. The bar grows slowly, and is first detected by the CNN with the image ID “4” (corresponding to $z = 0.27$), after which the bar remains throughout the remaining snapshots. It is interesting to note that the CNN is yet to detect the bar at ID “5” ($z = 0.37$) despite it being similar in length. It is likely that the more well defined disc in ID “4” – resulting in a relatively more prominent bar – led to the classification. Determining exactly when a bar is formed is thus difficult. The CNN classifications in Figure 4.11 suggest that the bar formed between $z = 0.37$ and $z = 0.27$, however it could be argued through visual inspection that the bar formed much earlier. It is also worth noting that the CNN does not distinguish between strong and weak bars, however it is likely that the CNN is biased towards strong bars as they are designed to extract the dominant features of a given image.

Nevertheless, we can still examine the creation and possible destruction of bars in EAGLE. We define bar creation and destruction as follows:

- Bar creation: The sample is classified as unbarred in the current snapshot t and barred in the next snapshot $t + 1$
- Bar destruction: The sample is classified as barred in the current snapshot t and unbarred in the next snapshot $t + 1$

Figure 4.12 shows the number of bars created and destroyed between snapshots (note that, as per the above definitions, the initial snapshot at $z = 1$ is excluded). We can see that there is a general downward trend in both the number of bars created and destroyed per snapshot. However, bar creation rises from $z = 0.62$ to $z = 0.36$, peaking at $z = 0.36$. The difference between the number of bars created and the number of bars destroyed is also greatest within this redshift range, which can be thought of as an epoch

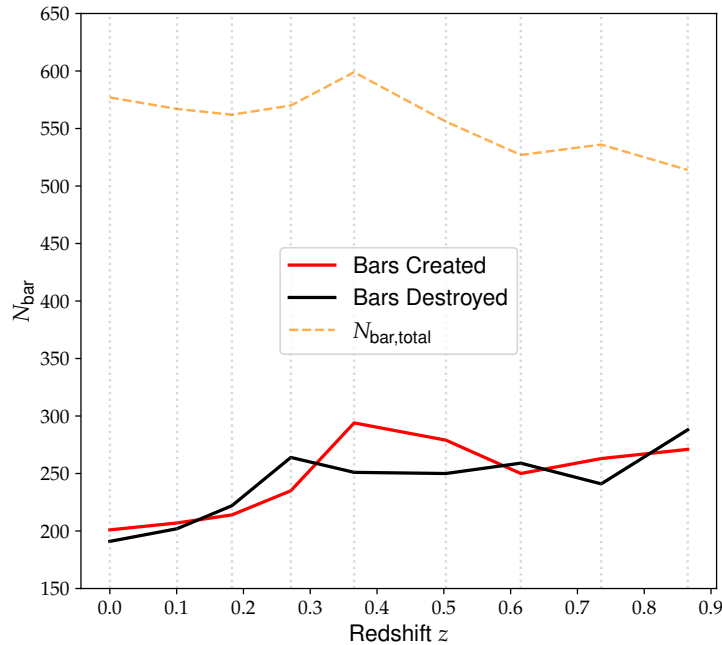


Figure 4.12: Plot of the number of bars created and the number of bars destroyed between each snapshot (left vertical axis), along with the total number of bars $N_{\text{bar,total}}$ detected for each snapshot (right vertical axis), for all 2,146 galaxies present in all snapshots.

of greatest net bar formation. Indeed, 50% more bars are created at $z = 0.36$ compared to at $z = 0$, and from $z = 0.62$ to $z = 0.36$ the total number of bars increases by 20%. At $z = 0.27$ and $z = 0.18$ more bars are being destroyed than created, leading to a drop in the total number of bars, before bars created once again outnumber bars destroyed in the final two snapshots. It is also curious to note that, solely for these 2,146 samples, the bar fraction is, on average, lower at around 26% to 28%. Given that the average stellar mass of these samples grows from $10^{10.56} M_{\odot}$ to $10^{10.81} M_{\odot}$ from $z = 1$ to $z = 0$, this is consistent with the lower overall fraction for intermediate mass samples as in Figure 4.4.

On average, 246 bars are created per snapshot and 240 are destroyed (roughly 11% of all samples). It is inevitable that some of these transition events are caused by CNN misclassification. Given the maximum network accuracy of 82% for barred galaxies in Figure 4.2, this suggests that 18% of detected bars may be erroneous, corresponding to roughly 100 bars per snapshot. Thus 20% to 25% of all bar creation and destruction events may be artificial, resulting instead from CNN misclassification. This is less than the 50% change in bar creation events from $z = 0.36$ to $z = 0$, and the number of bars created and destroyed per snapshot is well above the level of misclassification. The trends in Figure 4.12 nevertheless hint that the bar creation rate is not constant but also evolves with redshift. It has been posited that bars are transient features; simulations have shown that they are capable of being destroyed and regenerated, and that galaxies may well play host to more than one bar over their lifetime (Berentzen et al., 2004; Bournaud et al., 2005). We thus investigate the incidence of multiple bar creation and/or destruction events throughout the lifetime of each sample.

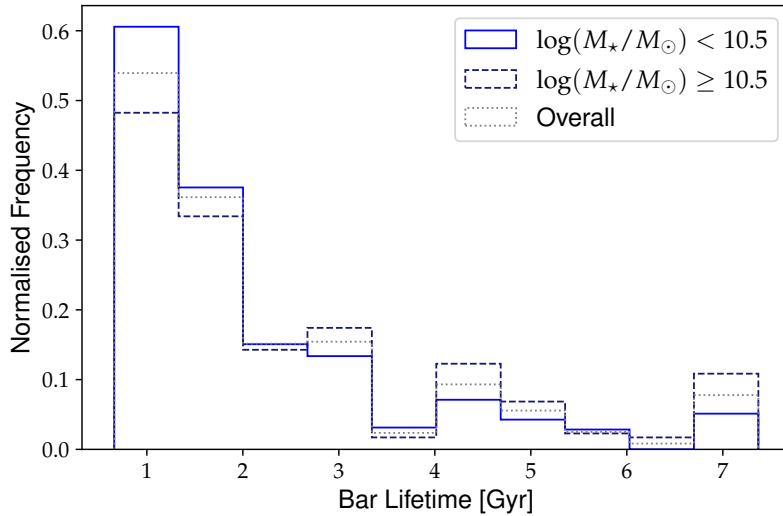


Figure 4.13: Histogram of bar lifetimes for low-mass galaxies and high-mass galaxies.

Of the 2,146 samples, 35.1% either did not form a bar or remained barred across all 10 snapshots, 35.8% formed one bar, 22.1% formed two bars and 7% formed three or more bars, where bar formation is indicated via a transition from unbarred to barred in consecutive snapshots. Less than 1% of samples remained barred throughout all ten snapshots. 65% of all 2,146 samples experienced at least one bar destruction event, and 29% experiencing two or more destruction events. Here bar destruction, like creation, is defined as a change in classification from barred to unbarred in consecutive snapshots. In total, 2,214 bars were created, 1,072 of which persisted for at least two consecutive snapshots.

This motivated the analysis of bar lifetimes, as determined by the difference in lookback times for the initial and final snapshots of the bar’s existence (excluding snapshots at the very start and end of the simulation). Figure 4.13 shows the distribution of bar lifetimes for low-mass and high-mass galaxies. The mean bar lifetime was found to be 2.24 Gyr, with a slightly shorter lifetime of 2.01 Gyr for low-mass galaxies, and 2.46 Gyr for high-mass galaxies. Note in this context, low mass galaxies are those with $\log(M_*/M_\odot) < 10.5$ for all consecutive snapshots in which it is classified as barred, and high mass are those with $\log(M_*/M_\odot) \geq 10.5$. The overall mean bar lifetime is slightly higher than the 1-2 Gyr lifetime found in simulations by Bournaud et al. (2005). It is extremely difficult to obtain estimates of bar lifetimes from observations, however the analysis of accretions timescales, such as in Elmegreen et al. (2009), has yielded similar ages.

It is important to note that this analysis is restricted only to galaxies present for all ten snapshots, which accounted for 54% of all unique galaxies present. If this were not the case, the above fractions would be skewed towards one or less bar formation events due to the shorter timescales for galaxies that are lost to mergers. It is also worth noting that for the samples that undergo high numbers of creation and destruction events, these events are more likely to be artificial, having been caused by CNN misclassifications. Similarly, the peak at around 1 Gyr in Figure 4.13 is likely inflated due to CNN misclassifications, especially in cases of multiple, short-lived bars. As a result, actual bar lifetimes may be higher than our value 2.24 Gyr. Figure 4.14 gives an example of a galaxy that underwent two bar formation events: one between $z = 1$ and $z = 0.87$, and the other between $z = 0.5$ and $z = 0.37$). By examining the sequence of images, the initial bar is clearly destroyed, after which a new bar begins to gradually form, likely driven by gas accretion (Bournaud & Combes, 2002).

4.4.3 Mechanisms for Bar Formation and Destruction

One major factor affecting the formation and evolution of galaxies is environment, which covers everything from ram pressure stripping, tidal interactions and fly-bys, to galaxy merging. Galaxy mergers play key roles in the growth and evolution of galaxies (Barnes & Hernquist, 1992; Cole et al., 2000; Conselice, 2014) and their dynamical evolution, especially through changes in angular momentum (Lagos et al., 2018a) and bulge growth (Aguerre et al., 2001). Mergers also affect star formation, gas flows and AGN activity (Perret et al., 2014; Capelo et al., 2015; Pearson et al., 2019). Mergers may also play a role in the formation and evolution of galaxy bars. Previous simulations have shown that mergers can promote the formation of a bar (Peirani et al., 2009; Cavanagh & Bekki, 2020), however they can also weaken existing bars and potentially destroy them (Ghosh et al., 2021). Another external mechanism responsible for inducing bar formation is that of tidal interactions, such as those induced by galaxy-galaxy interactions (Noguchi, 1987; Elmegreen et al., 1990; Miwa & Noguchi, 1998; Moetazedian et al., 2017; Peschken & Łokas, 2019) and flybys (Lang et al., 2014; Zana et al., 2018). Bars can also form spontaneously via secular evolution as a result of global, non-axisymmetric instabilities (Hohl, 1971; Combes & Sanders, 1981; Sellwood & Wilkinson, 1993; Athanassoula, 2003; Kim et al., 2016). Altogether, these various mechanisms are also capable of destroying bars. This can occur by tidal stripping, gas infall and/or buckling due to vertical instabilities, (Raha et al., 1991; Bournaud et al., 2005; Martinez-Valpuesta et al., 2006; Łokas, 2019; Xiang et al., 2021), with some physical processes capable of reforming bars after their destruction (Bournaud & Combes, 2002; Berentzen et al., 2004; Bournaud et al., 2005).

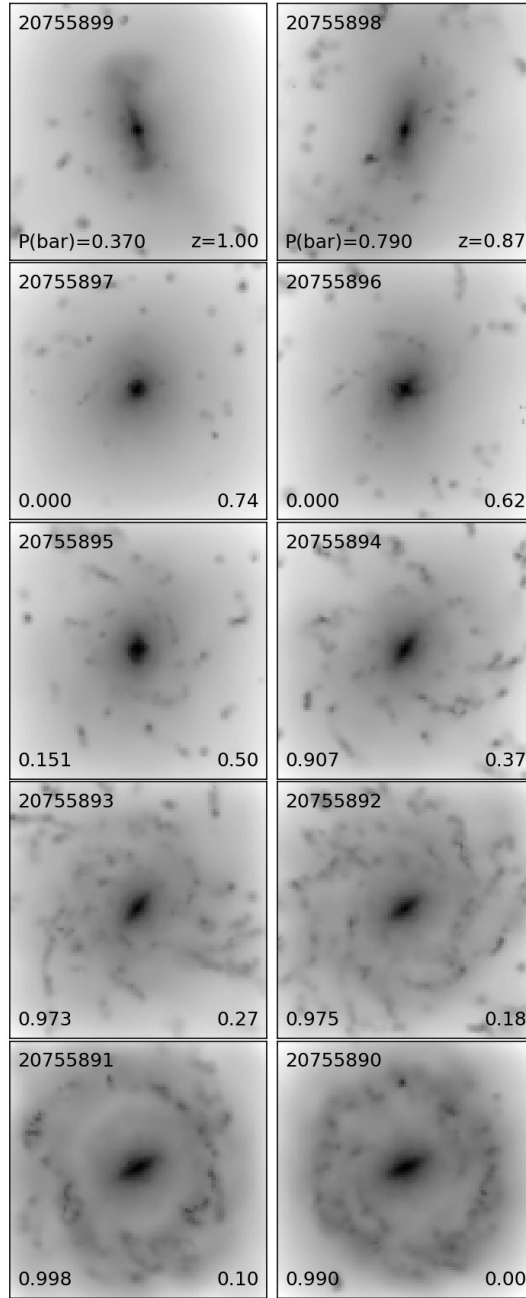


Figure 4.14: Evolution of an example sample over all ten snapshots, illustrating both the destruction and regeneration of a bar.

For all the 2,146 galaxies present for all snapshots, there were a grand total of 2,214 bar creation events, and 2,168 bar destruction events. Of the 2,214 bar creation events, 529 (24%) also corresponded to a change in morphology (i.e. the morphological classification differed to that of the previous snapshot). Of the 2,168 bar destruction events, 673 (31%) saw a change in morphology. Figure 4.15 illustrates the changes in morphology corresponding to bar creation and destruction events. Of these 529 morphological transitions corresponding to bar creation, the most common were unbarred irregulars/miscellaneous transitioning into barred spirals (29%), unbarred spirals turning into barred irregulars (24%) and unbarred

S0s transitioning into barred spirals (18%). Of the 673 morphological transitions corresponding to bar destruction, the most common change was from barred spirals to unbarred irregulars (37%), followed by barred spirals to unbarred S0s (15%) and barred irregulars to unbarred S0s (14%). We note that further morphological transitions, especially those with very few samples, are more likely to correspond to CNN misclassifications.

The transition from unbarred irregulars to barred spirals is consistent with tidal interactions. That the reverse (barred spirals to unbarred irregulars) is the most common change in morphology associated with bar destruction is also consistent with the effects of tidal stripping and/or deformation, as well as major merging. The change from barred spiral to unbarred S0 is consistent with the faded spiral mechanism. Other transitions to unbarred S0s could be due to minor merging, or the destruction of the bar from bulge growth or gas infall (Vera et al., 2016; Kruk et al., 2018). Importantly, the majority of galaxies did not experience a change in morphology when forming a bar (76%) or losing a bar (69%).

We can examine the effects of mergers on both bar creation and destruction through analysing the merger histories of each sample, and identifying the merger ratios of mergers linked to the snapshots that correspond to bar creation or destruction events. In Figure 4.16, it can be seen that the merger ratios for bar creation and destruction share similar distributions, however there is a clear difference for the case of major merging (log merger ratio from -1 to 0). Here, major mergers are linked to up to twice the number of bar destruction events compared to bar creation events. Hence major merging is more strongly linked to bar destruction. This can also be seen by comparing the merger ratios for bar destruction to a random selection of merger ratios out of all mergers (black, dashed line in Figure 4.16). This is consistent with previous N-body simulations from Cavanagh & Bekki (2020), which found that while mergers are capable

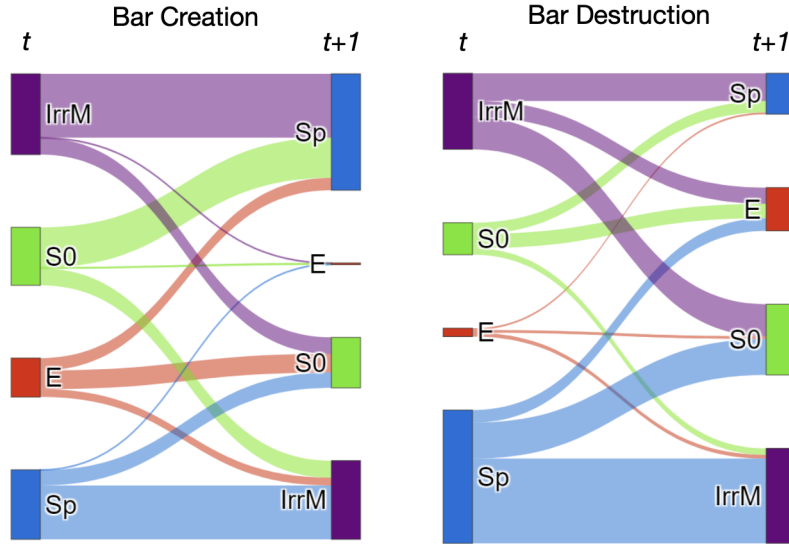


Figure 4.15: Sankey diagrams summarising the changes in morphology between two snapshots as a result of bar creation (left) and bar destruction (right). This is restricted to only those samples for which the classified morphological type at snapshot $t + 1$ differed to that at snapshot t , which comprises 24% of the total bar creation events and 31% of bar destruction events. The size of the nodes and paths is proportional to the number of samples, and both plots are scaled to the same relative size.

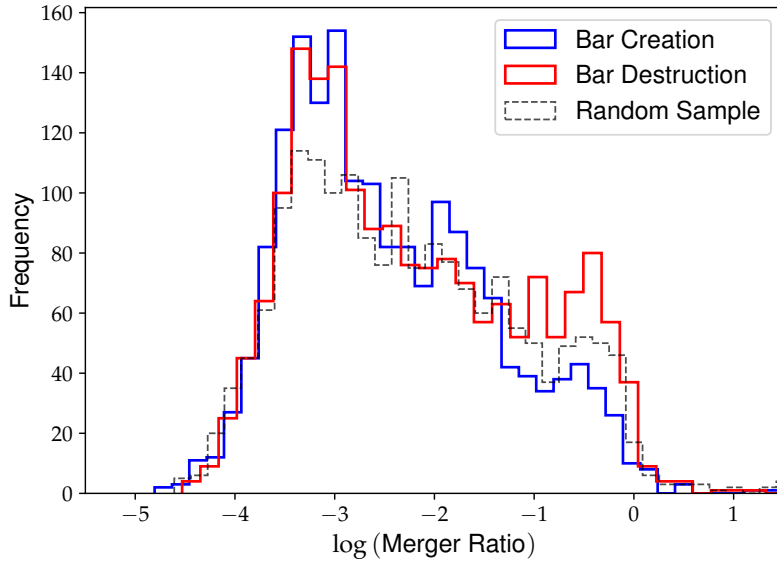


Figure 4.16: Frequency distribution of merger ratios for mergers corresponding to times of bar creation (blue) and bar destruction (red). The dashed, black line corresponds to a random sample of merger ratios.

of creating bars, major merging is far more likely to result in the destruction of a bar.

From Figure 4.16, it can be seen that minor merging occurs with both creation and destruction, with a slight peak in bar creation at $\log(\text{merger ratio}) = -2$, but all ratios are close to random. Smaller merger ratios (less than -3) are linked to accretion events. As for these accretion events, there is no noticeable difference in the distribution of merger ratios for bar destruction compared to bar creation. Crucially, however, is the fact that both distributions are higher than the random merger ratio sample. Although we cannot directly demonstrate an association, Figure 4.16 hints that accretion events are more strongly linked to both bar destruction and creation events, compared to a random sample of merger events. This comes as no surprise given the important role of accretion for the growth of bars (Bournaud & Combes, 2002; Athanassoula et al., 2013), as well as its capacity to weaken the bar and/or destroy it (Seo et al., 2019; Ghosh et al., 2021).

4.4.4 Bar Strength

As aforementioned in the previous subsection, our CNNs do not distinguish between strong and weak bars. The notion of bar strength is difficult to objectively define. Some studies have likened bar strength to the bar axis ratio (Martin, 1995), to the ellipticity of the bar (Laurikainen & Salo, 2002) and others with the length of the bar (Hoyle et al., 2011). Indeed, there are several approaches to quantitatively measure bar strength, which necessarily depend on how strength is defined. These range from measuring brightness or flux ratios of the bar in relation to the interbar region of the disc (Elmegreen & Elmegreen, 1985; Rozas et al., 1998), to measuring the difference between the amplitudes of the $m = 0$ and $m = 2$ Fourier components (Aguerri et al., 1998; Abraham & Merrifield, 2000; Das et al., 2008; Aguerri et al., 2009; Garcia-Gómez et al., 2017). Some studies instead suggest weak and strong bars are extremes on continuum of bar strength (Géron et al., 2021), with weaker bars sharing more in common with unbarred galaxies

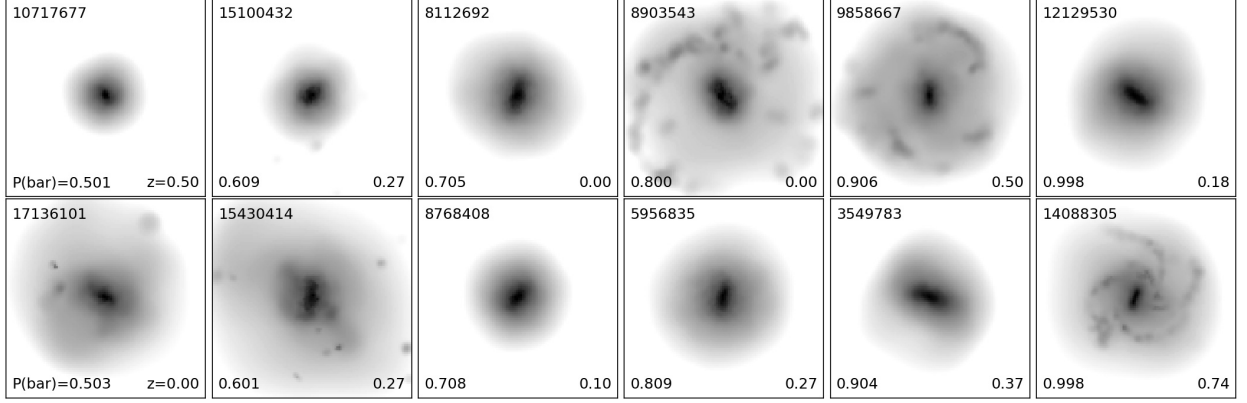


Figure 4.17: A random selection of barred galaxies with classification confidences from 0.5 to 1.

than barred galaxies (Abraham et al., 1999; Cuomo et al., 2019). In simulations, both Algorry et al. (2017) and Rosas-Guevara et al. (2019) find that roughly half of the identified barred galaxies in EAGLE and TNG100 respectively are strongly barred, with the other half weakly barred.

Bar strength can nevertheless be considered as an intuitive quality, however subjective that may be. Of the crowdsourced classifications of the Galaxy Zoo 2 project, strongly barred galaxies tend to correspond to those with majority vote-fractions ($p_{\text{bar}} > 0.5$), while weakly barred galaxies have lower vote fractions (Masters et al., 2012; Skibba et al., 2012; Willett et al., 2013). We can attempt to see whether our CNN confidences, i.e. the probability that the image is barred, exhibit similar characteristics. Figure 4.17 shows a random selection of galaxies with the probability of being barred, P_{bar} , varying from random (0.5) to near-certain. On visual inspection, the near-random classifications (0.5, 0.6) are indeed inconclusive; the examples in Figure 4.17 are likely prolate bulges. Yet, as P_{bar} increases, it can be argued that the presence of a bar becomes more clearer, in the sense that the bars itself become relatively more prominent. After all, a high classification confidence means that the CNN is more certain that it has detected the presence of a bar. It does not necessarily imply a stronger bar, only a more easily detectable bar.

We again stress that the CNNs do not directly distinguish between weak and strong bars, and therefore the probability of a sample being barred can only act as an effective proxy for bar strength (in an analogous manner to the Galaxy Zoo 2 vote fractions in Masters et al. 2012). While it is impossible to delineate a hard threshold for defining a weak and strong bar in terms of our CNN classification confidences, we argue that these confidences are useful in a qualitative sense to distinguish bars that are well defined and prominent against those that are harder to detect the presence of (if any at all). This can be useful for identifying follow-up candidates in surveys for further visual classification, e.g. samples with low classification probabilities. Of course, these confidences are all ultimately relative to the original training data used to train the CNN.

4.4.5 CNN Evaluation

With the continued development of large-scale cosmological simulations, in addition to future observational studies expected to probe billions of galaxies, there is a need for techniques for automated analysis in order to process such large amounts of data efficiently. The primary advantage of automated classification

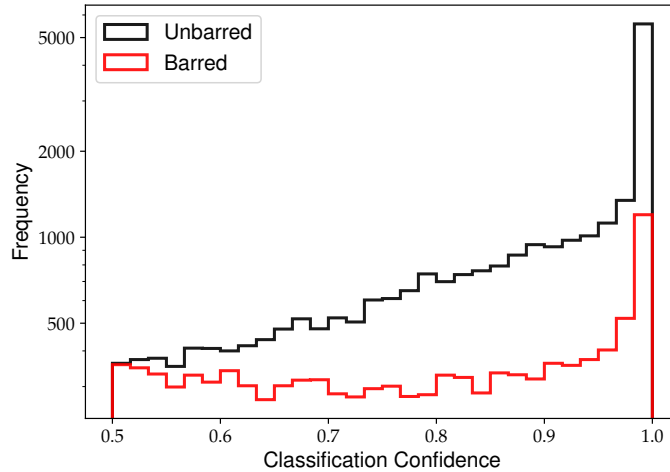


Figure 4.18: Frequency histograms of the classification confidences – as classified by the bar CNN – for barred (red) and unbarred (black) galaxies. Note the scale is logarithmic.

techniques, particularly CNNs, is their ability to generalise and scale, enabling otherwise intractable amounts of data to be processed within far more practical limits. The previous work by [Algorry et al. \(2017\)](#) studied a selection of only 269 discs in EAGLE. Our CNN approach allows us to analyse all EAGLE galaxies within a single snapshot, across multiple snapshots, limited only by our mass range criteria. This approach is uniquely well suited to analyse bar creation and destruction, changes in morphology, as well as bar lifetimes. This does not rule out the use of analytical bar detection techniques, such as ellipse fitting or Fourier analysis, to further inspect galaxies and confirm the presence of a bar. Such methods are useful for verifying low P_{bar} galaxies, as well as galaxies which undergo multiple episodes of bar formation.

CNNs are also highly versatile, and can be easily adapted for different classification tasks. While previous studies have applied CNNs trained on simulated data for use in classifying observational data ([Ghosh et al., 2020](#); [Eriksen et al., 2020](#)), this work achieves the opposite. The reason we use synthetic SDSS-like g -band images is since these are the images that our CNNs are trained on. While it is possible to adapt a CNN to classify images in other photometric bands, this requires performing a model adaptation step in the form of transfer learning. Recent examples of this in astronomy include transfer learning between different surveys ([Domínguez Sánchez et al., 2019](#)) and modifying general-purpose image classifiers ([Ackermann et al., 2018](#)). For the purpose and scope of this work, restricting classifications to the g -band avoids the need for large-scale transfer learning.

It is also important to note several limitations to the use of CNNs, especially for those reliant on labelled training data (i.e. supervised learning). The most crucial caveat is that supervised CNNs can only ever be as good as the data they are trained on. In the case of most large-scale, labelled observation datasets such as NA10 and GZ2, this upper limit is reflective on the limits of visual classification. CNNs are also prone to overfitting to the training the data, which can adversely affect performance when applied to broader data. Figure 4.18 shows the distribution of classification confidences for all barred and unbarred classifications. A long, smooth tail in both distributions is desirable compared to a narrow spike at high confidence, the latter of which is indicative of overfitting. The spread in classification confidences is also indicative of the difficulty in classifying bars.

4.5 Conclusions

In this work, we used CNNs trained to classify galaxy morphologies to study the evolution of barred galaxies in the EAGLE simulation. This was done by classifying a total of 35,082 synthetic g -band images of over 3,000 galaxy samples tracked over the redshift range $z = 0$ to $z = 1$. Our key conclusions are summarised follows:

- (i) We have found that the overall simulated bar fraction in the g -band is roughly constant at 32% to 34% from $z = 0$ to $z = 0.5$, before exhibiting a general decline to 26% at $z = 1$. The values for the bar fractions at $z = 0$ and beyond are generally consistent with previous observational studies. The bar fraction is highest in spiral galaxies (49% at $z = 0$), followed by irregular/miscellaneous samples (26%), lenticulars (22%) and ellipticals (3%).
- (ii) The bar fraction is higher in low-mass galaxies ($10^{10}M_{\odot}$ to $10^{10.5}M_{\odot}$). For S0 galaxies, bars are most common in intermediate-mass galaxies between $10^{10.5}$ and $10^{11}M_{\odot}$, while for all other morphological types, bars are most common in low-mass galaxies.
- (iii) From $z = 0.5$ to $z = 0$, the fraction of spiral galaxies decreased from 49% to under 40%, while the fraction of S0 galaxies rose from 25% to 37%. The fraction of ellipticals and irregulars did not vary significantly. Although the bar fraction in spirals and S0s both rise with decreasing redshift, the bar fraction in S0s remains lower than that for spirals. This suggests that the growth in S0s may be driven by unbarred spirals transitioning into S0s, or that the transition from spiral to S0 tends to destroy the bar.
- (iv) We have found that, in general, barred spirals are mostly similar to unbarred spirals. Barred spirals have greater values of $\kappa_{\text{co,rot}}$, slightly higher star formation rates, higher ellipticities and are less massive. Barred S0s have lower gas fractions than unbarred S0s, and also have lower star formation rates. Barred S0s also have significantly higher D/T ratios, and are thus more strongly disc dominated compared to their unbarred counterparts.
- (v) Bars have been shown to undergo episodes of creation, destruction and regeneration, with some galaxies forming multiple bars over the redshift range $z = 1$ to $z = 0$. Of the 2,146 samples present for all ten snapshots, 35.8% formed one bar, 22.1% formed two bars and 7% formed three or more bars.
- (vi) We have determined that, of the bars that persisted for at least two consecutive snapshots, the mean bar lifetime is 2.24 Gyr, with a slightly shorter mean lifetime of 2.01 Gyr for bars in low mass galaxies, and a longer mean lifetime of 2.46 Gyr for bars in high mass galaxies.
- (vii) We have shown that rate of bar creation and bar destruction is not constant, with both declining for decreasing z . Bar creation peaks at $z = 0.36$. The majority of bar creation and destruction events corresponded with no change in morphology. Of the events which did result in a change in morphology, the most common transition for bar creation was unbarred irregular/miscellaneous to barred spiral, and barred spiral to unbarred irregular/miscellaneous for bar destruction.

- (viii) We have found that major merging is more strongly linked to bar destruction as opposed to bar creation. In addition, accretion is more strongly linked to both bar creation and destruction. This is since there are more instances of accretion linked to both a bar creation or destruction event compared to a random sample.

The redshift evolution of the S0 fraction in COSMOS

Lenticular galaxies, also known as S0 galaxies, are a type of galaxy morphology that is considered to be an intermediary between ellipticals and spirals. The study of their formation and evolution, and how their prevalence varies over time, is of key interest to astronomers. Recent observational studies have reached conflicting results regarding whether the fraction of S0s has changed with redshift. Understanding how this fraction changes with redshift can reveal important insights into the long-term evolution of galaxies and the physical processes that drive morphological transitions. While many studies have examined S0s in dense environments such as galaxy clusters, comparatively fewer studies have examined S0s in the field environment. Thus far, our CNNs have successfully classified images of both real and simulated galaxies. Yet in order to study the redshift evolution of lenticulars in observations, our models must be capable of classifying images of more distant, high-redshift galaxies. Given that our models are trained on low-redshift images, this represents a significant and challenging domain shift due to poorer overall image qualities at higher redshifts. In this Chapter, we apply techniques of transfer learning to adapt and enhance our existing models, enabling them to classify high-redshift galaxies in COSMOS. With these new, augmented models, we study the redshift evolution of the S0 fraction, including its dependence on stellar mass. We find that the increase in the fraction of high-mass S0s occurs at higher redshifts compared to that of low-mass S0s. Furthermore, we find evidence for two distinct populations of S0s, including a significant population of star-forming S0s that cannot entirely be accounted for by the predominant “faded spiral” S0 formation pathway alone.

*This chapter is based on the manuscript “The redshift evolution of the S0 fraction for $z < 1$ in COSMOS”, which was published on February 14, 2023 in the journal *Monthly Notices of the Royal Astronomical Society*, Volume 520, Issue 4, pp. 5885–5902. There are some minor textual and formatting changes compared to the published version.*

Abstract

Lenticular (S0) galaxies are galaxies that exhibit a bulge and disk component, yet lack any clear spiral features. With features considered intermediary between spirals and ellipticals, S0s have been proposed to be a transitional morphology, however their exact origin and nature is still debated. In this work, we study the redshift evolution of the S0 fraction out to $z \sim 1$ using deep learning to classify F814W (*i*-band) HST-ACS images of 85,378 galaxies in the Cosmological Evolution Survey (COSMOS). We classify galaxies into four morphological categories: elliptical (E), S0, spiral (Sp), and irregular/miscellaneous (IrrM). Our deep learning models, initially trained to classify SDSS images with known morphologies, have been successfully adapted to classify high-redshift COSMOS images via transfer learning and data augmentation, enabling us to classify S0s with superior accuracy. We find that there is an increase in the fraction of S0 galaxies with decreasing redshift, along with a corresponding reduction in the fraction of spirals. We find a bimodality in the mass distribution of our classified S0s, from which we find two separate S0s populations: high-mass S0s, which are mostly red and quiescent; and low-mass S0s, which are generally bluer and include both passive and star-forming S0s, the latter of which cannot solely be explained via the faded spiral formation pathway. We also find that the S0 fraction in high-mass galaxies begins rising at higher z than in low-mass galaxies, implying that high-mass S0s evolved earlier.

5.1 Introduction

Understanding how galaxy morphologies vary over cosmic timescales is a crucial first step towards understanding the physical drivers of galaxy formation and evolution (Dressler, 1980; Larson et al., 1980; Bundy et al., 2005; Papovich et al., 2005; Driver et al., 2009; Ilbert et al., 2010; Kovač et al., 2010; Oesch et al., 2010; Conselice, 2014; Ferreira et al., 2022a). Observed galaxy populations, past and present, are also important constraints for cosmological simulations (Guo et al., 2011; Somerville & Davé, 2015; Vogelsberger et al., 2020). Of key interest is the nature and evolution of lenticular (S0) galaxies, a galaxy morphology that has long been established as an intermediate type between ellipticals and spirals (Hubble, 1936). The prevalence of lenticular galaxies, and the manner in which this fraction varies across both cosmic time and in different environments, is important to investigating their main formation pathways, and the physical processes that govern their subsequent evolution.

Multiple mechanisms and physical processes have been proposed to explain the formation of S0s (Deeley et al., 2020). These can be grouped into two main pathways. The first of these is the "faded spiral" pathway, which involves the transition from a spiral to a S0 galaxy (Dressler, 1980; Fasano et al., 2000; Poggianti et al., 2001; Moran et al., 2007b; Laurikainen et al., 2010; Johnston et al., 2014; D'Onofrio et al., 2015; Bait et al., 2017; Rizzo et al., 2018). This is believed to be dominant formation pathway in dense environments (Coccato et al., 2022), where there are many supporting factors such as the influence of tidal interactions and mergers (Bekki & Couch, 2011; Deger et al., 2018), ram pressure stripping (Gunn & Gott, 1972) and gas starvation (Larson et al., 1980; Bekki et al., 2002). All of these are capable of stripping gas from spirals and quenching further star formation activity (Barr et al., 2007). Similarly, in isolated, low density environments, S0s can naturally emerge via the faded spiral pathway as a result of the passive consumption, and eventual depletion, of in-situ gas needed to sustain star formation (Rizzo et al., 2018).

The second major pathway involves the formation of S0s as a result of mergers. Simulations have demonstrated that S0s can emerge as the remnants of major merging (Bekki, 1998; Borlaff et al., 2014; Querejeta et al., 2015; Tapia et al., 2017; Eliche-Moral et al., 2018), and there is observational evidence in support of major merging as a significant factor in the evolution of massive, early type galaxies (Prieto et al., 2013; Robotham et al., 2014), which agrees well with hierarchical models of galaxy formation and evolution (Somerville & Davé, 2015). In addition, several other mechanisms have been proposed to account for S0 formation, including accretion through successive minor merging (Diaz et al., 2018), AGN activity (van den Bergh, 2009), disc instabilities (Saha & Elmegreen, 2018), as well as through internal secular evolution and bulge growth (Kormendy & Kennicutt, 2004; Laurikainen et al., 2006, 2010). Stellar mass is also believed to play a key role in S0 formation, with studies suggesting that there is a difference in formation pathways between low-mass S0s, which are mostly faded spirals, and high-mass S0s, which instead likely formed via a different pathway, such as merging (Kannappan et al., 2009; Bellstedt et al., 2017; Fraser-McKelvie et al., 2018; Domínguez Sánchez et al., 2020).

It is well established that S0s are more prevalent in dense environments, whereby the fraction of S0s typically increases at the expense of a decreasing fraction of spirals (Dressler, 1980; Fasano et al., 2000; Cappellari et al., 2011; Mishra et al., 2019). However, the extent to which the S0 fraction varies over redshift is less certain. Some studies have posited that there has been little to no evolution of the relative fraction of S0s with redshift (van der Wel et al., 2007; Holden et al., 2009; Vulcani et al., 2011), while others have found evidence for an increasing S0 fraction with decreasing z in both cluster environments (Dressler et al., 1997; Desai et al., 2007; Poggianti et al., 2009; Just et al., 2010), as well as in the field (Oesch et al., 2010; Huertas-Company et al., 2015).

Studies into the evolution of galaxy morphology are contingent on the availability of classifications. However, the classification of galaxies by visual inspection is a slow, time-consuming task. Recent citizen science efforts have been vital in enabling visual classification to effectively scale to larger datasets (Willett et al., 2013; Masters et al., 2021; Walmsley et al., 2021), however the speed of the classification remains unchanged. As such, automated classification of galaxy morphologies is paramount in enabling vast volumes of data to be analysed quickly and efficiently, especially data at scales that render visual classification infeasible (Beck et al., 2018). Deep learning models, such as convolutional neural networks (CNNs) (LeCun et al., 2015), have been utilised to rapidly and efficiently analyse images of nearby, low-redshift galaxies with great success. Recent applications not only include morphological classification (Dieleman et al., 2015; Domínguez Sánchez et al., 2018; Cavanagh et al., 2021; Vega-Ferrero et al., 2021), but also include similarity-based clustering (Martin et al., 2020; Cheng et al., 2021a; Walmsley et al., 2022), as well as morphological segmentation (Hausen & Robertson, 2020). The advantage of a deep learning approach lies in its efficiency and versatility, the latter of which is especially pertinent for CNNs, which are especially adept at classifying images via feature extraction (Simonyan & Zisserman, 2015). While the majority of studies employing CNNs for classification have focused on readily available datasets of nearby, low-redshift galaxies, such as Galaxy Zoo (Willett et al., 2013), fewer studies have used a deep learning approach to classify high-redshift galaxies. The morphological classification of high-redshift galaxies is far more challenging due to the inherent limitations in image quality, and the lack of any sufficient large-scale dataset of known morphologies with which to train a CNN.

The Cosmic Evolution Survey (COSMOS) (Scoville et al., 2007) is a deep, multi-wavelength survey

aimed at investigating the evolution of galaxies. In this work, we develop and adapt several deep learning classifiers, initially trained on *g*-band SDSS images of galaxies from the [Nair & Abraham \(2010a\)](#) morphological catalogue (hereafter NA10), to classify COSMOS images via transfer learning. Transfer learning refers to a family of machine learning techniques designed to leverage the pre-existing capabilities of an initial, pretrained model in order to perform some task in a related domain (see [Weiss et al. 2016](#) for a review). Recent studies in astronomy have used this technique to adapt models trained on simulated galaxies to instead classify real galaxies ([Ghosh et al., 2020](#)) (and vice-versa; [Cavanagh et al. 2022](#)), in addition to adapting models to classify images from different surveys ([Domínguez Sánchez et al., 2018](#)).

The images we classify are *i*-band images from the Hubble Space Telescope Advanced Camera for Surveys (HST/ACS) ([Koekemoer et al., 2007](#); [Massey et al., 2010](#)). Although COSMOS extends to redshifts beyond $z > 3$, we limit our analysis to samples with $z < 1$. This is primarily to ensure that the images are at a reasonable resolution suitable for classification with our CNNs, and also to limit the adverse effects of *k*-correction and PSF-smoothing. Despite the HST imaging being deeper in surface brightness, the high-redshift COSMOS galaxies appear significantly noisier than the low-redshift SDSS galaxies, which is a barrier to any direct application of a deep learning model. We overcome this barrier through the use of data augmentation to augment our existing SDSS images with artificial noise such that they mirror the characteristic noise and quality of the COSMOS images. We then use these augmented images to fine-tune our initial models. The key advantage of this approach is that the classifications utilise the existing known morphologies, which is significant since these include a specific category for S0s. We find that this transfer learning approach allows us to classify S0s with a significantly higher accuracy compared to an otherwise direct application.

The structure of our paper is as follows. In Section 5.2, we briefly summarise our sample selection process, and how these images were then preprocessed and classified by our models. We further describe our CNN model, including its architecture, and also describe the data augmentation and transfer learning procedures. Lastly, we describe how we obtain final classifications for each of the COSMOS images. In Section 5.3, we present our results on morphological classifications and the redshift evolution of each morphology, focusing on the growth in the S0 fraction. We discuss the physical properties of our classified S0s, and show that they comprise two distinct populations. We also provide some example classifications, including those on which our 3-class and 4-class models disagree. In Section 5.4, we discuss our results and implications for the formation and evolution of S0s. We further evaluate our deep learning approach and use of data augmentation. We also examine the model’s performance on our NA10 test set not just for comparison, but also to judge the impact of the noise augmentation. Finally, we conclude this paper with a summary of our key results in Section 5.5.

5.2 Methods

5.2.1 Datasets

5.2.1.1 COSMOS Sample Selection

The sample of galaxies used for this work consists of a subsample of galaxies from the COSMOS2020 catalogue ([Weaver et al., 2022](#)). We select all galaxies with $z < 1$ using the LePHARE best redshift ([Arnouts](#)

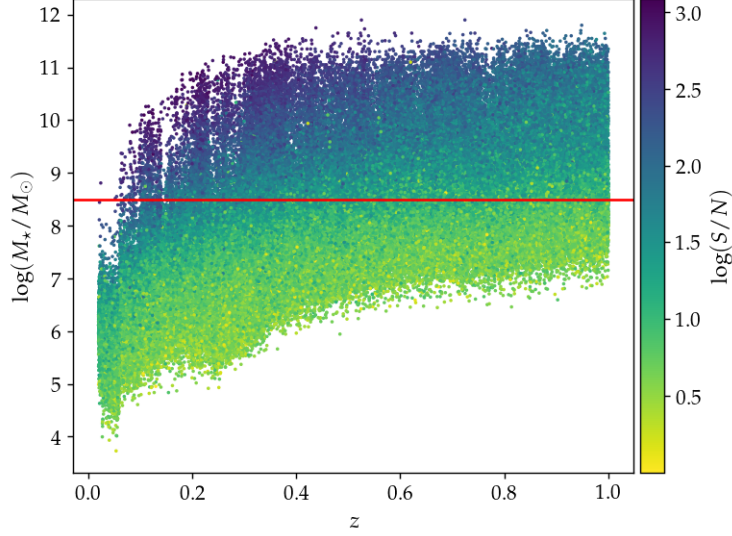


Figure 5.1: Full COSMOS sample for redshifts $z < 1$ with redshift on the x -axis and stellar mass on the y -axis. Samples are shaded according to their signal-to-noise ratio $\log(S/N)$. The solid, red line denotes the minimum mass considered for this study.

& Ilbert, 2011), and apply a lower mass cut of $\log(M_*/M_\odot) = 8.5$ with no upper limit. Figure 5.1 displays the mass distribution (LEPHARE best mass) of the $z < 1$ COSMOS2020 sample, illustrating our lower mass limit. Samples are coloured according to signal-to-noise, which is defined as the ratio of F814W flux to flux error (Leauthaud et al., 2007). Figure 5.1 illustrates the importance of a lower mass limit, which is necessary to discard samples with poor signal-to-noise. As a result, our sample subsequently consists of galaxies with $\log(S/N) > 1$. Importantly, we note that there are relatively few high-mass galaxies at redshifts below $z \approx 0.2$. Furthermore, signal-to-noise degrades with redshift, which has the greatest impact on low-mass galaxies. Values for galaxy properties are sourced from COSMOS2020.

Having established our dataset, we then obtained HST-ACS (Koekemoer et al., 2007) image cutouts for each sample from the NASA/IPAC Infrared Science Archive¹. These are monochrome i -band (F814W filter) images with a pixel scale of $0.03''$. All cutouts were obtained with an angular size corresponding to the same physical scale of 50 kpc. This is to match the scale of the SDSS images used to train the original model (see next subsection for details). Consequently, the angular sizes of the cutouts vary with redshift, resulting in images with a range of pixel sizes. We calculated the desired angular diameter in pixels for each sample with redshift z using ASTROPY (The Astropy Collaboration et al., 2013), in particular the cosmology package using the cosmological parameters from WMAP9 (Hinshaw et al., 2013). Not all samples in our subsample returned a cutout from the image server, however we were able to obtain a total of 85,378 images; corresponding to about 94% of the original subsample. All the images were then resized to a size of 100x100 pixels using the PILLOW package (Van Kemenade et al., 2022) using the default bi-cubic interpolation setting, and then linearly normalised so that each pixel in each cutout has a value strictly between 0 and 1 inclusive, where 1 is the brightest pixel. This is the final size and format of the images prior to being inputted to the convolutional neural network.

¹<https://irsa.ipac.caltech.edu/data/COSMOS/overview.html>

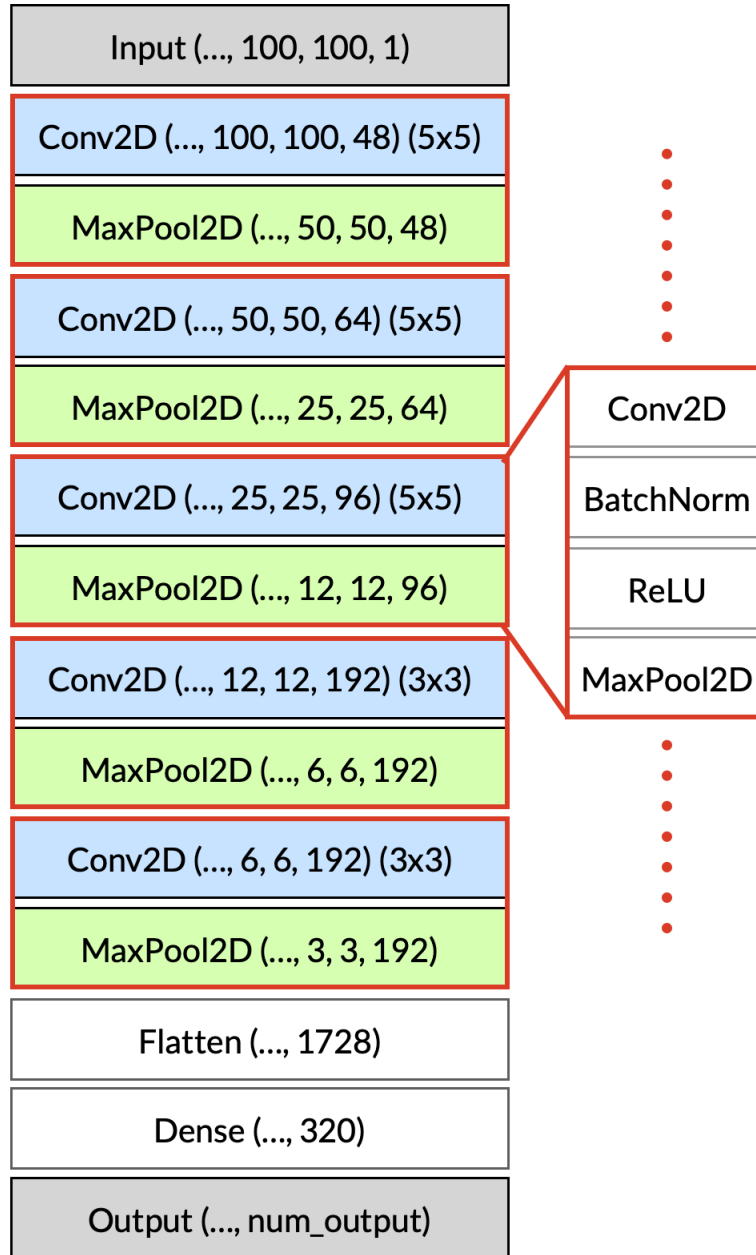


Figure 5.2: Outline of our convolutional neural network model. The tuples in parentheses indicate the output shape of each layer. Ellipses (...) denote the batch size. There are five blocks (shown with red borders) of alternating 2D convolution (Conv2D) and 2D max pooling (MaxPool2D) layers. These have output shapes of the form (batch_size, width, height, channels), where channels denotes the number of feature maps (and hence the number of convolutional filters). Furthermore, the kernel size for each Conv2D layer is shown in parentheses to the right of the output shape. Each convolution is immediately followed by both a batch normalisation layer, and then a ReLU activation layer (see the expanded block). In total, the model contains 1.28 million trainable parameters.

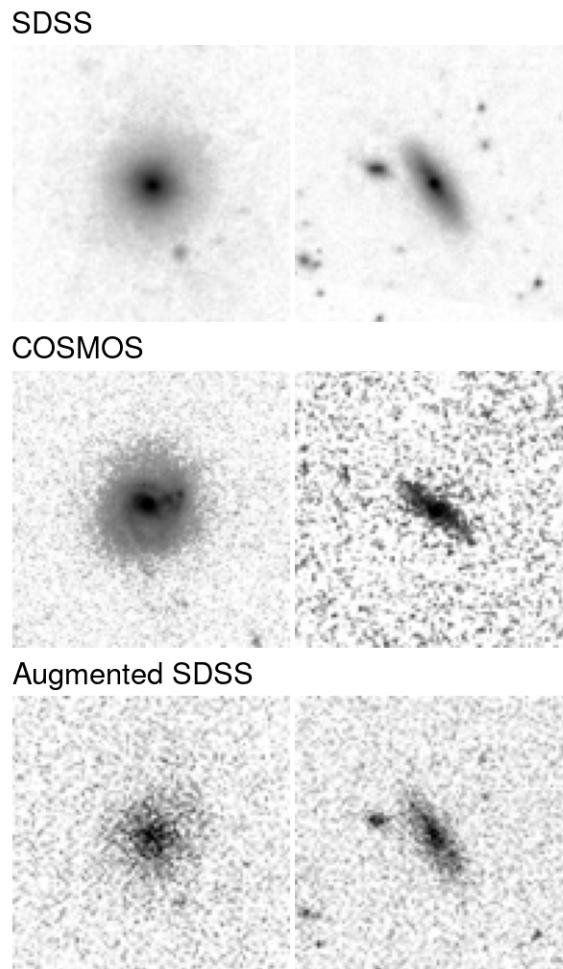


Figure 5.3: Example of our noise augmentation procedure. The top row shows some example SDSS images. The middle row shows some visually similar COSMOS images. The bottom row shows the same SDSS images as in the top row, but this time after having been modified with additional, artificial noise.

5.2.1.2 The Training Data

The training data used to train the initial models consists of 14,034 g -band SDSS DR7 images (Abazajian et al., 2009) of galaxies from the Nair & Abraham (2010a) (NA10) morphological catalogue. Each image shares the same physical scale of $50\text{kpc} \times 50\text{kpc}$. In particular, the catalogue consists of galaxies brighter than 16 mag (extinction-corrected g -band) between $z \approx 0.01$ and $z = 1$. Rather than sampling a complete volume, the NA10 sampling is instead statistically more representative for detailed morphological studies and, as such, over-emphasises massive galaxies. Each galaxy is visually classified according to numerical, Hubble T-Types. We group these T-Types together into broader elliptical, lenticular, spiral and irregular/miscellaneous morphological categories (see Cavanagh et al. 2021 for exact details). We do not place any restrictions on the physical sizes of the galaxies themselves. The training data underwent a similar data processing regime as described for the COSMOS sample. We further partitioned the full training data into separate training and test sets according to an 80:20 split. The training set is used to train the model, while the test set is reserved for final evaluation.

5.2.2 The CNN Models

Convolutional neural networks are specialised neural networks that are especially well-suited to analysing image data due to their capacity for abstract feature extraction. To process the input data and extract relevant features, CNNs use successive convolutional layers. These layers linearly convolve their given inputs with several filters, or kernels. The weights for these filters are initially random, but are progressively tweaked and optimised as the network is trained. It is in this manner that the CNN learns to extract features automatically. The use of learnable filters as opposed to pre-determined, hard-coded filters is one of the key advantages of CNNs, making them ideal models for general-purpose image classification across many disciplines. For a full treatment of CNNs, their operation and their training, see [Haykin \(2009\)](#); [LeCun et al. \(2015\)](#); [Goodfellow et al. \(2016\)](#).

The models developed for this study are adapted from our previous work ([Cavanagh et al., 2021](#)) (hereafter C21). In particular, we develop two models: one designed to classify galaxies as either elliptical, lenticular or spiral (3-class model); and another model which adds a fourth category for irregular/miscellaneous galaxies (4-class model). We train these models on 100x100 pixel sized g-band SDSS images that are each labelled with their known morphologies from the NA10 morphological catalogue. Our models are developed and trained in PYTHON using the TENSORFLOW ([Abadi et al., 2016](#)) and KERAS ([Chollet et al., 2015](#)) libraries. We utilise a similar data processing and training procedure as in C21, however the major difference is that this work utilises a new model architecture, as outlined in Figure 5.2. This new architecture underwent extensive hyperparameter tuning using OPTUNA ([Akiba et al., 2019](#)), a general-purpose optimisation framework. We find that this new architecture enables higher classification accuracies in both the 3-class and 4-class cases compared to that originally used in C21. Figure 5.2 shows some of these optimised hyperparameters including the number of convolutional filters, convolutional kernel sizes and the number of nodes in the dense layer. Our model uses batch normalisation layers ([Ioffe & Szegedy, 2015](#)) after each convolutional layer; these layers serve to better regularise the model and improve model robustness. We use ReLU activation functions throughout the model, with softmax activation in the output layer in order to output probabilities for each class. These probabilities are also referred to as classification confidences (or simply confidence) and we will use these terms interchangeably. We use the Adam optimiser ([Kingma & Ba, 2014](#)) with an initial learning rate of 8×10^{-4} , and we use categorical cross-entropy as our loss function. The model contains a total of 1.28 million trainable parameters.

5.2.3 Model Development and Evaluation

The specific models that we use to classify our sample of COSMOS images are adapted from models that have been pretrained on SDSS images from the NA10 catalogue (these are referred to hereafter as the base models). To perform this model adaptation, we use a combination of transfer learning and extensive data augmentation. In particular, we train new ensembles of 3-class and 4-class models. In each case, the models are initialised with the weights of the pretrained base model. These models are then retrained on SDSS images that have been injected with artificial Gaussian noise. This artificial noise is designed to resemble the characteristic range of noise levels in the images from our COSMOS subsample. The purpose of this data augmentation is to train the models to better classify noisy images. Full technical details regarding the methodology and results of this technique are given in Section 5.6, where testing

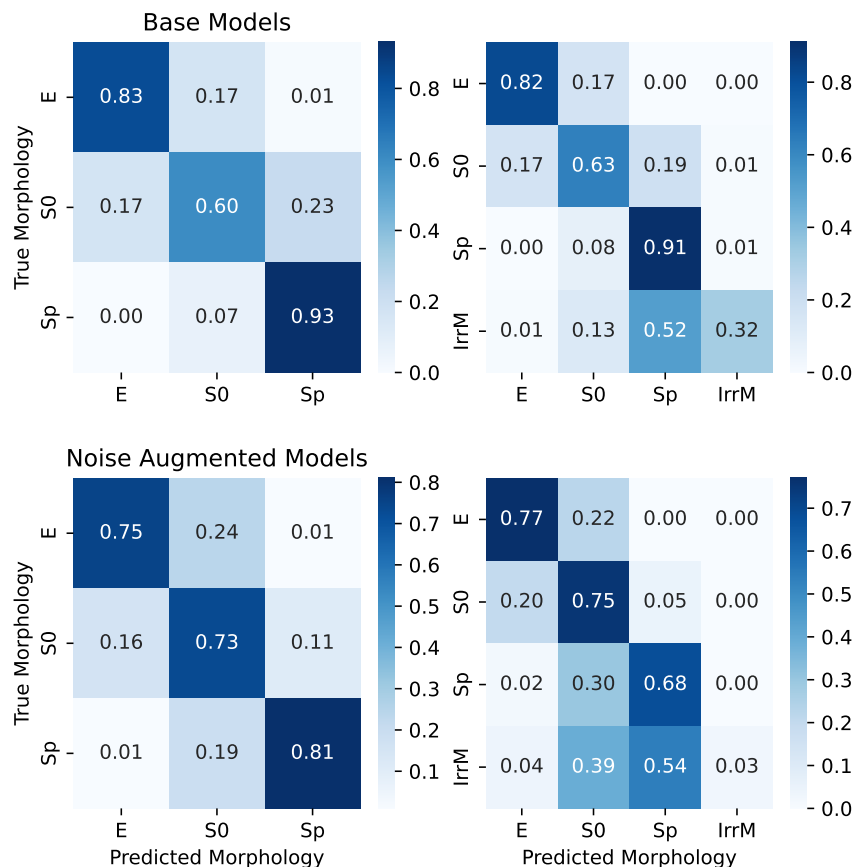


Figure 5.4: Confusion matrices for the base 3-class and 4-class models (top row), as well as the noise-augmented 3-class and 4-class models (bottom row), based on final evaluation with the holdout NA10 test set. Rows denote the true morphologies, while columns denote the predicted morphologies. The entries i, j denote the fractions of samples with true morphology i that are classified as the predicted morphology j . The diagonal entries correspond to recall (a.k.a. per-class accuracy). The same test set is used for both models, with the augmented models tested on augmented images. For the noise augmented models, the predictions (and hence accuracies) are based on the mean confidences of each of the five constituent ensemble models.

demonstrates that models trained with this technique perform considerably better at classifying noisy images. Due to the high variance associated with the noise augmentation and transfer learning, our final 3-class and 4-class models are model ensembles, each consisting of five independent models. The final classification is thus based on the output classification probabilities averaged across each individual model in the ensemble.

Figure 5.4 shows the confusion matrices for our original base 3-class and 4-class models, as well as for our final, augmented model ensembles. The most immediate impact of the noise augmentation is that the per-class accuracies for the first three categories (E, S0 and Sp) are smoothed. The S0 classification accuracy increases from 60% to 73% in the 3-class model, and from 63% to 75% in the 4-class model. The noise augmentation also results in a notable fall in spiral accuracy, particular in the 4-class model, with up to 30% of true spirals instead classified as S0s. Curiously, in the converse, only 5% of true S0s were misclassified as spirals. It is important to keep in mind that the base models had a broad spread in

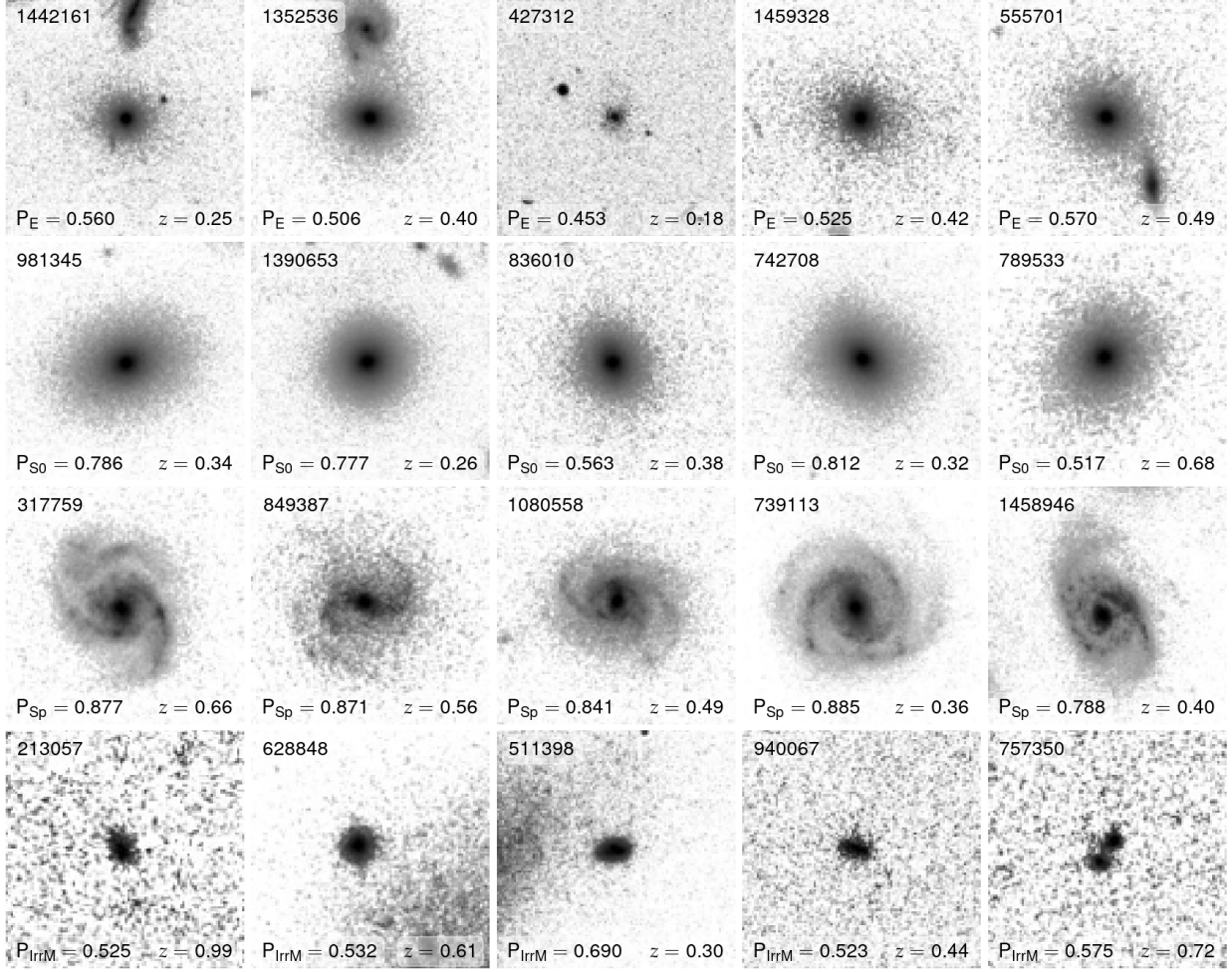


Figure 5.5: Random selection of COSMOS galaxies classified into each of the four morphological categories by our 4-class model. Each image is annotated with its COSMOS2020 ID (top-left), its classification confidence (bottom-left), and its redshift z (bottom-right).

classifications; 17% of true S0s were misclassified as elliptical, with up to 23% misclassified as spirals. This spread is reduced for the noise augmented models. Across the board, however, more samples are predicted to be S0s. We discuss these classification accuracies, their implications, and possible reasons for the shift in overall accuracies in further detail in Section 5.4.

5.3 Results

5.3.1 Morphological Classification

We classified all 85,378 images with our noise augmented 3-class and 4-class model ensembles. Since the 4-class model performed the best at classifying S0 galaxies (higher recall), it will be the primary focus of analysis for the remainder of the paper. Figure 5.5 shows a random selection of COSMOS galaxies classified into each of the four morphological classes by our 4-class model. Ellipticals are mostly fuzzy, featureless spheroids, while S0s have more pronounced discs and, in general, better defined edges. Spirals

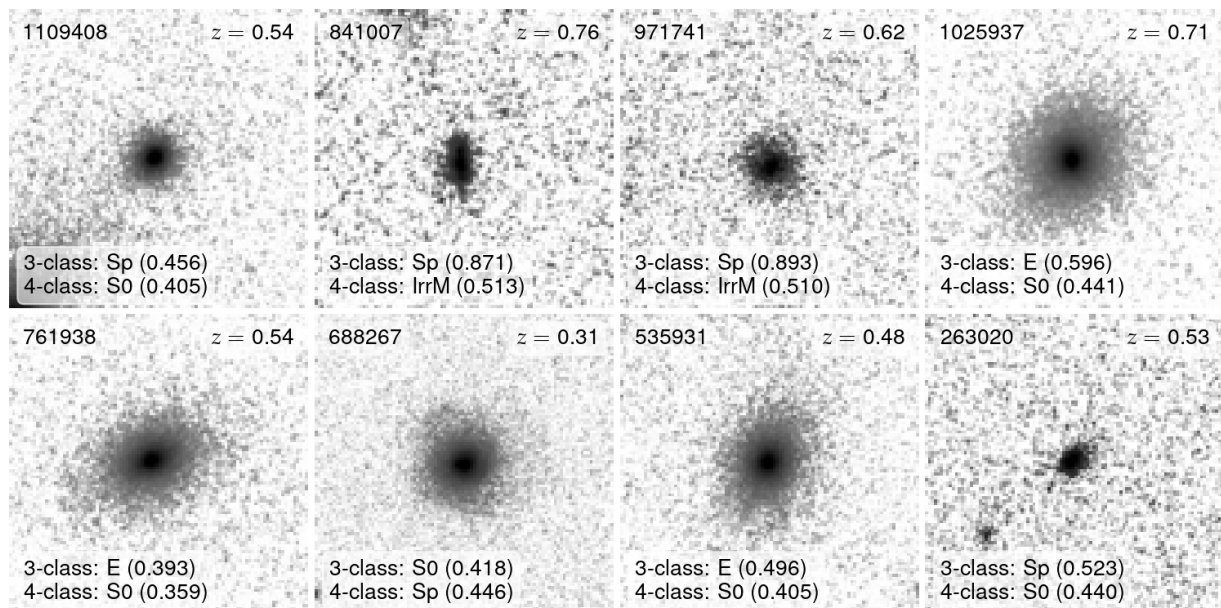


Figure 5.6: Random selection of COSMOS galaxies that have been classified differently by the 3-class and 4-class models.

all exhibit spiral arms, with ID 849387 in particular illustrating how noise corrupts the appearance of spiral arms. The irregulars/miscellaneous is a broad category that, in the case of the NA10 training data, includes proper irregulars, along with disrupted/interacting galaxies and all other galaxies with unclear morphology. The classified irregulars in COSMOS mainly manifest as small blobs from which it is hard to resolve any clear internal structure. It is likely that some of these, such as ID 628848, may in fact be low mass early type galaxies.

Except for irregulars, the vast majority of galaxies (over 96%) were assigned the same classification by our 3-class and 4-class model. However, some galaxies were classified differently, with 3-class spiral to 4-class lenticular being the most common classification change. Figure 5.6 shows a random selection of galaxies that were given different classifications. Importantly, of those that changed from spiral in the 3-class model to lenticular in the 4-class model (or vice versa), or from elliptical to lenticular, the classification confidences in both models remain low. These are examples of galaxies which the models have difficulty classifying. We will discuss the limitations of our models in more detail in Section 5.4.3.

5.3.2 Morphological Fractions

Based on our classifications, we are able to determine the fraction of each morphology as a function of redshift for $z < 1$. As described in Section 5.2.2, these classifications are based on the class corresponding to the highest mean probability across all individual models in the ensemble. Figure 5.7 shows the evolution of the morphological fractions with redshift for each morphology, with the shaded regions denoting 1σ standard error based on the classification accuracies of Figure 5.4. For reference at $z \approx 0$, we include the corresponding morphological fractions from the NA10 test set, as classified by our initial CNNs. In both the 3-class and 4-class model, we see that there is a gradual albeit sustained rise in the S0 fraction as redshift

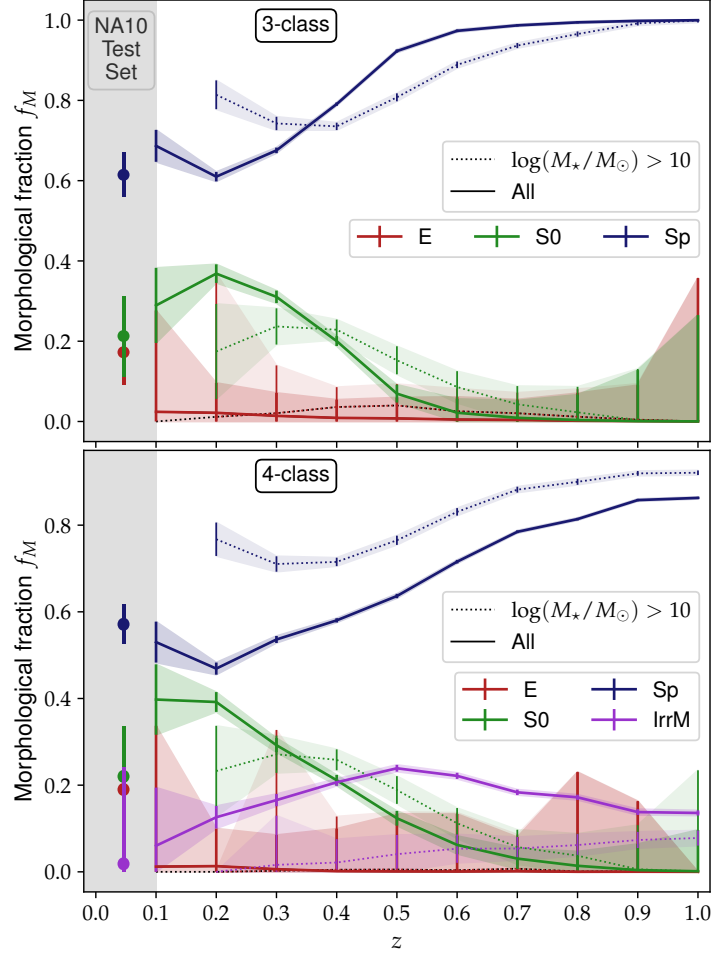


Figure 5.7: The redshift evolution of the morphological fractions f_M for each morphological class: ellipticals (red), lenticular (green), spiral (blue) and irregular/miscellaneous (mauve). Solid lines indicate the fractions for all galaxies, while the dotted lines denote only the galaxies with stellar masses $\log(M_*/M_\odot) > 10$. The top panel shows the results from the 3-class ensemble, while the 4-class ensemble results are shown in the bottom panel. Shaded regions denote 1σ standard error based on the classification accuracies for each class.

decreases, from less than 1% at $z \approx 1$ to around 40% at $z \approx 0.1$. The fraction of spirals correspondingly decreases with decreasing redshift. Both models detect an extremely low number of ellipticals, while the 4-class model predicts a modest fraction of irregular/miscellaneous samples that peak at around 23% at $z \approx 0.5$. Compared to the NA10 test set, the COSMOS dataset shows relatively more S0s and much fewer ellipticals at low redshifts. The fraction of spirals at $z \approx 0.1$ also agree well with the overall fraction in NA10; so too does the low fraction of irregulars.

Of the samples with masses $\log(M_*/M_\odot) > 10$, both models predict the S0 fraction to be slightly lower, and the 4-class model implies a higher fraction of spirals. Approximately 12% of samples with $\log(M_*/M_\odot) > 10$ at $z \approx 1$ are classified as irregular/miscellaneous; this fraction all but vanishes as redshift decreases. Consequently, irregulars are mostly below $10^{10} M_\odot$ at low redshifts. Importantly, as shown in Figure 5.1, the total number of samples in COSMOS decreases significantly at low redshifts. This is especially apparent for galaxies above $10^{10} M_\odot$, resulting in much greater uncertainties. There is also a

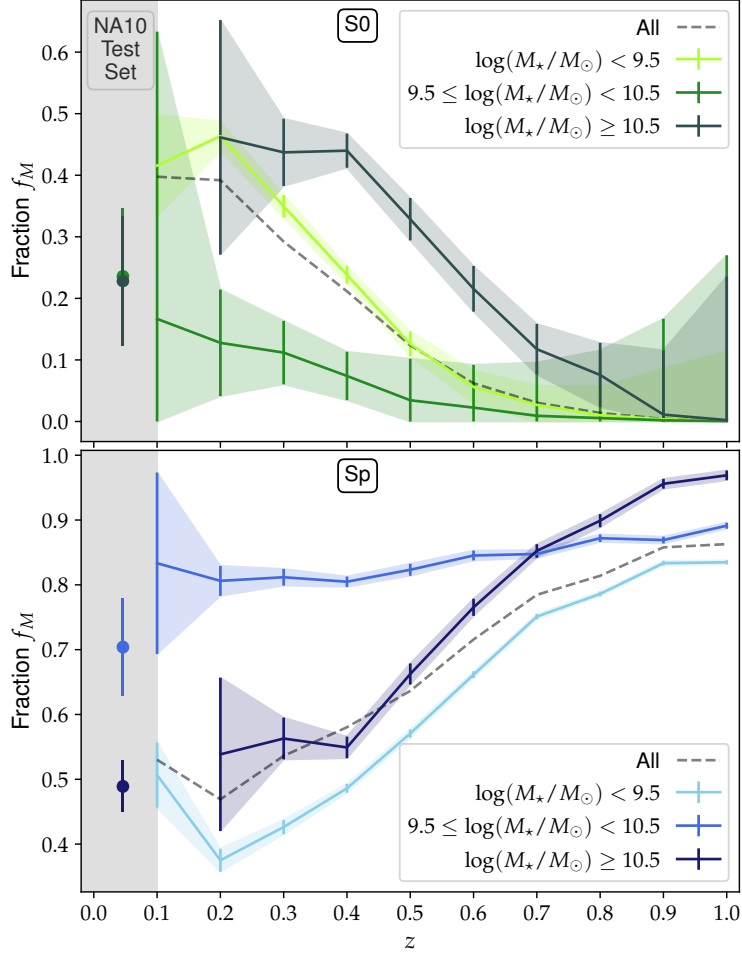


Figure 5.8: The redshift evolution of the morphological fractions f_M of S0 (top) and Sp (bottom) galaxies in the low mass, intermediate mass and high mass ranges. Fractions are based on the classifications of the 4-class ensemble models.

slight dip in the fraction of S0s in the 3-class model that appears to agree well with NA10, however this reduction could also be due to decrease in the number of COSMOS galaxies.

In Figure 5.7, we also saw that the S0 fraction for $\log(M_*/M_\odot) > 10$ samples began its rise at higher redshifts. We investigate this in more detail in Figure 5.8, which shows the S0 and spiral fractions for samples in three different mass ranges, as classified by the 4-class ensemble. Figure 5.8 shows that the redshift evolution of the S0 fraction varies significantly for different mass ranges. For high mass samples with $\log(M_*/M_\odot) \geq 10.5$, the rise in the S0 fraction occurs at higher redshifts, at around $z = 0.9$. For low-mass samples ($\log(M_*/M_\odot) < 9.5$), the rise is initially more gradual but begins to steadily increase at a similar rate past $z \approx 0.6$. For intermediate mass samples, the rise is much less pronounced, with a final fraction of less than 20%; less than half of the fraction for the low-mass and high-mass galaxies. These results imply that S0 prevalence depends strongly on stellar mass, favouring both low and high mass galaxies, with a dearth in intermediate galaxies.

Figure 5.8 also shows the redshift evolution of the fraction of spirals. At high redshifts, spirals are dominant across all mass ranges, but as redshift decreases both the low-mass and high-mass spiral fraction

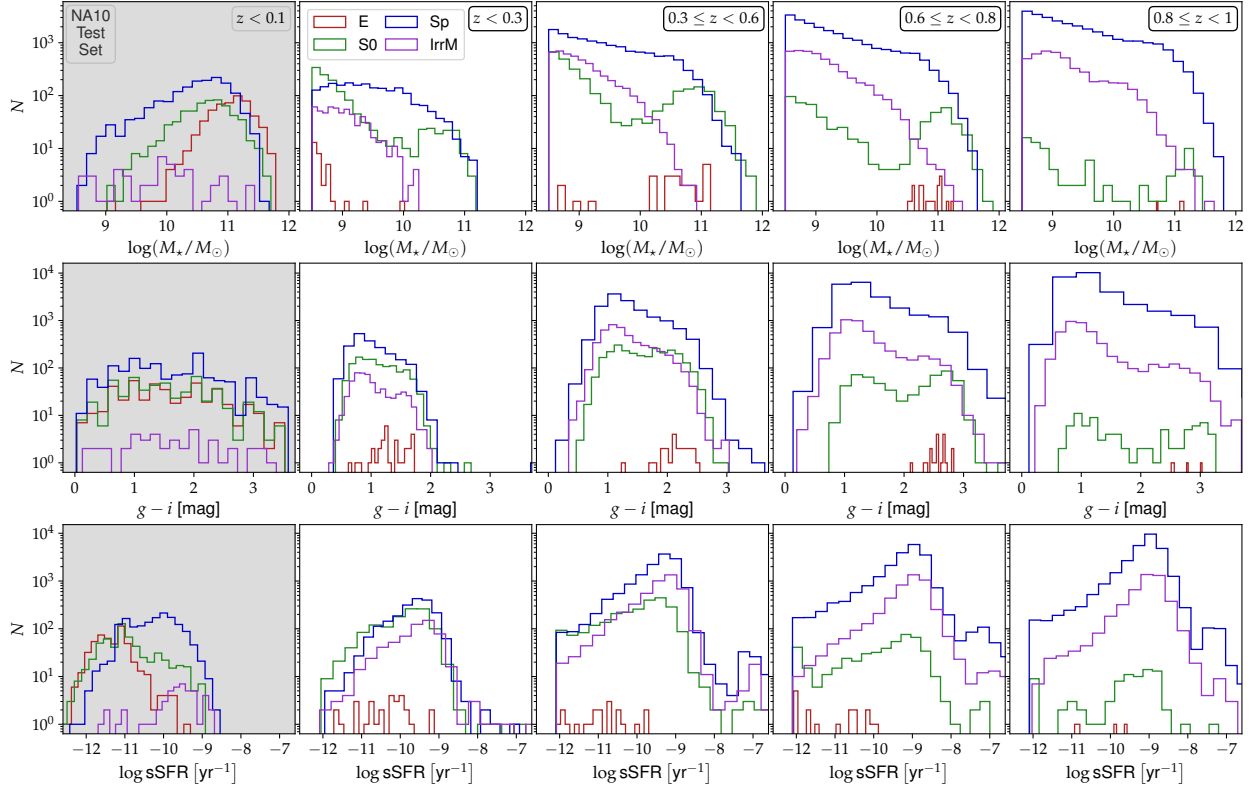


Figure 5.9: Histograms showing the distributions of stellar mass $\log(M_*/M_\odot)$ (top row), colour $g-i$ (middle row) and specific star formation rate (sSFR) (bottom row), for samples classified as elliptical, S0, spiral and irregular over different redshift ranges. The leftmost, shaded column shows the corresponding distributions for samples in the NA10 test set.

fall sharply, while the intermediate-mass spiral fraction remains relatively flat, remaining above 80% at all redshift ranges. The reduction in the spiral fraction mirrors the increase in the S0 fraction. As with Figure 5.7, Figure 5.8 also gives a $z \approx 0$ reference in terms of the corresponding fractions from the NA10 test set, albeit only for intermediate and high mass ranges due to low numbers of low mass samples. For S0s, the NA10 test set fractions for the intermediate and high-mass ranges are similar, however both have significantly higher uncertainties. Given that there is a dearth of high-mass ($> 10^{10.5} M_\odot$) samples for $z < 0.2$ in COSMOS, it's unclear whether the high-mass fraction continues to remain above 40%, or if it instead decreases. For spirals, the fractions agree well, with a slightly higher fraction of intermediate-mass spirals in COSMOS, although this is only tentative given the greater range of uncertainty at low redshift.

5.3.3 Physical Properties

The results in Figures 5.7 and 5.8 indicate that the S0 fraction and its evolution varies with mass. More crucially, these results hint at a potential double-peak mass distribution for S0s. In particular, Figure 5.8 shows that there are more S0s at the low and high mass range, but comparatively fewer S0s at intermediate masses. To investigate this – and see how S0s compare with the other morphologies – Figure 5.9 explores the distribution of masses for ellipticals, lenticulars, spirals and irregulars, along with two other markers of morphology: rest frame $g-i$ colour, and specific star formation rates (sSFR), as sourced from the

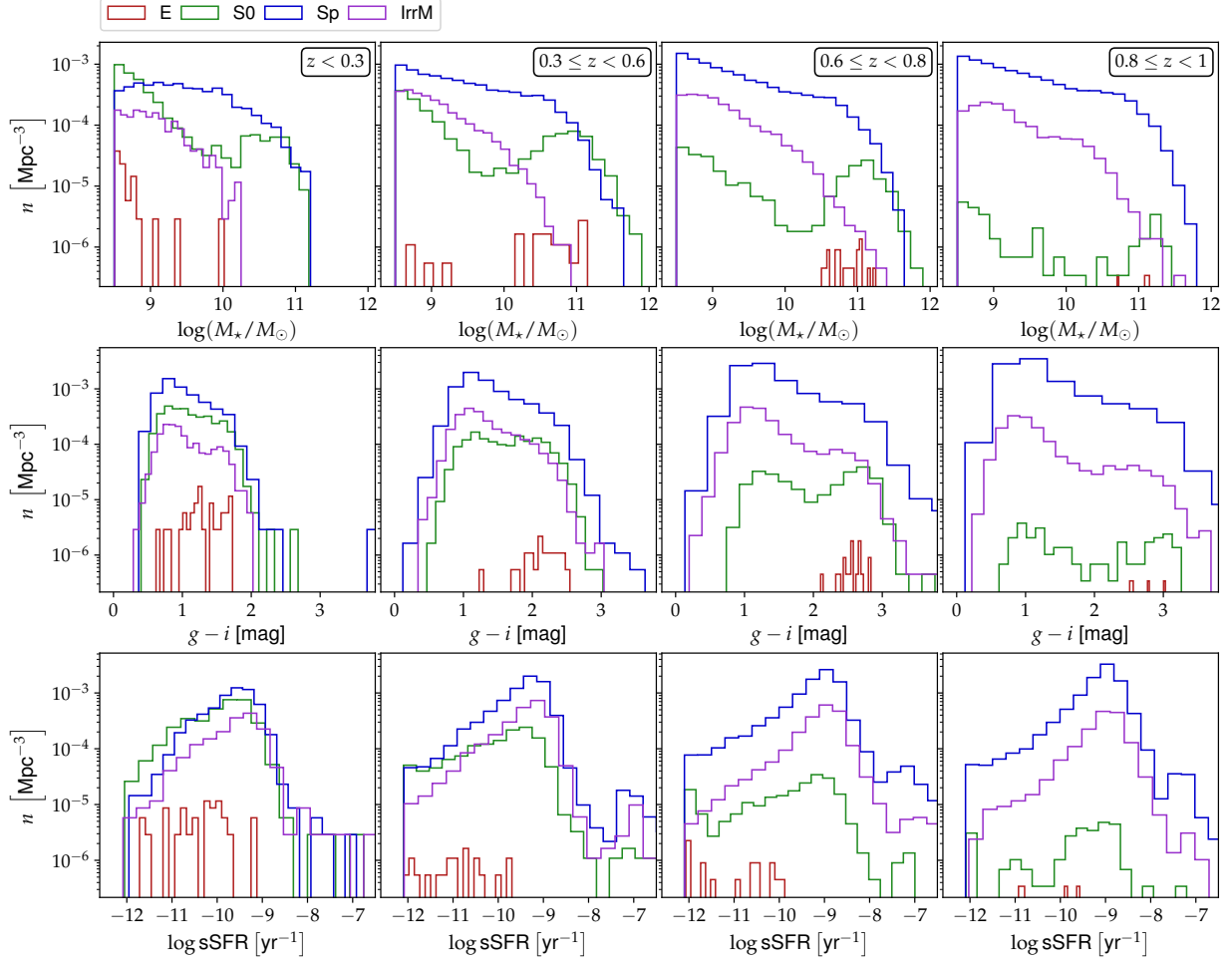


Figure 5.10: Number density histograms showing the distributions of stellar mass $\log(M_*/M_\odot)$, colour $g - i$ and sSFR for samples classified as elliptical, S0, spiral and irregular over different redshift ranges.

COSMOS2022 catalogue (Weaver et al., 2022). Inspecting Figure 5.9, it can be seen that the mass distribution of S0s indeed exhibits a double peak, with a distinct population of high mass S0s. This second peak is most prominent at higher redshifts and gradually disappears as redshift decreases, likely as a result of the drop in high-mass galaxies in the overall sample. In contrast, the number of low-mass S0s continues to grow in number. This population of high mass S0s appears to correspond to the secondary peaks at redder colours and lower specific star formation rates in the rightmost columns of Figure 5.9. As the relative number of high-mass S0s decrease with redshift, so too do these peaks disappear. The mass range corresponding to the fewest number of S0s (the “valley” of the double-peak distribution) fluctuates slightly about $10^{10} M_\odot$ across the four redshift ranges, but does not vary substantially. Although the mass distributions of each morphology clearly differ for $z < 0.3$, there is only a very slight variation in colour and star formation, with a slightly longer tail of low sSFR S0s. We note that the drop in the secondary, high-mass peak at lower redshifts may be consequence of COSMOS probing smaller volumes at lower redshifts, hence encountering much fewer high-mass samples (see Figure 5.1).

Figure 5.9 also shows the corresponding physical property distributions in the NA10 test set. The major difference is that the majority of samples in the NA10 test set are high mass samples, while the

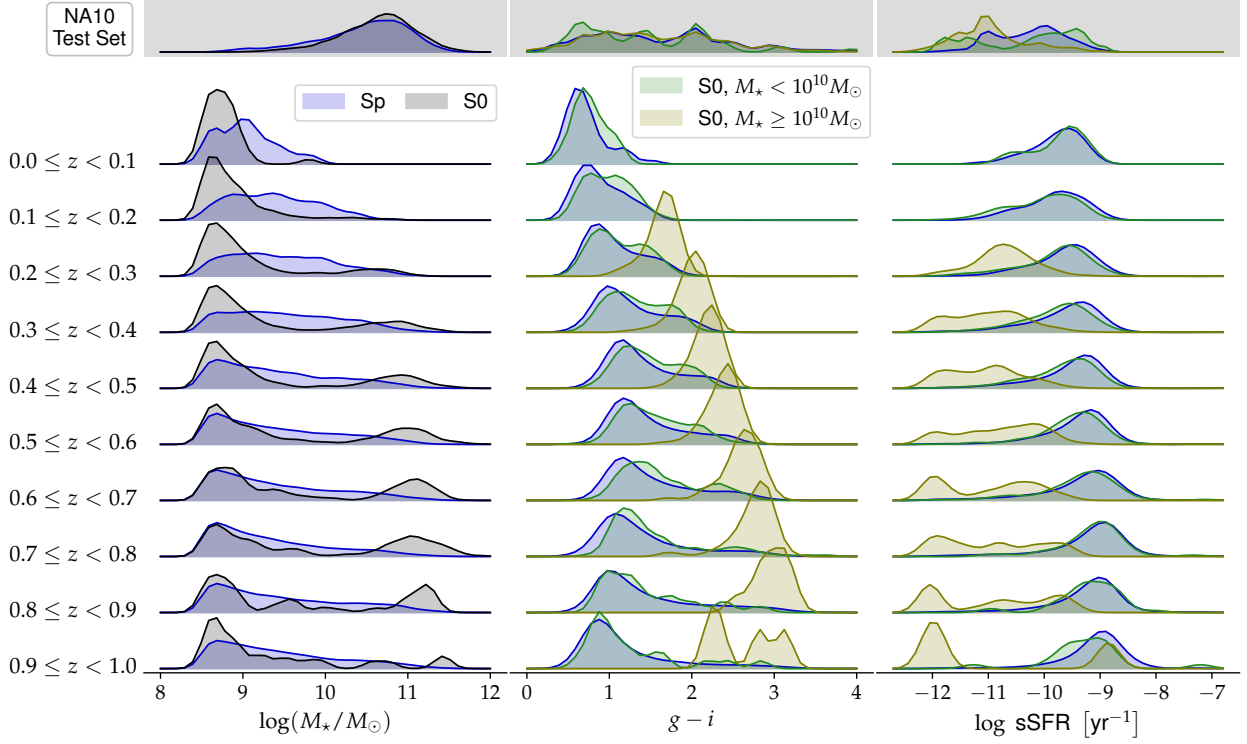


Figure 5.11: Ridgeplots showing the probability density distributions of stellar mass $\log(M_*/M_\odot)$, colour $g-i$ and sSFR for lenticulars and spirals in finer $0.1z$ redshift increments from $0 \leq z < 0.1$ to $0.9 \leq z < 1.0$. The columns for colour and sSFR split the S0s into samples with mass $< 10^{10} M_\odot$ in green and $\geq 10^{10} M_\odot$ in olive.

reverse is true for COSMOS. Even with this in mind, there is little evidence to support the existence of a peak in low-mass S0s in the NA10 test; instead, most S0s have masses above $10^{10} M_\odot$. In NA10, we can see that morphology is well separated by sSFR, but harder to disentangle in terms of $g-i$ colour. That said, the colours and specific star formation rates of NA10 are not directly comparable to that of COSMOS as these are separate surveys, each utilising different instruments, and hence different techniques for measuring such properties.

We can further examine the change in the populations of each morphology by looking at the number density distributions. Figure 5.10 shows the same mass, colour and sSFR distributions as in Figure 5.9, albeit in terms of number density. Here we can clearly see the rapid growth in S0s from high redshift to low redshift, as well as a gradual reduction in the number density of spirals. There are an extremely low number of ellipticals present in both Figures 5.9 and 5.10. We note that those that are classified at intermediate redshifts are generally massive, red and quiescents, while at low redshifts there is a significant jump in less massive ellipticals with a wider spread in colour and star formation rates. That said, there are simply too few ellipticals to draw any physical conclusions.

To better illustrate the bimodal distributions of S0 properties, Figure 5.11 investigates the redshift evolution in the overall frequency density distributions of mass, colour and sSFR in finer redshift bins. As in Figure 5.10, we again see that the double-peak mass distribution has mostly declined in prominence by $z \approx 0.3$, likely due to the reduction in massive galaxies in the COSMOS sample at low redshifts. For colour and sSFR, Figure 5.11 divides S0s into two separate distributions for samples with $M_* < 10^{10} M_\odot$

(low-mass) and samples with $M_\star \geq 10^{10} M_\odot$ (high-mass). The secondary peak of redder samples in the $g - i$ colour distribution consists almost exclusively of high-mass samples. Low-mass S0s are bluer, but not quite as blue as spirals. Both spirals and S0s have long tails. In the case of star formation, Figure 5.11 suggests that low-mass S0s and spirals are near indistinguishable at low-redshifts, and remain closely coupled throughout all redshift ranges, with S0s exhibiting a slightly longer tail. On the other hand, high mass S0s have lower rates of star formation and are clearly separated from the distribution low-mass S0s. Figure 5.11 thus shows that the secondary peaks in the colour and sSFR distributions observed in Figure 5.9 correspond to the high mass samples.

5.4 Discussion

5.4.1 Evolution of the S0 Fraction

Our results support an increase in the S0 fraction with decreasing redshift. In particular, this comes the expense of spiral galaxies, which decrease in prevalence. This result is consistent with previous observational studies in both dense and sparse environments (Desai et al., 2007; Poggianti et al., 2009; Oesch et al., 2010; Huertas-Company et al., 2015). However, a complete picture that fully describes the formation and evolution of S0s remains to be determined. Studies have demonstrated that there exist multiple mechanisms responsible for S0 formation, such as through a morphological transition from spirals via the faded spiral scenario, D’Onofrio et al. 2015; Rizzo et al. 2018; Deeley et al. 2020), which is the dominant formation pathway in dense environments and for low-mass S0s. Another formation scenario is that of major merging and/or accretion (Querejeta et al., 2015; Tapia et al., 2017; Diaz et al., 2018; Eliche-Moral et al., 2018), which is the preferred explanation for the formation of high-mass S0s (Fraser-McKelvie et al., 2018). S0s can also form passively through secular evolution and/or bulge growth (Kormendy & Kennicutt, 2004; Laurikainen et al., 2010), as well as through the passive consumption of gas (Rizzo et al., 2018). S0s also exhibit a wide range of physical and dynamical properties (Graham et al., 2018), including properties that cannot be explained by faded spirals alone (Williams et al., 2010; Méndez-Abreu et al., 2018; Deeley et al., 2021). The exact nature and physical origins of S0s remains a topic of current debate, for it remains unclear whether S0s are more closely related to ellipticals or spirals (Cortesi et al., 2013), or if their nature is simply that of a transitional morphology (Bremer et al., 2018; Phillipps et al., 2019). However, the consensus that has emerged from recent studies is that multiple formation pathways are necessary to fully account for the diverse properties of observed S0s (Fraser-McKelvie et al., 2018; Deeley et al., 2020; Coccato et al., 2022).

Figures 5.7 and 5.8 show that the redshift evolution of the S0 fraction depends strongly on stellar mass. In particular, the fraction of high mass S0s above $10^{10.5} M_\odot$ starts rising at higher z compared to the fraction of low-mass S0s. This suggests that high-mass S0s begin to form earlier than low-mass S0s. The reasons as to why the timescales for S0 formation may vary with stellar mass is unclear. One explanation is that this difference is a consequence of different formation mechanisms. Recent studies have indicated that the formation pathways for S0s may depend on stellar mass (Johnston et al., 2022). In particular, Fraser-McKelvie et al. (2018); Domínguez Sánchez et al. (2020) find that the formation of high-mass S0s is likely driven by mergers and accretion, while low-mass S0s are more likely to have formed via the faded

spiral pathway (see also Barway et al. 2013). Simulations have demonstrated that low-mass S0s better resemble a spiral progenitor rather than merger remnants (Bellstedt et al., 2017).

Méndez-Abreu et al. (2018) studied the evolution of high-mass S0s and suggested that their formation occurs at earlier epochs, and are largely driven by high-redshift dissipational processes (Johnston et al., 2020). The study by Oesch et al. (2010), which examined S0s in COSMOS, determined that secular evolution is also an important driver of morphological transitions away from spirals into S0s, especially for low-mass galaxies. However, we note that formation pathways are also strongly dependent on environment (Mishra et al., 2019), with the faded spiral pathway being especially dominant in dense environments such as clusters (Johnston et al., 2014; Coccato et al., 2022). Further studies are needed to ascertain the full extent of passive disc fading as a formation mechanism for S0s in low-density environments.

Figure 5.8 also showed that the S0 fraction in intermediate mass ($10^{9.5}M_{\odot}$ to $10^{10}M_{\odot}$) galaxies is not only the smallest, but also rises with the slowest rate. Furthermore, Figure 5.8 also illustrates that these intermediate-mass samples are overwhelmingly spiral galaxies. These results suggest that the formation pathways for S0s are more effective for low and high-mass galaxies.

5.4.2 A Bimodal Population

Based on our classifications and the results of Figures 5.9 and 5.11, we have unearthed tentative evidence supporting the existence of two distinct populations of S0s, distinguished by their bimodal mass distributions, and subsequent differences for both colour and specific star formation rate. Figure 5.12 illustrates these properties with both a galaxy colour-magnitude diagram and a plot of sSFR versus stellar mass. From the colour-magnitude diagram we can see two distinct populations of high-confidence S0 galaxies: a population of less luminous, bluer S0s; and a population of more luminous, redder S0s, with $M_g \approx 19$ serving as an approximate divider separating the two populations. While there do exist S0s in the sparse transition region between these populations ($-18 \lesssim M_g \lesssim -20$), these S0s tend to have much lower classification confidences ($P_{S0} < 0.5$) compared to those in the two hotspots. This transition region is instead largely dominated by spirals. The galaxies with the highest classification confidences $P_{S0} > 0.65$ are almost all concentrated in the redder, higher-luminosity population. Furthermore, we note that the low-luminosity samples have a greater spread in $g - i$ colour compared to the high-luminosity samples, which are instead predominantly red ($g - i \gtrsim 2$).

Figure 5.12 also illustrates sSFR, from which we can see that the majority of S0s have lower rates of star formation, and are mostly found below the galaxy main sequence. However, we also see evidence for star-forming S0s with star formation rates in line with spirals, as suggested by the frequency densities in Figure 5.11. In particular, these star-forming S0s are mostly low-mass S0s. There is a greater range of sSFR values for low-mass S0s compared to high-mass S0s, the latter of which are mostly quiescent, which is consistent with a merger-driven formation pathway for high-mass S0s. From Figure 5.12, we infer that the low-mass population includes both star-forming and passive S0s. This implies that there are multiple formation pathways for low-mass S0s.

We also note that there is a region of sparse sampling between $9 < \log(M_{\star}/M_{\odot}) < 10.5$ in which there are comparatively fewer galaxies. This region coincides with the valley in the S0 mass distribution at intermediate masses. Indeed, from Figure 5.8, we have established that the majority of these intermediate

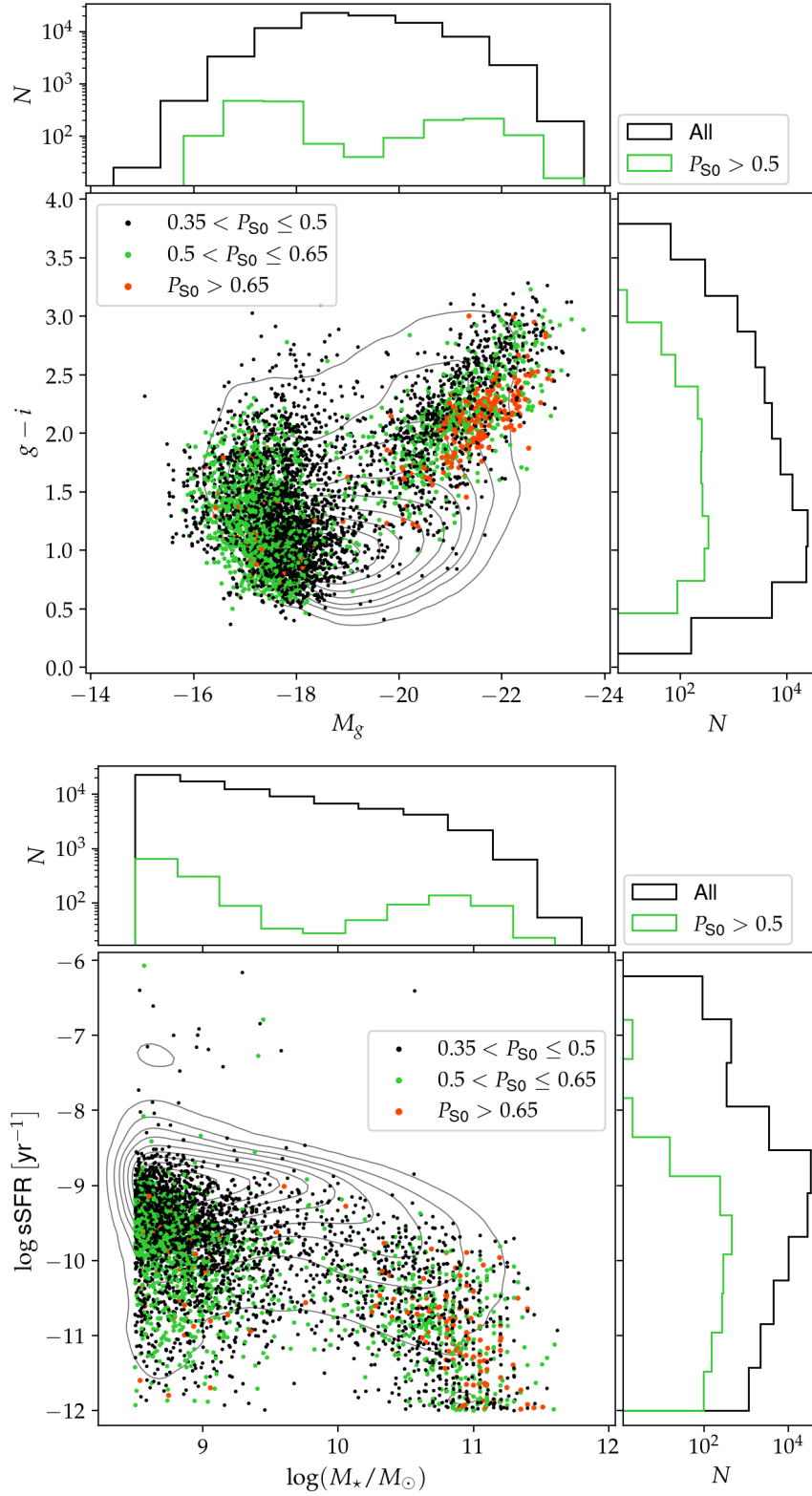


Figure 5.12: (Top) Absolute g magnitude M_g vs. colour $g - i$. (Bottom) Stellar mass $\log(M_*/M_\odot)$ vs. sSFR. Contours denote smoothed kernel density estimates of the distribution of the entire sample. Samples with $P_{S0} > 0.35$ are directly plotted and shaded according to their mean S0 output probability, as predicted by the augmented 4-class model ensemble.

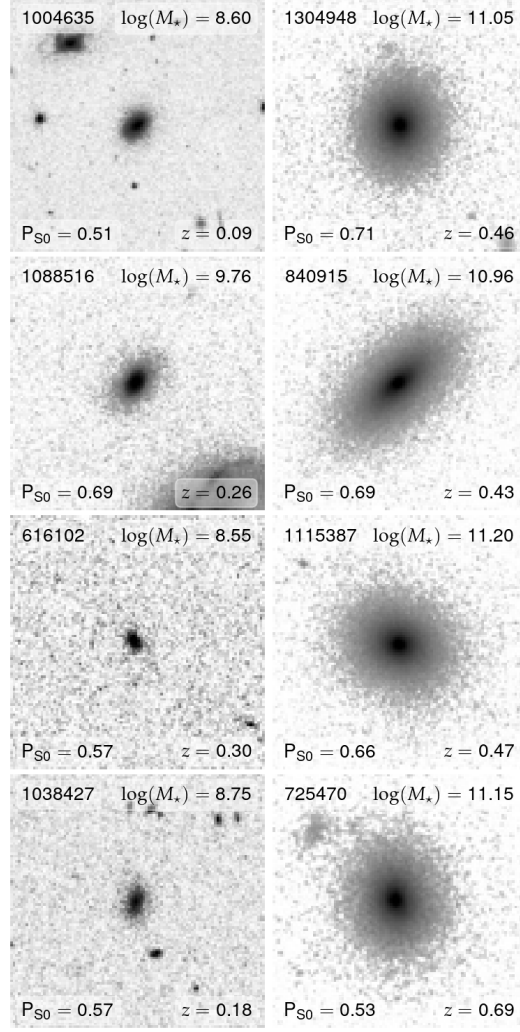


Figure 5.13: Random selection of example S0 galaxies classified with at least a confidence $P_{S0} > 0.5$. The left column shows a selection of low mass S0s, and the right row shows a selection of high mass S0s. Samples are annotated with their COSMOS ID in the top left, stellar mass in the top right in units of $\log(M_*/M_\odot)$, confidence P_{S0} in the bottom left, and redshift z in the bottom right.

mass galaxies are spirals, which is concordant with the lack of quiescent galaxies in this mass range, as seen in Figure 5.12. This explains the low S0 fraction for intermediate mass samples.

Figure 5.13 shows a random selection of low-mass and high-mass S0 galaxies, as sampled from the two subpopulations of high-mass quiescent S0s, and low-mass star-forming S0s. The high-mass examples are share the typical characteristics of an S0 galaxy; featureless, well-defined disc components with strong central bulges. By comparison, the low-mass examples are considerably more compact. Some such as 1088516 resemble dwarf lenticular galaxies, while others such as 616102 are harder to visually discern, with comparatively lower classification accuracies.

Recent work by [Tous et al. \(2020\)](#), based on machine learning classifications, also uncovered two subpopulations of S0s with distinct properties, some of which exhibited star formation rates more in-line with late type spirals. [Tous et al. \(2020\)](#) conclude that star-forming S0s may not be as rare as first thought.

Indeed, previous studies have searched for star-forming early types and concluded that they are rare, but most abundant at low stellar masses $M_\star < 10^9 M_\odot$ (Kannappan et al., 2009; Schawinski et al., 2009). Unlike passive S0s, star forming S0s cannot be readily explained by the faded spiral formation mechanism. It is believed that the star formation in such low-mass early types is consistent with starbursts triggered by minor merging (Wei et al., 2010) or accretion from gas-rich companions (Marino et al., 2011). Recent studies examining star-forming S0s (e.g. Rathore et al. 2022) have suggested that they could be quenched objects currently undergoing renewed star formation. The majority of star-forming S0s analysed by Rathore et al. (2022) had masses less than $10^{10.25} M_\odot$. Domínguez Sánchez et al. (2020) also found a bimodality in the S0 population, with high-mass galaxies exhibiting strong velocity dispersions with little to no metallicity gradients, while low-mass galaxies had flat velocity dispersion profiles and strong metallicity gradients. Cassata et al. (2007) also found evidence for an anomalous population of “blue spheroids”, which may be consistent with a mixture of spirals, dwarf ellipticals and S0s galaxies subject to strong bursts of star formation.

5.4.3 Our Deep Learning Approach

5.4.3.1 NA10 and COSMOS

The deep learning classifiers used to classify the images from our COSMOS sample were adapted from base models that were initially trained on SDSS g-band images with known morphologies from the NA10 dataset. As a baseline to aid in interpreting our results so far, Figure 5.14 shows the same diagrams as in Figure 5.12, but this time for samples from the NA10 test set, as classified by the base models. We caution that the values, especially sSFRs, are not directly comparable due to differences in the methods used to obtain them. Furthermore, NA10 samples a much higher mass range compared to our COSMOS sample. That said, Figure 5.14 offers an insight into the original capabilities of the model. In particular, even despite the poor sampling at lower masses, there is no indication of a secondary peak in mass. Most S0s are quiescent and clustered between 10^{10} and $10^{11} M_\odot$, with a peak around $10^{10.5} M_\odot$, however there are also several S0s with star formation rates in the main sequence, including samples with masses less than $10^{9.5} M_\odot$. However, there are too few low-mass samples in NA10 test set – and the entire NA10 sample as a whole – to clearly establish any bimodal mass distribution, nor the existence of low-mass, star forming S0s. On the other hand, the COSMOS sample is awash with low-mass samples. The fact that these models are able to identify S0s among these low-mass samples, despite there being so few in NA10, is significant, for if the models were simply trained to mimic NA10 then Figure 5.12 would look markedly different.

Both our models detected an extremely low number of ellipticals, to the point where we could not reliably establish either their properties or evolution. This is a significant issue given that ellipticals are plentiful in NA10, as shown in Figures 5.9. There are several factors that may account for this, not least the inherent differences between the NA10 dataset and low-redshift COSMOS data. Importantly, the NA10 dataset is not volume corrected and hence the fractions of ellipticals are not representative. We note that studies examining volume corrected SDSS galaxies have also obtained low fractions for ellipticals (Wilman & Erwin, 2012). Furthermore, the NA10 ellipticals are predominantly massive with stellar masses $\log(M_\star/M_\odot) > 10.5$, however COSMOS samples very few massive galaxies at low redshifts. Previous studies have also shown that the fraction of early types is considerably greater in dense environments

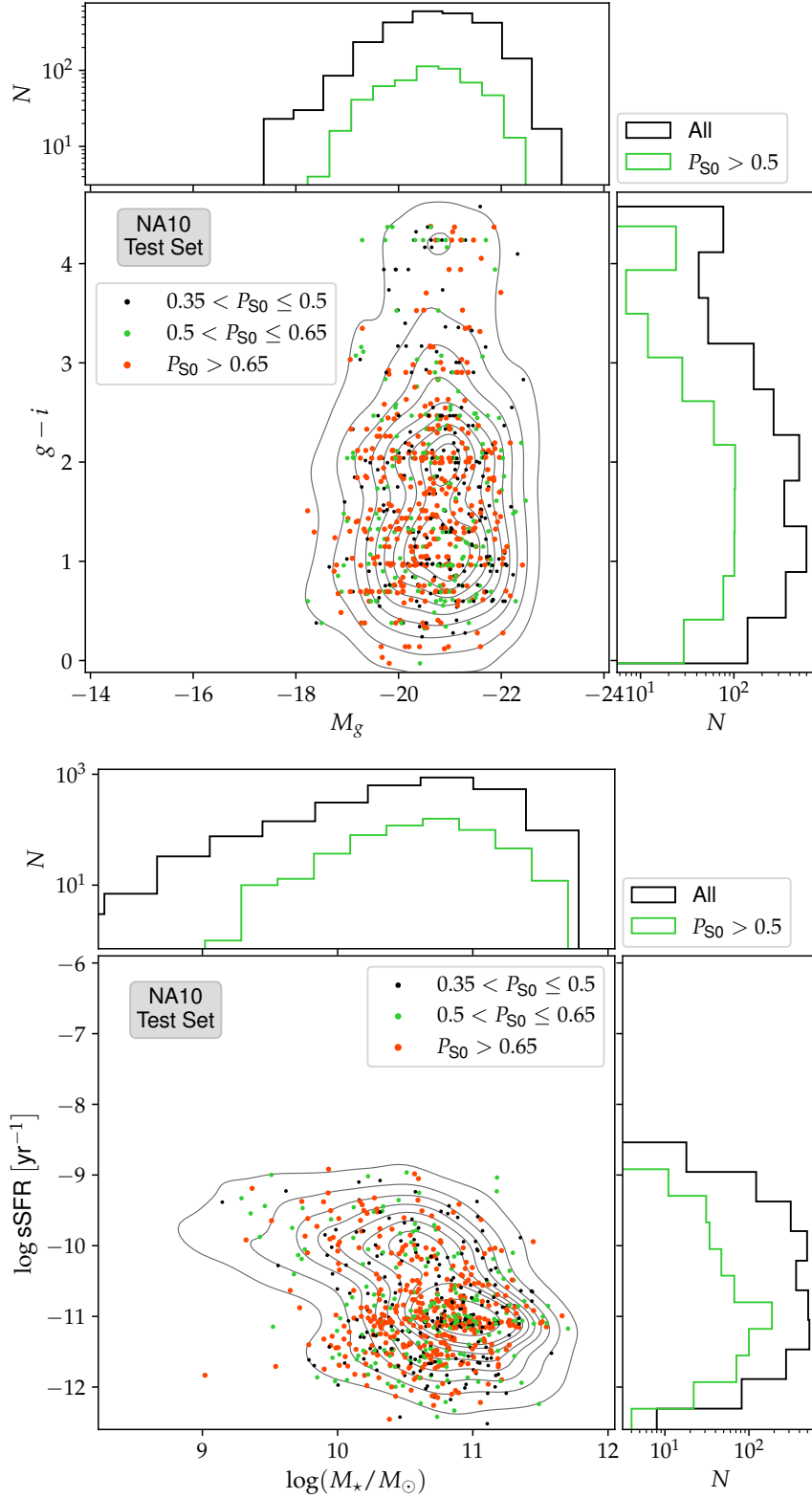


Figure 5.14: Absolute g magnitude M_g vs. colour $g-i$ (top) and stellar mass $\log(M_*/M_\odot)$ vs. $sSFR$ (bottom) for galaxies in the NA10 test set. As in Figure 5.12, the contours are drawn with respect to the full sample. Galaxies with $P_{S0} > 0.35$ are directly plotted and shaded according to their classification confidence, as predicted by the initial 4-class base model.

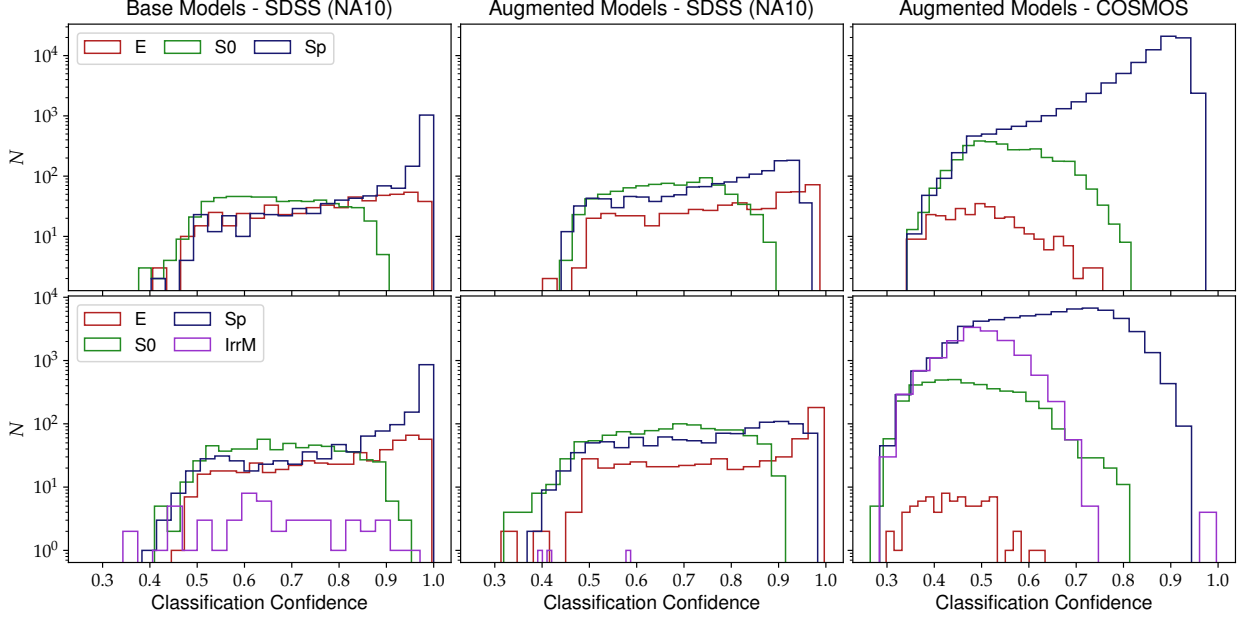


Figure 5.15: Distribution of the classification confidences of all predictions for each morphological class in the 3-class (top row) and 4-class (bottom row) models. The left column shows the initial, base models as evaluated on the original images from the NA10 test set. The middle column shows the results of the noise-augmented models, as evaluated on noisy images (also from the NA10 test set). The right column shows the noise augmented models as applied to the entire COSMOS dataset.

such as clusters rather than in the field (Fasano et al., 2000; Wilman et al., 2009; Poggianti et al., 2009; Kovač et al., 2010), with (Desai et al., 2007) finding little evolution in the fraction of ellipticals at low redshifts. In the COSMOS field environment, ellipticals should constitute at least 15% of the total number of galaxies (Oesch et al., 2010), increasing significantly for massive galaxies. We ought to be detectable at higher redshifts with our deep learning models, however this is where noise likely has a substantial impact.

5.4.3.2 The Impact of Noise on Classification Uncertainty

As aforementioned in Section 5.2, the data augmentation workflow is designed to add a random level of Gaussian noise to each image for each batch in each training epoch, such that the images collectively resemble the noise levels of the images in the COSMOS dataset (see Section 5.6 for further details on the exact procedure). As was observed in Figure 5.4, we found that the S0 accuracy increased, while the accuracy of the other classes decreased. These smoothed accuracies are consistent with the regularisation effect that comes from adding noise (Jin et al., 2016). In our 4-class model, irregulars were almost all misclassified as either S0 or spiral, and up to 30% of true spirals were misclassified as S0, which is likely reflective of the increased difficulty of extracting finer morphological features in noisy imaging.

We can examine the impact of the noise augmentation on a macro level by considering the distribution of classification confidences for each predicted class. Figure 5.15 shows the distribution of classification confidences for all samples classified into each morphological class, i.e. P_E for all samples classified as E, P_{S0} for all samples classified as S0, etc. A direct comparison can be made by examining the application

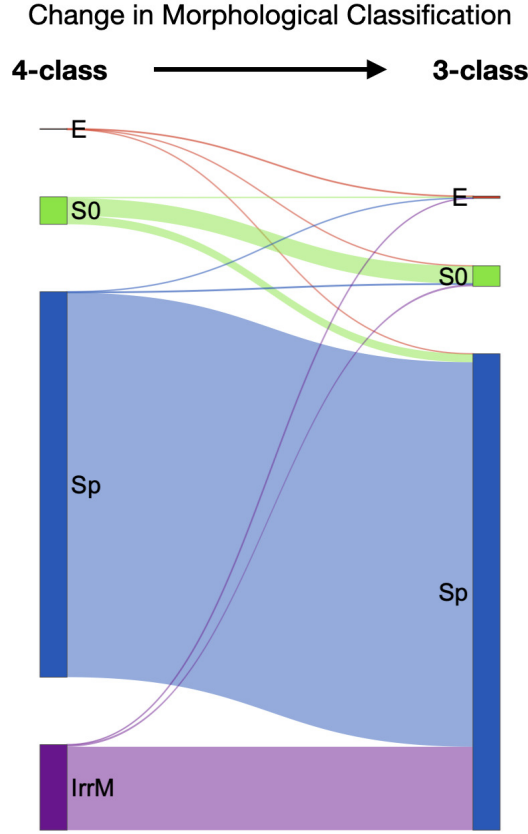


Figure 5.16: Sankey diagram showing the change in classifications for all galaxies when predicted by the 4-class model, to that predicted by the 3-class model. Vertical bands denote morphological categories, and lines denote flow. Both are sized linearly in proportion to the number of galaxies.

of the base model on the NA10 test set, as well as the noise-augmented models on noisy NA10 images; this is shown in the first two columns of Figure 5.15. Here, noisy NA10 images simply refers to images that underwent the same artificial noise injection as that used to train the noise-augmented models (as per Section 5.6). The most immediate difference is that irregulars all but vanish for the 4-class noise augmented model. In the base models, we see a very strong peak for spirals with confidences of around 1; such a peak of near-certain predictions is indicative of overfitting. The distributions for spirals are comparatively smoother and flatter for the noise augmented models, and the maximum confidence for spirals is less than 1. This is likely directly due to the noise impacting on the ability to discern spirals. We also see that S0s cannot be classified with a confidence greater than 0.9, reflecting the inherent uncertainty that the base models have in classifying S0s, as seen in Figure 5.4. Ellipticals maintain roughly the same distribution, with a slight peak in higher confidences for the 4-class noise augmented model. While most samples are classified with confidences greater than 0.5, there is a tail of samples with confidences below this, indicating that there are images in the NA10 test set that are difficult for the model to classify.

The last column in Figure 5.15 shows the distribution of prediction confidences for all 85,378 images in the COSMOS dataset. We can conclude that the S0s in COSMOS are harder to classify confidently; no samples have confidences beyond 0.8. Although the classification accuracies of the noise augmented models is higher for S0s, this comes at the cost of higher uncertainty, which is no surprise given the

fractions of spirals and ellipticals misclassified as S0s as seen in Figure 5.4. The ellipticals in COSMOS are classified with the lowest confidences, with $P_E < 0.65$ in the 4-class model. This is despite ellipticals being relatively easy to classify in NA10, even with the augmented models (see middle column of Figure 5.15). This suggests that either COSMOS contains a critical shortage of elliptical galaxies, or that ellipticals are much harder to accurately detect, even with a model trained specifically to be robust with respect to noise (especially if one accepts, as visual criteria for classification, that ellipticals are featureless spheroids). Furthermore, although our base models detect very few irregulars in the SDSS images, our augmented models are able to detect far more irregulars in COSMOS. This is likely a combination of factors, including that of increased noise paired with more limited image qualities and resolutions, however it is also possible that there are simply more inherently irregular galaxies in COSMOS.

It is also worth examining the degree to which the overall classifications differ between the 3-class and 4-class models. We saw individual examples of this in Figure 5.6. To illustrate the overall impact, Figure 5.16 shows the overall changes in classifications for galaxies in COSMOS initially classified as E, S0, Sp and IrrM by the 4-class model. The majority of classifications are unchanged, however there is a sizeable fraction of S0s that the 3-class model classifies as spirals. A degree of misclassification for S0s is expected given the confusion matrices in Figure 5.4, yet this is likely also exacerbated given the dominance of spirals in the 3-class model classifications. Importantly, the galaxies with different classifications were found to have substantially lower confidences (mean $P_{S0} \approx 0.4$, both models) than those whose classifications remained unchanged ($P_{S0} \approx 0.6$).

This study focuses exclusively on dealing with different noise levels since this has a substantial negative impact on CNN accuracy, as demonstrated Figure 5.17. However, differing noise levels is not the only consideration when adapting a model to classify images from different datasets. Other factors include the impacts of different angular and/or physical resolutions, as well as classifying images at different wavelengths, either due to different redshifts ranges or different observational bands. As our original training data spans a different redshift range to that of the COSMOS images, galaxy evolution is also another factor. Noise is a destructive process that directly impacts on the CNN’s ability to extract meaningful features, particularly in the first convolutional layer. Differing resolutions due to different PSFs can be easily characterised by a convolutional kernel, whereas noise cannot. Although the effects of different resolutions and wavelengths cannot be completely ignored, we argue that the impacts of these for our NA10 SDSS to COSMOS transfer learning is limited. This is due to the low input resolution of the CNN of 100×100 pixels, which is smaller than the average image pixel size of our COSMOS galaxies prior to data preprocessing (153×153 pixels). Hence, for an average image, the PSF is on the order of approximately $0.1''$ (Koekemoer et al., 2007), or around 3 pixels. This is further reduced to around 2 pixels when the images are downsized, which is well below the size of the first layer’s convolutional kernels (see Figure 5.2). For the original SDSS training images, the median PSF FWHM is around $1.3''$ (Abazajian et al., 2009) corresponding to approximately 0.78kpc at $z = 0.03$, or less than 2 pixels given our fixed 0.5 kpc physical pixel scale. The majority of the SDSS images were likewise downsized as part of the preprocessing. Furthermore, our model is restricted to single-band images, and we only classify samples up to an upper redshift limit of $z = 1$. This limits the impacts of both k-correction and evolution, which become increasingly significant for very high values of z .

It is difficult to determine the impact of these differing scales on our results, however these factors

cannot be ruled out in adversely impacting our model’s performance, as demonstrated with the reduction in overall classification confidences in Figure 5.15. We argue that noise has the most dominant impact, and its effects become progressively worse at higher redshifts, where resolution is increasingly limited. The combined impacts of noise, resolution, k-correction and evolution will ultimately place upper limits on the maximum redshift to which a deep learning approach based on images is suitable. Future studies utilising high-resolution JWST imaging will likely go beyond such existing limits, and enable more accurate deep learning classifications for very high-redshift galaxies (Ferreira et al., 2022a; Kartaltepe et al., 2022; Robertson et al., 2022).

5.5 Conclusion

In this work, we investigated the redshift evolution of the S0 fraction for $z < 1$ in COSMOS, through the application of convolutional neural networks to classify images of galaxies. Our transfer learning approach, as facilitated by data augmentation, enables us to adapt existing models trained on low-redshift SDSS images to classify high-redshift COSMOS images, all the while leveraging our existing images and labels of known morphologies. While previous studies have demonstrated it is possible to use transfer learning to adapt models to classify low-redshift images from different surveys (Domínguez Sánchez et al., 2019), ours is the first to show it is possible to adapt models to classify images across different redshift regimes. Our key findings are summarised as follows.

1. We have found that there is a sustained rise in the overall S0 fraction from $z = 1$ to $z = 0.1$ from less than 1% to around 40%. Furthermore, this rise comes at the expense of spiral galaxies, for which the fraction instead decreases.
2. The onset of the growth in the S0 fraction occurs at higher redshifts for high-mass ($\geq 10^{10.5} M_{\odot}$) samples at around $z = 0.9$, rising sharply to 45% by $z = 0.4$ where it remains stable. For low-mass samples ($< 10^{9.5} M_{\odot}$), the rise in the S0 fraction is delayed, reaching a comparable level by $z = 0.2$. Between $z = 0.5$ and $z = 0.7$, there are nearly twice as many high-mass S0s as there are low-mass S0s. We conclude that, in general, high-mass S0s evolved earlier than low-mass S0s. Previous studies have established that S0 formation pathways depend strongly on mass, and so it is likely that this difference is as a result of varying timescales for different S0 transformation processes.
3. At low redshifts, the S0 fraction in intermediate mass S0s ($10^{9.5} M_{\odot}$ to $10^{10.5} M_{\odot}$) is roughly half that of the fraction in high and low-mass samples. Overall, the S0 fraction is highest in the low mass and high mass range. On the other hand, the fraction of spirals is highest for intermediate masses (80%), while lower for low and high masses (around 50% and 55% respectively).
4. We have found a bimodal mass distribution in our classified S0 galaxies, such that they constitute two largely distinct subpopulations: high-mass S0s that are almost all red and quiescent; and low-mass S0s that are generally bluer with higher star formation. We have found that low-mass S0 population includes both passive S0s and star-forming S0s, the latter of which cannot be solely explained by the faded spiral mechanism. As such, we suggest that there are a range of physical processes responsible

for the formation and evolution of low-mass S0s. While faded spirals may explain passive low-mass S0s, other processes such as mergers may not completely suppress star formation. Further studies are necessary to definitively ascertain the differences in the formation of these two subpopulations.

5. We have demonstrated the effectiveness of transfer learning and data augmentation in adapting our initial models, pretrained on SDSS images of NA10 samples, to classify COSMOS images, which have significantly higher levels of noise. Through fine-tuning, we trained a new ensemble of models with artificially noisy SDSS images such that they replicate the characteristics of the COSMOS images. Importantly, this method allows us to utilise the known NA10 morphologies for training. We have found that this method led to a 10% increase in S0 accuracy on the NA10 test set. Even at moderate levels of noise, our augmented models dramatically outperform the initial base models, with overall classification accuracies around 70 to 80%, compared to less than 30% for the base models without noise augmentation. We thus conclude that transfer learning is crucial for CNNs to classify images from different surveys where there is a major difference in noise levels.

5.6 Appendix A: Noise Augmentation and Transfer Learning

Transfer learning is a general technique for domain adaptation, in which an existing model trained on one domain (or dataset) can instead be adapted to create a new model applicable to a different, yet related, domain (see [Weiss et al. 2016](#) for a review). The key motivation behind this approach is to leverage the pre-existing capabilities of existing models. Common present-day examples include taking large pretrained models such as Inception ([Szegedy et al., 2015](#)) that have been trained on very large, general-purpose datasets (such as ImageNet [Deng et al. 2009](#)) and then adapting these for image classification tasks, such as galaxy classification ([Ackermann et al., 2018](#)). Transfer learning has also been utilised in astronomy to adapt models to classify images in different astronomical surveys using solely the known morphologies from the original, base survey ([Domínguez Sánchez et al., 2019](#)). Not only is this technique efficient, but it is especially useful when there is insufficient labelled data to effectively train a full model from scratch. This work utilises transfer learning together with data augmentation to classify high-redshift COSMOS images.

5.6.1 Data Augmentation with Artificial Noise

There are several challenges that arise when shifting to a different observation space. Foremost among them is the issue of noise, resulting in degraded image quality and ultimately impacting the ability to classify the morphologies of more distant galaxies. Noise is one of several issues, including PSF smoothing and k-correction, that impact the ability to classify high-redshift samples, especially for distant galaxies. This work is focused solely on addressing noise, in particular by adapting our pretrained models to classify noisy images. We achieve this through adding artificial noise to our existing training data of SDSS images in order to replicate the characteristic noise of the COSMOS images ([Koekemoer et al., 2007](#); [Leauthaud et al., 2007](#)). This allows us to simply adapt the model using the existing images and their labels (see also [Vega-Ferrero et al. 2021](#)). This noise is applied on-demand during training through the use of a custom

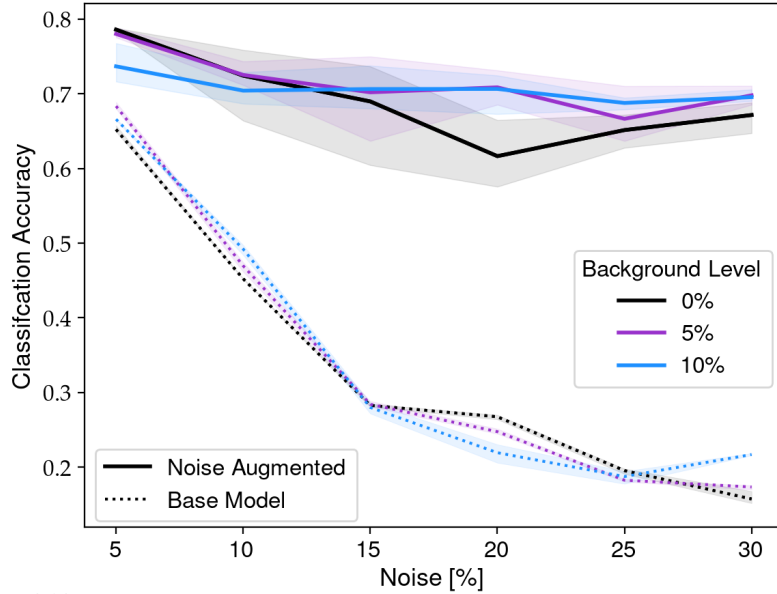


Figure 5.17: Plot of the classification accuracy for models classifying a test dataset at different, fixed levels of artificial noise and background. Shaded regions denote the full range of accuracies and losses for each of the three trials.

KERAS preprocessing layer, which applies a random level of additive Gaussian noise to the input images per minibatch for every epoch of training.

In general, corrupting real-valued inputs with additive Gaussian noise is a known regularisation technique that is especially effective at improving model robustness and reducing a model’s susceptibility to adversarial inputs, i.e. inputs with random perturbations intended to “fool” the network (Jin et al., 2016). For our purposes, we wish for our models to remain effective at classifying noisy images. Our noise augmentation process is tightly controlled so that the artificial noise is representative of the typical noise of the images in our COSMOS dataset.

The exact procedure involves adding Gaussian noise directly to the normalised pixel values. Importantly, the amount of noise added is not fixed; instead, the mean and standard deviation for the Gaussian noise is chosen uniformly within a range of predetermined bounds. To determine suitable bounds, we first analysed the background noise of random batches of COSMOS images at varying redshifts, obtaining approximate Gaussian fits. These bounds were then further refined through a visual comparison of the artificially noisy SDSS images with the actual COSMOS images. The values we obtained range from between -0.02 to 0.1 inclusive for the mean (background) and between 0 and 0.3 inclusive for the standard deviation (noise). This compares well with background levels of 0 to 10% of the peak flux, and a noise threshold of 0.1% to 40% of the peak flux from Leauthaud et al. (2007). Since these values are chosen uniformly there is no bias for morphology. With this custom noise augmentation layer, it is possible to apply arbitrary levels of noise within these bounds to a given SDSS image. This is important firstly to avoid overfitting (by ensuring that images are not all given identical levels of noise), and secondly to ensure that the augmented images collectively reflect the range of inherent image qualities and signal-to-noise values across our COSMOS dataset. Figure 5.3 showed an example of this procedure in action. After the noise is added, the resulting pixel values are clipped to between 0 and 1 inclusive. We stress that such noise is

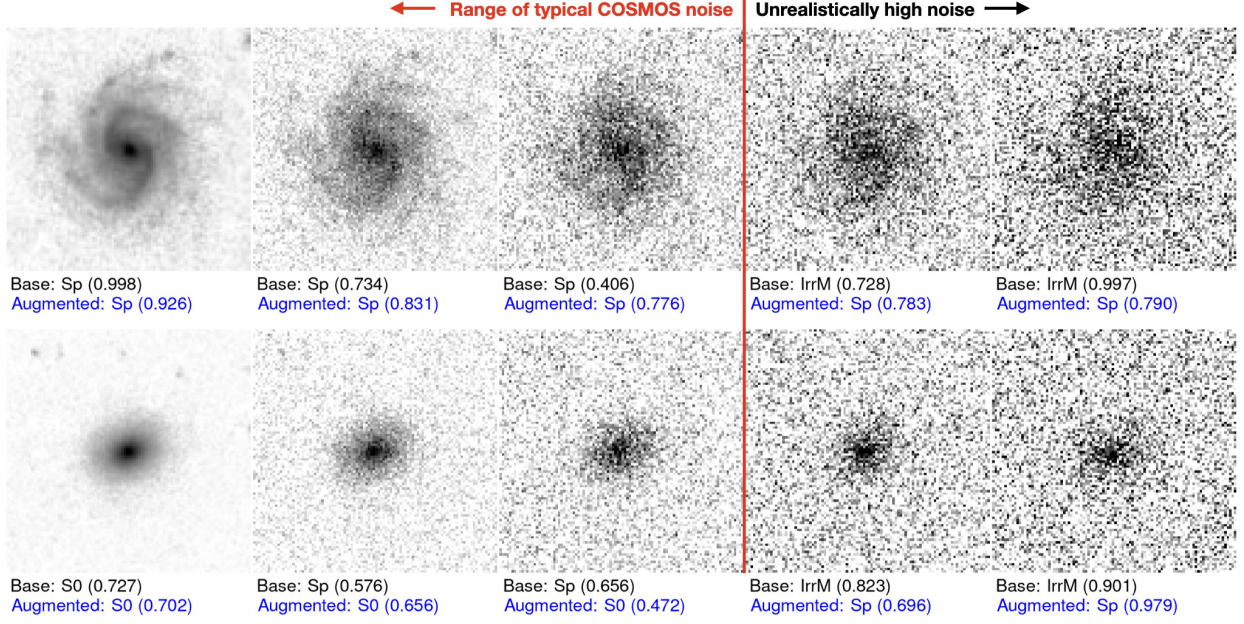


Figure 5.18: Morphological classifications for an example spiral and S0 galaxy from the NA10 test set when subject to increasing levels of artificial Gaussian noise. Classifications are obtained using the original 4-class base model, and the final noise-augmented 4-class ensemble model. For first column shows the original image with no noise, while the next columns show the same image with noise levels of 15%, 30%, 40% and 50%, with the vertical red line marking the limit of typical COSMOS noise levels. Classification confidences are shown in parentheses.

deliberately artificial and is not intended to be a 100% accurate reflection of the physical noise levels. The primary purpose of the noise augmentation is to help adapt the CNN to better classify degraded images.

5.6.2 The Impact of Artificial Noise

To analyse the impact of this noise augmentation on the overall classification accuracies of our CNNs, we tested the procedure using different, discrete Gaussian noise and background levels. We performed three trials for each fixed noise and background level. In each trial, an augmented model is both trained and tested with noise-augmented images. To compare performance, the base model is also tested with the same noise-augmented images as with the augmented model. Figure 5.17 shows a dramatic difference between the initial models and the noise-augmented models when tasked with classifying the augmented images. The initial models struggle to classify noisy images, with accuracies no better than random even at relatively modest levels of noise. The noise-augmented models have superior performance, with accuracies as high as 70% at the maximum typical noise level of 30%. The noise augmentation process is also highly variable, with the accuracies of the three individual trials differing by around 10%. Background levels have little impact on accuracy.

We further tested the models using unrealistically high levels of noise; that is, noise levels higher than the bounds we obtained from analysing the COSMOS images, and henceforth higher than the levels used to train the noise-augmented models. Figure 5.18 compares the classifications of the original 4-class base models with those of the final ensemble of augmented models (see section below). In the case of the

example spiral galaxy, both models correctly identify its morphological type throughout the range of typical COSMOS noise levels, however the classification confidence for the augmented ensemble is higher. In the case of the example S0 galaxy, the base model incorrectly classifies the noisy images as spirals, while the augmented model correctly identifies them as an S0 (albeit with relatively lower confidence levels). Both models degrade with unrealistically high levels of noise. It is nevertheless clear that models not explicitly adapted to classify noisy images will struggle.

5.6.3 Transfer Learning

To commence transfer learning, we first require pretrained base models. These are the 3-class and 4-class models trained on SDSS images from the NA10 catalogue with the new architecture as described in the previous subsection. There are several methods for conducting transfer learning, such as partially or completely freezing layers of a model (essentially keeping the existing parameters as they are) while tweaking other layers, or fine-tuning the entire model. We utilise the latter, fine-tuning approach. We instantiate the model with its existing weights, then (re)train it with the new noise augmentation layers, albeit with a much lower learning rate (reduced by a factor of 10 to 8×10^{-5}). Note that the model architecture and total number of trainable parameters remain unchanged. The training data also remains identical, including the dedicated test set used for evaluation, which is also subject to the noise augmentation process to ensure fair evaluation. Since the noise augmentation process is highly variable, we train ensembles of five 3-class models and five 4-class models. The final classification of a given image is obtained through averaging the output probabilities across each of the five individual models. The predicted class is henceforth defined as the class with the highest mean output probability. We trained these models using the SDSS images from the NA10 dataset, with 80% of the sample constituting the training set, and 20% set aside as a test set for final evaluation. To fairly evaluate the noise augmented models, the images in the test set are also augmented with a random amount of noise.

Morphokinematics and morphological segmentation

Morphology is an inherent property of galaxies. Throughout the previous Chapters, we have classified galaxy morphologies, examined their evolution over time, and investigated their physical properties, including stellar mass, luminosity, colour and star formation rates. However, physical properties are not the only markers of morphology. The kinematic properties of galaxies are also extremely important. Understanding the link between a galaxy’s morphology and its kinematic properties can yield vital insights into its potential formation mechanisms and evolutionary history, and also aid our fundamental understanding of morphology itself (Cappellari, 2016; van de Sande et al., 2021). Galaxies can also exhibit a large variety of finer morphological substructures, such as stellar bars and spiral arms (Buta, 2013). In particular, stellar bars are known to play a significant role in the evolution of galaxies, affecting all manner of processes from angular momentum redistribution to gas dynamics (Athanasoulas, 2005; Fanali et al., 2015). So far, our deep learning models have merely classified bars. Yet there is much more information that can be gleaned from examining the bar in more detail, such as its length, thickness and strength (Géron et al., 2021).

This Chapter explores additional applications of our deep learning approach, and is divided into two main sections: morphology and kinematics, and morphological segmentation. In the first section, we classify images of 5,536 galaxies from the SAMI DR3 survey (Croom et al., 2021a) with both our 4-class deep learning model (elliptical, lenticular, spiral, irregular/miscellaneous) and our galaxy bar model in order to study the kinematic properties of galaxies in terms of their predicted morphologies. This includes studying the kinematic differences between barred and unbarred galaxies, the first large-scale study to do so with SAMI galaxies. In the second section, we design and train a U-Net; a type of specialised deep learning model designed specifically for morphological segmentation. This model complements our existing suite of models, and allows us to study fine morphological structures in more detail. A typical U-Net can be thought of as an extension to a regular CNN, namely one that is turned back on itself such that its input shape and output shape are both the same size. We use this U-Net to extract pixel masks corresponding to the location of bars and spiral arms within images of galaxies. We train these U-Net using crowdsourced data from the Galaxy Zoo 3D survey (Masters et al., 2021). We then use these masks to estimate the lengths of stellar bars in barred galaxies from both our original NA10 dataset and the SAMI DR3 survey dataset. The content in this Chapter is expected to form the basis for future manuscripts.

6.1 Morphology and Kinematics

6.1.1 The Importance of Kinematics

The kinematic properties of galaxies can reveal important insights into the physical and environmental processes governing their structure and evolutionary history, as well as the processes that drive the assembly of galaxy mass throughout cosmic times (Yang et al., 2008; Cappellari et al., 2011; Falcón-Barroso et al., 2014). Galaxy kinematics are known to be intimately linked with morphology (de Zeeuw et al., 2002; Teklu et al., 2015; Cappellari, 2016; Thob et al., 2019). In general, late type galaxies tend to be rotationally supported, while early type galaxies are predominantly dispersion supported. Indeed, galaxy kinematics offer a means of studying the internal structure and dynamics of galaxies. As such, kinematic properties form the basis of several fundamental scaling laws (Faber & Jackson, 1976; Tully & Fisher, 1977; Fall & Romanowsky, 2018). They also exhibit broader scaling relations with other structural components, such as bulge and disk components (Kormendy, 1993; Kormendy & Kennicutt, 2004; Oh et al., 2020), as well as galaxy dark matter halos (Dutton et al., 2010; Wechsler & Tinker, 2018). Many modern morphologically classification systems directly incorporate kinematic information (Kalinova et al., 2017; van de Sande et al., 2017). For example, studies utilising integral-field spectroscopy have shown that early type galaxies consist of two subpopulations that can be separated based on their differing kinematic properties: fast rotators and slow rotators (Emsellem et al., 2011; Scott et al., 2014; Graham et al., 2018; van de Sande et al., 2021). Studies have also shown that the fraction of slow rotators may vary with environmental density (Brough et al., 2017); this kinematic-morphology density relation subsequently serves as an important constraint for cosmological simulations (Khochfar et al., 2011; Lagos et al., 2018b). Kinematics properties are important for studying the influence of stellar bars on the overall dynamics and evolution of galaxies (Aguerri et al., 2015; Cuomo et al., 2019; Roshan et al., 2021).

An important kinematic property is the spin parameter, denoted λ_R (originally introduced in Emsellem et al. 2007). This dimensionless parameter acts as a proxy for the baryon projected specific angular momentum and thus represents a luminosity-weighted ratio of ordered motion (i.e. rotation) to random motion. λ_R is not to be confused the classical, dynamical spin parameter λ , first introduced by Peebles (1971), but the former is a good approximator of the latter (Emsellem et al., 2007). λ_R can be readily obtained by integral field spectroscopy, and is typically presented as λ_{R_e} , the value of λ_R within one effective radius R_e (see Emsellem et al. 2011; van de Sande et al. 2021 for further details). The spin parameter is often used to demarcate slow rotators from fast rotators, though exact thresholds vary considerably between studies, with some thresholds also incorporating ellipticity ϵ (Emsellem et al., 2007, 2011; Cappellari, 2016).

It is also worth considering whether the spin parameter proxy λ_{R_e} can be used in a similar fashion to separate barred galaxies from unbarred galaxies. The previous study by Cervantes-Sodi et al. (2013), analysing the classical spin parameter λ , found that galaxies hosting longer stellar bars tend to have lower spin parameter values than galaxies hosting shorter bars, with unbarred galaxies having values in between. Other studies have suggested that the kinematic distinctions between barred and disc galaxies are far more subtle (Rawlings et al., 2020). Another important kinematic property is the velocity dispersion σ , which represents the statistical variation about the mean velocity and is thus a measure of random motion. σ is a useful quantity not just for studying the dynamical evolution of galaxies (Bezanson et al., 2012), but also for distinguishing morphological types. In particular, ellipticals and lenticulars have markedly

different kinematic properties, with lenticulars tending to be more rotationally supported than ellipticals (hence lower σ) (Moran et al., 2007a; Graham et al., 2018).

To study the differences in the kinematic properties of galaxies with different morphologies, we first classify the morphologies of 5,536 galaxies from the SAMI DR3 survey (Croom et al., 2021a). Although previous studies have examined disentangling morphology in terms of kinematics using the known SAMI DR3 morphologies (Rawlings et al., 2020; van de Sande et al., 2021), a similar analysis is yet to be done for barred galaxies. In this study, we classify both bars and general morphology using our combined suite of deep learning models (4-class model and bar model). We show that our models’ predicted morphologies are well separated when plotting λ_{R_e} in terms of both ϵ and σ . Furthermore, we also examine the ranges in these kinematic properties where bars are most common, and subsequently comment on the implications of these results for the formation and evolution of stellar bars.

6.1.2 The SAMI Dataset

The Sydney AAO Multi-object IFS (SAMI) survey (Croom et al., 2012; Bryant et al., 2015; Croom et al., 2021a) is a large-scale integral-field spectroscopy survey of several thousand nearby, low-redshift galaxies. Data from this survey has been analysed in a wide range of studies, including stellar kinematics and gas dynamics (Cortese et al., 2016; van de Sande et al., 2017), star formation rates (Medling et al., 2018), and the distributions of neutral hydrogen (Watts et al., 2023). It is therefore an ideal dataset to investigate the kinematic properties of galaxies of different morphological types. With the addition of our deep learning model to classify bars developed in Chapter 4, we can extend this analysis of morphology to also include barred and unbarred galaxies. Crucially, the SAMI dataset uses a completely independent classification scheme to that of NA10. In particular, both use numerical T-Types to classify morphologies, but the numerical scales and class definitions are completely different. The SAMI dataset is therefore an ideal test set with which to compare the classifications of our deep learning models (having been trained on NA10 T-Types) to these independent T-Types. Furthermore, directly applying our deep learning models to a completely different dataset serves as a test of their versatility and performance.

To this end, we classify 5,536 g-band HyperSuprime Cam (HSC) (Aihara et al., 2022) images of nearby, low-redshift galaxies from SAMI DR3 (Croom et al., 2021a). Specifically, these galaxies are from the Galaxy and Mass Assembly (GAMA) survey DR3 input catalogue¹ (Driver et al., 2011; Bryant et al., 2015). With a pixel scale of 0.168 arcseconds, the HSC imaging has a better image resolution than the SDSS (0.396 arcseconds). The image cutouts we use all have same, fixed size of 360×360 pixels. As such, each individual image no longer represents the same physical scale (as the galaxies are all located at slightly different redshifts). Although our models were trained on fixed-physical scale images, given that both the SAMI and NA10 galaxies are nearby, low-redshift galaxies with redshifts mostly less than $z \approx 0.1$, this is not a major issue since the physical scales do not vary considerably. Recall also in Chapter 2 that one of the common methods of data augmentation involves zooming. As with rotation (orientation should not matter), there is very little (at least from the point of the view of the CNN) to distinguish a small galaxy from an identical larger, slightly more distant galaxy if both happen to share the same pixel size. When it comes to applying CNNs, what ultimately does matter is that the input format (i.e., the input range

¹InputCatGAMADR3, available at <https://datacentral.org.au/>

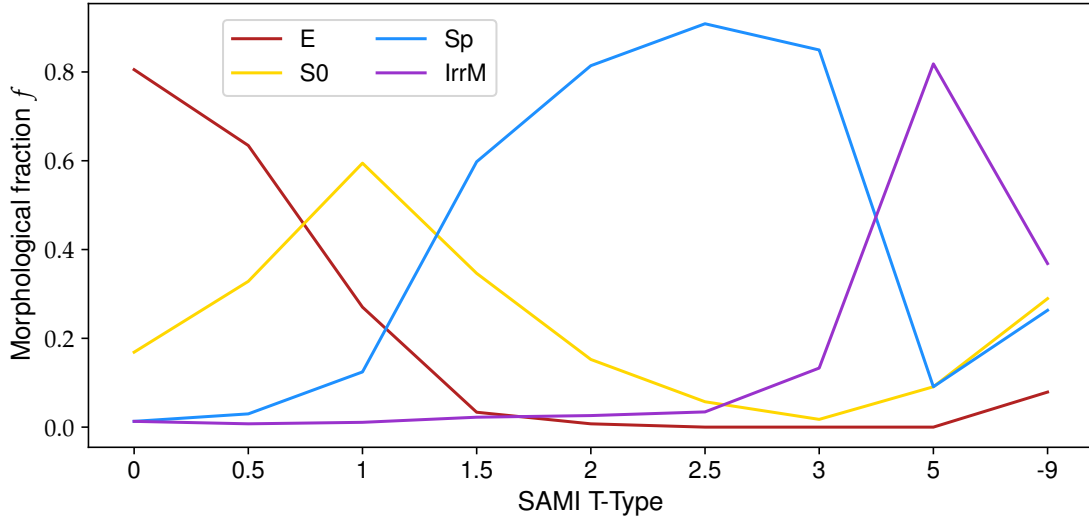


Figure 6.1: The morphological fractions of our predicted 4-class morphologies in each of the known SAMI T-Types. The SAMI T-Types are defined as 0 = E, 0.5 = E/S0, 1 = S0, 1.5 = S0/Early Spiral, 2 = Early Spiral, 2.5 = Early/Late Spiral, 3 = Late Spiral, 5 = Indeterminate, and -9 = No Agreement.

and input size) remains consistent with that used to train the original model. With that in mind, we resize all the HSC images to a pixel size of 100×100 , and apply the same normalisation as described in Section 2.4.1. A direct application is the best way to measure the real-world performance of our deep learning models, particularly in their ability to generalise. All images were classified by the same 4-class model developed and used in Chapter 3 (i.e. the C2 model), as well as the bar model developed and used in Chapter 4. Importantly, using the C2 model allows us to directly compare the SAMI results with results on the NA10 test set.

6.1.3 Comparing Classification Schemes

To see how our deep learning models perform, we compare our model’s predicted morphologies with the known SAMI morphologies, based on the SAMI T-Types. These T-Types are obtained from the VisualMorphologyDR3 catalogue from SAMI DR3 (Croom et al., 2021a), and the exact visual classification procedure is described in Cortese et al. (2016). Note that this catalogue contains classifications for only 3,068 galaxies. Although we classified all 5,536 galaxies from the input GAMA catalogue, we restrict our T-Type comparison analysis to only these 3,068 galaxies.

Figure 6.1 illustrates our morphology comparison, where we can see the change in the fractions of our predicted 4-class morphologies as we progress through the various SAMI T-Types. The trends in the predicted fractions agree well with the SAMI T-Types, with the highest fraction of ellipticals coinciding with SAMI T-Type 0, lenticulars with SAMI T-Type 1, spirals with SAMI T-Type 2.5, and irregular/miscellaneous with SAMI T-Type 5, a class that includes interacting galaxies, or galaxies that otherwise do not fit into any of the other T-Types. The fraction of spirals is slightly higher for the T-Types associated with late-type spirals compared to early-type spirals. It is interesting to note how the predicted fractions all collapse to below 40% for the No Agreement category (-9), with the deep learning model predicting a mix of different morphologies. This is to be expected given that this class contains galaxies

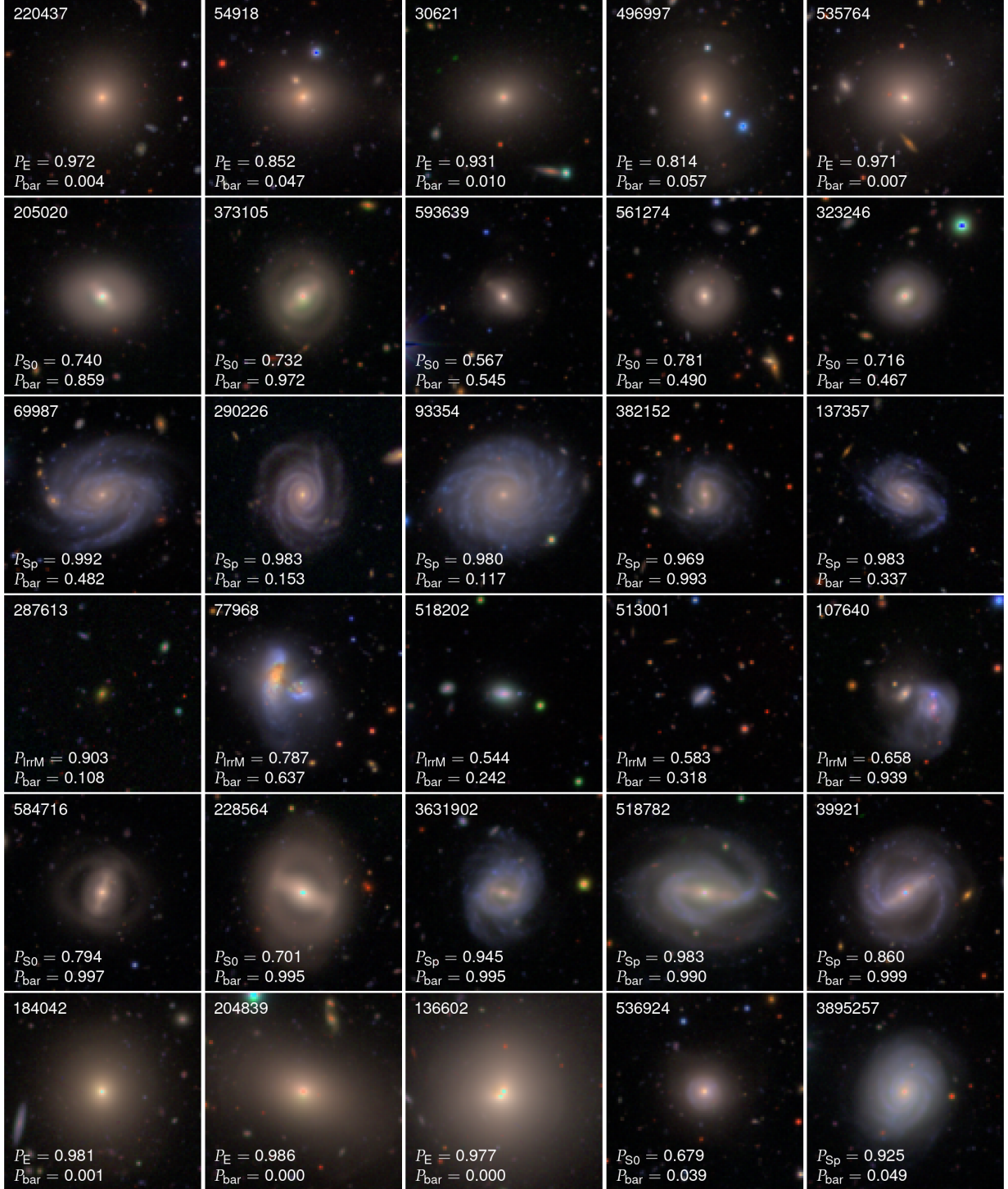


Figure 6.2: Random selection of galaxies classified by both our 4-class and bar deep learning models. From top to bottom, each row shows high-confidence examples of ellipticals, lenticulars, spirals, irregular/miscellaneous, along with highly-confident barred galaxies ($P_{\text{bar}} > 0.99$) and unbarred galaxies ($P_{\text{bar}} \approx 0$ for ellipticals, $P_{\text{bar}} \lesssim 0.05$ for S0 and Sp). Images are annotated with the SAMI Catalogue ID (CATID) in the top-left, with the confidence of the predicted morphological type and bar classification confidence shown in the bottom left. 100×100 pixel RGB cutouts are used for illustration.

where there was strong disagreement among the visual classifiers (see Cortese et al. 2016 for further details).

6.1.4 Morphological Classifications

Figure 6.2 shows a selection of galaxies classified into each of the four morphological classes, in addition to a selection of both highly-confident barred galaxies and highly-confident unbarred galaxies. Here we have annotated each image with both the confidence of its predicted 4-way morphology, as well as the classification confidence P_{bar} that the galaxy is barred. As in Chapter 4, a galaxy is defined as barred if $P_{\text{bar}} > 0.5$. As expected, ellipticals are predominantly featureless spheroids, while S0s exhibit a relatively clear disc structure. The first three example S0s are classified as barred; the first two with relatively high confidence, the third with fairly less confidence. The remaining two examples are on the cusp of being classified as barred, though this is likely as a result of the prominent bulges. Figure 6.2 contains two particularly noteworthy bar misclassifications, each involving galaxies that the 4-class model classified as irregular/miscellaneous. The first is galaxy 77963, which has been erroneously classified as barred, likely due to the long, thin shape of the interacting galaxy. The second galaxy, 107640, has also been incorrectly classified as barred with a very high confidence (possibly through misinterpreting the central bulge of the leftmost galaxy). Near-certain unbarred galaxies ($P_{\text{bar}} \approx 0$) are exclusively ellipticals, however Figure 6.2 also shows an example unbarred S0 and Sp, though we note the minimum P_{bar} for these classes is slightly higher at around 0.04 to 0.05.

It is also worth noting that the improved resolution of the HSC imaging compared to the SDSS imaging means that internal morphological substructures, such as bars, ansae and rings, are able to be resolved in much greater detail. However, some HSC images contain severe image artefacts and/or other distortions. It is prudent to highlight some of these degraded images, as this provides a good test of the robustness of our deep learning models on what are effectively edge cases that the models have not been explicitly trained to handle. Figure 6.3 shows a selection of images with severe visual artefacts, banding and/or

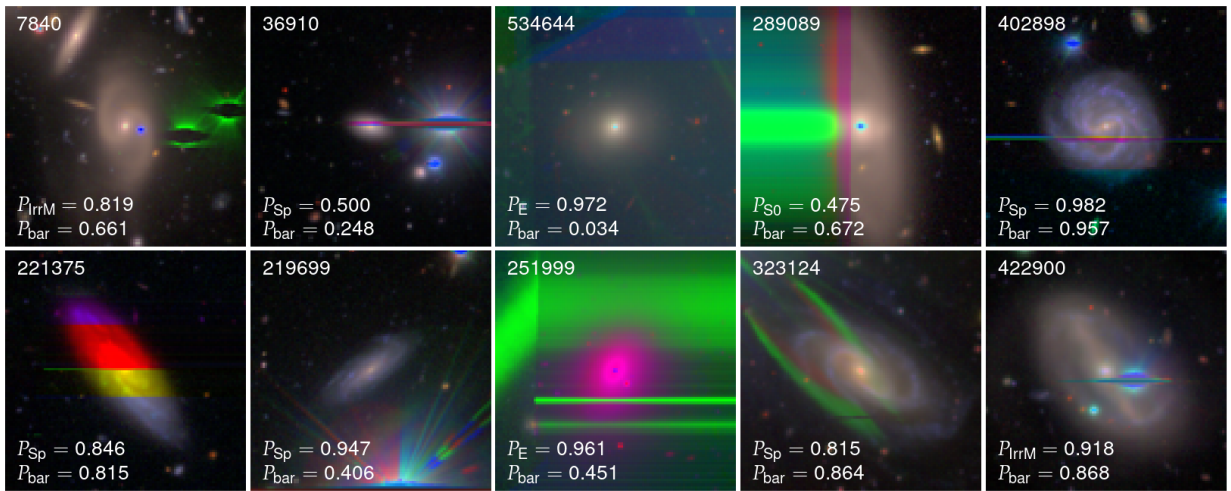


Figure 6.3: Selection of galaxy images with severe visual artefacts, banding or interference that have been classified by both our 4-class and bar deep learning models. Annotations are the same as in Figure 6.2.

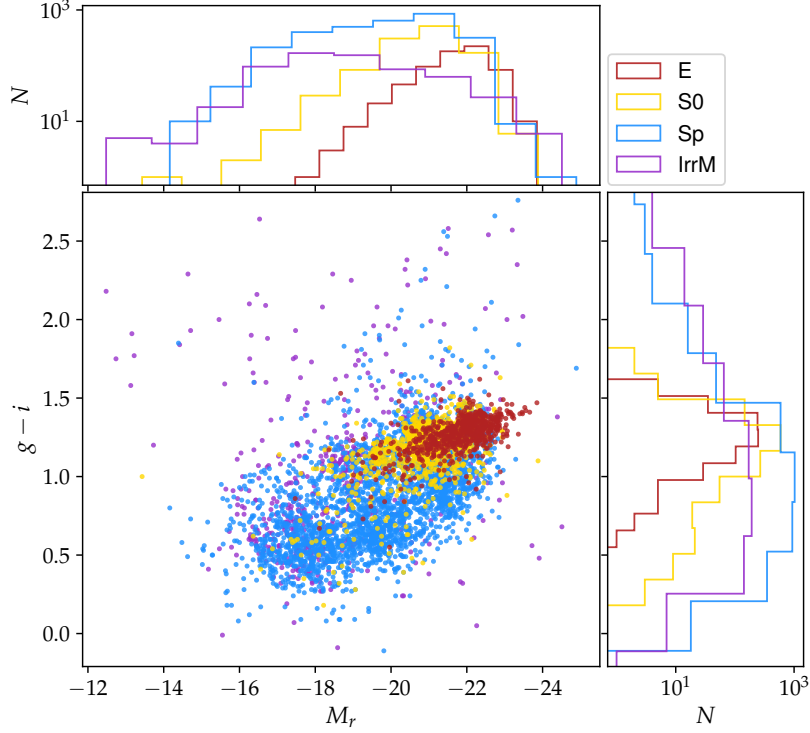


Figure 6.4: $g - i$ colour plotted against absolute magnitude M_r for SAMI galaxies from the input GAMA catalogue. Samples are shaded according to their predicted morphology, as classified by the 4-class model.

distortion, along with their classifications from both our 4-class and bar models. We note that some of the SDSS training images also contain visual artefacts, but none at this level of severity. In some cases, the morphological is unaffected and otherwise sensible. In cases where artefacts obscure large portions of the galaxy (36910 and 289089 in Figure 6.3), the model returns an irregular/miscellaneous classification. The bar model also appears to be influenced by the visual artefacts, particularly in the case of galaxy 25199 where the galaxy itself is completely spherical. Images such as these will no doubt present issues for visual classifiers let alone deep learning models. That said, the deep learning models are able to retain reasonable classifications in all but the most extreme of artefacts.

As a final check to ensure that our classifications are sensible, we can also examine the physical properties of our predicted morphologies. Figure 6.4 shows a colour-magnitude diagram. The individual morphologies are fairly well separated and spread out in a manner resembling bands. Ellipticals are mostly red and highly luminous, and cluster tightly together. Lenticulars are more spread out and somewhat less luminous than ellipticals, but also have a higher spread in colour. The majority of samples are classified as spiral, which dominate at bluer colours $g - i < 1$. Irregular/miscellaneous samples encompass the widest range of values, but they peak at lower luminosities than the other three classes. Our models' predicted morphologies are, for the most part, physically sensible. Coupled with the good level of agreement with the completely independent SAMI T-Type classification schema, it is clear that our deep learning models are able to perform well in a direct application. This serves as a testament to the inherent versatility of convolution neural networks and, on a much broader level, the fundamental ability for deep learning models to generalise.

6.1.5 Differentiating Morphology with Kinematics

As has been previously discussed, the spin parameter proxy λ_{R_e} is a key kinematic property of galaxies, which can be used to distinguish between different morphologies. This was demonstrated in [van de Sande et al. \(2021\)](#) with the existing, numerical SAMI T-Types from [Cortese et al. \(2016\)](#). We can conduct a similar analysis based on our deep learning classifications, allowing us to study the kinematic differences between ellipticals, lenticulars, spirals and irregular/miscellaneous galaxies. We do this by analysing λ_{R_e} in terms of both ellipticity ϵ and velocity dispersion σ using the samIDR3Stelkin catalogue (described in [van de Sande et al. 2017](#)) from SAMI DR3 ([Croom et al., 2021a](#)). The results are shown in Figure 6.5. In the plot of λ_{R_e} in terms of ϵ , we can see that ellipticals primarily occupy the bottom-left corner with ellipticities ϵ almost all below 0.5. This lack of ordered rotation is consistent with a merger-driven evolutionary history. While most ellipticals have very low spin parameters λ_{R_e} , there is nevertheless a degree of scatter with some ellipticals exhibiting λ_{R_e} values over 0.5, though it is possible that these outliers are simply misclassified lenticulars. Taking the definition of slow rotators to be galaxies with $\lambda_{R_e} < 0.1$ (as originally used in [Emsellem et al. 2007](#)), we can see that ellipticals are by far the dominant morphological type, along with several lenticulars and a handful of spirals. This is to be expected given that slow rotators are known to comprise around 14% to 20% of the total early-type galaxy population ([Khochfar et al., 2011](#); [Brough et al., 2017](#)). The fraction of slow rotators is also known to increase strongly with increasing stellar mass ([Emsellem et al., 2011](#); [Lagos et al., 2018b](#)). Figure 6.5 also shows that S0 galaxies exhibit a larger range of values, with a notably flat ϵ distribution, while also peaking at a slightly higher λ_{R_e} than ellipticals. Spirals generally have higher ellipticities and spin parameters. There are few irregular/miscellaneous galaxies, and these are clustered throughout the entire sample.

We can obtain an even clearer separation between morphological types by examining λ_{R_e} with respect to σ , as shown in the bottom panel of Figure 6.5. The predicted morphologies behave as expected, with ellipticals having higher velocity dispersions than spirals, with lenticulars located in between. There is less elliptical-spiral overlap with σ compared to ϵ , suggesting that for some fixed λ_{R_e} , σ is a more reliable indicator of morphology. Also notable is the more well-defined boundary between the populations of spirals and lenticulars in the λ_{R_e}, σ plot compared to the λ_{R_e}, ϵ plot. We note that there are significantly fewer irregular/miscellaneous samples in Figure 6.5 compared to Figure 6.4. This is likely a consequence of the selection criteria for the kinematic catalogue; over 1,000 of the SAMI galaxies with kinematic maps were excluded as their effective radius was below the minimum required to ensure accurate measurements ([van de Sande et al., 2017, 2021](#)), therefore excluding many irregular galaxies.

As discussed in Chapter 5, there are two main pathways for the formation of lenticulars: faded spirals, and mergers/accretion. These scenarios should result in S0s with properties that resemble spirals and ellipticals respectively ([Rizzo et al., 2018](#)). However, in terms of kinematic properties, it can be argued from the results of Figure 6.5 that S0s constitute their own distinct population. Our result showing that S0s have lower λ_{R_e} than spirals is consistent with the previous analysis by [Croom et al. \(2021b\)](#), however [Croom et al. \(2021b\)](#) also found that disc fading alone is insufficient to completely account for this difference. Indeed, previous studies examining S0s have concluded that they cannot be accounted for through any one pathway alone, and are more consistent with a mix of multiple different processes ([Fraser-McKelvie et al., 2018](#); [Deeley et al., 2020, 2021](#)).

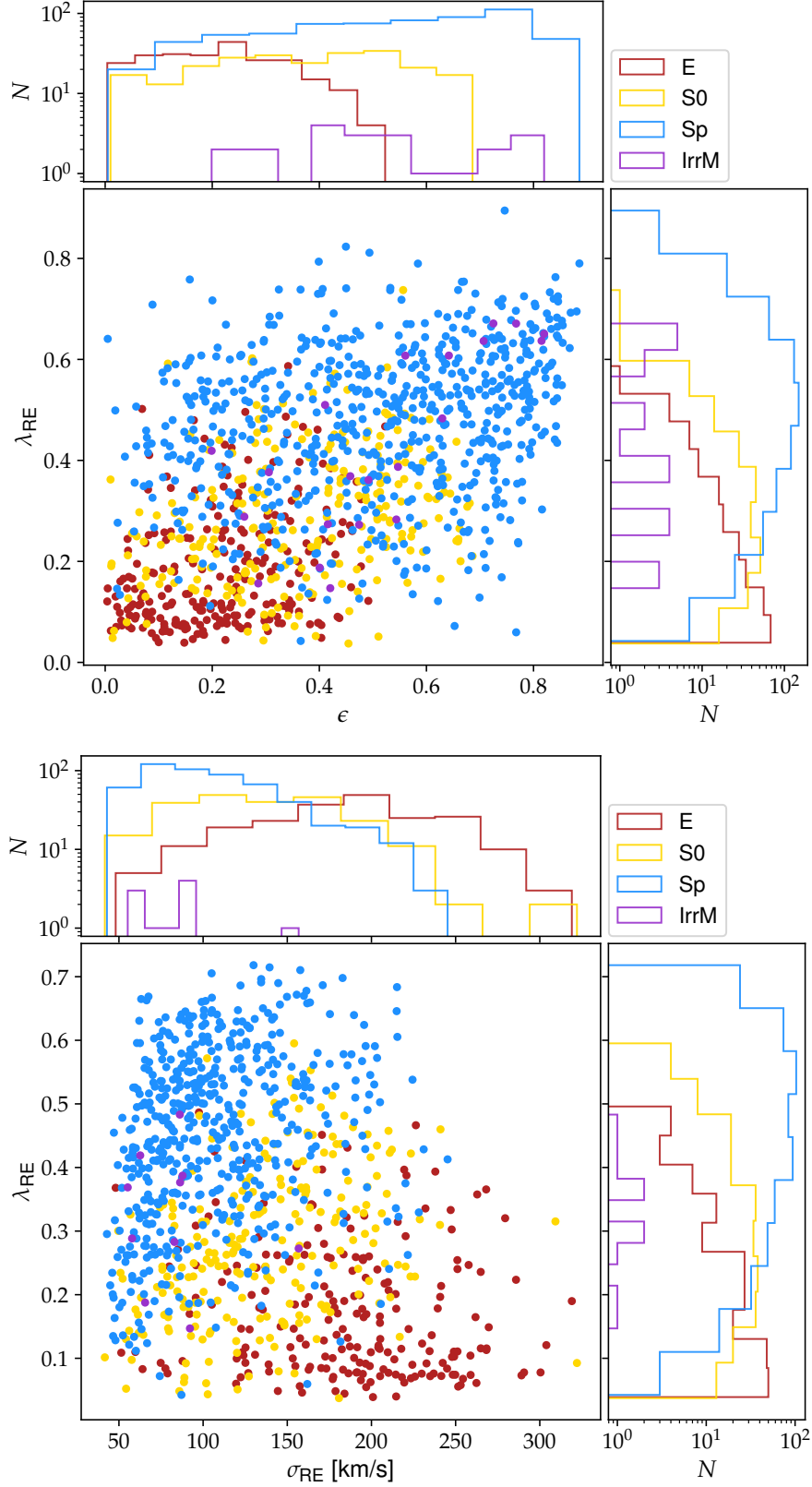


Figure 6.5: Scatter plots of the distribution of galaxies with spin parameter λ_{R_e} in terms of ellipticity ϵ (top) and velocity dispersion σ (bottom), coloured according to the predicted morphology as classified by our 4-class model. Note that the number of points is different in both plots (hence the slightly different histograms for λ_{R_e}); this is since values for ϵ and σ are cross-matched from different catalogues.

6.1.6 The Kinematics of Barred and Unbarred Galaxies

Having examined the kinematic differences between different galaxy morphologies, we now turn our attention to barred and unbarred galaxies. Figure 6.6 shows the same properties as in Figure 6.5, albeit all points are instead shaded by the classification confidence of being barred, P_{bar} , as predicted by our deep learning bar model. Here, barred and unbarred galaxies are defined as $P_{\text{bar}} \geq 0.5$ and $P_{\text{bar}} < 0.5$ respectively. We see that the number of bars depends strongly on ellipticity, with the highest numbers between $0.3 < \epsilon < 0.5$. Slow rotators ($\lambda_{R_e} < 0.1$) are predominantly unbarred; especially at low ellipticities ϵ and high velocity dispersion σ . Although bars are also most common in galaxies with lower σ , there are also many low- σ unbarred galaxies, implying that random motion is not the only way to suppress bar formation. As σ increases, the number of bars begins to depend more strongly on λ_{R_e} . For instance, at $\sigma \approx 200 \text{ km s}^{-1}$ almost all galaxies with $\lambda_{R_e} < 0.3$ are unbarred, while more galaxies with higher spin parameters are barred. This is consistent with the results of Figure 6.5, in which we observed a strong morphological dependence with increasing λ_{R_e} for galaxies with $150 \text{ km s}^{-1} \lesssim \sigma \lesssim 200 \text{ km s}^{-1}$ (transitioning from E to S0 to spiral). Comparing Figures 6.5 and 6.6 more directly, we can see that the bars are mostly present in spiral and lenticular galaxies, though there are several outliers such as barred ellipticals, which are likely the result of misclassifications. In addition to the predicted morphologies from our deep learning models, we can further examine the bar fractions across the known SAMI T-Types (as per Cortese et al. 2016). In particular, we obtain bar fractions of 17% for E/S0, 14% for S0s, 29% for S0/Sp, 64% for early spiral, 28% for early/late spiral and 58% for late spiral. For known SAMI ellipticals the bar fraction is less than 3%. These results compare well with previous studies that show bars are most common in early type spirals (Masters et al., 2011; Skibba et al., 2012; Vera et al., 2016), but we also detect a substantial fraction of bars in late-type spirals (see also Erwin 2018; Tawfeek et al. 2022).

If we treat P_{bar} as a proxy for bar strength (as was previously argued in Section 4.4.4), then the results of Figure 6.6 suggest that bar strength does not appear to vary significantly with λ_{R_e} ; high-confidence bars are common at both low- λ_{R_e} and high- λ_{R_e} regimes. Even though strong bars may be present among low λ_{R_e} galaxies, it is important to note that the vast majority are unbarred. In general, strong bars are also mostly confined to $\epsilon < 0.6$ and low σ , but there is a high degree of scatter within the overall barred population. Previous studies have established that strong bars tend to be preferentially found in early types (Erwin, 2005; Skibba et al., 2012; Vera et al., 2016), however this does not preclude the presence of strong bars in late type spirals (Erwin, 2018; Tawfeek et al., 2022). We again emphasise P_{bar} as a proxy for bar strength, rather than an absolute, objective measure. We especially note that there are many other aspects regarding the nature of bar strength (García-Gómez et al., 2017; Guo et al., 2019; Geron et al., 2021), as well as other factors that affect the outputs of the bar CNN itself. For example, we cannot rule out a high P_{bar} galaxy as simply being a galaxy that closely resembles one of the galaxies in the original bar model training set, rather than a galaxy that is inherently strongly barred.

There is yet to be a large-scale, systematic study of barred and unbarred galaxies in SAMI, let alone lenticulars. Examining both Figures 6.5 and 6.6 and inspecting the population of lenticulars, we can see that the barred lenticulars are mostly located in the spiral-to-lenticular transition region. Thus, focusing purely on kinematic properties, bars are more prevalent in more rotationally-supported (high λ_{R_e} , low σ) lenticulars compared to dispersion-supported (low λ_{R_e} , high σ) lenticulars. This particular distinction

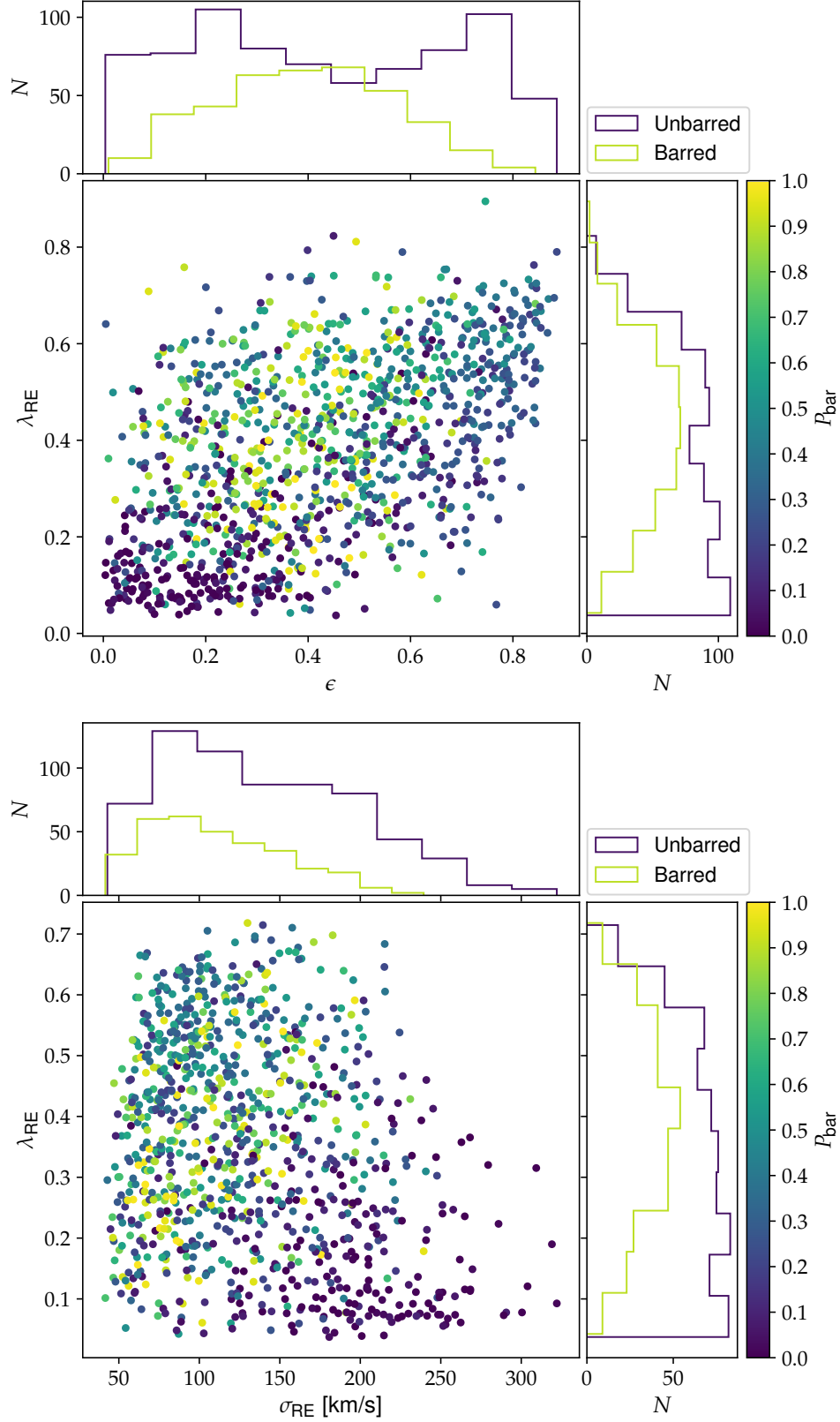


Figure 6.6: Scatter plots of the distribution of galaxies with spin parameter λ_{RE} in terms of ellipticity ϵ (top) and velocity dispersion σ (bottom). Points are coloured according to P_{bar} ; the confidence that a given sample is barred as classified by our bar deep learning model. For the histograms, barred and unbarred denote samples with $P_{\text{bar}} \geq 0.5$ and $P_{\text{bar}} < 0.5$ respectively.

has important ramifications for theories of bar formation in galaxies with significant bulges (c.f. [Kataria & Das 2018](#); [Kataria et al. 2020](#)). This is since galaxy bulges – key structural components of lenticular galaxies – are known to suppress the formation of stellar bars, especially those formed in isolation via secular evolution ([Kormendy & Kennicutt, 2004](#); [Sellwood, 2014](#)).

6.2 Morphological Segmentation

Deep learning techniques can be applied to variety of tasks, including classification, regression and clustering (see [Baron 2019](#); [Fluke & Jacobs 2020](#); [Huertas-Company & Lanusse 2023](#) for reviews on the diverse applications of deep learning in astronomy). One common task is that of image segmentation; that is, deconstructing an image into its constituent features that belong to certain object categories (this is also referred to as pixel-level classification). Common present-day applications that are heavily reliant on image segmentation include robotics, industrial automation, self-driving cars, medical imaging and mass surveillance (see [Minaee et al. 2020](#) for a recent survey). Studies in astronomy have also applied image segmentation to galaxies in order to study their structural components ([Hausen & Robertson, 2020](#); [Robertson et al., 2022](#)). In this section, we extend our existing suite of deep learning models by developing a specialised model capable of morphological segmentation: a U-Net. We use this U-Net to extract pixel masks corresponding to the locations of stellar bars and spiral arms in colour images of galaxies. Using these masks, we can estimate the lengths of the bars.

6.2.1 The U-Net Architecture

U-Nets are one of the most common deep learning models used to carry out image segmentation, having been initially conceived for use with biomedical imaging ([Ronneberger et al., 2015](#)). The term U-Net is used in a general sense to refer to deep learning models that all share a common layout and design, but there exist many sub-variants ([Minaee et al., 2020](#)). The classical architecture is based on a regular albeit truncated convolutional neural network. Starting from the input, the feature maps are first downscaled using the regular combination of alternating convolutional and pooling layers, until the features maps flattened and subsequently followed by one or more regular dense, fully-connected layers. This is the typical structure of a CNN as explored in Section 2.2; what follows ought to be the output layer. In a U-Net however, what instead follows is a series of alternating convolutional and *upsampling* layers. These layers are effectively a mirror image (in reverse) of the initial convolutional neural network. Instead of being downsampled, the feature maps are progressively upsampled such that final output ultimately consists of an image (or images) with the same width and height as the input image. The dense layers at the middle of a U-Net form a deliberate bottleneck, forcing the network to learn a compressed representation of the input data, which it then utilises to progressively construct the output image(s). Another defining feature of U-Nets is their use of residual connections (a.k.a. skip or shortcut connections). Residual connections are used to preserve a copy of a given layer’s output (referred to as identity mapping), which can later be added to the output of another layer further along the network. These shortcut connections are extremely valuable for they enable the construction of extremely deep neural networks ([He et al., 2015](#)). The use of identity mapping is also important for U-Nets as it helps to constrain the upsampling process.

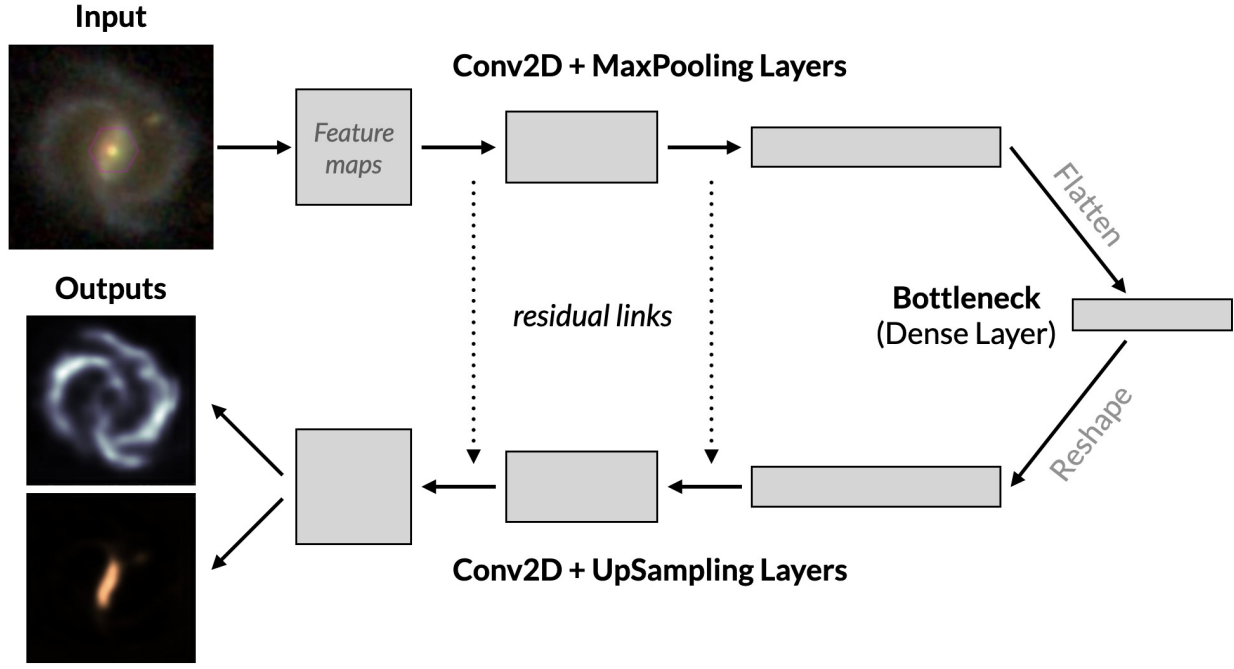


Figure 6.7: Diagram of a typical U-Net architecture. The first half consists of convolutional and max pooling (downsampling) layers; the second half is a mirror image but this time using upsampling layers. At the middle lies a forced bottleneck composed of dense layers. Note that the feature maps are first flattened, then processed by the bottleneck, and then the dense layer outputs are reshaped back into feature maps. Residual connections are used to connect feature maps from both halves together. In this example, the input is an RGB image of a galaxy, and the U-Net outputs two (monochrome) images corresponding to the pixel masks for spirals and bars (these are coloured for illustrative purposes using the bone and copper colormaps from matplotlib respectively).

Figure 6.7 displays a schematic of a simplified U-Net architecture. Our actual U-Net contains more layers than that shown in Figure 6.7, but nevertheless follows a very similar structure of alternating convolutions and max pooling, followed by a bottleneck dense layer, and then a series of alternating convolutions and upsampling layers. We also include two residual links, each connecting downsampled feature maps with their corresponding "mirror" feature maps on the upsampled half of the U-Net (as illustrated in Figure 6.7). The input to our model is a 128×128 pixel RGB image, and the output consists of two 128×128 pixel masks: one for bars, the other for spiral arms. We use this pixel size since all the feature maps are subsequently powers of 2, making it much easier to keep track of the downsampling and upsampling without extra padding or cropping.

There is one very importance difference between our U-Net compared to the typical U-Nets that are used for image segmentation. Such segmentation is usually a matter of classification; a U-Net decomposes an input image and outputs a label for each pixel corresponding to the category of the feature at that location (i.e. pixel-level classification). In our case – extracting pixel masks for stellar bars and spiral arms – the U-Net is not strictly being used purely for classification, but rather for regression. The Galaxy Zoo 3D masks are obtained through volunteers, each tasked with drawing pixel masks corresponding to the locations of bars and spirals. As such, there are invariably regions where the masks from multiple individual classifiers overlap. The degree of overlap can be considered as a proxy of confidence, and

therefore ought to be incorporated into the deep learning model. If the U-Net was configured solely for classification, then this information would be lost since the model would simply return a label for each pixel marked as a bar. Therefore, in order to properly account for overlapping regions, we set up the U-Net for regression. As such, we use a sigmoid activation function for the output layer, and define our loss function as the mean squared logarithmic error (MSLE). We found that using the MSLE as our loss function resulted in better convergence and stability than using mean squared error. Here the sigmoid activation function restricts the outputs to between 0 and 1. This is a different setup to the usual softmax activation and categorical cross-entropy loss otherwise used for the classification. Indeed, as our U-Net is performing a regression on the Galaxy Zoo 3D masks, the traditional notion of “accuracy” is meaningless. Instead, the key metric for training is strictly the MSLE loss.

6.2.2 Training the U-Net: Galaxy Zoo 3D

As with any deep learning model trained in a supervised manner, the training data must consist of inputs paired with desired outputs (denoted by labels). So far, our labels have simply been morphological classes. In order to train U-Nets, the labels must consist of entire images; in our case, pixel masks marking the locations of the bars and spirals. Fortunately, the recent Galaxy Zoo 3D survey (Masters et al., 2021) (hereafter GZ3D) provides such a dataset, with crowdsourced pixel masks for 29,831 galaxies sourced from the initial Mapping Nearby Galaxies at APO (MaNGA) survey target list (Bundy et al., 2015; Wake et al., 2017). All GZ3D data is publicly-available as a Value Added Catalogue in the SDSS Data Release 17 (Abdurro’uf et al., 2022). In addition to masks for bars and spirals, the complete GZ3D dataset also includes pixel masks for galaxy centres and foreground stars. These masks were obtained as part of the Zooniverse citizen science project, in which participants used a web browser interface to “paint” over images of galaxies in order to mark the locations and extent of the desired feature (bar, spiral, etc.). For the bars and spirals, the final masks are aggregated into “count masks” (Masters et al., 2021); that is, the values of each pixel denote the number of classifiers who included it as part of their bar/spiral mask. While such count masks are beneficial for conveying information about the confidence of certain masked regions (insofar as relatively more volunteer chose to highlight it), this does not come without its drawbacks. In particular, the number of volunteer classifiers for a given image vary widely; some with over 15 classifiers, some with only one. As such, one must define a confidence threshold to discard unreliable samples. In our case, we must also consider the conditions as imposed by the U-Net. As aforementioned in the previous section, we use the sigmoid activation function for our output layer, which allows us to directly incorporate the count masks. We also use the mean squared logarithmic error loss function, and hence our U-Net is trained via regression to approximate the values in the count masks as opposed to simply identifying bars and spiral regions. The mean squared logarithmic error loss function is also better suited to situations where there are large differences between values, such as in a typical GZ3D bar mask where only a tiny minority of pixels have nonzero values.

Unlike our deep learning models that have been developed prior to this point, our U-Net is designed to be trained on the RGB colour images provided as part of the GZ3D dataset, and is subsequently intended to be applied to colour images of galaxies. We select and prepare the training data for our U-Net as follows. We restrict our training data only to GZ3D galaxies whose bar or spiral masks have been drawn by at least three volunteer classifiers. This brings the total number of suitable galaxies down from the 29,831 total

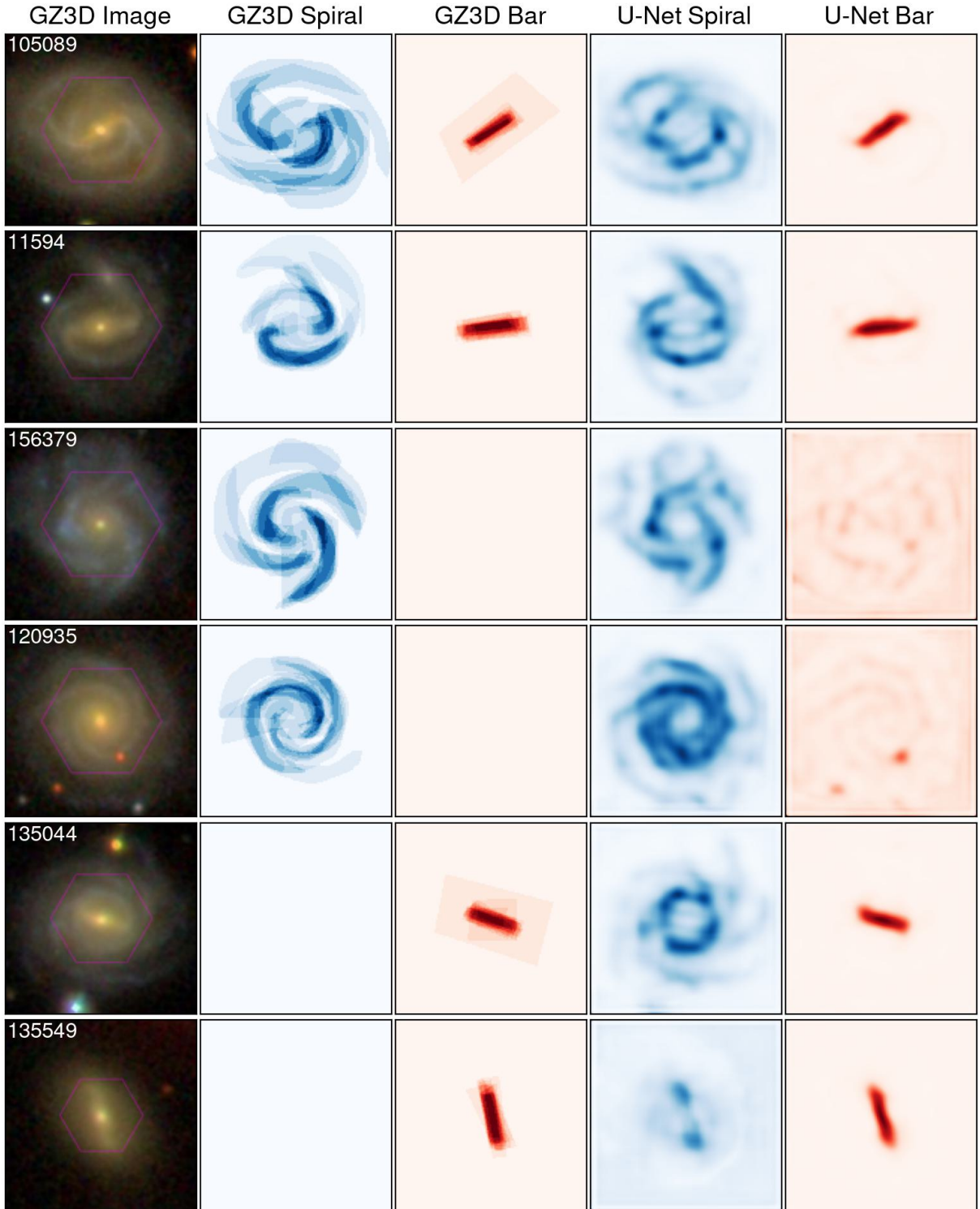


Figure 6.8: Application of the U-Net on a selection of galaxies from GZ3D test set. From left to right, the columns show the GZ3D SDSS image (annotated with the GZ3D ID), followed by the original GZ3D spiral and bar masks, and lastly the predicted spiral and bar masks correspond to the direct outputs of our U-Net. Spirals and bars are shaded with the Blues and Reds colourmaps from `matplotlib` respectively.

to 7,965 galaxies. Note that this includes galaxies with spiral masks but no bar masks (and vice-versa); the condition being that at least one of these masks has been drawn by a sufficiently high number of volunteers. The raw images have pixel sizes of 525×525 . We first apply a centre crop with a border width of 32 pixels; this serves to trim part of the background and better focus in on the galaxy, while subsequently enlarging the masks. We then resize the images along with their respective bar and spiral count masks to a final size of 128×128 pixels using Python’s `Pillow` package. Next, we rescale the count masks so that their pixels are all within the range 0 to 1 (this is to match the sigmoid activation used in the U-Net’s output layer). Rather than normalise each mask individually, we used a simple linear scaling across all masks. This preserves the relative volunteer counts across each image, hence allowing thresholds to be used after the model. The primary aim of training the U-Net is to recreate the GZ3D masks as accurately as possible; to that end we have kept the original count values as-is (except for the necessary resizing and rescaling). That said, it is possible that other, nonlinear scaling methods could be used to standardise the different volunteer counts, or alternatively make use of fractions rather than raw count values. These considerations will be a focus of future experimentation and refinement. We concatenate each spiral and bar mask together to form a single “target output” array of shape $128 \times 128 \times 2$; these will serve as the labels for training our deep learning model. We partition the images and labels into train and test sets using a ratio of 85:15, which was found to perform better than a ratio of 80:20. We implemented and trained the U-Net using the TensorFlow and Keras frameworks. Note that the mean squared logarithmic error is the sole metric used for the final evaluation.

6.2.3 Extracting Bars and Spiral Masks

Figure 6.8 illustrates the results of the trained U-Net when applied to some example galaxies from the GZ3D test set. In general, there is a fairly high level of agreement between the GZ3D masks and our U-Net’s predicted masks. In the spiral-only example, the U-Net nevertheless outputs a mask for the “bar” (and vice-versa for the bar-only example). However, the pixel values for these phantom masks are very low and can be readily removed by implementing a threshold. One major difference between the GZ3D masks and our predicted masks is that our predicted masks are significantly smoother, which is a consequence of both the convolutions and upsampling process. It can be argued that this smoothing is beneficial, in that the masks are able to better trace the shapes of the spirals compared to the polygonal shapes used throughout many of the GZ3D masks.

We have trained our U-Net using the known masks from the GZ3D dataset. We now explore the application of our U-Net to new images of galaxies. To test the U-Net’s versatility, we apply it to extract bar and spiral masks for galaxies from the NA10 and SAMI datasets. We use RGB colour images resized to 128×128 pixels, sourced from the latest data release of the SDSS via SkyServer and the HSC Public Data Release 3 respectively (courtesy of the NAOJ/HSC collaboration). In particular, this allows us to compare the U-Net’s performance when tasked to classify different imaging; SDSS imaging for NA10, HSC imaging for SAMI. Figure 6.9 illustrates the application of our U-Net to a random selection of known barred galaxies from NA10 (i.e. galaxies explicitly flagged as barred in the NA10 morphological catalogue), as well as a random selection of predicted bars in SAMI (based on our deep learning predictions in Section 6.1.6). Overall, there is a remarkable level of consistency given that both datasets use completely different imaging and subsequently exhibit different backgrounds and colours. The spiral masks also

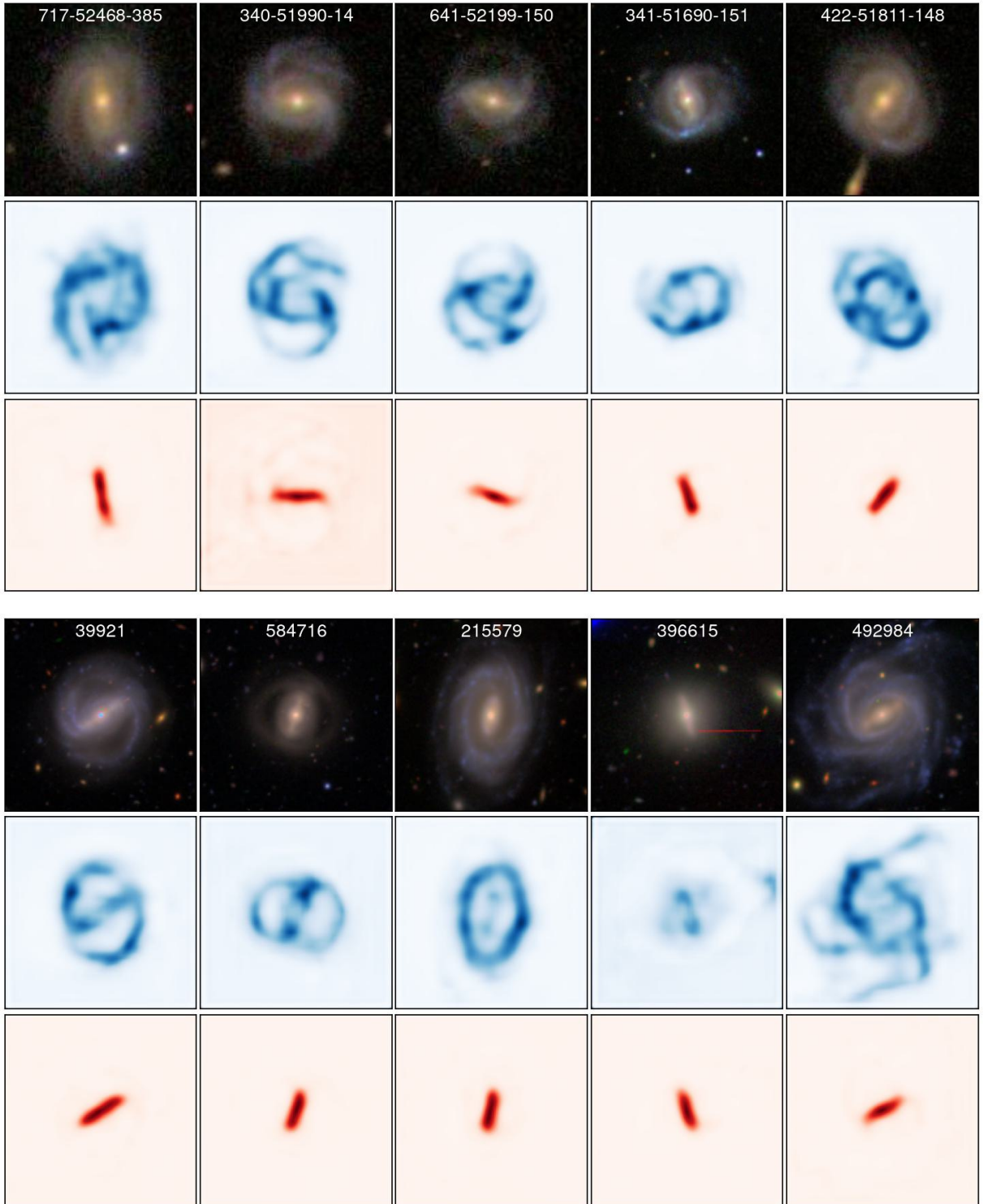


Figure 6.9: The application of our U-Net to a random selection of images of barred galaxies from the NA10 dataset (top three rows) and SAMI dataset (bottom three rows) respectively. Rows show the initial image, followed by the predicted spiral and bar masks as directly outputted by the U-Net. The example NA10 galaxies are annotated with their spID, while the SAMI galaxies are annotated with their catID.

manage to detect stellar rings, notably the faint ring accompanying the SAMI galaxy 584716. It is also encouraging that the bar masks are all centered and highlight a single region, despite some instances of visual interference such as in the NA10 galaxy 422-51811-148.

6.2.4 Automatic Bar Length Estimation

Having developed a U-Net capable of morphological segmentation, and having demonstrated its ability to extract accurate bar masks across multiple datasets with different imaging, we now turn our attention to analysing the physical lengths of stellar bars. Bar length, also denoted L_{bar} , is one of several phenomenological criteria often used to judge the “strength” of a stellar bar (Aguerri et al., 1998; Guo et al., 2019; Geron et al., 2021). Previous studies have shown that the bar length tends to scale with the mass of the host galaxy, and may also change depending on the morphology of the galaxy with larger bars in early-types and shorter bars in late-types (Erwin, 2005; Díaz-García et al., 2016; Erwin, 2019).

Traditional methods of estimating the lengths of stellar bars include ellipse fitting as well as the Fourier decomposition of the de-projected surface brightness profile (Ohta et al., 1990; Odewahn, 2004; Aguerri et al., 2009). Our U-Net offers a unique, deep learning approach for automatically estimating bar lengths requiring only the images of galaxies. We demonstrate this by applying our U-Net to extract bar masks for images of known barred galaxies from the NA10 dataset, as well as for barred from the SAMI dataset as predicted by our bar model in Section 6.1.6.

Our algorithmic approach to estimating bar lengths proceeds as follows. We first apply the U-Net to images of galaxies to extract the predicted bar masks. Recall that the U-Net was trained directly on the count masks, and so the total range of pixel values within a given mask will vary from galaxy to galaxy

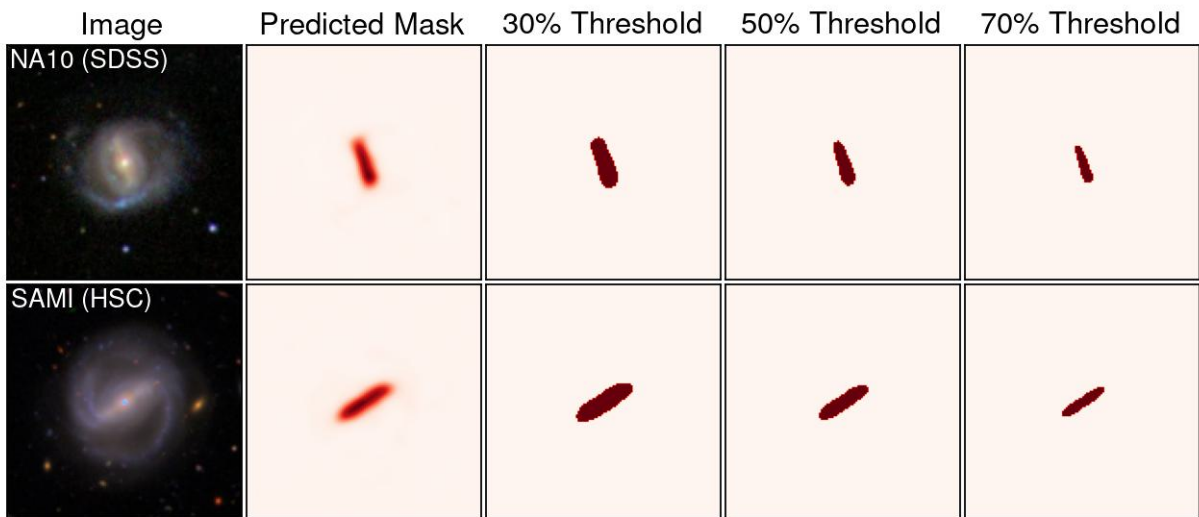


Figure 6.10: An illustration of the effect of applying different local thresholds to the pixel values of the output bar mask. The columns show the galaxy image, followed by its predicted bar mask as directly outputted by our U-Net. The next three columns show the resulting mask after applying thresholds of 30%, 50% and 70% (i.e. only pixels with values greater than 30%, 50% or 70% of the maximum pixel value are set to 1; the rest are set to 0). As the local threshold is fractional, it can be applied to any pixel mask regardless of its range of values.

(such as in spiral-only example galaxies in Figure 6.8). To account for this, we apply a *global threshold* in order to discard bar masks whose pixel values are too small. Specifically, a bar mask is chosen if its largest pixel is at least some fraction of the maximum pixel value across all bar masks for the given dataset. We set this global threshold at 10%.

Each bar mask is individually processed according to the following steps. We first apply an additional cutoff threshold to the predicted bar mask, converting it into binary integer mask. We refer to this second threshold as a *local threshold* in order to distinguish its role from the previous global threshold. The local threshold is instead defined as some fraction of the maximum pixel value in the current bar mask. In particular, we set the pixels equal to 0 if their value lies below this local threshold, and 1 otherwise. Using such a local threshold ensures that each bar mask will be treated the same, regardless of its range of pixel values. Figure 6.10 illustrates the impact of different local thresholds on an example SDSS image and HSC image from the NA10 and SAMI dataset respectively. Here, a threshold of 50% means that only pixels whose value is greater than half the value of the largest pixel (in the current mask) are set to 1, with all other pixels set to 0. We set the local threshold at 50%. Defining both these local and global thresholds as fractions allows our algorithm to automatically process bar masks for any arbitrary set of galaxy images, irrespective of the ranges of pixel values.

After processing the bar masks with the local threshold, we next utilise `scikit-image`'s `measure` module to automatically detect and measure the length of the bar ([van der Walt et al., 2014](#)). More specifically, we use the `label` function to detect and label connected pixel regions (i.e. neighbouring pixels that share the same value) in the resulting binary pixel bar mask. This is followed by the `regionprops` function, which enables us to subsequently estimate the bar length based on the length of the major axis of the fitted ellipse assigned to this connected region. This quantity is, of course, outputted in units of pixels.

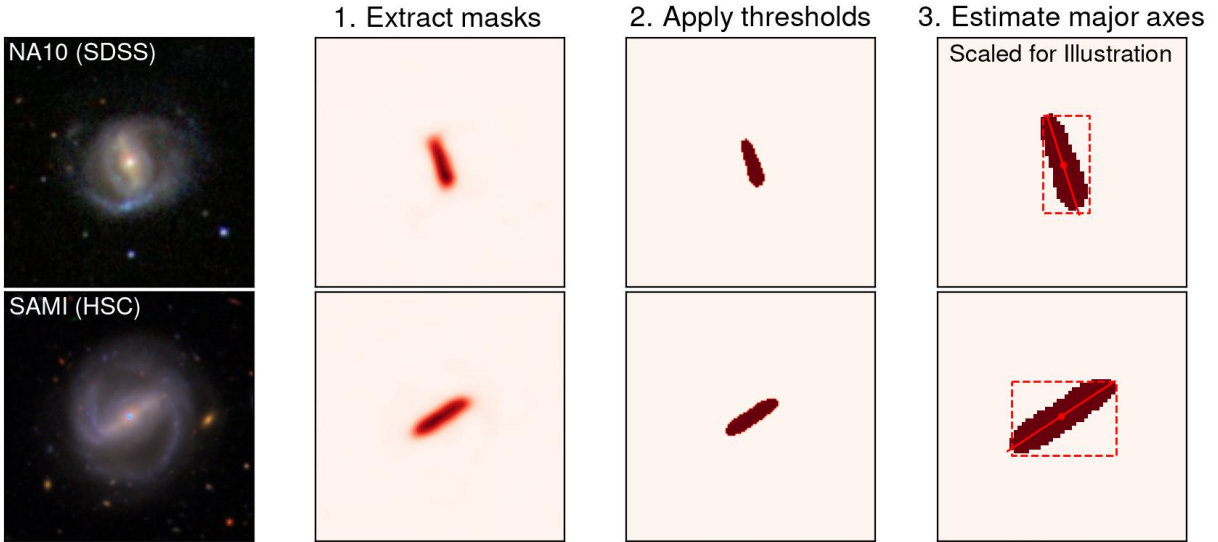


Figure 6.11: An overview of the steps involved in our approach to automatically estimate bar lengths. Columns show the original galaxy image, the U-Net's predicted bar mask, the binary integer mask after applying the local threshold, followed by a close-up plot of the bounding box and estimated major axes of the fitted ellipse using `scikit-image`'s `measure` module.

To convert it to a physical scale in kiloparsecs, we simply multiply the quantity by the per-pixel physical scale of the corresponding image. Figure 6.11 illustrates the full sequence of steps for two example barred galaxies.

It is possible that our U-Net returns multiple bar regions for a given image, such as through confusing a neighbouring galaxy or saturation artefact. This issue should, in theory, be avoided through using a suitably large local threshold. Nevertheless, we explicitly account for this possible edge case in our algorithm by returning the major axes of the largest connected region (on the presumption that this is most likely to be the actual bar). Based on our visual inspection of the U-Net’s performance on the GZ3D galaxies, the overwhelming majority of the predicted bar masks show just a single bar region, with the local threshold removing any extra “bars” in masks that contained them (usually cases with neighbouring galaxies).

Although configuring the U-Net for binary pixel classification would allow us to proceed directly to Step 3 in Figure 6.11, we note that this would still necessitate that a local threshold be applied to the training data, and subsequently prevent the U-Net from utilising the full gamut of information present in the count masks. Training the U-Net on the count masks thus allows us to define thresholds after its application, hence leveraging the different confidence levels inherent to the outputted pixel values. Either way, a threshold is inevitably required to define the “cutoff” of the bar in order to estimate its length.

NA10 (SDSS)



SAMI (HSC)

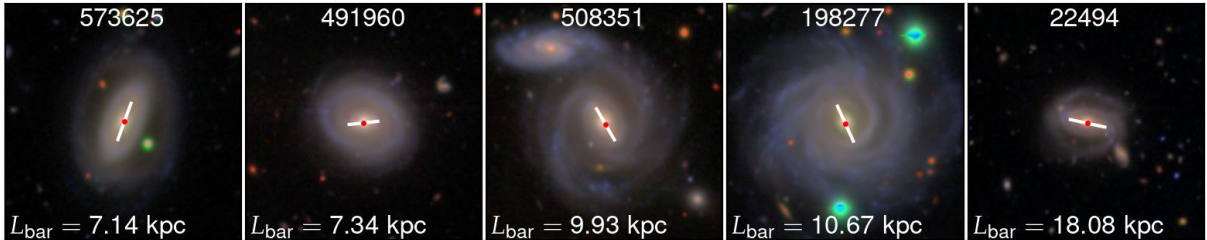


Figure 6.12: A random selection of example barred NA10 and SAMI galaxies ordered by their estimated bar length L_{bar} in kiloparsecs. The solid white lines and red dots denote the major axes and centres of the ellipses fitted to the predicted bar masks. The NA10 and SAMI galaxies are annotated with their spID and catID respectively. Note the SAMI images have variable physical pixel scales, while the NA10 images instead use a fixed physical pixel scale.

6.2.5 Bar Lengths in NA10 and SAMI

With both our U-Net and automatic bar length estimation algorithm implemented, we are in a position to examine the bar lengths of galaxies in both the NA10 and SAMI datasets, and establish its relation with various physical properties. We apply our U-Net only to galaxies from both datasets that have been previously classified as barred by our deep learning bar model (i.e. $P_{\text{bar}} \geq 0.5$). We note that the SDSS cutouts all share the same physical scale of 50kpc, while the physical scale varies for the HSC imaging (these images instead share the same pixel size). As such, to estimate the bar lengths in kiloparsecs we used *astropy* to calculate the angular diameter at the galaxy’s given spectroscopic redshift assuming a *Planck18* cosmology (Planck Collaboration et al., 2020), then applied the necessary unit conversions based on the fixed 0.168 arcsecond per pixel resolution of the HSC. Figure 6.12 shows a random selection of NA10 galaxies and SAMI galaxies ordered by their estimated bar length, and also illustrates the major axis length and orientation of the bar mask as measured by *scikit-image*, as per the final step in Figure 6.11. Overall, the estimated extent of the stellar bar shows a good level of agreement when overlaid over the original images, especially with respect to orientation.

Previous studies have examined the absolute bar size in terms of either the semi-major axis or the full length L_{bar} (Erwin, 2018, 2019; Geron et al., 2021). However, L_{bar} by itself is not necessarily the ideal quantity to meaningfully study the lengths of bars across different galaxies with different physical properties. After all, a large bar in a small galaxy may share the same, absolute length as a small bar in a large galaxy, but they are clearly sized differently in relation to their host galaxy. To take this into consideration, we can define a relative, normalised bar length by considering the dimensionless quantity L_{bar}/R_d where R_d is the scale radius of the galaxy disk. Examining how such relative bar lengths change with physical properties can reveal important insights into the nature of stellar bars (Combes & Elmegreen, 1993; Gadotti & de Souza, 2006; Hoyle et al., 2011; Guo et al., 2019; Fraser-McKelvie et al., 2020a). Figure 6.13 shows a series of plots for the relative bar length L_{bar}/R_d with respect to stellar mass, the bar classification confidence P_{bar} , redshift, λ_{R_e} , as well as morphology in terms of the NA10 T-Types (themselves based on Hubble T-Types), as well as the SAMI T-Types from Cortese et al. (2016). It is worth noting that the NA10 and SAMI datasets utilise different selection criteria, and hence they are not directly comparable (especially with regards to redshift).

Starting with stellar mass, we can see that NA10 and SAMI exhibit opposite trends, with the relative bar length decreasing with increasing mass in the case of NA10, while for SAMI there is a slight positive trend (albeit with significant variation). Curiously, the plots for both datasets appear to exhibit a localised peak for galaxies in a similar mass range of $10 < \log(M_*/M_\odot) < 10.5$. At very high masses above $10^{11}M_\odot$, the NA10 bars are significantly shorter, while the SAMI bars are at their highest length. Absolute bar lengths are well known to scale with increasing luminosity and mass (Kormendy, 1979; Erwin, 2005; Díaz-García et al., 2016). Indeed, our bar estimation pipeline does predict increasing overall L_{bar} for increasing M_* in both the NA10 and SAMI cases, as shown in Figure 6.14. However, when we normalise these lengths with respect to the size of the galaxy disk as shown in Figure 6.13, we see that the changes are less obvious. Indeed, Kruk et al. (2018) find that the disc scale radius increases with stellar mass, which may suggest that as a galaxy grows, so too does that bar, albeit not necessarily with respect to its disc (see also Díaz-García et al. 2016; Fraser-McKelvie et al. 2020a; Rosas-Guevara et al. 2022). It is argued in Erwin

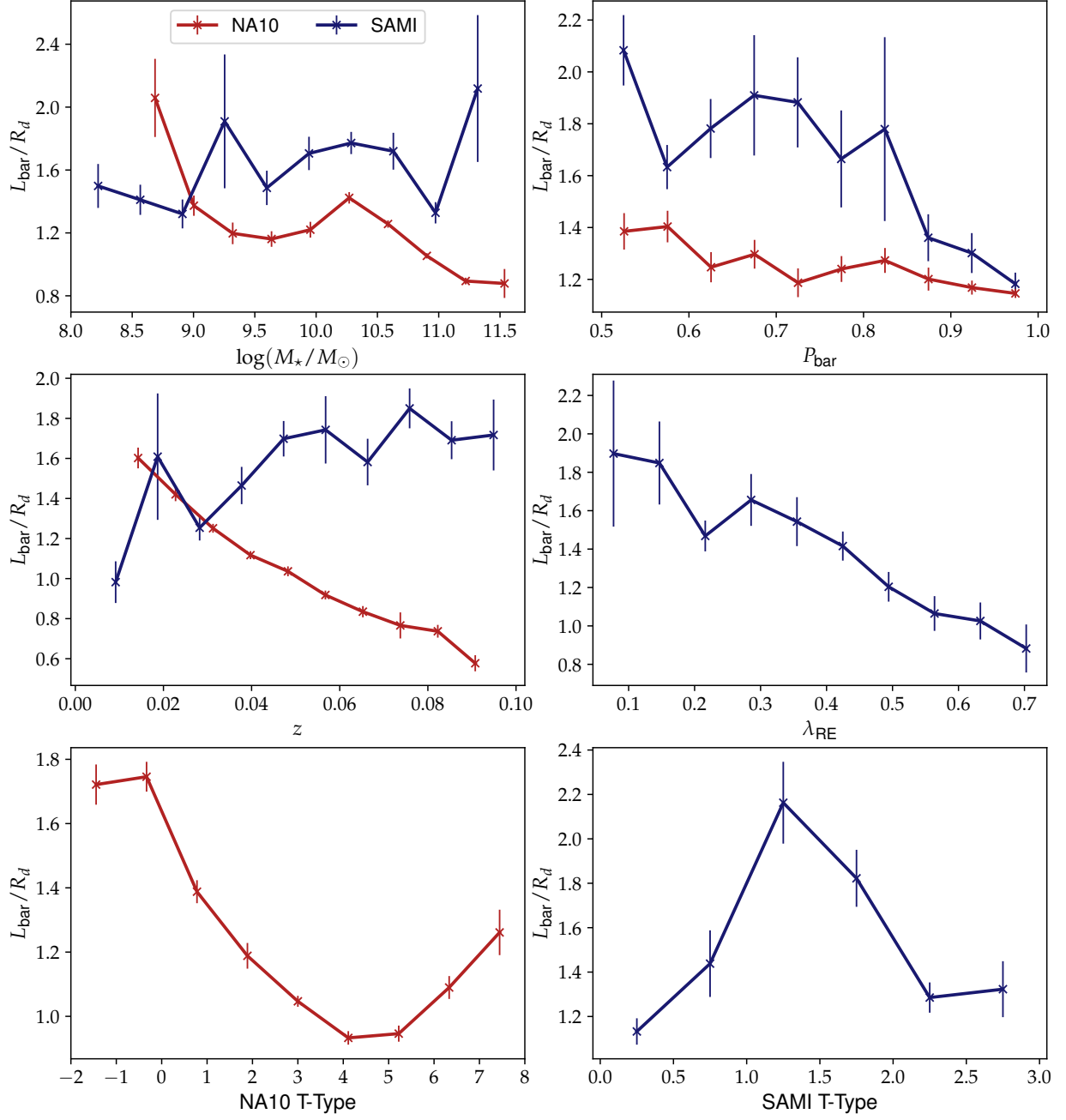


Figure 6.13: The bar length L_{bar}/R_d as functions of stellar mass, bar classification confidence P_{bar} , redshift z , the spin parameter proxy λ_{RE} , NA10 T-Types and SAMI T-Types. NA10 and SAMI galaxies are coloured in dark red and dark blue respectively. Error bars denote one standard error σ .

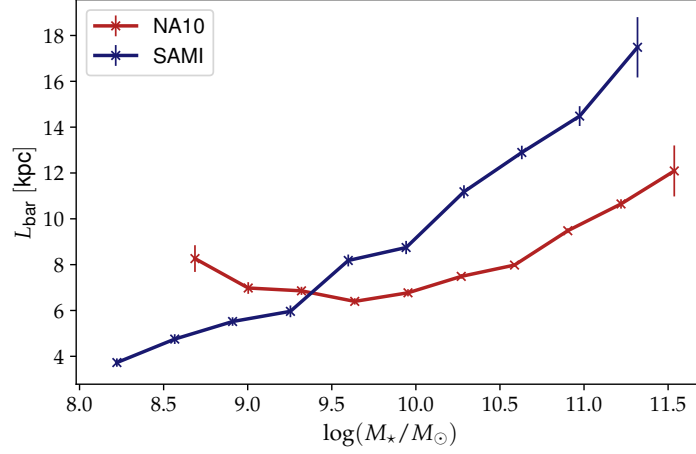


Figure 6.14: The absolute bar length L_{bar} as a function of stellar mass for the NA10 and SAMI galaxies, coloured dark red and dark blue respectively. Error bars denote one standard error σ .

(2019) that difficulties in detecting shorter bars may result in their underestimation in low mass galaxies, while Erwin (2018) makes the argument that limited image resolution also affects the ability to detect bars. This could be a reason as to why L_{bar}/R_d remains high for low mass galaxies in Figure 6.13, in the sense that the U-Net can only ever extract the largest bars while missing smaller bars.

Such considerations lead us to investigate whether the size of the bar L_{bar}/R_d changes with respect to the classification confidence P_{bar} , as based on our deep learning bar model. In both the NA10 and SAMI cases we see that the largest bars are all low-confidence barred galaxies, while higher-confidence barred galaxies exhibit shorter bars. We have referred in previous sections to P_{bar} as a potential proxy for bar strength, but it is first and foremost a measure of the CNN’s confidence. While confidence can be an indicator for bar strength (in a similar fashion to the Galaxy Zoo 2 bar vote fractions, as per Masters et al. 2012), it is unclear whether this can directly translate to bar length. In other words, longer bars are not necessarily stronger bars (Géron et al., 2021). The results in Figure 6.13 instead appear to have more to say regarding P_{bar} itself rather than bar strength. If we interpret P_{bar} as indicating how easy it is to visually identify a galaxy as barred, then it follows that bars in high P_{bar} galaxies are shorter than in low P_{bar} galaxies, since these bars are easier for the U-Net to subsequently extract. For low P_{bar} galaxies, the full extent of the bar may be less clear to distinguish, and hence the bar mask extracted by the U-Net could be larger than is actually the case, given the higher degree of uncertainty.

Figure 6.13 also shows L_{bar}/R_d in terms of redshift z and the spin parameter proxy λ_{R_e} . For NA10, we see a steady increase in relative bar length with decreasing redshift. For SAMI, we instead see an opposite and less dramatic trend, with bar sizes increasing with redshift up to $z \approx 0.05$, with the trend remaining relatively flat for redshifts thereafter. There are, of course, many factors that affect any potential changes in bar lengths with redshift; for instance, galaxies themselves evolve morphologically. Previous studies examining bars out to high redshifts have found that bar lengths (both absolute and normalised length) do not show any significant changes with redshift (Sheth et al., 2008; Kim et al., 2021), implying that bars scale proportionally with their host galaxy as both evolve over time. For the spin parameter proxy λ_{R_e} , we find that bars in low spin galaxies are longer than in high spin galaxies. This is consistent with the analysis

of bars based on the classical spin parameter in [Cervantes-Sodi et al. \(2013\)](#), in which low spin galaxies are more prone to self-interacting gravitational instabilities ([Combes & Sanders, 1981](#); [Sellwood & Wilkinson, 1993](#); [Athanasoula, 2003](#)), compared to the high ordered rotation found in galaxies with higher spin parameters. This dependence with λ_{R_e} can also be a consequence of morphology. As previously shown in [Figure 6.5](#), high λ_{R_e} galaxies are predominantly spirals, while low λ_{R_e} are dominated by lenticulars and ellipticals.

Concerning morphology, [Figure 6.13](#) shows the bar lengths as functions of two independent T-Type schema. For NA10, bars in late type galaxies are significantly shorter than bars in early type galaxies (with respect to the size of the galaxy), which is consistent with established trends from previous studies based on both N -body simulations and observations of nearby barred galaxies ([Elmegreen & Elmegreen, 1985](#); [Combes & Elmegreen, 1993](#); [Aguerri et al., 2009](#); [Erwin, 2019](#)). For SAMI we see a peak in bar length for S0 and S0/Sp T-Types, before dropping to less than half this value for the E/S0 morphological types, which could be due to the U-Net extracting bulges rather than bars. If we restrict the analysis strictly to T-Types including S0s and above then, as with NA10, the relative bar lengths are shorter in spirals and late-type galaxies. One explanation for late type galaxies having shorter bars is due to their higher gas content, which can potentially suppress bar formation and/or slow the rate of growth of the bar ([Bournaud et al., 2005](#); [Berentzen et al., 2007](#); [Athanasoula et al., 2013](#)). The previous study by [Hoyle et al. \(2011\)](#) found a similar dependence of bar size with colour, with redder, early-type galaxies hosting longer bars than bluer, late-type galaxies. [Erwin \(2005\)](#) also find bars are over twice as long in early-type galaxies as opposed to late-type discs.. It is worth noting that this bar size dependence also correlates with an increasing bar fraction in redder galaxies ([Masters et al., 2011](#); [Skibba et al., 2012](#); [Vera et al., 2016](#); [Cervantes Sodi, 2017](#)). However, the recent study by [Tawfeek et al. \(2022\)](#) is a notable exception having found a peak bar fraction for more massive, late-type galaxies (although this study did not examine bar size), while [Erwin \(2018\)](#) also finds that bars are similarly just as frequent in both gas-rich, bluer galaxies as gas-poor, redder galaxies.

Lastly, it is worth reiterating that this entire analysis is only made possible thanks to the deployment of a U-Net trained to extract bar pixel masks based solely on the image of a galaxy. In tandem with our existing suite of deep learning classifiers, in particular our bar model, we have successfully analysed bar masks and estimated the lengths of the stellar bars contained within them using an integrated and automated processing pipeline. There are several important caveats with this method. Firstly, the automatic bar estimation does rely on free parameters – the global and local thresholds – that need to be set. While we are confident that the values chosen in [Section 6.2.4](#) are suitable, other applications or datasets may warrant different threshold values. This further underscores the utility of having trained the U-Net on the GZ3D count masks rather than on masks subject pre-applied threshold, for this allows different thresholds to be applied after using the U-Net, ensuring the model’s versatility across arbitrary datasets. Another caveat is the ellipse fitting as conducted by `scikit-image`. Bars are ideally better suited to a more rectangular-weighted ellipse than a standard ellipse ([Athanasoula et al., 1990](#)), while `scikit-image` fits an ellipse with the same normalised second central moments as the identified pixel-connected region.

Conclusion

7.1 Thesis Summary

Morphology is a fundamental, innate property of galaxies, which is known to correlate with many important physical and kinematic properties, and also serves as an indicator of a galaxy’s evolutionary history. Understanding how the fraction of galaxies with different morphologies vary over cosmic timescales can offer crucial insights into the physical processes governing the formation and evolution of galaxies, ultimately enriching our understanding of the nature of the Universe itself. Traditional morphological classification via visual inspection is a laborious, time-consuming process that is ill-suited to the challenges posed by the sheer volume of data expected from future large-scale observational surveys. In this thesis, we leverage the power of modern deep learning to develop a suite of models capable of automatically classifying galaxies in an extremely rapid, consistent and accurate manner using only their images as inputs. We apply this suite of deep learning models to classify galaxies in several datasets from both cosmological simulations and observational surveys, and thus study the evolution of galaxy morphology. These deep learning models are inherently versatile and easily adaptable via transfer learning, and are well positioned to undertake future studies into galaxy morphology across a breadth of datasets.

A summary of each Chapter is provided below, with an emphasis on the key outcomes of this thesis that collectively represent a significant, original contribution to the wider fields of galaxy morphology and applied deep learning in astronomy.

- In Chapter 1, we introduced the fundamentals of galaxy morphology and morphological classification, with a focus on barred galaxies and lenticular galaxies. We discussed the limitations of traditional techniques for morphological classifications, especially their issues with scalability. This was followed by an overview of deep learning, where we outlined why it is well suited for the aims of our thesis, and provided an overview of its recent applications in astronomy. With this Chapter, we set the stage for the overarching focus of our thesis: the application of deep learning models to classify galaxy morphologies in large-scale datasets in order to study the evolution of galaxy morphology.
- In Chapter 2, we discussed the key theory, principles and methodology underpinning our deep learning approach. This included details on the typical structures of convolutional neural networks, along with details on how our specific C2 and C3 model architecture were optimised and refined

with hyperparameter tuning. We discussed how our models are trained, including our dataset partitioning, preprocessing and use of data augmentation. With the aid of visualisation techniques, we illustrated how convolutional neural networks process images galaxies, and how they represent morphology internally through visualising the latent space of our CNN model. We lastly discussed how neural networks can be adapted via transfer learning to classify different datasets, with a focus on images with dramatically different noise levels. The major outcomes of this Chapter included a thorough overview of our deep learning approach, an intuitive understanding of how convolutional neural networks work, and a detailed overview of the various methods and techniques used throughout all subsequent Chapters.

- In Chapter 3, we explored the development of our core deep learning models, based on a new convolutional neural network architecture capable of performing direct multi-class morphological classification into up to four classes. We trained and evaluated these models on single-band g-band images of galaxies with known morphologies from the NA10 morphological catalogue. We found that our new model architecture outperforms previous architectures, including some that were specifically designed for classifying galaxies. Our multi-class models are also especially advantageous over existing binary classifier models, which typically only classify galaxies into early and late types. Importantly, it is also demonstrated in this Chapter that our models, which analyse single-band, monochrome imaging, have comparable performance to previous studies which analysed colour images. This is a yet another crucial strength of our approach, for it enables our models to be applied to wider datasets where colour imaging is not readily available, such as high-redshift images. We successfully reproduced the properties of galaxies with known morphologies using the predicted morphologies from our models, demonstrating that the model's predictions are indeed physically sensible. We further assessed our model by applying it to an independent sample of galaxies from the Galaxy Zoo dataset.
- In Chapter 4, we applied our deep learning models to images of simulated galaxies from the EAGLE cosmological simulation. This included the development of an additional deep learning model capable of classifying galaxies as barred or unbarred, which we applied in order to study the evolution of barred galaxies across all EAGLE snapshots from redshift $z = 0$ to $z = 1$. This constitutes the first large-scale analysis of barred galaxies in EAGLE with deep learning, and analyses nearly a factor of 15 more galaxies than the sole previously known comparable study. We found that the bar fraction remains constant out to $z \approx 0.5$, before declining with redshift. Through classifying the morphologies of their host galaxies, we additionally found that bars are most prevalent in spiral galaxies. We analysed the properties of our classified barred galaxies, finding that barred lenticulars are more strongly disc dominated. Through tracking galaxies across multiple snapshots, we were able to detect events of bar creation, bar destruction and even bar regeneration, and we estimated the mean bar lifetime to be around 2.24 Gyr for galaxies hosting at least one bar throughout the snapshots. We determined that these rates of bar creation and destruction are not constant, instead falling over time with decreasing redshift. We also compared these predicted bar creation and destruction events with merger events, and found that major merging is more strongly aligned with incidences of bar destruction, while minor merging is more strongly aligned with bar creation.

- In Chapter 5, we explored the application of our deep learning models to classify images of galaxies from COSMOS. We developed a custom, specialised data augmentation routine in order to inject our existing NA10 training data with artificial noise such that the images resembled the characteristic noise levels of COSMOS. In this manner, we successfully adapted our models via transfer learning on this “noise-augmented” data, resulting in new, augmented models that are better suited to classifying noisy images. We also devised a new, more efficient model architecture specifically for this transfer learning application. We applied ensembles of these new augmented models to study the redshift evolution of the fraction of lenticular galaxies in COSMOS out to redshifts $z < 1$, and established that the fraction of lenticulars has increased over time with decreasing redshift. This constitutes the first systematic study of lenticular galaxies in the general field environment with deep learning. Furthermore, we determined that the rise in the S0 fraction depends on mass, with the S0 fraction rising at higher redshifts for high-mass galaxies, and for lower redshifts for low and intermediate-mass galaxies. This may be a consequence of different timescales for S0 formation processes. Through examining the properties of our classified S0s, we identified the presence of two subpopulations of S0s: a population of high-mass, red, quiescent S0s; and a population of comparatively less massive, bluer star-forming S0s. This latter population cannot be accounted for based solely on the “faded spiral” formation pathway. Hence, there must be other formation mechanisms at play for low-mass S0s that do not completely suppress star formation.
- In Chapter 6, we highlighted several additional applications of our suite of deep learning models. In the first major application, we classified the morphologies of galaxies from the SAMI DR3 dataset, and studied the kinematic properties of galaxies with different morphologies. Crucially, this study constitutes the first systematic analysis of the differences between barred and unbarred galaxies in SAMI. We also compared our predicted morphologies to the known SAMI T-Types, which use a completely independent classification system to the T-Types originally used to train our models, finding a high level of agreement. We showed that morphology is well separated when examining the spin parameter proxy λ_{R_e} as functions of both ellipticity ϵ and velocity dispersion σ . Having successfully applied our models directly to high-resolution HSC imaging of the SAMI galaxies, this Chapter also demonstrates the versatility of our deep learning approach.

Also in Chapter 6, we further extend our suite of deep learning models by developing a new model capable of image segmentation; a U-Net. In particular, we have demonstrated the ability of this U-Net to perform physically meaningful morphological segmentation of galaxies into masks corresponding to stellar bars and spiral arms. This culminating in the development of an automated algorithm to estimate the lengths of stellar bars, based on these predicted bar masks. We applied this U-Net to examine bar lengths in barred galaxies from the both the NA10 and SAMI datasets. This constitutes the first known usage of U-Nets for the purposes of studying stellar bars, and represents a novel, scalable technique for automatically studying barred galaxies.

7.2 Outlook and Future Work

Throughout this thesis, we have seen how our deep learning models can be applied in many settings, from simulations to observations, from low-redshift imaging to high-redshift imaging, and from classification to morphological segmentation. Indeed, deep learning techniques are inherently generic, and there is immense potential for its application to solve myriad problems throughout modern astronomy at large. This is especially pertinent with the advent of large-scale surveys and the subsequent challenges associated with extremely high quantities of data. In this section, we outline some of these potential applications and feasible future projects that directly follow from the foundations laid in this thesis.

7.2.1 The High-Redshift Universe

The advent of the James Webb Space Telescope offers an immense opportunity to study the morphological evolution of galaxies out to extremely high redshifts, offering unique insights into the physical processes governing the early universe. While it has long been thought that high-redshift galaxies are predominantly clumpy, preliminaries studies based on high-resolution JWST imaging has instead shown an incredibly diverse range of morphologies, including more regular morphological types compared to previous Hubble Space Telescope (HST) imaging (Ferreira et al., 2022a; Fudamoto et al., 2022; Kartaltepe et al., 2022; Jacobs et al., 2022; Robertson et al., 2022). Indeed, studying high-redshift galaxies is difficult due to the issues with resolution, image quality and galaxy evolution. While we have demonstrated in Chapter 5 that transfer learning can successfully enable models to classify much noisier images (compared to the images they were originally trained on), there is nevertheless an upper limit in redshift beyond which images are too degraded (especially with respect to COSMOS’s HST imaging). The vastly improved imaging capabilities of the JWST have pushed beyond the limits set by the HST, obtaining high-resolution images that are well suited for analysis with deep learning. However, there is still the issue of galaxy evolution. At very high redshifts, galaxies exhibit markedly different morphologies. There is hence a limit beyond which models trained on images of nearby galaxies with present-day morphologies are no longer suitable. One way to address this is to supplement the training data with labelled, high-redshift data, or simply to increase the diversity of morphologies present in the training set. This is important given that the success of deep learning models lie in their ability generalise to unseen data. Another method is to explicitly model evolution as a means of data augmentation by rendering images of present-day galaxies as they would have looked like at higher redshifts (see Domínguez Sánchez et al. 2019).

It is expected that with the advent of high-resolution galaxy imaging, more and more studies will utilise deep learning techniques. Our current suite of deep learning models developed and applied in this thesis are well positioned to capitalise on this trend. Furthermore, our models can be readily adapted or retrained with high-resolution imaging. We thus expect to continue applying our models not just to classify future JWST imaging, but also to classify high-resolution imaging from many other deep field observational surveys, such as the Dark Energy Camera Legacy Survey (DeCALs; Dey et al. 2019). We also expect our models to continue to be applied to current and future high-resolution simulations. Potential applications include continuing our analysis in Chapter 4 by potentially studying the morphological evolution of galaxies across multiple snapshots up to higher and higher redshifts.

7.2.2 Expanded Inputs: Multi-Band Imaging & Kinematic Maps

For the majority of this thesis, our deep learning models have analysed single-band, monochromatic images of galaxies. However, it is important to keep in mind that the inputs to a convolutional neural network are arbitrary; there is nothing to preclude using additional input channels (as we have already seen in Section 6.2 with our use of three-channel, RGB colour inputs). One could, provided there is sufficient data, train a CNN to classify multi-band images, such as by using any arbitrary combination of u , g , r , i and z SDSS band filters (or indeed all five together). Our choice of using single-band imaging for our core deep learning models was largely motivated by practicality, making the models much easier to apply to simulations (where one can create mock images in the relevant band), as well as observational datasets where colour imaging is not readily available (e.g. COSMOS). Indeed, we found that the accuracy of our models for multi-class classification on single-band imaging is comparable to previous studies that used deep learning to classify colour imaging (c.f. [Barchi et al. 2020](#)). Interestingly, the study by [Bottrell et al. \(2019\)](#) found only small gains in accuracy when transitioning from single-band imaging to multi-band imaging, arguing that colour is not strictly necessary to maintain a high classification accuracy. That said, there is still a wealth of morphological information contained in colour imaging (or multi-band imaging more generally) that can potentially be leveraged for future deep learning models. This is especially useful for better classifying irregulars, which are predominantly blue, as well as better distinguishing stellar bars, which are more easily seen in infrared imaging. Indeed, retraining my existing suite of models on multi-band imaging is a focus for future work. However, such use of multi-band imaging comes with the caveats of more limited applicability, and higher computational requirements (especially memory usage) for model development and training.

Another potential extension is to train a deep learning model to classify images of a galaxy’s kinematic maps. For example, a model could be trained to classify galaxies using their velocity maps and/or dispersion maps, possibly even in tandem with single-band and/or multi-band imaging. Such a model could be used to determine whether it is possible to differentiate galaxies into different morphological types by purely analysing their kinematic information (in a similar manner to how our models have differentiated galaxies into different morphological types simply by analysing their images, as shown in Figure 2.5). Such a model could even combine this kinematic information with the actual images of galaxies into a single unified input (e.g. one channel for a g -band image, the other for a velocity map) to see whether this could lead to higher classification accuracies. A deep learning model trained to classify morphologies based on kinematic maps would be an especially useful tool for studying lenticular galaxies, particularly with regards to their kinematic differences with ellipticals, as well as for studying the differences between barred and unbarred galaxies. Such a model could be trained on the kinematic maps from integral field spectroscopy (IFS) surveys such as SAMI ([Croom et al., 2021a](#)) or MaNGA ([Bundy et al., 2015](#)), or even mock IFS data from high-resolution cosmological simulations (see [Harborne et al. 2020](#)). This would open up many opportunities for studying the large-scale kinematic properties of different galaxy morphologies.

7.2.3 The Nature of Barred Galaxies

Our U-Net developed in Chapter 6 has immense potential to reveal critical insights into the nature of bars, as we have seen through our preliminary analysis of bar lengths in Section 6.2.5. However, performing

morphological decomposition can enable us to estimate the mass fraction of bars, i.e. the mass of the bar with respect to its host galaxy. This quantity is difficult to measure, requiring detailed stellar mass maps (Zibetti et al., 2009) and some technique to outline the extent of the bar. In the latter case, our U-Net automatically does this for us by directly outputting a pixel mask, which can later be overlaid atop a stellar mass map to estimate the mass fraction of the stellar bar. While studies have examined the sizes of bars (Díaz-García et al., 2016; Erwin, 2019), the mass fractions of bars remain relatively unexplored.

U-Nets could also be used to perform morphological decompositions for barred galaxies in simulations. This could enable more detailed studies into the shapes of bars, such as their ellipticities and even triaxialities, as well as their kinematic properties. A future application could explore how these bar properties vary with respect to the properties of the host galaxies (including morphology), over the lifetime of the bar (which we have shown in Section 4.4.2 to be possible to quantify based on our deep learning bar model classifications), and indeed across different redshift ranges throughout the duration of the simulation. Although we made the argument in Chapter 4 that our bar model is better suited to large-scale cosmological simulations, where it is possible to study the statistics of a large population of barred galaxies, a U-Net opens up the opportunity to extract masks of bars in high-resolution zoom-in simulations (where the internal dynamics of galaxies are resolved at much finer scales). U-Nets, and deep learning applications in general, clearly have much potential for improving our understanding of the nature of bars and the role they play in governing the evolution of their host galaxies.

7.2.4 The Fundamental Structure of Galaxy Morphology

As we explored in Chapter 2, representation learning is a key, fundamental aspect of deep learning. Indeed, visualising the latent spaces of convolutional neural networks allows us to see how they structure and organise morphology. Section 2.5.2 demonstrated that it is possible to use deep learning models in order to study the underlying features of a given dataset, whether in a supervised manner with known classes as in Figure 2.5, or in an unsupervised manner without any classes. Such studies can be useful for analysing the fundamental structure of galaxy morphology (Cheng et al., 2021a). Our existing suite of deep learning models have been applied to datasets containing images of observed and simulated galaxies, and thus they are in an excellent position to conduct analyses of the underlying features of galaxy morphology across this wide range of datasets. It would be of great interest to compare the distribution of SAMI T-Types and NA10 T-Types on the latent manifolds of galaxy morphology as determined by CNNs, as this may reveal a more fundamental definition of morphology in terms of its principal component axes.

7.2.5 Architectural Refinements & The Future of Deep Learning in Astronomy

It is anticipated that our suite of deep learning models will undergo continued refinement and extensions, potentially resulting in highly specialised models tailored for specific domains, such as our noise-augmented models developed in Chapter 5 to classify COSMOS imaging, or the development of new and/or refined model architectures. Model development and training is a largely technical endeavour, and the limiting factors for investing in such improvements ultimately come down to time and computational resources. This is unavoidable given the experimental nature of deep learning and the large amount of computational time needed to optimise models with hyperparameter tuning. However, it is important to

note that the actual training of the final model itself consumes much less time than that typically spent on tuning. Of course, our existing suite of models can also be readily adapted with transfer learning, and we have shown that they are capable of being applied to a variety of datasets to classify images of both real galaxies in addition to simulated galaxies.

With the advent of large-scale surveys such as the JWST, LSST and SKA, there is immense potential for a deep learning approach for morphological classification (Ivezić et al., 2019; Koribalski et al., 2020; Ferreira et al., 2022a; Kartaltepe et al., 2022). Our models are well positioned to tackle the challenges of morphological classification and, as we have demonstrated in Chapter 6, morphological segmentation. Furthermore, although our suite of deep learning models have currently classified images of galaxies in the optical regime, the same core methods can also be adopted to train models capable of classifying images of galaxies in the radio regime (see Wu et al. 2019).

Finally, it is worth noting that the field of deep learning itself is rapidly evolving, with new models, methods and design practices emerging on a near daily basis. Deep learning also encompasses a vast range of techniques targeting specific tasks and problems, including classification, regression, clustering and segmentation. These capabilities are reflected in the broad range of applications of deep learning to astronomy (Baron, 2019; Fluke & Jacobs, 2020), and it is expected that these applications will only grow into the future (Huertas-Company & Lanusse, 2023). While convolutional neural networks remain the dominant deep learning model for image classification, they are certainly not the only model that has shown great success. In particular, recent studies have applied transformers, a type of model originally developed for natural language processing, to instead classify images. Such models are called vision transformers (Dosovitskiy et al., 2021), and they have quickly emerged as a strong alternative to convolutional neural networks. Other recently-developed deep learning models include diffusion models (more specifically, denoising diffusion probabilistic models; Ho et al. 2020), a type of latent variable model that has seen tremendous success at image generation, image upscaling and image denoising, the latter two being particularly useful for observational surveys. It is anticipated that these emerging state-of-the-art deep learning methods, along with well established methods such as CNNs and U-Nets, will continue see more and more applications in astronomy, ultimately enhancing our understanding of the cosmos.

Model Availability

Our core deep learning models are publicly available on GitHub in the following repositories:

- <https://github.com/Mythcell/c2>

This repository contains the 3-class, 4-class and bar models based on the C2 architecture, which were developed in Chapter 3 and subsequently used in Chapters 4 and 6.

- <https://github.com/Mythcell/c3>

This repository contains the 3-class and 4-class models based on the C3 architecture, which were developed and applied in Chapter 5.

Important note: The C2 and C3 models were originally trained using different versions of TensorFlow, Python and CUDA, summarised below:

C2: TensorFlow 2.2, Python 3.6, CUDA 10.2

C3: TensorFlow 2.7, Python 3.9, CUDA 11.3

All of the models have been verified to work with the latest version of TensorFlow (version 2.11 at the time of writing), and should remain compatible with future versions of TensorFlow 2 and its associated Python and CUDA requirements. Note that C3 requires at least TensorFlow 2.3 due to its usage of the built-in RandomRotation and RandomFlip layers, which are not present in version 2.2.

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